

10th MATHEMATICS

L^AT_EX Material 2026-2027

Name : _____

Class : _____

School Name : _____

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(L^AT_EX, HTML, MD, JS, CSS, Python, PHP, Macro,
Word, Excel, PowerPoint, MySQL Programmer)

PREFACE

(Author's Note)

(Numbers and letters, they are known as the two eyes of living beings.)

*"An equation for me has no meaning,
unless it expresses a thought of God."*

- Srinivasa Ramanujan

My Dear Beloved Students,

Welcome to the world of mathematics! It gives me immense joy and pride to place this **L^AT_EX Material** in your hands. Mathematics is not just a subject of numbers and formulas; it is a beautiful pathway that teaches us logical thinking, problem-solving, and perseverance.

I have crafted this guide specifically for 10th Standard Mathematics students with a singular, high-aiming goal: to make your learning journey smooth, understandable, and highly rewarding in your board examinations. Inside this material, complex theorems, intricate equations, and geometry proofs have been broken down into simple, step-by-step formats with neatly aligned tables and clear diagrams. It is designed to take the fear out of math and replace it with confidence.

To you, the students, I want to say this: Do not fear mistakes. Every problem you struggle with and eventually solve makes your mind sharper. With consistent practice and a clear understanding of the concepts, scoring top marks is absolutely within your reach.

May this material serve as a strong foundation for your success. I pray for your bright future and grand victories!

Best Wishes for a Successful Year,

**Y. Seenivasan, M.Sc., B.Ed.
Mathematics Teacher - 8489880553**

Guidelines for Success: Notes to Students...

- **Understand, Don't Memorize:** Always try to understand the core concept behind a formula or theorem before applying it. Treat math like a puzzle to be solved.
- **Practice Makes Perfect:** Mathematics requires the use of pen and paper. Practice writing out theorems (like Thales, Pythagoras, and ABT) step-by-step multiple times.
- **Master the Basics:** Give special attention to 1-mark Multiple Choice Questions (MCQs) and 2-mark problems. A strong grip on these will boost your total score immensely.
- **Step Marks Matter:** In 5-mark questions, write down the given data, the formula, and follow logical steps. Clear steps earn valuable marks even if the final answer is slightly off.
- **Stay Confident:** Believe in your hard work. Deeply focus on your daily practice, and success will surely follow!

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NOTES AND ROUGH WORK

CHAPTER - 1. RELATIONS AND FUNCTIONS

PUBLIC EXAMINATION IMPORTANT QUESTIONS-2026 2 MARK IMPORTANT QUESTIONS

1. If $A = \{1, 3, 5\}$ and $B = \{2, 3\}$ then (i) find $A \times B$ and $B \times A$.

JUL-21

solution

$$\begin{aligned} A \times B &= \{1, 3, 5\} \times \{2, 3\} \\ &= \{(1, 2), (1, 3), (3, 2), (3, 3), (5, 2), (5, 3)\} \quad \dots (1) \\ B \times A &= \{2, 3\} \times \{1, 3, 5\} \\ &= \{(2, 1), (2, 3), (2, 5), (3, 1), (3, 3), (3, 5)\} \quad \dots (2) \end{aligned}$$

2. If $A \times B = \{(3, 2), (3, 4), (5, 2), (5, 4)\}$ then find A and B .

APR-23, PTA-4

solution

$$\begin{aligned} A &= \{\text{Set of all First co-ordinates of elements of } A \times B\}. \\ A &= \{3, 5\} \\ B &= \{\text{Set of all Second co-ordinates of elements of } A \times B\}. \\ B &= \{2, 4\} \\ A &= \{3, 5\} \text{ and } B = \{2, 4\}. \end{aligned}$$

3. Find $A \times B$, $A \times A$ and $B \times A$

- (i) $A = \{2, -2, 3\}$ and $B = \{1, -4\}$ **JUL-25** (ii) $A = \{m, n\}$; $B = \phi$

PTA-1

solution

$$\begin{aligned} \text{(i)} \quad A &= \{2, -2, 3\} \text{ and } B = \{1, -4\} \\ A \times B &= \{2, -2, 3\} \times \{1, -4\} \\ &= \{(2, 1), (2, -4), (-2, 1), (-2, -4), (3, 1), (3, -4)\} \\ A \times A &= \{2, -2, 3\} \times \{2, -2, 3\} \\ &= \{(2, 2), (2, -2), (2, 3), (-2, 2), (-2, -2), (-2, 3), \\ &\quad (3, 2), (3, -2), (3, 3)\} \\ B \times A &= \{1, -4\} \times \{2, -2, 3\} \\ &= \{(1, 2), (1, -2), (1, 3), (-4, 2), (-4, -2), (-4, 3)\} \\ \text{(ii)} \quad A &= \{m, n\}, B = \phi \\ A \text{ or } B &\text{ are null sets } A \times B \\ A &= \phi \text{ (or) } B = \phi \\ A \times B &= \phi, B \times A = \phi \\ A \times A &= \{m, n\} \times \{m, n\} \\ &= \{(m, m), (m, n), (n, m), (n, n)\} \end{aligned}$$

4. Let $A = \{1, 2, 3\}$ and $B = \{x \mid x \text{ is a prime number less than } 10\}$. Find $A \times B$ and $B \times A$. **APR-22, 23**

solution

$$\begin{aligned} A &= \{1, 2, 3\}; \quad B = \{2, 3, 5, 7\} \\ A \times B &= \{(1, 2), (1, 3), (1, 5), (1, 7), (2, 2), (2, 3), \\ &\quad (2, 5), (2, 7), (3, 2), (3, 3), (3, 5), (3, 7)\} \\ B \times A &= \{(2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), \\ &\quad (5, 1), (5, 2), (5, 3), (7, 1), (7, 2), (7, 3)\} \end{aligned}$$

5. If $B \times A = \{(-2, 3), (-2, 4), (0, 3), (0, 4), (3, 3), (3, 4)\}$ find A and B .

JUL-24, APR-23

solution

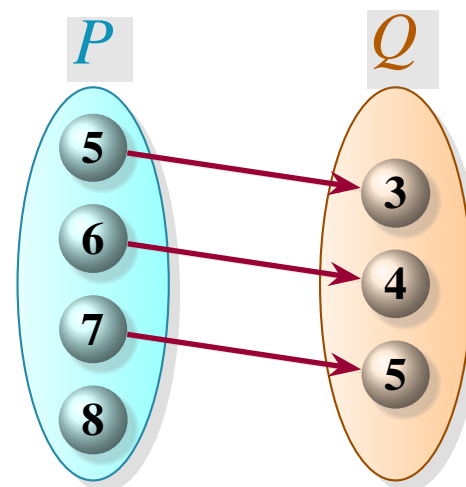
$$\begin{aligned} B &= \{\text{Set of all First co-ordinates of elements of } B \times A\}. \\ B &= \{-2, 0, 3\} \\ A &= \{\text{Set of all Second co-ordinates of elements of } B \times A\}. \\ A &= \{3, 4\} \\ A &= \{3, 4\} \text{ and } B = \{-2, 0, 3\}. \end{aligned}$$

6. The arrow diagram shows (Fig.1.10) a relationship between the sets P and Q . Write the relation in (i) Set builder form

(ii) Roster form

(iii) What is the domain and range of R . **APR-22, JUL-25**

solution



(i) Set builder form:

$$\mathbb{R} = \{(x, y) \mid y = x - 2, x \in P, y \in Q\}$$

(ii) Roster form $\mathbb{R} = \{(5, 3), (6, 4), (7, 5)\}$

(iii) Domain of $\mathbb{R} = \{5, 6, 7\}$;

Range of $\mathbb{R} = \{3, 4, 5\}$

7. Let $A = \{1, 2, 3, 4, \dots, 45\}$ and R be the relation defined as "square is of a number" on A . Write R as a subset of $A \times A$. Also, find the domain and range of R . **JUL-21**

solution

$$A = \{1, 2, 3, 4, \dots, 45\}$$

Relation is "is square of" and $A \rightarrow A$ on A

$$A \times A = \{(1, 1), (1, 2), (1, 3), (1, 4) \dots (45, 45)\}$$

The square of '1' is $1 \in A$ and $(1, 1) \in A \times A$

The square of 2 is $4 \in A$ and $(4, 2) \in A \times A$

The square of 3 is $9 \in A$ and $(9, 3) \in A \times A$

The square of 4 is $16 \in A$ and $(16, 4) \in A \times A$

The square of 5 is $25 \in A$ and $(25, 5) \in A \times A$

The square of 6 is $36 \in A$ and $(36, 6) \in A \times A$

The square of 7 is $49 \notin A$.

$$R = \{(1, 1), (4, 2), (9, 3), (16, 4), (25, 5), (36, 6)\}$$

$$\text{Domain of } R = \{1, 4, 9, 16, 25, 36\}$$

$$\text{Range of } R = \{1, 2, 3, 4, 5, 6\}$$

8. A Relation R is given by the set $\{(x, y)/y = x + 3, x \in \{0, 1, 2, 3, 4, 5\}\}$. Determine its domain and range. **JUL-23, PTA-5, MAR-26**

solution

$$\{(x, y)/y = x + 3, x \in \{0, 1, 2, 3, 4, 5\}\}.$$

$$\text{When } x = 0, y = 0 + 3 = 3$$

$$\text{When } x = 1, y = 1 + 3 = 4$$

$$\text{When } x = 2, y = 2 + 3 = 5$$

$$\text{When } x = 3, y = 3 + 3 = 6$$

$$\text{When } x = 4, y = 4 + 3 = 7$$

$$\text{When } x = 5, y = 5 + 3 = 8$$

$$\text{Relation } R = \{(0, 3), (1, 4), (2, 5), (3, 6), \\ (4, 7), (5, 8)\}$$

$$\text{Domain} = \{0, 1, 2, 3, 4, 5\}$$

$$\text{Range} = \{3, 4, 5, 6, 7, 8\}$$

9. Represent each of the given relations by (a) an arrow diagram, (b) a graph and (c) a set in roster form, wherever possible. **JUL-22**

(i) $\{(x, y) \mid x = 2y, x \in \{2, 3, 4, 5\}, y \in \{1, 2, 3, 4\}\}$

(ii) $\{(x, y) \mid y = x + 3, x, y \text{ are natural numbers } < 10\}$

solution

(i) Set-Builder form:

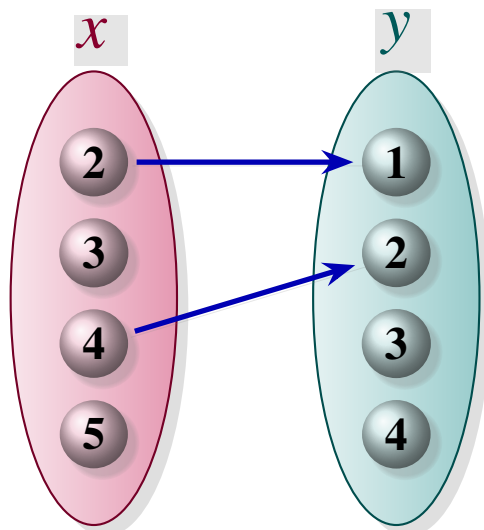
$$\{(x, y)/x = 2y, x \in \{2, 3, 4, 5\}, y \in \{1, 2, 3, 4\}\}$$

$$x = 2y$$

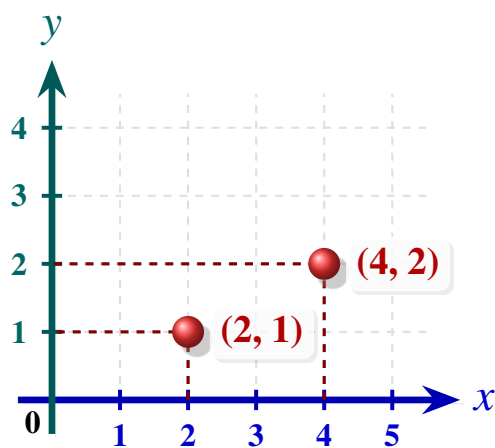
$$\text{When } y = 1, x = 2(1) = 2 \in x$$

When $y = 2, x = 2(2) = 4 \in x$
 When $y = 3, x = 2(3) = 6 \notin x$
 When $y = 4, x = 2(4) = 8 \notin x$
 Relation $R = \{(2, 1), (4, 2)\}$

(a) Arrow diagram



(b) Graph

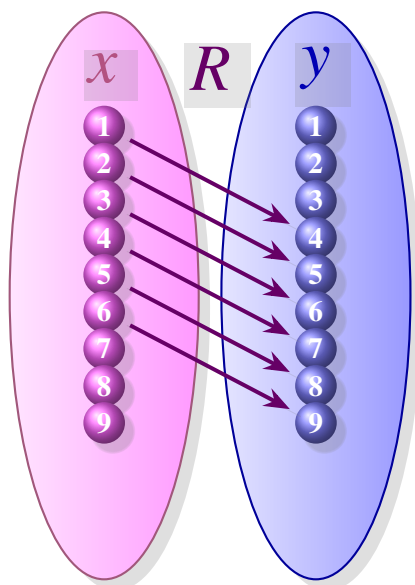
(c) Roster form $R = \{(2, 1), (4, 2)\}$

(ii) $\{(x, y) \mid y = x + 3, x, y \text{ are natural numbers} < 10\}$

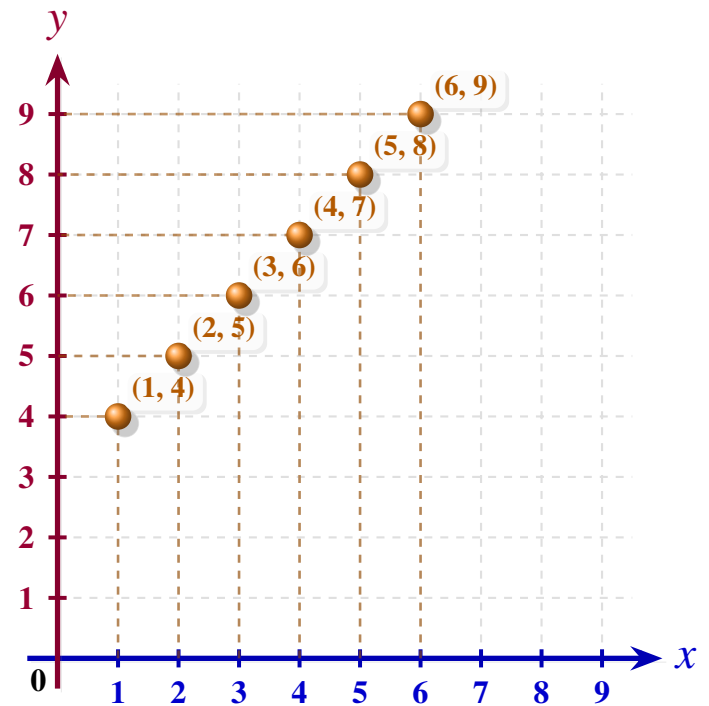
When $x = 1, y = 1 + 3 = 4$
 When $x = 2, y = 2 + 3 = 5$
 When $x = 3, y = 3 + 3 = 6$
 When $x = 4, y = 4 + 3 = 7$
 When $x = 5, y = 5 + 3 = 8$
 When $x = 6, y = 6 + 3 = 9$
 When $x = 7, y = 7 + 3 = 10$ is not possible
 x and y are less than 10.

$\therefore$ Relation $R = \{(1, 4), (2, 5), (3, 6), (4, 7), (5, 8), (6, 9)\}$

(a) Arrow diagram



(b) Graph



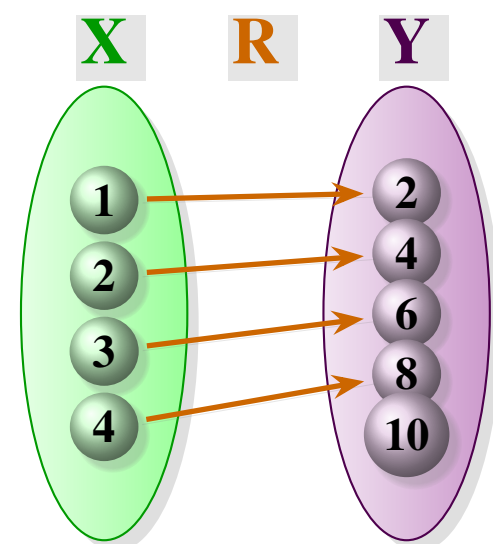
(c) Roster Form

$R = \{(1, 4), (2, 5), (3, 6), (4, 7), (5, 8), (6, 9)\}$

10. Let $X = \{1, 2, 3, 4\}$ and $Y = \{2, 4, 6, 8, 10\}$ and $R = \{(1, 2), (2, 4), (3, 6), (4, 8)\}$. Show that R is a function and find its domain, co-domain and range?

solution

All elements in X have only one image in Y . R is a function.



Domain $X = \{1, 2, 3, 4\}$;
 Co-domain $Y = \{2, 4, 6, 8, 10\}$;
 Range of $f = \{2, 4, 6, 8\}$.

11. A relation $f : X \rightarrow Y$ is defined by $f(x) = x^2 - 2$ where, $X = \{-2, -1, 0, 3\}$ and $Y = \mathbb{R}$.

(i) List the elements of f

(ii) Is f a function?

PTA-1

solution

(i) $f(-2) = (-2)^2 - 2 = 2$
 $f(-1) = (-1)^2 - 2 = -1$
 $f(0) = (0)^2 - 2 = -2$
 $f(3) = (3)^2 - 2 = 7$
 $f = \{(-2, 2), (-1, -1), (0, -2), (3, 7)\}$

(ii) Each element in the domain of f has a unique image. f is a function.

12. Given $f(x) = 2x - x^2$, find (i) $f(1)$ (ii) $f(x+1)$ (iii) $f(x) + f(1)$

JUL-24

solution

(i) x Put in 1
 $f(1) = 2(1) - (1)^2 = 2 - 1 = 1$
 (ii) Put x in $x + 1$,
 $f(x+1) = 2(x+1) - (x+1)^2$

$$= 2x + 2 - (x^2 + 2x + 1)$$

$$= -x^2 + 1$$

- (iii) $f(x) + f(1) = (2x - x^2) + 1 = -x^2 + 2x + 1$
 $[f(x) + f(1) \neq f(x + 1)]$
 general $f(a + b)$ is not equal to $f(a) + f(b)$

13. Given the function $f : x \rightarrow x^2 - 5x + 6$, evaluate
 (i) $f(-1)$ (ii) $f(2a)$ (iii) $f(2)$ (iv) $f(x - 1)$

solution

- (i) Put x in 1,
 $f(1) = 2(1) - (1)^2 = 2 - 1 = 1$
 (ii) Put x in $x + 1$,
 $f(x + 1) = 2(x + 1) - (x + 1)^2$
 $= 2x + 2 - (x^2 + 2x + 1)$
 $= -x^2 + 1$
 (iii) $f(x) + f(1) = (2x - x^2) + 1 = -x^2 + 2x + 1$
 $f(x) + f(1) \neq f(x + 1)$.
 general $f(a + b)$ is not equal to $f(a) + f(b)$

14. Let $f(x) = 2x + 5$. If $x \neq 0$ then find $\frac{f(x+2) - f(2)}{x}$.

solution

$$f(x) = 2x + 5, \quad x \neq 0$$

$$\frac{f(x+2) - f(2)}{x} = \frac{[2(x+2) + 5] - [2(2) + 5]}{x}$$

$$= \frac{2x + 4 + 5 - 9}{x} = \frac{2x + 9 - 9}{x}$$

$$= \frac{2x}{x} = 2$$

15. A function f is defined by $f(x) = 3 - 2x$. Find x such that $f(x^2) = (f(x))^2$.

solution

$$f(x) = 3 - 2x$$

$$f(x^2) = [f(x)]^2$$

$$3 - 2x^2 = (3 - 2x)^2$$

$$3 - 2x^2 = 9 - 12x + 4x^2$$

$$6x^2 - 12x + 6 = 0$$

$$6(x^2 - 2x + 1) = 0$$

$$(x - 1)^2 = 0 \implies x = 1.$$

16. A plane is flying at a speed of 500 km per hour. Express the distance 'd' travelled by the plane as function of time t in hours.

solution

Let the distance be 'd'

$$\text{Speed} = 500 \text{ km/hr,}$$

$$\text{Time} = 't' \text{ hours}$$

$$\text{Distance} = \text{Speed} \times \text{time}$$

$$d(t) = 500 \times t$$

17. If $A = \{-2, -1, 0, 1, 2\}$ and $f : A \rightarrow B$ is an onto function defined by $f(x) = x^2 + x + 1$ then find B .

solution

$$A = \{-2, -1, 0, 1, 2\}$$

$$f(x) = x^2 + x + 1.$$

$$f(-2) = (-2)^2 + (-2) + 1 = 1$$

$$f(-1) = (-1)^2 + (-1) + 1 = 1$$

$$f(0) = 0^2 + 0 + 1 = 1$$

$$f(1) = 1^2 + 1 + 1 = 3$$

$$f(2) = 2^2 + 2 + 1 = 7$$

f is an onto function, range of Co-domain. $f = B$

$$B = \{1, 3, 7\}.$$

18. Let f be a function from $\mathbb{R}$ to $\mathbb{R}$ defined by $f(x) = 3x - 5$. Find the values of a and b given that $(a, 4)$ and $(1, b)$ belong to f . **PTA-6**

solution

$$f(x) = 3x - 5$$

$$f = \{(x, 3x - 5) \mid x \in \mathbb{R}\}$$

$(a, 4)$ means the image of a is 4.

$$f(a) = 4$$

$$3a - 5 = 4$$

$$\implies \boxed{a = 3}$$

$(1, b)$ means the image of 1 is b .

$$f(1) = b$$

$$3(1) - 5 = b \implies \boxed{b = -2}$$

19. Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(m) = m^2 + m + 3$ is one-one function.

solution

$$f : \mathbb{N} \rightarrow \mathbb{N}$$

$$f(m) = m^2 + m + 3$$

when $m = 1$, $f(1) = (1)^2 + 1 + 3 = 5$
 when $m = 2$, $f(2) = (2)^2 + 2 + 3 = 9$
 when $m = 3$, $f(3) = (3)^2 + 3 + 3 = 15$ and so on.

A function for which every element of the range of the function exactly one element of the domain.

That is one-to-one function.

20. Let $A = \{1, 2, 3, 4\}$ and $B = \mathbb{N}$. Let $f : A \rightarrow B$ be defined by $f(x) = x^3$ then,

- (i) find the range of f (ii) identify the type of function

solution

$$A = \{1, 2, 3, 4\}, B = \mathbb{N}$$

(i) $f : A \rightarrow B$

$$f(x) = x^3$$

$$f(1) = (1)^3 = 1 \in \mathbb{N}$$

$$f(2) = (2)^3 = 8 \in \mathbb{N}$$

$$f(3) = (3)^3 = 27 \in \mathbb{N}$$

$$f(4) = (4)^3 = 64 \in \mathbb{N}$$

Range of $f = \{1, 8, 27, 64\}$

- (ii) f is a function from A to B all elements having different images in B .
 $\therefore f : A \rightarrow B$ is a function. It is one-to-one and into function and also called a cubic function.

21. Find $f \circ g$ and $g \circ f$ when $f(x) = 2x + 1$

APR-25

solution

$$f(x) = 2x + 1, \quad g(x) = x^2 - 2$$

$$(f \circ g)(x) = f(g(x)) = f(x^2 - 2)$$

$$= 2(x^2 - 2) + 1 = 2x^2 - 3$$

$$(g \circ f)(x) = g(f(x)) = g(2x + 1)$$

$$= (2x + 1)^2 - 2 = 4x^2 + 4x - 1.$$

$$f \circ g = 2x^2 - 3, \quad g \circ f = 4x^2 + 4x - 1.$$

$$f \circ g \neq g \circ f.$$

22. If $f(x) = 3x - 2$, $g(x) = 2x + k$ and if $f \circ g = g \circ f$, then find the value of k .

solution

$$f(x) = 3x - 2, \quad g(x) = 2x + k$$

$$(f \circ g)(x) = f(g(x)) = f(2x + k)$$

$$= 3(2x + k) - 2 = 6x + 3k - 2$$

$$\begin{aligned}
 (f \circ g)(x) &= 6x + 3k - 2. \\
 (g \circ f)(x) &= g(3x - 2) = 2(3x - 2) + k \\
 (g \circ f)(x) &= 6x - 4 + k. \\
 f \circ g &= g \circ f \\
 6x + 3k - 2 &= 6x - 4 + k \\
 6x - 6x + 3k - k &= -4 + 2 \\
 2k &= -2 \implies \boxed{k = -1}
 \end{aligned}$$

23. Find k if $f \circ f(k) = 5$ where $f(k) = 2k - 1$.

APR-23, PTA-4

solution

$$\begin{aligned}
 f \circ f(k) &= f(f(k)) = f(2k - 1) \\
 &= 2(2k - 1) - 1 = 4k - 3 \\
 f \circ f(k) &= 4k - 3 \\
 f \circ f(k) &= 5 \\
 4k - 3 &= 5 \implies \boxed{k = 2}.
 \end{aligned}$$

24. Using the functions f and g given below, find $f \circ g$ and $g \circ f$. Check whether $f \circ g = g \circ f$.

(i) $f(x) = x - 6, g(x) = x^2$ **JUL-23**

(ii) $f(x) = \frac{2}{x}, g(x) = 2x^2 - 1$

(iii) $f(x) = \frac{x+6}{3}, g(x) = 3 - x$

(iv) $f(x) = 3 + x, g(x) = x - 4$

(v) $f(x) = 4x^2 - 1, g(x) = 1 + x$

MDL-19

solution

(i) $f(x) = x - 6, g(x) = x^2$

$$\begin{aligned}
 f \circ g &= f[g(x)] = f[x^2] = x^2 - 6 \\
 g \circ f &= g[f(x)] = g[x - 6] = (x - 6)^2 \\
 &= x^2 - 12x + 36 \\
 \therefore f \circ g &\neq g \circ f
 \end{aligned}$$

(ii) $f(x) = \frac{2}{x}, g(x) = 2x^2 - 1$

$$\begin{aligned}
 f \circ g &= f[g(x)] = f[2x^2 - 1] = \frac{2}{2x^2 - 1} \\
 g \circ f &= g[f(x)] = g\left[\frac{2}{x}\right] = 2\left(\frac{2}{x}\right)^2 - 1 \\
 &= 2\left(\frac{4}{x^2}\right) - 1 = \frac{8}{x^2} - 1 = \frac{8 - x^2}{x^2} \\
 \therefore f \circ g &\neq g \circ f
 \end{aligned}$$

(iii) $f(x) = \frac{x+6}{3}, g(x) = 3 - x$

$$\begin{aligned}
 f \circ g &= f[g(x)] = f(3 - x) = \frac{3 - x + 6}{3} = \frac{9 - x}{3} \\
 g \circ f &= g[f(x)] = g\left(\frac{x+6}{3}\right) = 3 - \left(\frac{x+6}{3}\right) \\
 &= \frac{9 - x - 6}{3} = \frac{3 - x}{3} \quad \therefore f \circ g \neq g \circ f
 \end{aligned}$$

(iv) $f(x) = 3 + x, g(x) = x - 4$

$$\begin{aligned}
 f \circ g &= f[g(x)] = f[x - 4] = 3 + x - 4 = x - 1 \\
 g \circ f &= g[f(x)] = g[3 + x] = 3 + x - 4 = x - 1 \\
 \therefore f \circ g &= g \circ f
 \end{aligned}$$

(v) $f(x) = 4x^2 - 1, g(x) = 1 + x$

$$\begin{aligned}
 f \circ g &= f[g(x)] = f(1 + x) = 4(1 + x)^2 - 1 \\
 &= 4(1 + 2x + x^2) - 1 \\
 &= 4 + 8x + 4x^2 - 1 \\
 &= 4x^2 + 8x + 3 \\
 g \circ f &= g[f(x)] = 1 + 4x^2 - 1 \\
 &= 4x^2 \\
 \therefore f \circ g &\neq g \circ f
 \end{aligned}$$

25. Find the value of k , such that $f \circ g = g \circ f$

(i) $f(x) = 3x + 2, g(x) = 6x - k$ (ii) $f(x) = 2x - k, g(x) = 4x + 5$

solution

(i) $f(x) = 3x + 2, g(x) = 6x - k$

$$\begin{aligned}
 f \circ g &= f[g(x)] = f(6x - k) \\
 &= 3(6x - k) + 2 \\
 &= 18x - 3k + 2 \\
 g \circ f &= g[f(x)] = 6(3x + 2) - k \\
 &= 18x + 12 - k \\
 f \circ g &= g \circ f \\
 \therefore 18x - 3k + 2 &= 18x + 12 - k \\
 3k - k &= -12 + 2 \\
 2k &= -10 \\
 \boxed{k} &= -5
 \end{aligned}$$

(ii) $f(x) = 2x - k, g(x) = 4x + 5$

$$\begin{aligned}
 f \circ g &= f[g(x)] = f[4x + 5] \\
 &= 2(4x + 5) - k \\
 &= 8x + 10 - k \\
 g \circ f &= g[f(x)] = g(2x - k) = 4(2x - k) + 5 \\
 &= 8x - 4k + 5 \\
 f \circ g &= g \circ f \\
 \therefore 8x + 10 - k &= 8x - 4k + 5 \\
 4k - k &= 5 - 10 \\
 3k &= -5 \\
 \boxed{k} &= \frac{-5}{3}
 \end{aligned}$$

26. If $f(x) = 2x - 1, g(x) = \frac{x+1}{2}$, show that $f \circ g = g \circ f = x$

solution

$$\begin{aligned}
 f(x) &= 2x - 1, g(x) = \frac{x+1}{2} \\
 f \circ g &= f[g(x)] = f\left(\frac{x+1}{2}\right) \\
 &= 2\left(\frac{x+1}{2}\right) - 1 = x + 1 - 1 = x \\
 g \circ f &= g[f(x)] = g(2x - 1) = \frac{2x - 1 + 1}{2} \\
 &= \frac{2x}{2} = x \\
 \therefore f \circ g &= g \circ f = x \quad \text{Hence proved.}
 \end{aligned}$$

27. If $f(x) = x^2 - 1, g(x) = x - 2$ find a , if $g \circ f(a) = 1$.

PTA-2

solution

$$\begin{aligned}
 f(x) &= x^2 - 1, g(x) = x - 2 \\
 f(a) &= a^2 - 1 \\
 (g \circ f)(a) &= 1 \\
 g[f(a)] &= 1 \\
 g[a^2 - 1] &= 1 \\
 a^2 - 1 - 2 &= 1 \\
 a^2 - 3 &= 1 \\
 a^2 &= 1 + 3 = 4 \\
 \boxed{a} &= \pm 2
 \end{aligned}$$

28. Let $f(x) = x^2 - 1$. Find (i) $f \circ f$ (ii) $f \circ f \circ f$

solution

(i) $f \circ f = f[f(x)] = f(x^2 - 1)$

$$\begin{aligned}
 &= (x^2 - 1)^2 - 1 = x^4 - 2x^2 + 1 - 1 = x^4 - 2x^2 \\
 \text{(ii) } f \circ f \circ f &= f[f[f(x)]] = f[f(x^2 - 1)] = f[x^4 - 2x^2] \\
 &= (x^4 - 2x^2)^2 - 1 = x^8 - 4x^6 + 4x^4 - 1
 \end{aligned}$$

29. If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x^5$ and $g(x) = x^4$ then check if f, g are one-one and $f \circ g$ is one-one? **PTA-6**

solution

$$\begin{aligned} f(x) &= x^5, g(x) = x^4 \\ f(x) &= x^5 \\ \therefore f(x) &\text{ is one-one function} \\ g(1) &= 1; \quad g(-1) = 1 \\ g(x) &\text{ is not one-one function} \\ f \circ g &= f[g(x)] = f(x^4) = (x^4)^5 = x^{20} \\ f \circ g &\text{ is also One - One function as } x \text{ is mapped} \\ &\text{ with different value of } f \circ g. \end{aligned}$$

30. Given that $f(x) = \begin{cases} \sqrt{x-1} & x \geq 1 \\ 4 & x < 1 \end{cases}$. Find
(i) $f(0)$ (ii) $f(3)$ (iii) $f(a+1)$ in terms of a . (Given that $a \geq 0$)

solution

$$\begin{aligned} f(x) &= \begin{cases} \sqrt{x-1} & x \geq 1 \\ 4 & x < 1 \end{cases} \\ \text{(i)} \quad f(0) &= 4 \quad [\because 0 < 1] \\ \text{(ii)} \quad f(3) &= \sqrt{3-1} = \sqrt{2} \quad [\because 3 \geq 1] \\ \text{(iii)} \quad f(a+1) &= \sqrt{a+1-1} = \sqrt{a} \quad [\because \text{Given } a \geq 0 \\ &\quad [\because a+1 \geq 1] \end{aligned}$$

31. Find the domain of the function $f(x) = \sqrt{1 + \sqrt{1 - \sqrt{1 - x^2}}}$

solution

$$\begin{aligned} f(x) &= \sqrt{1 - \sqrt{1 - \sqrt{1 - x^2}}} \\ f(x) &= \sqrt{1 - t} \\ &\text{where } t = \sqrt{1 - \sqrt{1 - x^2}} \\ 1 - t &\geq 0 \\ t &\leq 1 \\ \sqrt{1 - \sqrt{1 - x^2}} &\leq 1 \\ \text{Squaring } 1 - \sqrt{1 - x^2} &\leq 1 \\ -\sqrt{1 - x^2} &\leq 0 \\ \sqrt{1 - x^2} &\geq 0 \\ 1 - x^2 &\geq 0 \\ x^2 &\leq 1 \\ \Rightarrow x &\in [-1, 1] \text{ i.e., } \{-1, 0, 1\} \end{aligned}$$

32. If $f(x) = \frac{x-1}{x+1}, x \neq -1$ show that $f(f(x)) = -\frac{1}{x}$, provided $x \neq 0$

solution

$$\begin{aligned} f(x) &= \frac{x-1}{x+1}, x \neq -1 \\ f[f(x)] &= f\left[\frac{x-1}{x+1}\right] \\ &= \frac{\frac{x-1}{x+1} - 1}{\frac{x-1}{x+1} + 1} = \frac{\frac{x-1-(x+1)}{x+1}}{\frac{x-1+(x+1)}{x+1}} \\ &= \frac{x-1-x-1}{x-1+x+1} = \frac{-2}{2x} = -\frac{1}{x} \\ &\text{Hence proved.} \end{aligned}$$

CHAPTER - 1. RELATIONS AND FUNCTIONS

5 MARK IMPORTANT QUESTIONS

1. Let $A = \{x \in \mathbb{N} \mid 1 < x < 4\}$, $B = \{x \in \mathbb{W} \mid 0 \leq x < 2\}$ and $C = \{x \in \mathbb{N} \mid x < 3\}$.

Then verify that

- (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$
(ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

APR-24

solution

$$\begin{aligned} \text{(i)} \quad A &= \{x \in \mathbb{N} \mid 1 < x < 4\} = \{2, 3\} \\ B &= \{x \in \mathbb{W} \mid 0 \leq x < 2\} = \{0, 1\} \\ C &= \{x \in \mathbb{N} \mid x < 3\} = \{1, 2\} \\ A \times (B \cup C) &= (A \times B) \cup (A \times C) \\ B \cup C &= \{0, 1\} \cup \{1, 2\} \\ &= \{0, 1, 2\} \\ A \times (B \cup C) &= \{2, 3\} \times \{0, 1, 2\} \\ &= \{(2, 0), (2, 1), (2, 2), (3, 0), \\ &\quad (3, 1), (3, 2)\} \dots (1) \\ A \times B &= \{2, 3\} \times \{0, 1\} \\ &= \{(2, 0), (2, 1), (3, 0), (3, 1)\} \\ A \times C &= \{2, 3\} \times \{1, 2\} \\ &= \{(2, 1), (2, 2), (3, 1), (3, 2)\} \\ (A \times B) \cup (A \times C) &= \{(2, 0), (2, 1), (3, 0), (3, 1)\} \\ &\quad \cup \{(2, 1), (2, 2), (3, 1), (3, 2)\} \\ (A \times B) \cup (A \times C) &= \{(2, 0), (2, 1), (2, 2), (3, 0), \\ &\quad (3, 1), (3, 2)\} \dots (2) \end{aligned}$$

From (1) and (2),

$$A \times (B \cup C) = (A \times B) \cup (A \times C) \text{ is verified.}$$

$$\begin{aligned} \text{(ii)} \quad A \times (B \cap C) &= (A \times B) \cap (A \times C) \\ B \cap C &= \{0, 1\} \cap \{1, 2\} = \{1\} \\ A \times (B \cap C) &= \{2, 3\} \times \{1\} = \{(2, 1), (3, 1)\} \dots (1) \\ A \times B &= \{2, 3\} \times \{0, 1\} = \{(2, 0), \\ &\quad (2, 1), (3, 0), (3, 1)\} \\ A \times C &= \{2, 3\} \times \{1, 2\} \\ &= \{(2, 1), (2, 2), (3, 1), (3, 2)\} \\ (A \times B) \cap (A \times C) &= \{(2, 0), (2, 1), (3, 0), (3, 1)\} \cap \\ &\quad \{(2, 1), (2, 2), (3, 1), (3, 2)\} \\ (A \times B) \cap (A \times C) &= \{(2, 1), (3, 1)\} \dots (2) \end{aligned}$$

From (1) and (2),

$$A \times (B \cap C) = (A \times B) \cap (A \times C) \text{ is verified.}$$

2. Given $A = \{1, 2, 3\}$, $B = \{2, 3, 5\}$, $C = \{3, 4\}$ and $D = \{1, 3, 5\}$, check if $(A \cap C) \times (B \cap D) = (A \times B) \cap (C \times D)$ is true?

solution

$$\begin{aligned} \text{Given } A &= \{1, 2, 3\}, B = \{2, 3, 5\}, C = \{3, 4\}, D = \{1, 3, 5\} \\ A \cap C &= \{3\} \\ B \cap D &= \{3, 5\} \\ (A \cap C) \times (B \cap D) &= \{3\} \times \{3, 5\} \\ &= \{(3, 3), (3, 5)\} \dots (1) \\ A \times B &= \{(1, 2), (1, 3), (1, 5), (2, 2), \\ &\quad (2, 3), (2, 5), (3, 2), \\ &\quad (3, 3), (3, 5)\} \\ C \times D &= \{(3, 1), (3, 3), (3, 5), (4, 1), \\ &\quad (4, 3), (4, 5)\} \\ (A \times B) \cap (C \times D) &= \{(3, 3), (3, 5)\} \dots (2) \end{aligned}$$

From (1) and (2),

$$\therefore (A \cap C) \times (B \cap D) = (A \times B) \cap (C \times D)$$

Hence it is true.

3. Let $A = \{x \in \mathbb{W} \mid x < 2\}$, $B = \{x \in \mathbb{N} \mid 1 < x \leq 4\}$ and $C = \{3, 5\}$. Verify that

$$(i) A \times (B \cup C) = (A \times B) \cup (A \times C)$$

JUL-23, PTA-2

$$(ii) A \times (B \cap C) = (A \times B) \cap (A \times C)$$

JUL-21, PTA-5, MAR-26

$$(iii) (A \cup B) \times C = (A \times C) \cup (B \times C)$$

solution

$$(i) \text{ Given } A = \{x \in \mathbb{W} \mid x < 2\} \Rightarrow A = \{0, 1\}$$

$$B = \{x \in \mathbb{N} \mid 1 < x \leq 4\} \Rightarrow B = \{2, 3, 4\}$$

$$C = \{3, 5\}$$

$$A \times (B \cup C) = (A \times B) \cup (A \times C)$$

$$B \cup C = \{2, 3, 4, 5\}$$

$$A \times (B \cup C) = \{0, 1\} \times \{2, 3, 4, 5\}$$

$$= \{(0, 2), (0, 3), (0, 4), (0, 5), (1, 2), (1, 3), (1, 4), (1, 5)\} \dots (1)$$

$$A \times B = \{0, 1\} \times \{2, 3, 4\}$$

$$= \{(0, 2), (0, 3), (0, 4), (1, 2),$$

$$(1, 3), (1, 4)\}$$

$$A \times C = \{0, 1\} \times \{3, 5\}$$

$$= \{(0, 3), (0, 5), (1, 3), (1, 5)\}$$

$$(A \times B) \cup (A \times C) = \{(0, 2), (0, 3), (0, 4), (0, 5),$$

$$(1, 2), (1, 3), (1, 4), (1, 5)\} \dots (2)$$

From (1) and (2),

$$A \times (B \cup C) = (A \times B) \cup (A \times C)$$

Hence verified.

$$(ii) A \times (B \cap C) = (A \times B) \cap (A \times C)$$

$$A = \{0, 1\}, B = \{2, 3, 4\}, C = \{3, 5\}$$

$$B \cap C = \{3\}$$

$$A \times (B \cap C) = \{0, 1\} \times \{3\}$$

$$= \{(0, 3), (1, 3)\} \dots (1)$$

$$A \times B = \{(0, 2), (0, 3), (0, 4), (1, 2), (1, 3),$$

$$(1, 4)\}$$

$$A \times C = \{(0, 3), (0, 5), (1, 3), (1, 5)\}$$

$$(A \times B) \cap (A \times C) = \{(0, 3), (1, 3)\} \dots (2)$$

From (1) and (2)

$$A \times (B \cap C) = (A \times B) \cap (A \times C)$$

Hence verified.

$$(iii) (A \cup B) \times C = (A \times C) \cup (B \times C)$$

$$A = \{0, 1\}, B = \{2, 3, 4\}, C = \{3, 5\}$$

$$(A \cup B) = \{0, 1, 2, 3, 4\}$$

$$(A \cup B) \times C = \{0, 1, 2, 3, 4\} \times \{3, 5\}$$

$$= \{(0, 3), (0, 5), (1, 3), (1, 5), (2, 3), (2, 5), (3, 3),$$

$$(3, 5), (4, 3), (4, 5)\} \dots (1)$$

$$A \times C = \{0, 1\} \times \{3, 5\}$$

$$= \{(0, 3), (0, 5), (1, 3), (1, 5)\}$$

$$B \times C = \{2, 3, 4\} \times \{3, 5\}$$

$$= \{(2, 3), (2, 5), (3, 3), (3, 5),$$

$$(4, 3), (4, 5)\}$$

$$(A \times C) \cup (B \times C) = \{(0, 3), (0, 5), (1, 3), (1, 5), (2, 3),$$

$$(2, 5), (3, 3), (3, 5), (4, 3), (4, 5)\} \dots (2)$$

From (1) and (2),

$$(A \cup B) \times C = (A \times C) \cup (B \times C)$$

Hence verified.

4. Let A = The set of all natural numbers less than 8, B = The set of all prime numbers less than 8, C = The set of even prime number. Verify that

$$(i) (A \cap B) \times C = (A \times C) \cap (B \times C)$$

$$(ii) A \times (B - C) = (A \times B) - (A \times C).$$

APR-22, PTA-1**JUL-20, 24, APR-25****solution**

A is the set of all natural No less than 8

$$A = \{1, 2, 3, 4, 5, 6, 7\}$$

B set of all prime No less than 8

$$B = \{2, 3, 5, 7\}$$

C set of all even prime No

$$C = \{2\}$$

$$(i) (A \cap B) \times C = (A \times C) \cap (B \times C)$$

$$A \cap B = \{2, 3, 5, 7\}$$

$$(A \cap B) \times C = \{2, 3, 5, 7\} \times \{2\}$$

$$= \{(2, 2), (3, 2), (5, 2), (7, 2)\} \dots (1)$$

$$A \times C = \{1, 2, 3, 4, 5, 6, 7\} \times \{2\}$$

$$= \{(1, 2), (2, 2), (3, 2), (4, 2),$$

$$(5, 2), (6, 2), (7, 2)\}$$

$$B \times C = \{2, 3, 5, 7\} \times \{2\}$$

$$= \{(2, 2), (3, 2), (5, 2), (7, 2)\}$$

$$(A \times C) \cap (B \times C) = \{(2, 2), (3, 2), (5, 2), (7, 2)\} \dots (2)$$

From (1) and (2)

$$(A \cap B) \times C = (A \times C) \cap (B \times C)$$

Hence verified.

$$(ii) A \times (B - C) = (A \times B) - (A \times C)$$

$$B - C = \{2, 3, 5, 7\} - \{2\} = \{3, 5, 7\}$$

$$A \times (B - C) = \{1, 2, 3, 4, 5, 6, 7\} \times \{3, 5, 7\}$$

$$= \{(1, 3), (1, 5), (1, 7), (2, 3), (2, 5),$$

$$(2, 7), (3, 3), (3, 5), (3, 7), (4, 3),$$

$$(4, 5), (4, 7), (5, 3), (5, 5), (5, 7),$$

$$(6, 3), (6, 5), (6, 7), (7, 3), (7, 5),$$

$$(7, 7)\} \dots (1)$$

$$A \times B = \{1, 2, 3, 4, 5, 6, 7\} \times \{2, 3, 5, 7\}$$

$$= \{(1, 2), (1, 3), (1, 5), (1, 7), (2, 2),$$

$$(2, 3), (2, 5), (2, 7), (3, 2), (3, 3),$$

$$(3, 5), (3, 7), (4, 2), (4, 3), (4, 5),$$

$$(4, 7), (5, 2), (5, 3), (5, 5), (5, 7),$$

$$(6, 2), (6, 3), (6, 5), (6, 7), (7, 2),$$

$$(7, 3), (7, 5), (7, 7)\}$$

$$A \times C = \{1, 2, 3, 4, 5, 6, 7\} \times \{2\}$$

$$= \{(1, 2), (2, 2), (3, 2), (4, 2), (5, 2), (6, 2),$$

$$(7, 2)\}$$

$$(A \times B) - (A \times C) = \{(1, 3), (1, 5), (1, 7), (2, 3), (2, 5),$$

$$(2, 7), (3, 3), (3, 5), (3, 7), (4, 3),$$

$$(4, 5), (4, 7), (5, 3), (5, 5), (5, 7),$$

$$(6, 3), (6, 5), (6, 7), (7, 3), (7, 5),$$

$$(7, 7)\} \dots (2)$$

From (1) and (2),

$$A \times (B - C) = (A \times B) - (A \times C)$$

Hence verified.

5. If $X = \{-5, 1, 3, 4\}$ and $Y = \{a, b, c\}$, then which of the following relations are functions from X to Y ?

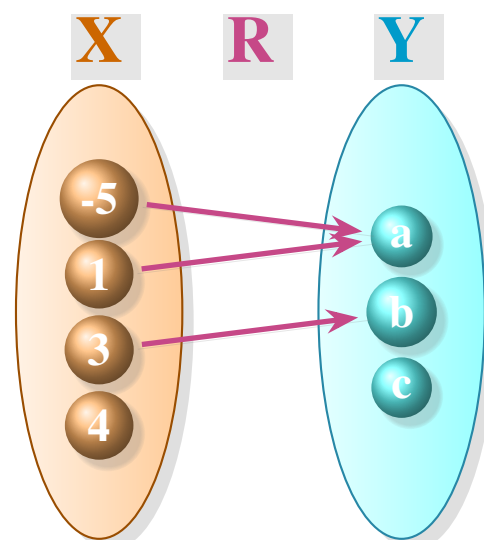
$$(i) R_1 = \{(-5, a), (1, a), (3, b)\} \quad \text{MAR-26}$$

$$(ii) R_2 = \{(-5, b), (1, b), (3, a), (4, c)\}$$

$$(iii) R_3 = \{(-5, a), (1, a), (3, b), (1, c), (1, b)\}$$

solution

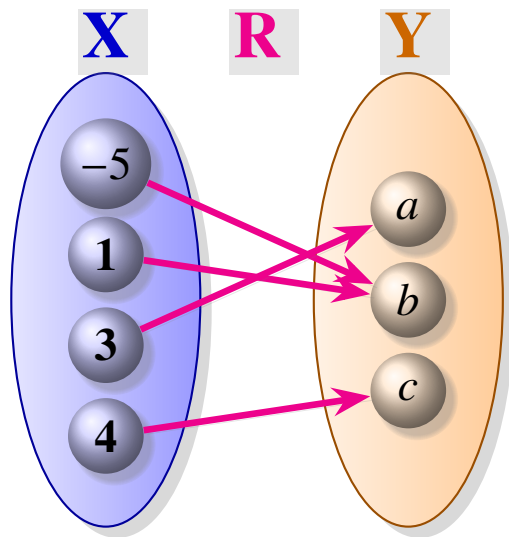
$$(i) R_1 = \{(-5, a), (1, a), (3, b)\}$$

Arrow Diagram.

R_1 is not a function as $4 \in X$ does not have an image in Y .

(ii) $R_2 = \{(-5, b), (1, b), (3, a), (4, c)\}$

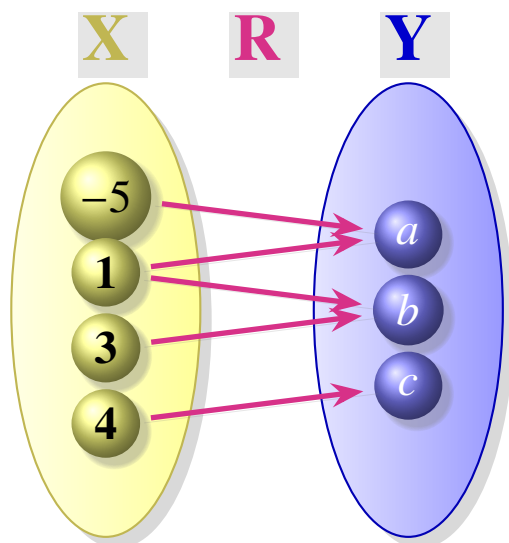
Arrow diagram



R_2 is a function each element X has an unique image in Y .

(iii) $R_3 = \{(-5, a), (1, a), (3, b), (4, c), (1, b)\}$

Arrow Diagram.



R_3 is not a function $1 \in X$ has two images $a \in Y$ and $b \in Y$.

6. A function f is defined by $f(x) = 2x - 3$

(i) find $\frac{f(0)+f(1)}{2}$ (ii) find x such that $f(x) = 0$.

(iii) find x such that $f(x) = x$.

(iv) find x such that $f(x) = f(1 - x)$

solution

$$\begin{aligned} f(x) &= 2x - 3 \\ \text{(i)} \quad \frac{f(0) + f(1)}{2} &= \frac{[2(0) - 3] + [2(1) - 3]}{2} \\ &= \frac{0 - 3 + 2 - 3}{2} = \frac{2 - 6}{2} \\ &= \frac{-4}{2} \\ \frac{f(0) + f(1)}{2} &= -2 \end{aligned}$$

(ii) Given $f(x) = 0$

$$\therefore 2x - 3 = 0$$

$$2x = 3$$

$$x = \frac{3}{2}$$

(iii) Given $f(x) = x$

$$2x - 3 = x$$

$$2x - x = 3 \implies x = 3$$

(iv) Given $f(x) = f(1 - x)$

$$2x - 3 = 2(1 - x) - 3$$

$$2x - 3 = 2 - 2x - 3$$

$$2x + 2x = 2 - 3 + 3$$

$$4x = 2$$

$$x = \frac{2}{4} = \frac{1}{2}$$

7. The data in the adjacent table depicts the length of a person forehead and their corresponding height. Based on this data, a student finds a relationship between the height (y) and the forehead length (x) as $y = ax + b$, where a, b are constants.

(i) Check if this relation is a function. (ii) Find a and b .

(iii) Find the height of a person whose forehead length is 40 cm

(iv) Find the length of forehead of a person if the height is 53.3 inches.

PTA-4

Length x of Forehead (in cm)	Height y (in inch)
35	56
45	65
50	69.5
55	74

solution

(i) Yes, Relation

$R = \{(35, 56), (45, 65), (50, 69.5), (55, 74)\}$ is a function.

(ii) Given $y = ax + b$

$$x = 35, y = 56$$

$$56 = 35a + b \quad \dots (1)$$

$$x = 45, y = 65$$

$$65 = 45a + b \quad \dots (2)$$

$$\text{Solve (1) and (2)} \quad a = 0.90 \quad \text{and} \quad b = 24.5$$

(iii) Given, forehead length is 40 cm

$$x = 40, y = ax + b$$

$$y = (0.90)(40) + 24.5 = 60.5$$

$\therefore$ Height of woman is 60.5 inches

(iv) Given height is 53.3 inches

$$y = 53.3, x = ?$$

$$53.3 = 0.9x + 24.5$$

$$53.3 - 24.5 = 0.9x$$

$$x = \frac{28.8}{0.9} = 32$$

Length of forehead is 32 cm.

8. Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 5, 8, 11, 14\}$ be two sets. Let $f : A \rightarrow B$ be a function given by $f(x) = 3x - 1$. Represent this function

PTA-3, JUL-20, 25

(i) by arrow diagram

(ii) in a table form

(iii) as a set of

ordered pairs

(iv) in a graphical form

solution

$$A = \{1, 2, 3, 4\}; B = \{2, 5, 8, 11, 14\};$$

$$f(x) = 3x - 1$$

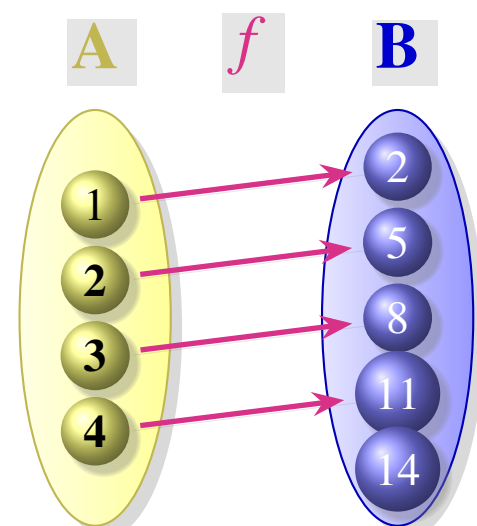
$$f(1) = 3(1) - 1 = 3 - 1 = 2$$

$$f(2) = 3(2) - 1 = 6 - 1 = 5$$

$$f(3) = 3(3) - 1 = 9 - 1 = 8$$

$$f(4) = 3(4) - 1 = 12 - 1 = 11$$

(i) Arrow diagram



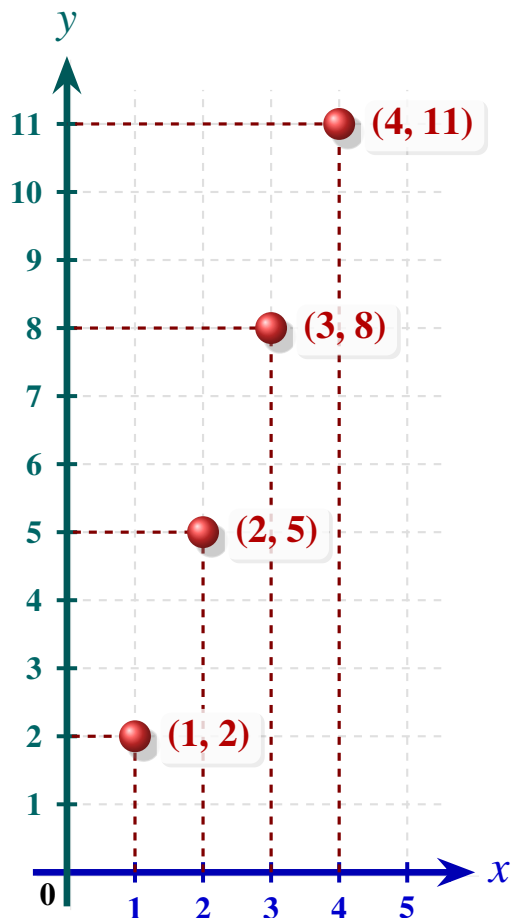
(ii) Table form

x	1	2	3	4
$f(x)$	2	5	8	11

(iii) Set of ordered pairs

$$f = \{(1, 2), (2, 5), (3, 8), (4, 11)\}$$

(iv) Graphical form



9. Let f be a function $f : \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(x) = 3x + 2, x \in \mathbb{N}$

PTA-3

- (i) Find the images of 1, 2, 3 (ii) Find the pre-images of 29, 53
(ii) Identify the type of function

solution

The function $f : \mathbb{N} \rightarrow \mathbb{N}$ $f(x) = 3x + 2$

(i) If $x = 1$, $f(1) = 3(1) + 2 = 5$

If $x = 2$, $f(2) = 3(2) + 2 = 8$

If $x = 3$, $f(3) = 3(3) + 2 = 11$

The images of 1, 2, 3 are 5, 8, 11

(ii) If x is the pre-image of 29, then

$$f(x) = 29.$$

$$3x + 2 = 29 \implies 3x = 27 \implies x = 9.$$

If x is the pre-image of 53,

$$f(x) = 53. \text{ Hence } 3x + 2 = 53$$

$$3x = 51 \implies x = 17.$$

Thus the pre-images of 29 and 53 are 9 and 17 respectively.

(iii) Since different elements of $\mathbb{N}$ have different images in the co-domain, the function f is one-one function.

The co-domain of f is $\mathbb{N}$.

But the range of $f = \{5, 8, 11, 14, 17, \dots\}$ is a subset of $\mathbb{N}$.

Therefore f is not an onto function.

f is an into function.

f is one-one and into function.

10. If the function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$f(x) = \begin{cases} 2x + 7; & x < -2 \\ x^2 - 2; & -2 \leq x < 3 \\ 3x - 2; & x \geq 3 \end{cases}$$

,find the values of

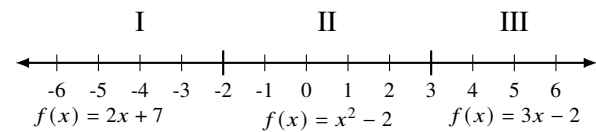
(i) $f(4)$

(ii) $f(-2)$

(iii) $f(4) + 2f(1)$

(iv) $\frac{f(1)-3f(4)}{f(-3)}$

solution



(i) $x = 4$ lie in the III Interval

$$f(x) = 3x - 2; f(4) = 3(4) - 2 = 10$$

(ii) $x = -2$ lies in the II Interval

$$f(x) = x^2 - 2; f(-2) = (-2)^2 - 2 = 2$$

(iii) From (i), $f(4) = 10$.

To find $f(1)$, $x = 1$ lies in II Interval

$$f(x) = x^2 - 2 \implies f(1) = 1^2 - 2 = -1$$

$$f(4) + 2f(1) = 10 + 2(-1) = 8$$

(iv) $f(1) = -1$ and $f(4) = 10$.

To Find $f(-3)$, $x = -3$, lies I Interval

$$f(x) = 2x + 7$$

$$f(-3) = 2(-3) + 7 = 1$$

$$\frac{f(1) - 3f(4)}{f(-3)} = \frac{1 - 3(10)}{1} = -31$$

11. Let $f : A \rightarrow B$ be a function defined by $f(x) = \frac{x}{2} - 1$, where $A = \{2, 4, 6, 10, 12\}$, $B = \{0, 1, 2, 4, 5, 9\}$. Represent f by

(i) set of ordered pairs

(ii) a table

(iii) an arrow diagram

(iv) a graph

APR-23, MDL-19

solution

$$f : A \rightarrow B, f(x) = \frac{x}{2} - 1$$

$$A = \{2, 4, 6, 10, 12\}, B = \{0, 1, 2, 4, 5, 9\}$$

$$f(2) = \frac{2}{2} - 1 = 2 - 2 = 0$$

$$f(4) = \frac{4}{2} - 1 = 2 - 1 = 1$$

$$f(6) = \frac{6}{2} - 1 = 3 - 1 = 2$$

$$f(10) = \frac{10}{2} - 1 = 5 - 1 = 4$$

$$f(12) = \frac{12}{2} - 1 = 6 - 1 = 5$$

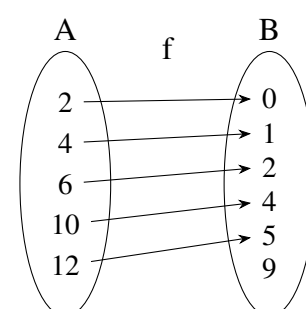
(i) Set of ordered pairs

$$f = \{(2, 0), (4, 1), (6, 2), (10, 4), (12, 5)\}$$

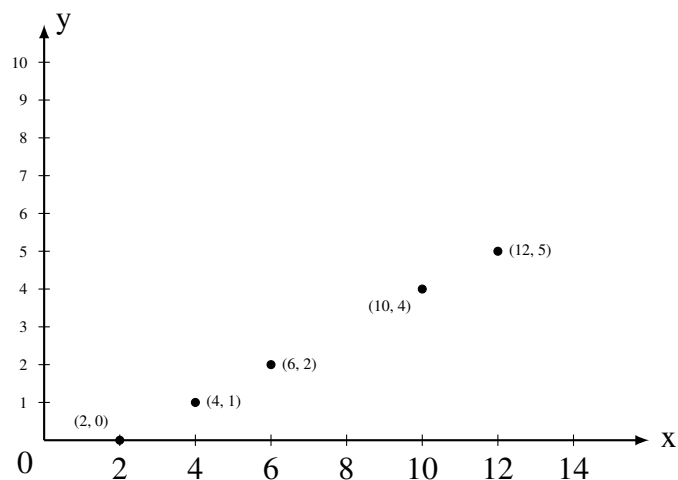
(ii) Table

x	2	4	6	10	12
$f(x)$	0	1	2	4	5

(iii) Arrow Diagram



(iv) Graph



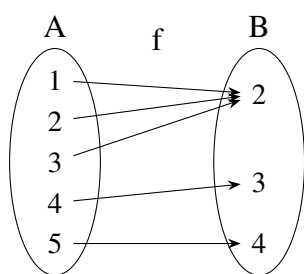
12. Represent the function $f = \{(1, 2), (2, 2), (3, 2), (4, 3), (5, 4)\}$ through

- (i) an arrow diagram (ii) a table form (iii) a graph

solution

Function $f = \{(1, 2), (2, 2), (3, 2), (4, 3), (5, 4)\}$

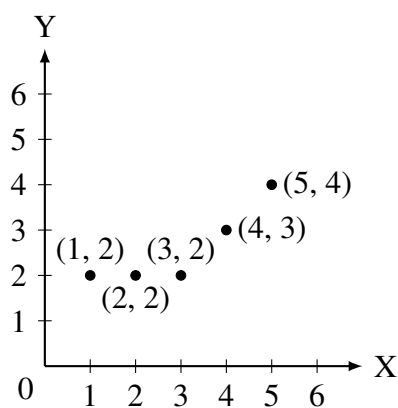
(i) Arrow diagram



(ii) Table form

x	1	2	3	4	5
$f(x)$	2	2	2	3	4

(iii) Graphical Form

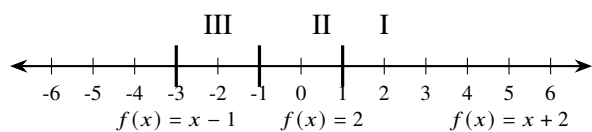


13. If the function f is defined by $f(x) = \begin{cases} x+2; & x > 1 \\ 2; & -1 \leq x \leq 1 \\ x-1; & -3 < x < -1 \end{cases}$ find the

values of

- (i) $f(3)$ (ii) $f(0)$ (iii) $f(-1.5)$ (iv) $f(2) + f(-2)$

solution



$$f(x) = \begin{cases} x+2 & \text{if } x > 1 \\ 2 & \text{if } -1 \leq x \leq 1 \\ x-1 & \text{if } -3 < x < -1 \end{cases}$$

(i) $f(3) = 3 + 2 = 5$ $[\because 3 > 1]$

(ii) $f(0) = 2$ $[\because -1 \leq 0 \leq 1]$

(iii) $f(-1.5) = -1.5 - 1 = -2.5$ $[\because -3 < -1.5 < -1]$

(iv) $f(2) + f(-2) = (2 + 2) + (-2 - 1) = 4 - 3 = 1$ $[\because 2 > 1 \text{ and } -3 < -2 < -1]$

14. A function $f : [-5, 9] \rightarrow \mathbb{R}$ is defined as follows:

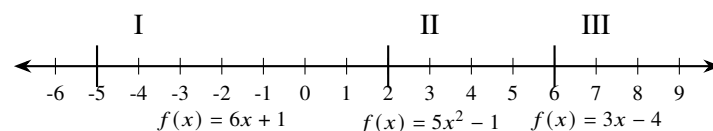
JUL-24, PTA-4

$$f(x) = \begin{cases} 6x + 1; & -5 \leq x < 2 \\ 5x^2 - 1; & 2 \leq x < 6 \\ 3x - 4; & 6 \leq x \leq 9 \end{cases}$$

Find (i) $f(-3) + f(2)$ (ii) $f(7) - f(1)$ (iii) $2f(4) + f(8)$

(iv) $\frac{2f(-2) - f(6)}{f(4) + f(-2)}$

solution



$$f(x) = \begin{cases} 6x + 1 & \text{if } -5 \leq x < 2 \\ 5x^2 - 1 & \text{if } 2 \leq x < 6 \\ 3x - 4 & \text{if } 6 \leq x \leq 9 \end{cases}$$

(i) $f(-3) + f(2) = [6(-3) + 1] + [5(2)^2 - 1]$
 $= (-18 + 1) + (20 - 1)$
 $= -17 + 19 = 2$ $[\because -5 \leq -3 < 2$
 $2 \leq 2 < 6]$

(ii) $f(7) - f(1) = [3(7) - 4] - [6(1) + 1]$
 $= (21 - 4) - (6 + 1)$
 $= 17 - 7 = 10$ $[\because 6 \leq 7 \leq 9$
 $-5 \leq 1 < 2]$

(iii) $2f(4) + f(8) = 2[5(4)^2 - 1] + [3(8) - 4]$
 $= 2[80 - 1] + [24 - 4]$
 $= 158 + 20 = 178$ $[\because 2 \leq 4 < 6$
 $6 \leq 8 \leq 9]$

(iv) $\frac{2f(-2) - f(6)}{f(4) + f(-2)}$
 $f(-2) = 6(-2) + 1 = -12 + 1 = -11$ $[\because -5 \leq -2 < 2]$
 $f(6) = 3(6) - 4 = 18 - 4 = 14$ $[\because 6 \leq 6 \leq 9]$
 $f(4) = 5(4)^2 - 1 = 80 - 1 = 79$ $[\because 2 \leq 4 < 6]$
 $\therefore \frac{2f(-2) - f(6)}{f(4) + f(-2)} = \frac{2(-11) - 14}{79 - 11}$
 $= \frac{-22 - 14}{68} = \frac{-36}{68} = -\frac{9}{17}$

15. The distance S an object travels under the influence of gravity in time t seconds is given by $S(t) = \frac{1}{2}gt^2 + at + b$ where, (g is the acceleration due to gravity), $a > 0, b$ are constants. Verify whether the function $S(t)$ is one-one or not.

PTA-3

solution

$$S(t) = \frac{1}{2}gt^2 + at + b \quad (a, b \text{ are constants})$$

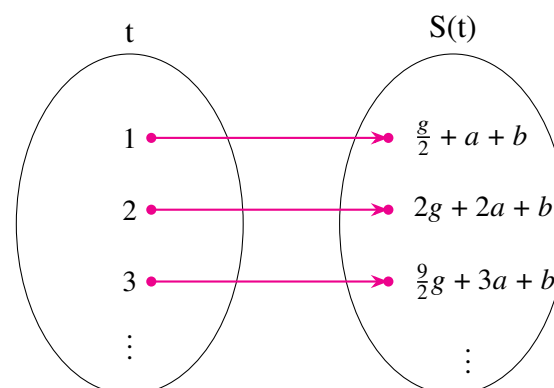
t = time in seconds

Let $t = 1, 2, 3 \dots$ Seconds

When $t = 1 \Rightarrow S(1) = \frac{g}{2} + a + b$

When $t = 2 \Rightarrow S(2) = 2g + 2a + b$

When $t = 3 \Rightarrow S(3) = \frac{9}{2}g + 3a + b$ and so on.



All the elements of t , having different images in $S(t)$
Hence it is an One-to-one function.

16. The function 't' which maps temperature in Celsius (C) into temperature in Fahrenheit (F) is defined by $t(C) = F$ where $F = \frac{9}{5}C + 32$. Find,

PTA-1

- (i) $t(0)$ (ii) $t(28)$ (iii) $t(-10)$
 (iv) the value of C when $t(C) = 212$ (v) the temperature when the Celsius value is equal to the Fahrenheit value.

solution

$$\text{Given } t(C) = F = \frac{9C}{5} + 32$$

- (i) $t(0) = \frac{9(0)}{5} + 32 = 32^\circ F$
 (ii) $t(28) = \frac{9(28)}{5} + 32 = 50.4 + 32 = 82.4^\circ F$
 (iii) $t(-10) = \frac{9(-10)}{5} + 32 = -18 + 32 = 14^\circ F$
 (iv) When $t(c) = 212$,

$$\begin{aligned} 212 &= \frac{9C}{5} + 32 \\ \frac{9C}{5} &= 180 \\ C &= 180 \times \frac{5}{9} \\ C &= 100^\circ C \end{aligned}$$

- (v) Celsius value = Fahrenheit value,

$$\begin{aligned} C &= \frac{9C}{5} + 32 \\ 5C &= 9C + 160 \\ -4C &= 160 \\ C &= -40^\circ \end{aligned}$$

17. If $f(x) = 2x + 3$, $g(x) = 1 - 2x$ and $h(x) = 3x$. Prove that $f \circ (g \circ h) = (f \circ g) \circ h$

solution

$$\begin{aligned} f(x) &= 2x + 3, g(x) = 1 - 2x, h(x) = 3x \\ (f \circ g)(x) &= f(g(x)) = f(1 - 2x) \\ &= 2(1 - 2x) + 3 = 5 - 4x \\ \text{LHS, } (f \circ g) \circ h(x) &= (f \circ g)(h(x)) \\ &= (f \circ g)(3x) \\ &= 5 - 4(3x) = 5 - 12x \quad \dots (1) \\ (g \circ h)(x) &= g(h(x)) = g(3x) \\ &= 1 - 2(3x) = 1 - 6x \\ \text{RHS, } f \circ (g \circ h)(x) &= f(1 - 6x) = 2(1 - 6x) + 3 \\ &= 5 - 12x \quad \dots (2) \\ \text{From (1) and (2)} \\ (f \circ g) \circ h &= f \circ (g \circ h) \end{aligned}$$

18. Find x if $gff(x) = fgg(x)$, given $f(x) = 3x + 1$ and $g(x) = x + 3$

MDL-19

solution

$$\begin{aligned} gff(x) &= g[f\{f(x)\}] \\ g[f(3x + 1)] &= g[3(3x + 1) + 1] \\ &= g(9x + 4) = [(9x + 4) + 3] \\ &= 9x + 7 \dots (1) \\ fgg(x) &= f[g\{g(x)\}] \\ &= f[g(x + 3)] = f[(x + 3) + 3] \\ f(x + 6) &= [3(x + 6) + 1] \\ &= 3x + 19 \dots (2) \\ 9x + 7 &= 3x + 19. \\ x &= 2 \end{aligned}$$

19. Consider the functions $f(x), g(x), h(x)$ as given below. Show that $(f \circ g) \circ h = f \circ (g \circ h)$ in each case.

- (i) $f(x) = x - 1, g(x) = 3x + 1$ and $h(x) = x^2$

JUL-25,MAR-26

- (ii) $f(x) = x^2, g(x) = 2x$ and $h(x) = x + 4$

- (iii) $f(x) = x - 4, g(x) = x^2$ and $h(x) = 3x - 5$

PTA-2

solution

$$\begin{aligned} \text{(i)} \quad f(x) &= x - 1, g(x) = 3x + 1, h(x) = x^2 \\ f \circ g &= f[g(x)] = f[3x + 1] \\ &= (3x + 1) - 1 = 3x \\ (f \circ g) \circ h &= (f \circ g)[h(x)] = (f \circ g)[x^2] = 3x^2 \\ g \circ h &= g[h(x)] = g[x^2] = 3x^2 + 1 \\ f \circ (g \circ h) &= f[g(h(x))] = f[g(x^2)] = f[3x^2 + 1] \\ &= (3x^2 + 1) - 1 = 3x^2 \\ \therefore (f \circ g) \circ h &= f \circ (g \circ h) \\ \text{Hence proved.} \\ \text{(ii)} \quad f(x) &= x^2, g(x) = 2x, h(x) = x + 4 \\ f \circ g &= f[g(x)] = f[2x] \\ &= f(2x) = (2x)^2 = 4x^2 \\ (f \circ g) \circ h &= (f \circ g)[h(x)] = (f \circ g)[x + 4] \\ &= 4(x + 4)^2 \\ &= 4(x^2 + 8x + 16) = 4x^2 + 32x + 64 \\ g \circ h &= g[h(x)] = g[x + 4] = 2(x + 4) \\ f \circ (g \circ h) &= f[g \circ h(x)] = f[2(x + 4)] \\ &= [2(x + 4)]^2 \\ &= 4(x + 4)^2 = 4(x^2 + 8x + 16) \\ &= 4x^2 + 32x + 64 \\ \therefore (f \circ g) \circ h &= f \circ (g \circ h) \\ \text{Hence proved.} \end{aligned}$$

- (iii) $f(x) = x - 4, g(x) = x^2, h(x) = 3x - 5$

$$\begin{aligned} f \circ g &= f[g(x)] = f(x^2) = x^2 - 4 \\ (f \circ g) \circ h &= (f \circ g)[h(x)] = (f \circ g)[3x - 5] \\ &= (3x - 5)^2 - 4 = 9x^2 - 30x + 25 - 4 \\ &= 9x^2 - 30x + 21 \\ g \circ h &= g[h(x)] \\ &= g(3x - 5) = (3x - 5)^2 = 9x^2 - 30x + 25 \\ f \circ (g \circ h) &= f[g(h(x))] = f[g(3x - 5)] = f[(3x - 5)^2] \\ &= 9x^2 - 30x + 25 - 4 = 9x^2 - 30x + 21 \\ \therefore (f \circ g) \circ h &= f \circ (g \circ h) \\ \text{Hence proved.} \end{aligned}$$

20. If $f(x) = x^2, g(x) = 3x$ and $h(x) = x - 2$, Prove that $(f \circ g) \circ h = f \circ (g \circ h)$

solution

$$\begin{aligned} f(x) &= x^2, g(x) = 3x, h(x) = x - 2 \\ f \circ g &= f[g(x)] = f(3x) \\ &= (3x)^2 = 9x^2 \\ (f \circ g) \circ h &= (f \circ g)[h(x)] = (f \circ g)[x - 2] \\ &= 9(x - 2)^2 \\ g \circ h &= g[h(x)] = g[x - 2] = 3(x - 2) \\ f \circ (g \circ h) &= f[g(h(x))] \\ &= f[3(x - 2)] = [3(x - 2)]^2 = 9(x - 2)^2 \\ \therefore (f \circ g) \circ h &= f \circ (g \circ h) \\ \text{Hence proved.} \end{aligned}$$

21. The functions f and g are defined by $f(x) = 6x + 8; g(x) = \frac{x-2}{3}$

- (i) Calculate the value of $gg\left(\frac{1}{2}\right)$
 (ii) Write an expression for $gf(x)$ in its simplest form.

solution

$$\begin{aligned} f(x) &= 6x + 8, g(x) = \frac{x-2}{3} \\ \text{(i)} \quad g\left(\frac{1}{2}\right) &= \frac{\frac{1}{2} - 2}{3} = \frac{1 - 4}{2 \times 3} = -\frac{3}{6} = -\frac{1}{2} \end{aligned}$$

$$\begin{aligned}\therefore gg\left(\frac{1}{2}\right) &= g\left[g\left(\frac{1}{2}\right)\right] \\ &= g\left(-\frac{1}{2}\right) = \frac{-\frac{1}{2}-2}{3} = \frac{-1-4}{2 \times 3} = -\frac{5}{6}\end{aligned}$$

(ii) $gf(x) = g[f(x)] = g[6x+8]$

$$\begin{aligned}&= \frac{6x+8-2}{3} = \frac{6x+6}{3} \\ &= \frac{6(x+1)}{3} = 2(x+1)\end{aligned}$$

22. Write the domain of the following real functions

PTA-6

(a) $f(x) = \frac{2x+1}{x-9}$ **PTA-6**

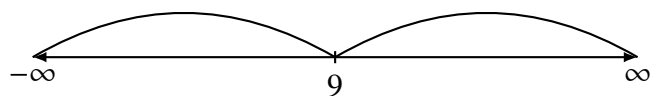
(b) $p(x) = \frac{-5}{4x^2+1}$

(c) $g(x) = \sqrt{x-2}$ **PTA-6**

(d) $h(x) = x+6$

solution

(i) $f(x) = \frac{2x+1}{x-9}$
If $x-9=0$
 $x=9$



$$(-\infty, 9) \cup (9, \infty)$$

$$(x/x \neq 9) \implies R - \{9\}$$

(ii) $p(x) = \frac{-5}{4x^2+1}$

All Real Values of x

$$x \in R$$

(iii) $g(x) = \sqrt{x-2}$

$g(x)$ is defined real only when $x \geq 2$

(iv) $h(x) = x+6$

$h(x)$ is defined for all real values of ' x '.

$$x \in R.$$

CHAPTER - 2. NUMBERS AND SEQUENCES

PUBLIC EXAMINATION IMPORTANT QUESTIONS-2026

2 MARK IMPORTANT QUESTIONS

1. If the Highest Common Factor of 210 and 55 is expressible in the form $55x - 325$, find x .

solution

Using Euclid's Division Algorithm

$$210 = 55 \times 3 + 45$$

$$55 = 45 \times 1 + 10$$

$$45 = 10 \times 4 + 5$$

$$10 = 5 \times 2 + 0$$

The remainder is zero.

last divisor 5 is the Highest Common

H.C.F. is expressible in the form $55x - 325 = 5$.

$$55x = 330$$

$$x = 6$$

2. Find the greatest number that will divide 445 and 572 leaving remainders 4 and 5 respectively.

solution

The remainders are 4, 5 respectively

$$445 - 4 = 441, 572 - 5 = 567.$$

The H.C.F. of 441 and 567

Using Euclid's Division Algorithm

$$567 = 441 \times 1 + 126$$

$$441 = 126 \times 3 + 63$$

$$126 = 63 \times 2 + 0$$

H.C.F. of 441, 567 = 63

The required number is 63.

3. A man has 532 flower pots. He wants to arrange them in rows such that each row contains 21 flower pots. Find the number of completed rows and how many flower pots are left over. **APR-25**

solution

Given number of flower pots = 532

By Euclid's Division Lemma

$$a = bq + r, 0 \leq r < |b|$$

$$532 = 21 \times 25 + 7, 0 \leq r < |21|$$

$$\begin{array}{r} 25 \\ 21 \overline{) 532} \\ \underline{42} \\ 112 \\ \underline{105} \\ 7 \end{array}$$

$\therefore$ Number of completed rows = 25

Number of flower pots left over = 7

4. Prove that the product of two consecutive positive integers is divisible by 2. **PTA-1**

solution

Let the 2 consecutive positive integers be $x, x+1$

$\therefore$ Product of 2 integers = $x(x+1)$

$$= x^2 + x$$

Case (i)

If x is an even number

Let $x = 2k$

$$\therefore x^2 + x = (2k)^2 + 2k$$

$$= 2k(2k+1), \text{ divisible by 2}$$

Case (ii)

If x is an odd number,

$$\text{Let } x = 2k + 1$$

$$x^2 + x = (2k + 1)^2 + (2k + 1)$$

$$= 4k^2 + 4k + 1 + 2k + 1$$

$$= 4k^2 + 6k + 2$$

$$= 2(2k^2 + 3k + 1) \text{ divisible by 2}$$

Product of 2 consecutive positive integers is divisible by 2.

5. Find the largest number which divides 1230 and 1926 leaving remainder 12 in each case. **JUL-22**

solution

To find the largest number which divides 1230 and 1926 leaving remainder 12.

HCF of $1230 - 12 = 1218$ and $1926 - 12 = 1914$.

To find HCF of 1218 and 1914. Using Euclid's Division 1914 and 1218

$$1914 = 1218 \times 1 + 696$$

$$1218 = 696 \times 1 + 522$$

$$696 = 522 \times 1 + 174$$

$$522 = (174) \times 3 + 0$$

$$\text{HCF} = 174$$

The required largest number = 174.

6. If d is the Highest Common Factor of 32 and 60, find x and y satisfying $d = 32x + 60y$.

solution

Given d is the HCF of 32 and 60

$$\therefore 60 = 32 \times 1 + 28 \quad \dots\dots\dots (1)$$

$$32 = 28 \times 1 + 4 \quad \dots\dots\dots (2)$$

$$28 = (4) \times 7 + 0$$

$$\therefore \boxed{d = 4}$$

$$4 = 32 - (28 \times 1) \quad [\text{From (2)}]$$

$$= 32 - 1(60 - 32) \quad [\text{From (1)}]$$

$$= 32 - (1 \times 60 - 1 \times 32)$$

$$= 32 - 1 \times 60 + 1 \times 32$$

$$= 32(1 + 1) + 60(-1)$$

$$d = 32(2) + 60(-1)$$

$$\Rightarrow 32x + 60y = 32(2) + 60(-1)$$

$$\therefore \boxed{x = 2, \quad y = -1}$$

7. Is $7 \times 5 \times 3 \times 2 + 3$ a composite number?. Justify your Answer. **PTA-3**

solution

Yes, the given number is a composite number.

$$7 \times 5 \times 3 \times 2 + 3 = 3 \times (7 \times 5 \times 2 + 1) = 3 \times 71$$

Since the given number can be factorized in terms of two primes. it is a composite number.

8. ' a ' and ' b ' are two positive integers such that $a^b \times b^a = 800$. Find ' a ' and ' b '. **APR-24, JUL-25**

solution

$$a^b \times b^a = 800$$

2	800
2	400
2	200
2	100
2	50
5	25
5	5
	1

$$800 = 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 5 = 2^5 \times 5^2$$

$$a^b \times b^a = 2^5 \times 5^2$$

$$\boxed{a = 2, b = 5} \text{ (or) } \boxed{a = 5, b = 2}$$

9. If m, n are natural numbers, for what values of m , does $2^n \times 5^m$ ends in 5 ? **JUL-20**

solution

$$\text{Let } x = 2^n \times 5^m.$$

Since m and n are natural numbers and 2^n is even.

For any value of m , $2^n \times 5^m$ will not be ended in 5.

$\therefore$ No value of m will make x true.

10. Find the HCF of 252525 and 363636.

solution

Given numbers are 252525, 363636

By using prime factorization

5	252525	2	363636
5	50505	2	181818
	10101	3	90909
		3	30303
			10101

$$\therefore 252525 = 5 \times 5 \times \times 10101$$

$$363636 = 2 \times 2 \times 3 \times 3 \times 10101$$

$$\therefore \text{HCF} = 10101$$

11. If $13824 = 2^a \times 3^b$ then find a and b . **APR-2**

solution

$$\text{Given } 2^a \times 3^b = 13824$$

2	13824
2	6912
2	3456
2	1728
2	864
2	432
2	216
2	108
2	54
3	27
3	9
3	3
	1

$$2^a \times 3^b = 2^9 \times 3^3$$

$$\therefore \boxed{a = 9, \quad b = 3}$$

12. Find the least number that is divisible by the first ten natural numbers.

JUL-22, 23

solution

The required number is the LCM of $(1, 2, 3, \dots, 10)$.

2	1,2,3,4,5,6,7,8,9,10
2	1,1,3,2,5,3,7,4,9,5
2	1,1,3,1,5,3,7,2,9,5
3	1,1,3,1,5,3,7,1,9,5
3	1,1,1,1,5,1,7,1,3,5
5	1,1,1,1,5,1,7,1,1,5
7	1,1,1,1,1,1,7,1,1,1
	1,1,1,1,1,1,1,1,1,1

$$\text{L.C.M} = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7$$

$$= 2520$$

13. Find the least positive value of x such that $67 + x \equiv 1 \pmod{4}$ **PTA-2**

solution

$$67 + x \equiv 1 \pmod{4}$$

$$67 + x - 1 = 4n, \text{ for some integer } n$$

$$66 + x = 4n$$

$$x = 2 \Rightarrow 68 \text{ divisible by 4}$$

$66 + x$ is a multiple of 4.

The least positive value of x must be 2.

Since 68 is the nearest multiple of 4 more than 66.

14. Solve $8x \equiv 1 \pmod{11}$.

solution

$8x \equiv 1 \pmod{11}$ can be written as

$8x - 1 = 11k$, for some integer k .

$$x = \frac{11k + 1}{8}$$

Put $k = 5, 13, 21, 29, \dots$ $11k + 1$ is divisible by 8.

$$x = \frac{11 \times 5 + 1}{8} = 7$$

$$x = \frac{11 \times 13 + 1}{8} = 18$$

The solutions are 7, 18, 29, 40, $\dots$

15. Compute x , such that $10^4 \equiv x \pmod{19}$

MDL-19

solution

$$10^2 = 100 \equiv 5 \pmod{19}$$

$$10^4 = (10^2)^2 \equiv 5^2 \pmod{19}$$

$$10^4 \equiv 25 \pmod{19}$$

$$10^4 \equiv 6 \pmod{19} \quad (25 \equiv 6 \pmod{19})$$

$$x = 6.$$

16. Find the number of integer solutions of $3x \equiv 1 \pmod{15}$.

JUL-21, PTA-5

solution

$3x \equiv 1 \pmod{15}$ can be written as

$$3x - 1 = 15k \text{ for some integer } k$$

$$3x = 15k + 1$$

$$x = \frac{15k + 1}{3}$$

$$x = 5k + \frac{1}{3}$$

Since $5k$ is an integer, $5k + \frac{1}{3}$ cannot be an integer.
So there is no integer solution.

17. Find the least positive value of x such that

(i) $71 \equiv x \pmod{8}$ **MAR-26**

solution

$$71 \equiv x \pmod{8} \quad 8 \overline{)71}$$

$$71 = (8 \times 8) + 7 \quad \underline{-64}$$

$$71 \equiv 7 \pmod{8} \quad 7$$

Comparing $71 \equiv 7 \pmod{8}$ with $71 \equiv x \pmod{8}$.
The Least positive value of x is 7.

18. Find the least positive value of x such that $5x \equiv 4 \pmod{6}$ **PTA-2**

solution

$$5x \equiv 4 \pmod{6}$$

$$5x - 4 = 6k \text{ for some integer } k.$$

$$x = \frac{6k + 4}{5}$$

Put $k = 1, 6, 11, 16, \dots$ then $6k + 4$ is divisible by 5

$$x = \frac{6(1) + 4}{5} = \frac{10}{5} = 2$$

$$x = \frac{6(6) + 4}{5} = \frac{40}{5} = 8$$

$$x = \frac{6(11) + 4}{5} = \frac{70}{5} = 14$$

$$x = \frac{6(16) + 4}{5} = \frac{100}{5} = 20$$

The solutions are 2, 8, 14, 20, $\dots$

19. What is the time 100 hours after 7 a.m.?

solution

Starting from 7 o' clock 100 hours. Use modulo 24.

$$7 \text{ o' clock a.m.} + 100 \pmod{24} = 7 \text{ o' clock} + 4 \text{ hrs}$$

$$\text{since } = 100 \pmod{24}$$

$$= 4 \pmod{24}$$

$$= 11 \text{ o' clock a.m.}$$

100 hrs after 7 o' clock a.m. is 11 o' clock a.m.

20. What is the time 15 hours before 11 p.m.?

solution

Starting from 11 o' clock 13 hours. Use modulo 12.

$$11 \text{ o' clock p.m.} - 15 \pmod{12} = 11 \text{ o' clock} - 3 \text{ hrs}$$

$$= 15 \pmod{12}$$

$$= 3 \pmod{12}$$

$$= 8 \text{ o' clock a.m.}$$

15 hrs before 11 o' clock p.m. is 8 o' clock p.m.

21. Today is Tuesday. My uncle will come after 45 days. In which day my uncle will be coming?

solution

Today is Tuesday

Day after 45 days = ?

When divide 45 by 7, remainder is 3.

$$45 \equiv 3 \pmod{7}.$$

$$\begin{array}{r} 6 \\ 7 \overline{)45} \\ \underline{-42} \\ 3 \end{array}$$

The 3rd day from Tuesday is Friday. My Uncle will come Friday.

22. The general term of a sequence is defined as

$$a_n = \begin{cases} n(n+3) & ; n \in \mathbb{N} \text{ is odd} \\ n^2 + 1 & ; n \in \mathbb{N} \text{ is even} \end{cases}$$

Find the eleventh and eighteenth terms.

solution

To find a_{11} , since 11 is odd, Put $n = 11$ in $a_n = n(n+3)$

The eleventh term

$$a_{11} = 11(11+3) = 121 + 33 = 154,$$

To find a_{18} , since 18 is even Put $n = 18$ in $a_n = n^2 + 1$

The eighteenth term

$$a_{18} = 18^2 + 1 = 324 + 1 = 325.$$

23. Find a_8 and a_{15} whose n^{th} term is $a_n = \begin{cases} \frac{n^2-1}{n+3} & ; n \text{ is even, } n \in \mathbb{N} \\ \frac{n^2}{2n+1} & ; n \text{ is odd, } n \in \mathbb{N} \end{cases}$

solution

To find a_8 , since $n = 8$ is even, put $n = 8$ in

$$a_n = \frac{n^2 - 1}{n + 3}$$

To find a_8 , since $n = 8$ is even, put $n = 8$ in $a_n = \frac{n^2 - 1}{n + 3}$

$$a_8 = \frac{8^2 - 1}{8 + 3} = \frac{64 - 1}{11} = \frac{63}{11}$$

To find a_{15} , since $n = 15$ is odd

$$\text{put } n = 15 \text{ in } a_n = \frac{n^2}{2n + 1}$$

$$a_{15} = \frac{15^2}{2(15) + 1} = \frac{225}{30 + 1} = \frac{225}{31}$$

$$\therefore a_8 = \frac{63}{11} \text{ and } a_{15} = \frac{225}{31}$$

24. Find the indicated terms of the sequences whose n^{th} terms are given by (i) $a_n = \frac{5n}{n+2}$; a_6 and a_{13} (ii) $a_n = -(n^2 - 4)$; a_4, a_{11} .

solution

$$\begin{aligned} \text{i) } a_n &= \frac{5n}{n+2}, \quad a_6, a_{13} = ? \\ a_6 &= \frac{5(6)}{6+2} = \frac{30}{8} = \frac{15}{4} \\ a_{13} &= \frac{5(13)}{13+2} = \frac{65}{15} = \frac{13}{3} \end{aligned}$$

$$\begin{aligned} \text{ii) } a_n &= -(n^2 - 4); \quad a_4, a_{11} = ? \\ a_4 &= -(16 - 4) = -12 \\ a_{11} &= -(121 - 4) = -117 \end{aligned}$$

25. Check whether the following sequences are in A.P. or not? $3\sqrt{2}, 5\sqrt{2}, 7\sqrt{2}, 9\sqrt{2}, \dots$

solution

$$\begin{aligned} t_2 - t_1 &= 5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2} \\ t_3 - t_2 &= 7\sqrt{2} - 5\sqrt{2} = 2\sqrt{2} \\ t_4 - t_3 &= 9\sqrt{2} - 7\sqrt{2} = 2\sqrt{2} \end{aligned}$$

The difference between consecutive terms are equal.

The terms of the sequence $3\sqrt{2}, 5\sqrt{2}, 7\sqrt{2}, 9\sqrt{2}, \dots$ are in A.P.

26. Write an A.P. whose first term is 20 and common difference is 8.

solution

First term $= a = 20$; common difference $= d = 8$

The A.P is $a, a + d, a + 2d, a + 3d, \dots$

$20, 20 + 8, 20 + 2(8), 20 + 3(8) \dots$

The required A.P. is $20, 28, 36, 44, \dots$

27. Find the number of terms in the A.P. $3, 6, 9, 12, \dots, 111$.

JUL-21

solution

First term $a = 3$;

Common difference $d = 6 - 3 = 3$;

last term $l = 111$

$$\begin{aligned} n &= \left(\frac{l - a}{d} \right) + 1 \\ n &= \left(\frac{111 - 3}{3} \right) + 1 \\ &= \left(\frac{108}{3} \right) + 1 \\ &= 36 + 1 = 37 \end{aligned}$$

The A.P. contain 37 terms.

28. Find the 19th term of an A.P. $-11, -15, -19, \dots$

JUL-22, MDL-19

solution

The A.P is $-11, -15, -19, \dots$

First term $a = -11$

Common difference $d = t_2 - t_1 = -15 - (-11)$

$= -15 + 11$

$d = -4$

$\therefore n^{\text{th}}$ term of an A.P. is $t_n = a + (n - 1)d$

19th term (t_{19}) $= -11 + (19 - 1)(-4)$

$= -11 + 18(-4)$

$= -11 + (-72) = -83$

$\therefore$ 19th term of $-11, -15, -19, \dots$ is -83 .

29. Which term of an A.P. $16, 11, 6, 1, \dots$ is -54 ?

JUL-24, APR-22

solution

First term $a = 16$;

Common difference $d = 11 - 16 = -5$;

last term $l = -54$

$$\begin{aligned} n &= \left(\frac{l - a}{d} \right) + 1 \\ n &= \left(\frac{-54 - 16}{-5} \right) + 1 \\ &= \left(\frac{-70}{-5} \right) + 1 \\ &= 14 + 1 = 15 \end{aligned}$$

-54 is a 15th term of A.P $16, 11, 6, \dots$

30. Find the middle term(s) of an A.P. $9, 15, 21, 27, \dots, 183$.

PTA-1

solution

Given A.P is $9, 15, 21, 27, \dots, 183$

$a = 9, d = 6, l = 183$

$$\begin{aligned} n &= \frac{l - a}{d} + 1 \\ &= \frac{183 - 9}{6} + 1 \\ &= \frac{174}{6} + 1 \\ &= 29 + 1 \\ n &= 30 \end{aligned}$$

$$\begin{aligned} \therefore \text{Middle terms are } \frac{30}{2}, \frac{30}{2} + 1 \\ &= 15^{\text{th}}, 16^{\text{th}} \end{aligned}$$

$$\begin{aligned} t_{15} &= a + 14d & t_{16} &= a + 15d \\ &= 9 + 14(6) & &= 9 + 15(6) \\ &= 9 + 84 & &= 9 + 90 \\ t_{15} &= 93 & t_{16} &= 99 \end{aligned}$$

The middle terms are 93, 99.

31. If $3 + k, 18 - k, 5k + 1$ are in A.P. then find k .

JUL-21, PTA-3

solution

Given $3 + k, 18 - k, 5k + 1$ are in A.P.

$t_1 = 3 + k, \quad t_2 = 18 - k, \quad t_3 = 5k + 1$

$$\begin{aligned} t_2 - t_1 &= t_3 - t_2 \\ (18 - k) - (3 + k) &= (5k + 1) - (18 - k) \\ 18 - k - 3 - k &= 5k + 1 - 18 + k \\ 15 - 2k &= 6k - 17 \\ 15 + 17 &= 6k + 2k \\ 32 &= 8k \\ k &= \frac{32}{8} = 4 \end{aligned}$$

32. In a theatre, there are 20 seats in the front row and 30 rows were allotted. Each successive row contains two additional seats than its front row. How many seats are there in the last row?

PTA-4

solution

A.P form is $20, 20 + 2, 20 + 2 + 2 \dots$

$a = 20, d = 22 - 2 = 2, n = 30$

$$\begin{aligned} t_n &= a + (n - 1)d \\ t_{30} &= a + 29d \\ &= 20 + 29(2) \\ &= 20 + 58 \\ &= 78 \end{aligned}$$

$\therefore$ The no. of seats in 30th row 78

33. In a winter season let us take the temperature of Ooty from Monday to Friday to be in A.P. The sum of temperatures from Monday to Wednesday is 0°C and the sum of the temperatures from Wednesday to Friday is 18°C . Find the temperature on each of the five days.

solution

Let temperature from Monday to Friday respectively

$$a, a + d, a + 2d, a + 3d, a + 4d$$

(i) Given $a + (a + d) + (a + 2d) = 0$

$$3a + 3d = 0$$

$$a + d = 0$$

$$a = -d$$

(ii) Given $(a + 2d) + (a + 3d) + (a + 4d) = 18$

$$3a + 9d = 18$$

$$-3d + 9d = 18$$

$$6d = 18$$

$$d = 3$$

$$a = -3$$

The temperature of each of the 5 days $-3^\circ\text{C}, 0^\circ\text{C}, 3^\circ\text{C}, 6^\circ\text{C}, 9^\circ\text{C}$.

34. Find the sum of $0.40 + 0.43 + 0.46 + \dots + 1$.

solution

Given $a = 0.40, l = 1, d = 0.43 - 0.40 = 0.03$

$$\begin{aligned} n &= \left(\frac{l - a}{d} \right) + 1 \\ &= \left(\frac{1 - 0.40}{0.03} \right) + 1 \\ n &= 21 \end{aligned}$$

Sum of first n terms of an A.P.

$$\begin{aligned} S_n &= \frac{n}{2} [a + l] \\ n &= 21 \\ S_{21} &= \frac{21}{2} [0.40 + 1] \\ &= 14.7 \end{aligned}$$

The sum of 21 terms of the given series is 14.7.

35. Find the sum of the $6 + 13 + 20 + \dots + 97$.

solution

Given series $6 + 13 + 20 + \dots + 97$

$a = 6, d = 13 - 6 = 7, l = 97$

$$\begin{aligned} \therefore n &= \frac{l - a}{d} + 1 \\ &= \frac{97 - 6}{7} + 1 \\ &= \frac{91}{7} + 1 \\ &= 13 + 1 \\ n &= 14 \\ \therefore S_n &= \frac{n}{2} (a + l) \\ S_{14} &= \frac{14}{2} (6 + 97) \\ &= 7 \times 103 \\ S_{14} &= 721 \end{aligned}$$

36. Find the sum of all odd positive integers less than 450.

solution

To find the sum : $1 + 3 + 5 + 7 + \dots + 449$

$a = 1, d = 3 - 1 = 2, l = 449$

$$\begin{aligned} \therefore n &= \frac{l - a}{d} + 1 \\ &= \frac{449 - 1}{2} + 1 \\ &= \frac{448}{2} + 1 \\ &= 224 + 1 \\ n &= 225 \end{aligned}$$

$$\begin{aligned} \therefore S_n &= \frac{n}{2} [a + l] \\ S_{225} &= \frac{225}{2} [450] \\ &= 225 \times 225 \\ S_{225} &= 50,625 \end{aligned}$$

37. Find the geometric progression whose first term and common ratios are given by $a = -7, r = 6$ **PTA-5**

solution

The general form of Geometric progression is $a, ar, ar^2, \dots$

$$a = -7, ar = -7 \times 6 = -42,$$

$$ar^2 = -7 \times 6^2 = -252$$

The required Geometric Progression is $-7, -42, -252, \dots$

38. Find the 8th term of the G.P. $9, 3, 1, \dots$ **JUL-23, 24**

solution

To find the 8th term of G.P use the n^{th} term formula $t_n = ar^{n-1}$

First term $a = 9$, common ratio $r = \frac{t_2}{t_1} = \frac{3}{9} = \frac{1}{3}$

$$t_8 = 9 \times \left(\frac{1}{3} \right)^{8-1} = 9 \times \left(\frac{1}{3} \right)^7 = \frac{1}{243}$$

The 8th term of the G.P. is $\frac{1}{243}$.

39. In a G.P. $729, 243, 81, \dots$ find t_7 . **JUL-25**

solution

$$n^{\text{th}} \text{ term of G.P.} = ar^{n-1}$$

$$a = 729$$

$$r = \frac{t_2}{t_1} = \frac{243}{729}$$

$$r = \frac{1}{3}$$

$$\begin{aligned} t_7 &= ar^{7-1} = ar^6 = 729 \left(\frac{1}{3} \right)^6 \\ &= 729 \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \end{aligned}$$

$$t_7 = 1$$

40. Find x so that $x + 6, x + 12$ and $x + 15$ are consecutive terms of a Geometric Progression.

solution

Since in G.P Common Ratios are equal

$$\frac{t_2}{t_1} = \frac{t_3}{t_2}$$

$$\frac{x + 12}{x + 6} = \frac{x + 15}{x + 12}$$

$$(x + 12)^2 = (x + 6)(x + 15)$$

$$x^2 + 24x + 144 = x^2 + 6x + 15x + 90$$

$$x^2 + 24x + 144 - x^2 - 6x - 15x - 90 = 0$$

$$24x - 21x + 144 - 90 = 0$$

$$3x + 54 = 0$$

$$3x = -54$$

$$x = \frac{-54}{3}$$

$$x = -18$$

41. Find the number of terms in the following G.P. $4, 8, 16, \dots, 8192$?

solution

$$a = 4, r = \frac{8}{4} = 2$$

$$n^{\text{th}} \text{ term of a G.P. } t_n = ar^{n-1}$$

$$8192 = 4r^{n-1}$$

$$r^{n-1} = \frac{8192}{4}$$

$$r^{n-1} = 2048 = 2^{11}$$

$$n - 1 = 11 \quad [\because r = 2]$$

$$n = 11 + 1$$

$$n = 12$$

2	2048
2	1024
2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

42. If a, b, c are in A.P. then show that $3^a, 3^b, 3^c$ are in G.P.

solution

Given a, b, c are in A.P.

$$\therefore t_2 - t_1 = t_3 - t_2$$

$$b - a = c - b$$

$$b + b = c + a$$

$$2b = a + c$$

$$\therefore 3^{2b} = 3^{a+c}$$

$$3^{b+b} = 3^{a+c}$$

$$3^b \cdot 3^b = 3^a \cdot 3^c$$

$$\frac{3^b}{3^a} = \frac{3^c}{3^b}$$

The $3^a, 3^b, 3^c$ are in G.P.

43. Find the sum of 8 terms of the G.P. $1, -3, 9, -27, \dots$

solution

First term $a = 1$,

Common ratio $r = \frac{-3}{1} = -3 < 1, n = 8$.

Sum to n terms of a G.P. is

$$S_n = \frac{a(r^n - 1)}{r - 1} \text{ if } r \neq 1$$

$$S_n = \frac{1((-3)^8 - 1)}{(-3) - 1}$$

$$= \frac{6561 - 1}{-4}$$

$$S_n = -1640$$

44. Find the sum $3 + 1 + \frac{1}{3} + \dots \infty$.

solution

$$a = 3, r = \frac{t_2}{t_1} = \frac{1}{3}$$

$$S_\infty = \frac{a}{1 - r}$$

$$= \frac{3}{1 - \frac{1}{3}}$$

$$S_\infty = \frac{9}{2}$$

45. Find the rational form of the number $0.6666\dots$

solution

$0.6666\dots = 0.6 + 0.06 + 0.006 + 0.0006 + \dots$ $0.6, 0.06, 0.006 \dots$ forms an G.P.

First term $a = 0.6$

common ratio $r = \frac{0.06}{0.6} = 0.1$. $-1 < r = 0.1 < 1$.

Using the infinite G.P. formula,

$$0.6666\dots = 0.6 + 0.06 + 0.006 + 0.0006 + \dots$$

$$= \frac{0.6}{1 - 0.1} = \frac{0.6}{0.9} = \frac{2}{3}$$

Thus the rational number equivalent of $0.6666\dots$ is $\frac{2}{3}$.

46. Find the sum to infinity of (i) $9 + 3 + 1 + \dots$

MAR-26

(ii) $21 + 14 + \frac{28}{3} + \dots$

solution

(i) $9 + 3 + 1 + \dots$ is a geometric series

$$a = 9, r = \frac{3}{9} = \frac{1}{3} < 1$$

$$S_\infty = \frac{a}{1 - r} = \frac{9}{1 - \frac{1}{3}}$$

$$= \frac{9}{\frac{2}{3}}$$

$$S_\infty = \frac{27}{2}$$

(ii) $21 + 14 + \frac{28}{3}, \dots$ is geometric series

$$a = 21, r = \frac{14}{21} = \frac{2}{3} < 1$$

$$\therefore S_\infty = \frac{a}{1 - r} = \frac{21}{1 - \frac{2}{3}}$$

$$= \frac{21}{1 - \frac{2}{3}}$$

$$= \frac{21}{\frac{1}{3}}$$

$$S_\infty = 63$$

47. Find the sum of first six terms of the G.P. $5, 15, 45, \dots$

solution

To find $5 + 15 + 45 + \dots$ upto 6 terms.

$$\text{Sum upto } n \text{ terms of a G.P. } S_n = \frac{a(r^n - 1)}{r - 1}$$

$$a = 5, r = \frac{15}{5} = 3$$

$$\therefore S_6 = \frac{5(3^6 - 1)}{3 - 1} = \frac{5(729 - 1)}{2}$$

$$= \frac{5 \times 728}{2} = 5 \times 364$$

$$= 1820$$

$$5 + 15 + 45 + \dots \text{ upto 6 terms} = 1820$$

48. Find the rational form of the number $0.\overline{123}$.

solution

$$0.\overline{123} = 0.123123123\dots$$

$$= 0.123 + 0.000123 + 0.000000123 + \dots$$

$$\frac{t_2}{t_1} = \frac{0.000123}{0.123} = \frac{0.123}{123} = 0.001$$

$\therefore$ It is a G.P. $a = 0.123$ and $r = 0.001$

$$\text{Sum of infinity} = \frac{a}{1 - r}$$

$$= \frac{0.123}{1 - 0.001} = \frac{0.123}{1 - \frac{1}{1000}}$$

$$= \frac{0.123}{\frac{999}{1000}} = \frac{123}{999}$$

$$\therefore 0.\overline{123} = \frac{41}{333}$$

49. Find the sum of (i) $1 + 3 + 5 + \dots$ to 40 terms (ii) $2 + 4 + 6 + \dots + 80$ (iii) $1 + 3 + 5 + \dots + 55$

PTA-6

solution

Sum of Odd Number Formula $1 + 3 + 5 + \dots + n = n^2$

$$(i) \quad 1 + 3 + 5 + \dots + 40 \text{ terms} = 40^2 = 1600$$

$$(ii) \quad 2 + 4 + 6 + \dots + 80$$

$$= 2(1 + 2 + 3 + \dots + 40)$$

$$= 2 \times \frac{40 \times (40 + 1)}{2} = 1640$$

$$(iii) \quad 1 + 3 + 5 + \dots + 55$$

$$\begin{aligned}\text{Formula, } n &= \frac{(l-a)}{d} + 1 \\ n &= \frac{(55-1)}{2} + 1 = 28. \\ 1 + 3 + 5 + \dots + 55 \\ &= (28)^2 = 784.\end{aligned}$$

50. If $1 + 2 + 3 + \dots + n = 666$ then find n .

PTA-2

solution

$$\begin{aligned}1 + 2 + 3 + \dots + n &= \frac{n(n+1)}{2}, \\ \frac{n(n+1)}{2} &= 666 \\ n^2 + n - 1332 &= 0 \\ (n+37)(n-36) &= 0 \\ n &= -37 \text{ or } n = 36 \\ n &\neq -37 \\ n &= 36.\end{aligned}$$

51. If $1 + 2 + 3 + \dots + k = 325$, then find $1^3 + 2^3 + 3^3 + \dots + k^3$.

solution

$$\begin{aligned}\text{Given } 1 + 2 + 3 + \dots + k &= 325 \\ \frac{k(k+1)}{2} &= 325 \\ \therefore 1^3 + 2^3 + 3^3 + \dots + k^3 \\ &= \left(\frac{k(k+1)}{2}\right)^2 \\ &= (325)^2 \\ &= 105625\end{aligned}$$

52. If $1^3 + 2^3 + 3^3 + \dots + k^3 = 44100$ then find $1 + 2 + 3 + \dots + k$.

solution

$$\begin{aligned}\text{Given } 1^3 + 2^3 + 3^3 + \dots + k^3 &= 44100 \\ \Rightarrow \left(\frac{k(k+1)}{2}\right)^2 &= 44100 \\ \Rightarrow \frac{k(k+1)}{2} &= \sqrt{44100} = 210 \\ \Rightarrow 1 + 2 + 3 + \dots + k &= 210\end{aligned}$$

53. Prove that $n^2 - n$ divisible by 2 for every positive integer n .

solution

Any positive integer is of the form $2q$ (or) $2q + 1$ for some integer q .

$$\begin{aligned}\text{i) } n^2 - n &= (2q)^2 - 2q \\ &= 2q(2q - 1) \\ &\text{which is divisible by 2.} \\ \text{ii) } n^2 - n &= (2q + 1)^2 - (2q + 1) \\ &= (2q + 1)(2q + 1 - 1) \\ &= 2q(2q + 1), \text{ which is divisible by 2.} \\ &\text{Hence proved.}\end{aligned}$$

54. Find the 12th term from the last term of the A.P $-2, -4, -6, \dots -100$.

solution

$$\begin{aligned}\text{Given A.P is } &-2, -4, -6, \dots, -100 \\ \text{To find : } t_{12} &\text{ from the last term} \\ a &= -100, \quad d = 2 \\ t_n &= a + (n-1)d \\ t_{12} &= a + 11d \\ &= -100 + 11(2) \\ &= -100 + 22 \\ t_{12} &= -78\end{aligned}$$

55. Show that 107 is of the form $4q + 3$ for any integer q .

solution

Using Euclids Lemma 107 is divided by 4,

$$107 = 4(26) + 3$$

This is of the form

$$a = bq + r$$

$$107 = 4q + 3 \text{ for } q = 26.$$

CHAPTER - 2. NUMBERS AND SEQUENCES

5 MARK IMPORTANT QUESTIONS-2026

1. Find the HCF of 396, 504, 636. **JUL-21, 25**

solution

To find H.C.F. of 396 and 504

Using Euclid's division algorithm

$$504 = 396 \times 1 + 108$$

The remainder is $108 \neq 0$

Again applying Euclid's division algorithm

$$396 = 108 \times 3 + 72$$

The remainder is $72 \neq 0$

Again applying Euclid's division algorithm

$$108 = 72 \times 1 + 36$$

The remainder is $36 \neq 0$

Again applying Euclid's division algorithm

$$72 = 36 \times 2 + 0$$

The remainder is zero.H.C.F.

of 396, 504 = 36. To find the H.C.F. of 636 and 36.

Using Euclid's division algorithm

$$636 = 36 \times 17 + 24$$

The remainder is $24 \neq 0$

Again Applying Euclid's algorithm

$$36 = 24 \times 1 + 12$$

The remainder is $12 \neq 0$

Again applying Euclid's division algorithm $24 = 12 \times 2 + 0$

The remainder is zero.H.C.F. of 636, 36 = 12

Highest Common Factor of 396, 504 and 636 is 12.

2. Use Euclid's Division Algorithm to find the Highest Common Factor (HCF) of

(i) 340 and 412

(ii) 867 and 255

(iii) 10224 and 9648

(iv) 84, 90 and 120

solution

(i) To find the H.C.F. of (340, 412),

$$412 > 340 = \text{H.C.F. (412, 340)}$$

Using Euclid's division algorithm

$$412 = 340 \times 1 + 72$$

The remainder $72 \neq 0$

Again applying Euclid's division algorithm

$$340 = 72 \times 4 + 52$$

The remainder $52 \neq 0$

Again applying Euclid's division algorithm

$$72 = 52 \times 1 + 20$$

The remainder $20 \neq 0$

Again applying Euclid's division algorithm

$$52 = 20 \times 2 + 12$$

The remainder $12 \neq 0$

Again applying Euclid's division algorithm

$$20 = 12 \times 1 + 8$$

The remainder $8 \neq 0$
 Again applying Euclid's Division Algorithm
 $12 = 8(1) + 4$
 The remainder $4 \neq 0$,
 Again applying Euclid's Division Algorithm
 $8 = 4(2) + 0$

The remainder $= 0$
 $\therefore$ H.C.F. (340, 412) = 4

(ii) 867 and $255, 867 > 255$

Applying repeatedly Euclid's Division
 $867 = 255(3) + 102$
 $255 = 102(2) + 51$
 $102 = 51(2) + 0$

The remainder $= 0$
 H.C.F. (867, 255) = 51

(iii) 10224 and 9648
 $10224 > 9648$

Applying repeatedly Euclid's Division
 $10224 = 9648(1) + 576$
 $9648 = 576(16) + 432$
 $576 = 432(1) + 144$
 $432 = 144(3) + 0$

The remainder $= 0$
 $\therefore$ H.C.F. (10224, 9648) = 144

(iv) $84, 90$ and 120
 Find the H.C.F. of 84 and 90 .

H.C.F. (90, 84)
 Applying Euclid's Division Algorithm
 $90 = 84(1) + 6$
 $84 = 6(14) + 0$
 Remainder $= 0$
 $\therefore$ H.C.F. (90, 84) = 6
 Finding H.C.F. (120, 6)
 $120 = 6(20) + 0$
 Remainder $= 0$
 $\therefore$ H.C.F. is 6.
 H.C.F. of $84, 90$ and 120 is 6.

3. If $p_1^{x_1} \times p_2^{x_2} \times p_3^{x_3} \times p_4^{x_4} = 113400$ where p_1, p_2, p_3, p_4 are primes in ascending order and x_1, x_2, x_3, x_4 are integers, find the value of p_1, p_2, p_3, p_4 and x_1, x_2, x_3, x_4 . **APR-25**

solution

2	113400
2	56700
2	28350
3	14175
3	4725
3	1575
3	525
5	175
5	35
7	7
	1

The number 113400 factorized

$$113400 = 2^3 \times 3^4 \times 5^2 \times 7^1$$

2, 3, 5, 7 are primes in Ascending Order.

$$p_1^{x_1} \times p_2^{x_2} \times p_3^{x_3} \times p_4^{x_4} = 2^3 \times 3^4 \times 5^2 \times 7^1$$

$p_1 = 2$;	$p_2 = 3$;	$p_3 = 5$;	$p_4 = 7$
$x_1 = 3$;	$x_2 = 4$;	$x_3 = 2$;	$x_4 = 1$.

4. Find the greatest number consisting of 6 digits which is exactly divisible by 24, 15, 36 ?

solution

$$\text{L.C.M.} = 3 \times 2 \times 2 \times 2 \times 5 \times 3 = 360$$

Greatest number of 6 digit is 999999

$$\text{L.C.M. of } 24, 15 \text{ and } 36 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 360$$

360	2777
	999999
	720
3	24,15,36
2	8,5,12
2	4,5,6
	2,5,3
	2799
	2520
	2799
	2520
	279

Dividing 999999 by 360 remainder is 279.

$\therefore$ Greatest number of 6 digit, divisible by 24, 15

$$36 = 999999 - 279 = 999720$$

The required number is $= 999720$

5. Kala and Vani are friends. Kala says, "Today is my birthday" and she asks Vani, "When will you celebrate your birthday?" Vani replies, "Today is Monday and I celebrated my birthday 75 days ago". Find the day when Vani celebrated her birthday.

solution

Let the numbers 0, 1, 2, 3, 4, 5, 6 to represent from Sunday to Saturday
 Vani says today is Monday.

Monday is 1. Since Vani's birthday was 75 days ago

Modulo 7, since a week contain 7 days.

$$-74 \pmod{7} \equiv -4 \pmod{7}$$

$$\equiv 7 - 4 \pmod{7} \equiv 3 \pmod{7}$$

(Since, $-74 - 3 = -77$ is divisible by 7)

$$1 - 75 \equiv 3 \pmod{7}$$

The day for the number 3 is Wednesday.

Vani's birthday on Wednesday.

6. Prove that $2^n + 6 \times 9^n$ is always divisible by 7 for any positive integer n .

solution

7 is divisible by 7

$$\therefore 7 \equiv 0 \pmod{7}$$

$$7 + 2 \equiv 0 + 2 \pmod{7}$$

[$\therefore$ Adding the constant 2]

$$9 \equiv 2 \pmod{7}$$

$$9^n \equiv 2^n \pmod{7}$$

$$6 \times 9^n \equiv 6 \times 2^n \pmod{7}$$

[$\therefore$ multiplying by the constant 6]

$$2^n + 6 \times 9^n \equiv 2^n + 6 \times 2^n \pmod{7}$$

[$\therefore$ Adding 2^n]

$$\equiv 2^n(1 + 6) \pmod{7}$$

$$\equiv 2^n \times 7 \pmod{7}$$

$$\equiv 2^n \times 0 \pmod{7}$$

$$\equiv 0 \pmod{7}$$

$\therefore 2^n + 6 \times 9^n$ is divided by 7

$2^n + 6 \times 9^n$ is always divisible by 7 any positive integer n .

7. Find the remainder when 2^{81} is divided by 17.

solution

$$2^5 \equiv 15 \pmod{17}$$

$$(2^5)^2 \equiv 15^2 \pmod{17}$$

$$\equiv 4 \pmod{17}$$

$$2^{10} \equiv 4 \pmod{17}$$

$$(2^{10})^4 \equiv 4^4 \pmod{17}$$

$$2^{40} \equiv 1 \pmod{17}$$

$$\begin{aligned}(2^{40})^2 &\equiv 1^2 \pmod{17} \\ 2 \cdot 2^{80} &\equiv 2 \times 1 \pmod{17} \\ 2^{81} &\equiv 2 \pmod{17}\end{aligned}$$

Remainder 2^{81} is divided by 17 is 2

8. The duration of flight travel from Chennai to London through British Airlines is approximately 11 hours. The airplane begins its journey on Sunday at 23:30 hours. If the time at Chennai is four and half hours ahead to that of London's time, then find the time at London, when will the flight lands at London Airport.

solution

Starting time from Chennai = 23.30 hrs

Travelling time = 11 hrs

We use modulo 24.

$$\begin{aligned}\therefore \text{Reaching time} &= 23.30 + 11 \pmod{24} \\ &= 34.30 \pmod{24} \\ &= 10.30 \pmod{24}\end{aligned}$$

$$\text{Since } 11 = \underset{\substack{\downarrow \\ \text{Sunday}}}{0} \times 24 + \underset{\substack{\downarrow \\ \text{Monday}}}{11}$$

It reaches London on Monday at 10.30 a.m

Chennai time = 4.30 hrs + London time

$$\begin{aligned}\text{London time} &= \text{Chennai time} - 4.30 \text{ a.m} \\ &= 10.30 - 4.30 = 6 \text{ a.m}\end{aligned}$$

$\therefore$ The flight will land at London Airport on Monday at 6 a.m

9. Find the first five terms of the following sequence.

$$a_1 = 1, a_2 = 1, a_n = \frac{a_{n-1}}{a_{n-2} + 3}; n \geq 3, n \in \mathbb{N}$$

solution

$$\text{Given } a_1 = 1, a_2 = 1, a_n = \frac{a_{n-1}}{a_{n-2} + 3}; n \geq 3, n \in \mathbb{N}$$

$$a_3 = \frac{a_{3-1}}{a_{3-2} + 3} = \frac{a_2}{a_1 + 3} = \frac{1}{1 + 3} = \frac{1}{4}$$

The fourth term a_4 n a_2 and a_3 .

$$a_4 = \frac{a_{4-1}}{a_{4-2} + 3} = \frac{a_3}{a_2 + 3} = \frac{\frac{1}{4}}{1 + 3} = \frac{\frac{1}{4}}{4} = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

The fifth term a_5 is

$$a_5 = \frac{a_{5-1}}{a_{5-2} + 3} = \frac{a_4}{a_3 + 3} = \frac{\frac{1}{16}}{\frac{1}{4} + 3} = \frac{1}{16} \times \frac{4}{13} = \frac{1}{52}$$

The first five terms of the sequence are

$$1, 1, \frac{1}{4}, \frac{1}{16}, \frac{1}{52}$$

10. If l^{th} , m^{th} and n^{th} terms of an A.P. are x, y, z respectively, then show that (i) $x(m - n) + y(n - l) + z(l - m) = 0$

$$(ii) (x - y)n + (y - z)l + (z - x)m = 0 \quad \text{APR-22}$$

solution

(i) Let a be the first term and d be the common difference

$$t_l = x, t_m = y, t_n = z$$

$$a + (l - 1)d = x \quad \dots(1)$$

$$a + (m - 1)d = y \quad \dots(2)$$

$$a + (n - 1)d = z \quad \dots(3)$$

$$\begin{aligned}x(m - n) + y(n - l) + z(l - m) &= a[(m - n) + (n - l) + (l - m)] \\ &\quad + d[(l - 1)(m - n) + (m - 1)(n - l) + (n - 1)(l - m)] \\ &= a[0] + d[lm - ln - m + n + mn - lm - n + l + ln - mn - l + m] \\ &= a(0) + d(0) = 0\end{aligned}$$

$$(ii) x - y = (l - m)d$$

$$y - z = (m - n)d$$

$$z - x = (n - l)d$$

$$(x - y)n + (y - z)l + (z - x)m$$

$$= [(l - m)n + (m - n)l + (n - l)m]d$$

$$= [ln - mn + lm - nl + nm - lm]d$$

$$(x - y)n + (y - z)l + (z - x)m = 0$$

11. In an A.P., sum of four consecutive terms is 28 and the sum of their squares is 276. Find the four numbers.

solution

The four terms in the form $(a - 3d), (a - d), (a + d)$ and $(a + 3d)$.

Sum of the four terms is 28,

$$a - 3d + a - d + a + d + a + 3d = 28$$

$$4a = 28 \implies a = 7$$

since sum of their squares is 276,

$$(a - 3d)^2 + (a - d)^2 + (a + d)^2 + (a + 3d)^2 = 276.$$

$$\begin{aligned}a^2 - 6ad + 9d^2 + a^2 - 2ad + d^2 + a^2 + 2ad \\ + d^2 + a^2 + 6ad + 9d^2 = 276\end{aligned}$$

$$4a^2 + 20d^2 = 276 \implies 4(7)^2 + 20d^2 = 276.$$

$$d^2 = 4 \implies d = \pm 2$$

If $d = 2$ then the four numbers are $7 - 3(2),$

$$7 - 2, 7 + 2, 7 + 3(2)$$

The four numbers are $\boxed{1, 5, 9, 13}$

If $a = 7, d = -2$ then the four numbers are $\boxed{13, 9, 5, 1}$

12. A mother divides ₹207 into three parts such that the amount are in A.P. and gives it to her three children. The product of the two least amounts that the children had ₹4623. Find the amount received by each child.

solution

The three Children be in the form of A.P.

$$a - d, a, a + d.$$

Since, sum of the amount is ₹207

$$(a - d) + a + (a + d) = 307$$

$$3a = 207 \implies \boxed{a = 69}$$

It is given that product of the two least amounts is 4623

$$(a - d)a = 4623$$

$$(69 - d)69 = 4623$$

$$\boxed{d = 2}$$

Amount given by the mother to three children are

$$\text{₹}(69 - 2), \text{₹}69, \text{₹}(69 + 2). \text{₹}67, \text{₹}69, \text{₹}71.$$

13. If nine times ninth term is equal to the fifteen times fifteenth term, show that six times twenty fourth term is zero. **JUL-22**

solution

$$\text{Given } 9(t_9) = 15(t_{15})$$

$$\text{To Prove : } 6(t_{24}) = 0$$

$$9(t_9) = 15(t_{15})$$

$$9(a + 8d) = 15(a + 14d)$$

$$3(a + 8d) = 5(a + 14d)$$

$$3a + 24d = 5a + 70d$$

$$2a + 46d = 0$$

$$2(a + 23d) = 0$$

Multiplying 3 on both sides,

$$6(a + 23d) = 0$$

$$6(t_{24}) = 0$$

Hence proved.

14. Find x, y and z , given that the numbers $x, 10, y, 24, z$ are in A.P.

solution

$$\text{Given } t_1 = x, t_2 = 10, t_3 = y, t_4 = 24, t_5 = z.$$

$x, 10, y, 24, z$ are in A.P.

Then difference between consecutive terms are equal.

$$t_2 - t_1 = t_3 - t_2 = t_4 - t_3$$

$$10 - x = 24 - y = \text{common difference 'd'}$$

$$t_4 - t_2 = 2d$$

$$24 - 10 = 2d$$

$$14 = 2d$$

$$\boxed{d = \frac{14}{2} = 7}$$

$$10 - x = 24 - y = 7$$

$$\text{If } 10 - x = 7; \quad x = 10 - 7 = 3$$

$$\text{If } 24 - y = 7; \quad y = 24 - 7 = 17$$

$$z - 24 = 7; \quad z = 24 + 7 = 31$$

$$\therefore \boxed{x = 3; y = 17; z = 31}$$

15. The sum of three consecutive terms that are in A.P. is 27 and their product is 288. Find the three terms.

solution

Let the 3 consecutive terms in an A.P. be

$$a - d, a, a + d$$

$$\text{i) Sum of 3 terms} = 27$$

$$a - d + a + a + d = 27$$

$$3a = 27$$

$$\boxed{a = 9}$$

$$\text{ii) Product of 3 terms} = 288$$

$$(a - d).a.(a + d) = 288$$

$$a(a^2 - d^2) = 288$$

$$9(81 - d^2) = 288$$

$$81 - d^2 = 32$$

$$d^2 = 49$$

$$\boxed{d = \pm 7}$$

$$\text{If } a = 9, d = 7 \implies \text{The Three terms are } \boxed{2, 9, 16}$$

$$\text{If } a = 9, d = -7 \implies \text{The Three terms are } \boxed{16, 9, 2}$$

16. The ratio of 6th and 8th term of an A.P. is 7:9. Find the ratio of 9th term to 13th term. **APR-22, MDL-19**

solution

$$\text{Given } \frac{t_6}{t_8} = \frac{7}{9}$$

$$\frac{a + 5d}{a + 7d} = \frac{7}{9}$$

$$9a + 45d = 7a + 49d$$

$$2a = 4d$$

$$\boxed{a = 2d} \quad \dots(1)$$

$$\frac{t_9}{t_{13}} = \frac{a + 8d}{a + 12d}$$

$$= \frac{2d + 8d}{2d + 12d} \quad (\text{from (1)})$$

$$= \frac{10d}{14d}$$

$$= \frac{5}{7}$$

$$\therefore \boxed{t_9 : t_{13} = 5 : 7}$$

17. The 13th term of an A.P. is 3 and the sum of first 13 terms is 234. Find the common difference and the sum of first 21 terms.

solution

$$\text{Given } 13^{\text{th}} \text{ term} = 3$$

$$t_{13} = a + 12d = 3 \quad \dots(1)$$

$$\text{Sum of first 13 terms} = 234$$

$$S_{13} = \frac{13}{2}[2a + 12d] = 234$$

$$2a + 12d = 36 \quad \dots(2)$$

$$\text{Solve (1) and (2) } a = 33, d = \frac{-5}{2}$$

$$\text{Common difference is } \frac{-5}{2}.$$

Sum of first 21 terms

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$S_{21} = \frac{21}{2} \left[2 \times 33 + (21 - 1) \times \left(-\frac{5}{2} \right) \right]$$

$$= \frac{21}{2}[66 - 50]$$

$$\boxed{S_{21} = 168}$$

18. Find the sum of all natural numbers between 300 and 600 which are divisible by 7. **MAR-26**

solution

The natural numbers between 300 and 600

which are divisible by 7 are 301, 308, 315, ..., 595.

The sum of all natural numbers between 300 and 600

$$301 + 308 + 315 + \dots + 595.$$

The terms of the above series are in A.P.

First term $a = 301$; common difference $\boxed{d = 7}$; Last term $l = 595$.

$$n = \left(\frac{l - a}{d} \right) + 1$$

$$= \left(\frac{595 - 301}{7} \right) + 1 = \left(\frac{294}{7} \right) + 1 = 42 + 1$$

$$\boxed{n = 43}$$

$$S_n = \frac{n}{2}[a + l],$$

$$S_{57} = \frac{43}{2}[301 + 595] = \frac{43}{2}[896] = 19264.$$

19. The houses of a street are numbered from 1 to 49. Senthil's house is numbered such that the sum of numbers of the houses prior to Senthil's house is equal to the sum of numbers of the houses following Senthil's house. Find Senthil's house number? **APR-23, PTA-2**

solution

Let Senthil's house number be x . **Hints:** $\boxed{S_{x-1} = S_{49} - S_x}$

$$\text{Given that } 1 + 2 + 3 + \dots + (x - 1) = (x + 1) + (x + 2) + \dots + 49$$

$$1 + 2 + 3 + \dots + (x - 1) = [1 + 2 + 3 + \dots + 49] - [1 + 2 + 3 + \dots + x]$$

$$\frac{x - 1}{2}[1 + (x - 1)] = \frac{49}{2}[1 + 49] - \frac{x}{2}[1 + x]$$

$$\frac{x(x - 1)}{2} = \frac{49 \times 50}{2} - \frac{x(x + 1)}{2}$$

$$\frac{x^2 - x + x^2 + x}{2} = 49 \times 25 \implies x^2 = 49 \times 25$$

$$x^2 = \sqrt{49 \times 25} \implies \boxed{x = 7 \times 5 = 35}$$

Senthil's House number is 35.

20. The sum of first n , $2n$ and $3n$ terms of an A.P. are S_1 , S_2 and S_3 respectively. Prove that $S_3 = 3(S_2 - S_1)$. **MDL-19**

solution

If S_1 , S_2 and S_3 are sum of first n , $2n$, $3n$ terms of an A.P

$$S_1 = \frac{n}{2}[2a + (n - 1)d], \quad S_2 = \frac{2n}{2}[2a + (2n - 1)d],$$

$$S_3 = \frac{3n}{2}[2a + (3n - 1)d]$$

$$S_2 - S_1 = \frac{2n}{2}[2a + (2n - 1)d] - \frac{n}{2}[2a + (n - 1)d]$$

$$= \frac{n}{2}[(4a + 2(2n - 1)d] - [2a + (n - 1)d]$$

$$S_2 - S_1 = \frac{n}{2} \times [2a + (3n - 1)d]$$

$$3(S_2 - S_1) = \frac{3n}{2}[2a + (3n - 1)d]$$

$$3(S_2 - S_1) = S_3.$$

Hence Proved

21. The 104th term and 4th term of an A.P. are 125 and 0. Find the sum of first 35 terms.

solution

$$\text{Given: } t_{104} = 125, \quad t_4 = 0$$

$$\text{To find: } S_{35}$$

$$a + 103d = 125$$

$$a + 3d = 0$$

$$100d = 125$$

$$d = \frac{5}{4}$$

$$a + 3 \left[\frac{5}{4} \right] = 0, \quad a + \frac{15}{4} = 0 \quad \boxed{a = -\frac{15}{4}}$$

$$\therefore S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{35} = \frac{35}{2} \left[-\frac{15}{2} + (34) \left(\frac{5}{4} \right) \right]$$

$$= \frac{35}{2} \left[-\frac{15}{2} + \frac{85}{2} \right]$$

$$= \frac{35}{2} \times 35$$

$$= 35 \times 17.5$$

$$\boxed{S_{35} = 612.5}$$

22. A man repays a loan of ₹65,000 by paying ₹400 in the first month and then increasing the payment by ₹300 every month. How long will it take for him to clear the loan?

solution

Amounts of repayment in successive months

$$400 + 700 + 1000 + \dots n \text{ months} = ₹65,000$$

$$a = 400, d = 300, S_n = 65,000$$

$$\frac{n}{2} [2a + (n-1)d] = 65,000$$

$$\frac{n}{2} [800 + (n-1)300] = 65,000$$

$$n[400 + (n-1)150] = 65,000$$

$$n[150n + 250] = 65,000$$

$$n[3n + 5] = 1,300$$

$$3n^2 + 5n - 1300 = 0$$

$$n = 20, -\frac{65}{3}$$

$$\begin{array}{c|c} 5 & -3900 \\ -65 & -60 \\ \hline 3 & 3 \end{array} \Rightarrow \frac{65}{3}, -20 \quad \boxed{\therefore n = 20}$$

A man take 20 months to clear the Loan.

23. If $S_1, S_2, S_3, \dots, S_m$ are the sums of n terms of m A.P.'s whose first terms are $1, 2, 3, \dots, m$ and whose common differences are $1, 3, 5, \dots, (2m-1)$ respectively, then Show that $S_1 + S_2 + S_3 + \dots + S_m = \frac{1}{2}mn(mn+1)$

solution

$$\text{First A.P. is } \Rightarrow a = 1, d = 1$$

$$S_1 = \frac{n}{2} [2 + (n-1)1]$$

$$= \frac{n}{2} [n+1]$$

$$\text{Second A.P. is } \Rightarrow a = 2, d = 3$$

$$S_2 = \frac{n}{2} [4 + (n-1)3]$$

$$= \frac{n}{2} [3n+1]$$

$$m^{\text{th}} \text{ A.P. is } \Rightarrow a = m, d = 2m-1$$

$$S_m = \frac{n}{2} [2m + (n-1)(2m-1)]$$

$$= \frac{n}{2} [2m + 2mn - 2m - n + 1]$$

$$= \frac{n}{2} [2mn - n + 1]$$

$$= \frac{n}{2} [(2m-1)n + 1]$$

$$\begin{aligned} S_1 + S_2 + S_3 + \dots + S_m &= \frac{n}{2}(n+1) + \frac{n}{2}(3n+1) + \dots \\ &\quad + \frac{n}{2}((2m-1)n+1) \\ &= \frac{n}{2}[(n+3n+\dots+(2m-1)n) \\ &\quad + (1+1+\dots m \text{ terms})] \\ &= \frac{n}{2}[(n(1+3+5+\dots+(2m-1)) + m)] \\ &= \frac{n}{2}[(n(m^2)) + m] = \frac{n}{2}[m(mn+1)] \\ &= \frac{1}{2}mn(mn+1) \\ &= \text{RHS} \\ &\text{Hence Proved} \end{aligned}$$

24. Find the sum $\left[\frac{a-b}{a+b} + \frac{3a-2b}{a+b} + \frac{5a-3b}{a+b} + \dots \text{ to 12 terms} \right]$.

solution

Given series is

$$\frac{a-b}{a+b} + \frac{3a-2b}{a+b} + \frac{5a-3b}{a+b} + \dots \text{ to 12 terms}$$

$$\text{First term} = \frac{a-b}{a+b}, \quad \text{Common difference} = \frac{2a-b}{a+b}$$

$$S_{12} = \frac{12}{2} \left[2 \left(\frac{a-b}{a+b} \right) + 11 \left(\frac{2a-b}{a+b} \right) \right] \quad (\because S_n = \frac{n}{2} [2a + (n-1)d])$$

$$= 6 \left[\frac{2a-2b+22a-11b}{a+b} \right]$$

$$= 6 \left[\frac{24a-13b}{a+b} \right]$$

$$S_{12} = \frac{6}{a+b} [24a-13b]$$

25. In a Geometric progression, the 4th term is $\frac{8}{9}$ and the 7th term is $\frac{64}{243}$. Find the Geometric Progression.

solution

$$\text{Fourth term, } t_4 = \frac{8}{9} \text{ gives } ar^3 = \frac{8}{9} \quad \dots(1)$$

$$\text{Seventh term, } t_7 = \frac{64}{243} \text{ gives } ar^6 = \frac{64}{243} \quad \dots(2)$$

$$\text{Divide (2) by (1)} \frac{ar^6}{ar^3} = \frac{\frac{64}{243}}{\frac{8}{9}}$$

$$r^3 = \frac{8}{27} \Rightarrow \boxed{r = \frac{2}{3}}$$

Sub the value of r in (1)

$$a \times \left[\frac{2}{3} \right]^3 = \frac{8}{9} \Rightarrow \boxed{a = 3}$$

The Geometric Progression is

$$a, ar, ar^2, \dots \Rightarrow 3, 2, \frac{4}{3}, \dots$$

26. The product of three consecutive terms of a Geometric Progression is 343 and their sum is $\frac{91}{3}$. Find the three terms.

solution

$$\text{Since the product of 3 consecutive terms is } \frac{a}{r}, a, ar.$$

$$\text{Product of the terms} = 343$$

$$\frac{a}{r} \times a \times ar = 343$$

$$a^3 = 7^3 \Rightarrow \boxed{a = 7}$$

$$\text{Sum of the terms} = \frac{91}{3}$$

$$7 \left(\frac{1+r+r^2}{r} \right) = \frac{91}{3}$$

$$3 + 3r + 3r^2 = 13r \Rightarrow 3r^2 - 10r + 3 = 0$$

$$(3r-1)(r-3) = 0 \Rightarrow \boxed{r = 3} \text{ or } r = \frac{1}{3}$$

$$\text{If } a = 7, r = 3 \text{ The three terms are } \boxed{\frac{7}{3}, 7, 21}.$$

$$\text{If } a = 7, r = \frac{1}{3} \text{ The three terms are } \boxed{21, 7, \frac{7}{3}}.$$

27. In a G.P. the 9th term is 32805 and 6th term is 1215. Find the 12th term.

solution

Given $t_9 = 32805$, $t_6 = 1215$, $t_{12} = ?$

$$a \cdot r^8 = 32805 \quad \dots\dots\dots(1)$$

$$a \cdot r^5 = 1215 \quad \dots\dots\dots(2)$$

$$(1) \div (2) \Rightarrow r^3 = \frac{32805}{1215}$$

$$\Rightarrow r^3 = 27$$

$$\Rightarrow \boxed{r = 3}$$

Sub. $r = 3$ in (2)

$$a \times 3^5 = 1215$$

$$a \times 243 = 1215$$

$$a = \frac{1215}{243}$$

$$\boxed{a = 5}$$

$$\therefore t_{12} = a \cdot r^{11} = 5 \times 3^{11}$$

28. In a G.P. the product of three consecutive terms is 27 and the sum of the product of two terms taken at a time is $\frac{57}{2}$. Find the three terms.

solution

Let the 3 consecutive terms of a G.P be $\frac{a}{r}, a, ar$

i) Product of three terms = 27

$$\frac{a}{r} \times a \times ar = 27$$

$$a^3 = 27$$

$$\therefore \boxed{a = 3}$$

ii) Sum of product of terms taken 2 at a time

$$= \frac{57}{2}$$

$$\frac{a}{r} \times a + a \times ar + ar \times \frac{a}{r} = \frac{57}{2}$$

$$a^2 \left[\frac{1}{r} + r + 1 \right] = \frac{57}{2}$$

$$9 \left[\frac{1+r^2+r}{r} \right] = \frac{57}{2}$$

$$\frac{1+r+r^2}{r} = \frac{57^{19}}{9 \times 2}$$

$$\frac{1+r+r^2}{r} = \frac{19}{6}$$

$$6r^2 + 6r + 6 = 19r$$

$$6r^2 - 13r + 6 = 0$$

$$r = \frac{3}{2}, \frac{2}{3}$$

$$\therefore a = 3, r = \frac{3}{2} \Rightarrow 3 \text{ terms are } \boxed{2, 3, \frac{9}{2}}$$

$$\therefore a = 3, r = \frac{2}{3} \Rightarrow 3 \text{ terms are } \boxed{\frac{9}{2}, 3, 2}$$

29. A man joined a company as Assistant Manager. The company gave him a starting salary of ₹60,000 and agreed to increase his salary 5% annually. What will be his salary after 5 years? **PTA-6**

solution

Initial salary for the first year = ₹60000

At the end of first year salary = $60000 \times \left(\frac{105}{100}\right)$

At the end of second year salary

$$= (60000) \left(\frac{105}{100}\right)^2$$

Continuing this way the salary increase will be

$$60000, 60000 \left(\frac{105}{100}\right), 60000 \left(\frac{105}{100}\right)^2 \dots$$

This form a G.P. with $a = 60000$ and $r = \frac{105}{100}$

Salary after 5 years (6th year) = $ar^{n-1} = ar^{6-1} = ar^5$

$$= 60000 \left(\frac{105}{100}\right)^5$$

$$= 60000 \times \frac{105}{100} \times \frac{105}{100} \times \frac{105}{100} \times \frac{105}{100} \times \frac{105}{100}$$

$\therefore$ After 5 years his salary = ₹76577

30. If a, b, c are three consecutive terms of an A.P. and x, y, z are three consecutive terms of a

G.P. then prove that $x^{b-c} \times y^{c-a} \times z^{a-b} = 1$.

solution

Given a, b, c are in A.P.

$$b - a = c - b$$

$$b + b = c + a$$

$$2b = c + a \quad \dots(1)$$

x, y, z are three consecutive terms of a G.P.

$$\therefore x = a$$

$$y = ar$$

$$z = ar^2$$

$$x^{b-c} \times y^{c-a} \times z^{a-b} = a^{b-c} \times (ar)^{c-a} \times (ar^2)^{a-b}$$

$$= a^{b-c} \times a^{c-a} \times r^{c-a} \times a^{a-b} \times r^{2(a-b)}$$

$$= a^{b-c+c-a+a-b} \times r^{c-a+2a-2b}$$

$$= a^0 \times r^{(a+c)-2b}$$

$$= 1 \times r^{2b-2b} \quad [\text{from (1) } a + c = 2b]$$

$$= 1 \times r^0$$

$$= 1 \times 1 = 1$$

$$= \text{RHS}$$

$$\therefore x^{b-c} \times y^{c-a} \times z^{a-b} = 1$$

Hence proved.

31. Find the sum to n terms of the series $5 + 55 + 555 + \dots$ **APR-23, PTA-4**

solution

$$5 + 55 + 555 + \dots + n \text{ terms} = 5[1 + 11 + 111 + \dots + n \text{ terms}]$$

Multiply and Divide by 9

$$= \frac{5 \times 9}{9} [1 + 11 + 111 + \dots + n \text{ terms}]$$

$$= \frac{5}{9} [9 + 99 + 999 + \dots + n \text{ terms}]$$

$$= \frac{5}{9} [(10 - 1) + (100 - 1) + (1000 - 1) + \dots + n \text{ terms}]$$

$$= \frac{5}{9} [(10 + 100 + 1000 + \dots + n \text{ terms}) - (1 + 1 + \dots + n \text{ terms})]$$

$$S_n = \frac{a(r^n - 1)}{r - 1}, a = 10, r = \frac{100}{10} = 10$$

$$S_n = na, a = 1, r = 1$$

$$= \frac{5}{9} \left[\frac{10(10^n - 1)}{(10 - 1)} - n \right]$$

$$5 + 55 + 555 + \dots = \frac{50(10^n - 1)}{81} - \frac{5n}{9}$$

32. Find the sum to n terms of the series

(i) $0.4 + 0.44 + 0.444 + \dots$ to n terms

(ii) $3 + 33 + 333 + \dots$ to n terms **JUL-23, 24, MAR-26**

solution

i) $0.4 + 0.44 + 0.444 + \dots$ to n terms

$$= \frac{4}{10} + \frac{44}{100} + \frac{444}{1000} + \dots \text{ to } n \text{ terms}$$

Multiply and Divide by 9

$$= \frac{4 \times 9}{9} \left[\frac{1}{10} + \frac{11}{100} + \frac{111}{1000} + \dots n \text{ terms} \right]$$

$$= \frac{4}{9} \left[\frac{9}{10} + \frac{99}{100} + \frac{999}{1000} + \dots n \text{ terms} \right]$$

$$= \frac{4}{9} \left[\left(1 - \frac{1}{10}\right) + \left(1 - \frac{1}{100}\right) + \left(1 - \frac{1}{1000}\right) + \dots + n \text{ terms} \right]$$

$$= \frac{4}{9} \left[(1 + 1 + 1 + \dots + n \text{ terms}) - \left(\frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \dots + n \text{ terms} \right) \right]$$

$$S_n = na, a = 1, r = 1$$

$$S_n = \frac{a(1-r^n)}{1-r}, a = \frac{1}{10}, r = \frac{100}{10} = \frac{1}{10}$$

$$= \frac{4}{9} \left[n - \frac{1}{10} \left(\frac{1 - \left(\frac{1}{10}\right)^n}{1 - \frac{1}{10}} \right) \right]$$

$$0.4 + 0.44 + 0.444 + \dots = \frac{4}{9} \left[n - \frac{1}{9} \left[1 - \left(\frac{1}{10}\right)^n \right] \right]$$

ii) $3 + 33 + 333 + \dots$ upto n terms

Multiply and Divide by 9

$$= \frac{3 \times 9}{9} (1 + 11 + 111 + \dots + n \text{ terms})$$

$$= \frac{3}{9} (9 + 99 + 999 + \dots + n \text{ terms})$$

$$= \frac{3}{9} [(10 - 1) + (100 - 1) + (1000 - 1) + \dots + n \text{ terms}]$$

$$= \frac{3}{9} [(10 + 100 + 1000 + \dots + n \text{ terms}) - [(1 + 1 + 1 + \dots + n \text{ terms})]]$$

$$S_n = \frac{a(r^n - 1)}{r - 1}, a = 10, r = \frac{100}{10} = 10$$

$$S_n = na, a = 1, r = 1$$

$$= \frac{3}{9} \left[10 \cdot \left(\frac{10^n - 1}{9} \right) - n \right]$$

$$= \frac{30}{81} (10^n - 1) - \frac{3n}{9}$$

$$3 + 33 + 333 + \dots = \frac{10}{27} (10^n - 1) - \frac{n}{3}$$

33. Find the sum of the Geometric series $3 + 6 + 12 + \dots + 1536$. **PTA-3**

solution

$$a = 3, r = \frac{6}{3} = 2$$

n^{th} term of the G.P., $t_n = ar^{n-1}$

$$1536 = 3(2)^{n-1}$$

$$\frac{1536}{3} = 2^{n-1}$$

$$512 = 2^{n-1}$$

$$2^9 = 2^{n-1}$$

$$n - 1 = 9$$

$$n = 9 + 1 = 10$$

$\therefore$ Number of terms in the series $n = 10$.

$\therefore$ Sum upto n terms $S_n = \frac{a(r^n - 1)}{r - 1}$

$$S_{10} = \frac{3(2^{10} - 1)}{2 - 1}$$

$$= \frac{3 \times (1024 - 1)}{1} = 3 \times 1023$$

$$3 + 6 + 12 + \dots + 1536 = 3069.$$

34. If $S_n = (x + y) + (x^2 + xy + y^2) + (x^3 + x^2y + xy^2 + y^3) + \dots + n$ terms then prove that **PTA-1**

$$(x - y)S_n = \left[\frac{x^2(x^n - 1)}{x - 1} - \frac{y^2(y^n - 1)}{y - 1} \right]$$

solution

Given

$$S_n = (x + y) + (x^2 + xy + y^2) + (x^3 + x^2y + xy^2 + y^3) + \dots + n \text{ terms}$$

$$(x - y)S_n = (x - y)(x + y) + (x - y)$$

$$(x^2 + xy + y^2) + (x - y)(x^3 + x^2y + xy^2 + y^3)$$

$$+ \dots + n \text{ terms}$$

$$(x - y)S_n = (x^2 - y^2) + (x^3 - y^3) + (x^4 - y^4)$$

$$+ \dots + n \text{ terms}$$

$$= (x^2 + x^3 + x^4 + \dots + n \text{ terms}) - (y^2 + y^3 + y^4 + \dots + n \text{ terms}),$$

Geometric series is

$$a = x^2, r = x \text{ \& } a = y^2, r = y \quad \left(\because S_n = a \cdot \frac{r^n - 1}{r - 1} \right)$$

$$\therefore (x - y)S_n = \frac{x^2(x^n - 1)}{x - 1} - \frac{y^2(y^n - 1)}{y - 1}$$

Hence proved.

35. Find the sum of (i) $5^2 + 10^2 + 15^2 + \dots + 105^2$ (ii) $15^2 + 16^2 + 17^2 + \dots + 28^2$

solution

$$(ii) \quad 5^2 + 10^2 + 15^2 + \dots + 105^2$$

$$= 5^2(1^2 + 2^2 + 3^2 + \dots + 21^2)$$

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$n = 21$$

$$= 25 \times \frac{25 \times (21 + 1)(2 \times 21 + 1)}{6}$$

$$= \frac{25 \times 21^7 \times 22^{11} \times 43}{6^2} = 25 \times 77 \times 43$$

$$5^2 + 10^2 + \dots + 105^2 = 82775$$

$$(iii) \quad 15^2 + 16^2 + 17^2 + \dots + 28^2$$

$$= (1^2 + 2^2 + 3^2 + \dots + 28^2) - (1^2 + 2^2 + 3^2 + \dots + 14^2)$$

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$n = 28,$$

$$14$$

$$= \frac{28^{14} \times 29 \times 57^{19}}{6^2} - \frac{14^7 \times 15^5 \times 29}{6^2}$$

$$= 29(14 \times 19 - 7 \times 5) = 29(266 - 35)$$

$$= 29 \times 231 = 6699$$

$$15^2 + 16^2 + 17^2 + \dots + 28^2 = 6699$$

36. Find the sum of $9^3 + 10^3 + \dots + 21^3$ **APR-24**

solution

$$9^3 + 10^3 + \dots + 21^3 = (1^3 + 2^3 + 3^3 + \dots + 21^3) - (1^3 + 2^3 + 3^3 + \dots + 8^3)$$

$$1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

$$n = 21, n = 8$$

$$= \left[\frac{21 \times (21 + 1)}{2} \right]^2 - \left[\frac{8 \times (8 + 1)}{2} \right]^2$$

$$= \left[\frac{21 \times 22^{11}}{2} \right]^2 - \left[\frac{8^4 \times 9}{2} \right]^2$$

$$= (231)^2 - (36)^2$$

$$= (231 + 36)(231 - 36) = 267 \times 195$$

$$9^3 + 10^3 + \dots + 21^3 = 52065$$

37. Find the sum of (i) $1 + 4 + 9 + 16 + \dots + 225$ (ii) $6^2 + 7^2 + 8^2 + \dots + 21^2$ **APR-25** (iii) $10^3 + 11^3 + 12^3 + \dots + 20^3$ **PTA-5**

solution

$$(i) \quad 1 + 4 + 9 + 16 + \dots + 225$$

$$= 1^2 + 2^2 + 3^2 + \dots + 15^2$$

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\boxed{n = 15}$$

$$= \frac{15^5 \times 16^8 \times 31}{\cancel{6}}$$

$$1 + 4 + 9 + 16 + \dots + 225 = 40 \times 31 = 1240$$

$$(ii) \quad 6^2 + 7^2 + 8^2 + \dots + 21^2$$

$$= (1^2 + 2^2 + 3^2 + \dots + 21^2) - (1^2 + 2^2 + \dots + 5^2)$$

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\boxed{n = 21}, \boxed{n = 5}$$

$$= \frac{21^7 \times 22^{11} \times 43}{\cancel{6}} - \frac{5 \times \cancel{6} \times 11}{\cancel{6}}$$

$$= 3311 - 55$$

$$6^2 + 7^2 + 8^2 + \dots + 21^2 = 3256$$

$$(iii) \quad 10^3 + 11^3 + 12^3 + \dots + 20^3$$

$$= (1^3 + 2^3 + \dots + 20^3) - (1^3 + 2^3 + \dots + 9^3)$$

$$1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

$$\boxed{n = 20}, \boxed{n = 9}$$

$$= \left(\frac{20^{10} \times 21}{2} \right)^2 - \left(\frac{9 \times 10^5}{2} \right)^2$$

$$= (210)^2 - (45)^2$$

$$= 44100 - 2025$$

$$10^3 + 11^3 + 12^3 + \dots + 20^3 = 42075$$

38. The sum of the cubes of the first n natural numbers is 2025, then find the value of n .

solution

Given sum of the squares of first ' n ' natural numbers = 285

$$\frac{n(n+1)(2n+1)}{6} = 285 \quad \dots\dots\dots(1)$$

Sum of their cubes = 2025

$$\left(\frac{n(n+1)}{2} \right)^2 = 2025$$

$$n \left(\frac{n+1}{2} \right) = 45 \quad \dots\dots\dots(2)$$

Sub (2) in (1)

$$\frac{n(n+1)}{2} \times \frac{2n+1}{3} = 285$$

$$45 \times \frac{2n+1}{3} = 285$$

$$2n+1 = \frac{285}{15} = 19$$

$$2n = 18$$

$$\therefore \boxed{n = 9}$$

39. Rekha has 15 square colour papers of sizes 10 cm, 11 cm, 12 cm, ..., 24 cm. How much area can be decorated with these colour papers?

JUN-23, 24, 25, PTA-1**solution**

Rekha has sides of 15 square Colour papers are 10 cm, 11 cm, 12 cm, 24 cm

$$10^2 + 11^2 + 12^2 + \dots + 24^2 = (1^2 + 2^2 + 3^2 + \dots + 24^2) - (1^2 + 2^2 + \dots + 9^2)$$

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\boxed{n = 24}, \boxed{n = 9}$$

$$= \frac{24^4 \times 25 \times 49}{\cancel{6}} - \frac{9^3 \times 10^5 \times 19}{\cancel{6}}$$

$$= 100 \times 49 - 15 \times 19$$

$$= 4900 - 285$$

$$= 4615 \text{ cm}^2$$

Rekha has decorated 4615 cm² area of square Colour papers.

40. Find the sum of the series $(2^3 - 1^3) + (4^3 - 3^3) + (6^3 - 5^3) + \dots$ to
(i) n terms (ii) 8 terms

solution

$$(i) \quad (2^3 - 1^3) + (4^3 - 3^3) + (6^3 - 5^3) + \dots = (2^3 + 4^3 + \dots + n^3 \text{ terms}) - (1^3 + 3^3 + \dots + n^3 \text{ terms})$$

$$(1^3 + 2^3 + 3^3 \dots + n^3) = \left[\frac{n(n+1)}{2} \right]^2$$

$$(1^3 + 3^3 + 5^3 \dots + (2n-1)^3) = n^2(2n^2 - 1)$$

$$= 2^3(1^3 + 2^3 + \dots + n^3 \text{ terms}) - (1^3 + 3^3 + \dots + n^3 \text{ terms})$$

$$= 8 \left[\frac{n(n+1)}{2} \right]^2 - n^2(2n^2 - 1)$$

$$= 2(n^2(n+1)^2) - n^2(2n^2 - 1)$$

$$= 2n^2(n^2 + 2n + 1) - n^2(2n^2 - 1)$$

$$= n^2(2n^2 + 4n + 2 - 2n^2 + 1)$$

$$= n^2(4n + 3)$$

$$\boxed{= 4n^3 + 3n^2}$$

$$(ii) \quad \text{When } n = 8, S_8 = 4(8^3) + 3(8^2)$$

$$= 8^2((4 \times 8) + 3)$$

$$= 64(32 + 3) = 64 \times 35$$

$$\boxed{S_8 = 2240}$$

41. A milk man has 175 litres of cow's milk and 105 litres of buffalow's milk. He wishes to sell the milk by filling the two types of milk in cans of equal capacity. Calculate the following

- (i) Capacity of a can (ii) Number of cans of cow's milk
(iii) Number of cans of buffalow's milk.

solution

Cow's milk = 175 lrs.

Buffalow's milk = 105 lrs.

Since milk in cans of equal capacity, To Find HCF

$$i) \quad \text{Capacity of a can} = \text{HCF}(175, 105)$$

$$= 7 \times 5$$

$$= 35 \text{ litres}$$

$$ii) \quad \text{Number of cans of Cow's milk} = \frac{175^{35}}{35_7} = 5$$

$$iii) \quad \text{Number of cans of buffalow's milk} = \frac{105^{21}}{35_7} = 3$$

42. A man saved ₹16500 in ten years. In each year after the first he saved ₹100 more than he did in the preceding year. How much did he save in the first year? **PTA-4**

solution

Given $S_n = ₹16500, d = ₹100, n = 10$ in A.P.

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$\frac{10}{2}[2a + 9(100)] = 16500$$

$$2a + 900 = \frac{16500}{5}$$

$$2a + 900 = 3300$$

$$2a = 2400$$

$$\therefore a = 1200$$

∴ He saved Rs. 1200 in First year.

43. Find the G.P. in which the 2nd term is $\sqrt{6}$ and the 6th term is $9\sqrt{6}$.

solution

Given $t_2 = \sqrt{6}$, $t_6 = 9\sqrt{6}$ in G.P.

G.P General Form $a, ar^2, ar^3, \dots ar^{(n-1)}$

$$a \cdot r = \sqrt{6}, \quad a \cdot r^5 = 9\sqrt{6}$$

$$\frac{a \cdot r^5}{ar} = \frac{9\sqrt{6}}{\sqrt{6}}$$

$$\therefore r^4 = 9$$

$$(r^2)^2 = 3^2$$

$$r = \sqrt{3}$$

$$\therefore a \times \sqrt{3} = \sqrt{6}$$

$$a = \frac{\sqrt{6}}{\sqrt{3}}$$

$$= \frac{\sqrt{2 \times 3}}{\sqrt{3}}$$

$$\therefore a = \sqrt{2}$$

∴ The G.P is $\sqrt{2}, \sqrt{6}, \sqrt{18}, \dots$ (or) $\sqrt{2}, \sqrt{6}, 3\sqrt{2}, \dots$

CHAPTER - 3. ALGEBRA**PUBLIC EXAM IMPORTANT QUESTIONS****2 MARK IMPORTANT QUESTIONS****CHAPTER - 3. ALGEBRA****2 MARK IMPORTANT QUESTIONS**

1. The father's age is six times his son's age. Six years hence the age of father will be four times his son's age. Find the present ages (in years) of the son and father.

solution

Let son's present age = x
 Let father's present age = y
 Given condition 1: $y = 6x \dots (1)$
 Given condition 2: $y + 6 = 4(x + 6) \dots (2)$
 Sub $y = 6x$ in (2):
 $6x + 6 = 4x + 24$
 $6x - 4x = 24 - 6$
 $2x = 18 \implies x = 9$
 $y = 6(9) = 54$
 Son's age = 9 yrs, Father's age = 54 yrs

2. Solve $2x - 3y = 6, x + y = 1$

solution

Equation 1: $2x - 3y = 6$
 Equation 2: $x + y = 1 \implies x = 1 - y$
 Sub x in Eq 1:
 $2(1 - y) - 3y = 6$
 $2 - 2y - 3y = 6$
 $-5y = 6 - 2$
 $-5y = 4 \implies y = -\frac{4}{5}$
 Sub y To Find x :
 $x = 1 - \left(-\frac{4}{5}\right)$
 $x = 1 + \frac{4}{5}$
 $x = \frac{9}{5}$

3. Find the LCM of the (i) $8x^4y^2, 48x^2y^4$ (ii) $x^3 - 27, (x - 3)^2, x^2 - 9$

solution

(i) $8x^4y^2 = 2^3 \cdot x^4 \cdot y^2$
 $48x^2y^4 = 2^4 \cdot 3 \cdot x^2 \cdot y^4$
 LCM = $2^4 \cdot 3 \cdot (\text{Highest powers of } x, y)$
 LCM = $16 \cdot 3 \cdot x^4 \cdot y^4 = 48x^4y^4$
 (ii) $x^3 - 27 = (x - 3)(x^2 + 3x + 9)$
 $(x - 3)^2 = (x - 3)^2$
 $x^2 - 9 = (x - 3)(x + 3)$
 LCM = $(x - 3)^2(x + 3)(x^2 + 3x + 9)$

4. Find the LCM of the (i) $9a^3b^2, 12a^2b^2c$ (ii) $p^2 - 3p + 2, p^2 - 4$ (iii) $(2x^2 - 3xy)^2, (4x - 6y)^3, 8x^3 - 27y^3$

solution

(i) $9a^3b^2 = 3^2 \cdot a^3 \cdot b^2$
 $12a^2b^2c = 2^2 \cdot 3 \cdot a^2 \cdot b^2 \cdot c$
 LCM = $2^2 \cdot 3^2 \cdot a^3 \cdot b^2 \cdot c = 36a^3b^2c$

(ii) $p^2 - 3p + 2 = (p - 1)(p - 2)$
 $p^2 - 4 = (p - 2)(p + 2)$
 LCM = $(p - 1)(p - 2)(p + 2)$
 (iii) $(2x^2 - 3xy)^2 = [x(2x - 3y)]^2 = x^2(2x - 3y)^2$
 $(4x - 6y)^3 = [2(2x - 3y)]^3 = 8(2x - 3y)^3$
 $8x^3 - 27y^3 = (2x)^3 - (3y)^3$
 $= (2x - 3y)(4x^2 + 6xy + 9y^2)$
 LCM = $8x^2(2x - 3y)^3(4x^2 + 6xy + 9y^2)$

5. Find the LCM of polynomials $a^2 + 4a - 12, a^2 - 5a + 6$ whose GCD is $a - 2$

PTA-6**solution**

Let $P(x) = a^2 + 4a - 12 = (a + 6)(a - 2)$
 Let $Q(x) = a^2 - 5a + 6 = (a - 3)(a - 2)$
 WKT LCM $\times$ GCD = $P(x) \times Q(x)$
 $\text{LCM} = \frac{P(x) \times Q(x)}{\text{GCD}}$
 $\text{LCM} = \frac{(a + 6)(a - 2)(a - 3)(a - 2)}{(a - 2)}$
 $\text{LCM} = (a + 6)(a - 3)(a - 2)$

6. Reduce the rational expression its lowest form (i) $\frac{x-3}{x^2-9}$.

(ii) $\frac{x^2-16}{x^2+8x+16}$ **MAR-26****solution**(i) Simplifying $\frac{x-3}{x^2-9}$:

$$\begin{aligned} \frac{x-3}{x^2-9} &= \frac{x-3}{x^2-3^2} \\ &= \frac{x-3}{(x-3)(x+3)} \quad [a^2 - b^2 = (a-b)(a+b)] \\ &= \frac{1}{x+3} \end{aligned}$$

(ii) Simplifying $\frac{x^2-16}{x^2+8x+16}$:

$$\begin{aligned} \frac{x^2-16}{x^2+8x+16} &= \frac{x^2-4^2}{x^2+2(4)(x)+4^2} \\ &= \frac{(x-4)(x+4)}{(x+4)^2} \quad [a^2 - b^2 = (a-b)(a+b)] \\ &= \frac{x-4}{x+4} \end{aligned}$$

7. Find the excluded values of the following expressions

(i) $\frac{x+10}{8x}$ (ii) $\frac{7p+2}{8p^2+13p+5}$ (iii) $\frac{x}{x^2+1}$ **solution**(i) Expression is undefined $x=0$.

$$8x = 0 \implies x = 0 \text{ (Excluded value)}$$

(ii) $8p^2 + 13p + 5 = 0$

$$8p^2 + 8p + 5p + 5 = 0$$

$$8p(p+1) + 5(p+1) = 0$$

$$(8p+5)(p+1) = 0$$

$$p = -\frac{5}{8}, \quad p = -1 \text{ (Excluded values)}$$

(iii) $x^2 + 1 = 0 \implies x^2 = -1$

No real excluded values.

8. Find the excluded values, if any of the following expressions.

(i) $\frac{t}{t^2-5t+6}$ **APR-22, JUL-25** (ii) $\frac{x^2+6x+8}{x^2+x-2}$ (iii) $\frac{x^3-27}{x^3+x^2-6x}$

solution

$$(i) \quad t^2 - 5t + 6 = 0$$

$$(t-2)(t-3) = 0 \implies \boxed{t = 2, 3}$$

$$(ii) \quad x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0 \implies \boxed{x = -2, 1}$$

$$(iii) \quad x^3 + x^2 - 6x = 0$$

$$x(x^2 + x - 6) = 0$$

$$x(x+3)(x-2) = 0 \implies \boxed{x = 0, -3, 2}$$

$$9. \text{ Simplify (i) } \frac{4x^2y}{2z^2} \times \frac{6xz^3}{20y^4} \quad \text{APR-24, MDL-19}$$

$$(ii) \quad \frac{p^2 - 10p + 21}{p-7} \times \frac{p^2 + p - 12}{(p-3)^2}$$

solution

$$(i) \quad \frac{4x^2y}{2z^2} \times \frac{6xz^3}{20y^4} = \left(\frac{4}{2} \cdot \frac{6}{20}\right) \frac{x^2 \cdot x \cdot y \cdot z^3}{z^2 \cdot y^4}$$

$$= \left(2 \cdot \frac{3}{10}\right) \frac{x^3 \cdot z}{y^3}$$

$$= \frac{6}{10} \frac{x^3 z}{y^3} = \frac{3x^3 z}{5y^3}$$

$$(ii) \quad \frac{p^2 - 10p + 21}{p-7} \times \frac{p^2 + p - 12}{(p-3)^2} = \frac{(p-7)(p-3)}{(p-7)} \times \frac{(p+4)(p-3)}{(p-3)^2}$$

$$= (p-3) \times \frac{(p+4)(p-3)}{(p-3)^2}$$

$$= \frac{(p-3)^2(p+4)}{(p-3)^2}$$

$$= p+4$$

$$10. \text{ Simplify } \frac{x^3 - y^3}{3x^2 + 9xy + 6y^2} \times \frac{x^2 + 2xy + y^2}{x^2 - y^2}$$

solution

$$\begin{aligned} x^3 - y^3 &= (x-y)(x^2 + xy + y^2) \\ 3(x^2 + 3xy + 2y^2) &= 3(x+2y)(x+y) \\ x^2 + 2xy + y^2 &= (x+y)^2 \\ x^2 - y^2 &= (x-y)(x+y) \\ \frac{x^3 - y^3}{3x^2 + 9xy + 6y^2} \times \frac{x^2 + 2xy + y^2}{x^2 - y^2} &= \frac{(x-y)(x^2 + xy + y^2)}{3(x+2y)(x+y)} \times \frac{(x+y)^2}{(x-y)(x+y)} \\ &= \frac{(x-y)(x+y)^2(x^2 + xy + y^2)}{3(x+2y)(x+y)^2(x-y)} \\ &= \frac{x^2 + xy + y^2}{3(x+2y)} \end{aligned}$$

$$11. \text{ Simplify } \frac{x+2}{4y} \div \frac{x^2 - x - 6}{12y^2}$$

solution

$$\begin{aligned} \frac{x+2}{4y} \div \frac{x^2 - x - 6}{12y^2} &= \frac{x+2}{4y} \times \frac{12y^2}{x^2 - x - 6} \\ x^2 - x - 6 &= (x-3)(x+2) \\ &= \frac{x+2}{4y} \times \frac{12y^2}{(x-3)(x+2)} \\ &= \frac{12y^2(x+2)}{4y(x-3)(x+2)} \\ &= \frac{12y^2}{4y(x-3)} = \frac{3y}{x-3} \end{aligned}$$

$$12. \text{ If a polynomial } p(x) = x^2 - 5x - 14 \text{ is divided by another polynomial } q(x) \text{ we get } \frac{x-7}{x+2}, \text{ find } q(x). \quad \text{PTA-2}$$

solution

$$\begin{aligned} \frac{p(x)}{q(x)} &= \frac{x-7}{x+2} \\ p(x) = x^2 - 5x - 14 &= (x-7)(x+2) \\ \frac{(x-7)(x+2)}{q(x)} &= \frac{x-7}{x+2} \\ q(x)(x-7) &= (x-7)(x+2)(x+2) \\ q(x) &= \frac{(x-7)(x+2)^2}{x-7} \\ q(x) &= (x+2)^2 \\ q(x) &= x^2 + 4x + 4 \end{aligned}$$

$$13. \text{ Simplify (i) } \frac{x(x+1)}{x-2} + \frac{x(1-x)}{x-2} \quad (ii) \quad \frac{x+2}{x+3} + \frac{x-1}{x-2}$$

$$(iii) \quad \frac{x^3}{x-y} + \frac{y^3}{y-x}$$

solution

$$(i) \quad \frac{x(x+1)}{x-2} + \frac{x(1-x)}{x-2} = \frac{x^2 + x + x - x^2}{x-2} = \frac{2x}{x-2}$$

$$(ii) \quad \frac{x+2}{x+3} + \frac{x-1}{x-2} = \frac{(x+2)(x-2) + (x-1)(x+3)}{(x+3)(x-2)}$$

$$= \frac{(x^2 - 4) + (x^2 + 2x - 3)}{x^2 + x - 6}$$

$$= \frac{2x^2 + 2x - 7}{x^2 + x - 6}$$

$$(iii) \quad \frac{x^3}{x-y} + \frac{y^3}{y-x} = \frac{x^3}{x-y} - \frac{y^3}{x-y} \quad [y-x = -(x-y)]$$

$$= \frac{x^3 - y^3}{x-y}$$

$$= \frac{(x-y)(x^2 + xy + y^2)}{x-y}$$

$$= x^2 + xy + y^2$$

$$14. \text{ Simplify (i) } \frac{(2x+1)(x-2)}{x-4} - \frac{(2x^2 - 5x + 2)}{x-4}$$

$$(ii) \quad \frac{4x}{x^2 - 1} - \frac{x+1}{x-1}$$

solution

$$\begin{aligned} \text{Given (i) } \frac{(2x+1)(x-2)}{x-4} - \frac{(2x^2 - 5x + 2)}{x-4} &= \frac{2x^2 - 4x + x - 2 - (2x^2 - 5x + 2)}{x-4} \\ &= \frac{2x^2 - 3x - 2 - 2x^2 + 5x - 2}{x-4} \\ &= \frac{2x - 4}{x-4} = \frac{2(x-2)}{x-4} \\ (ii) \quad \frac{4x}{x^2 - 1} - \frac{x+1}{x-1} &= \frac{4x}{(x-1)(x+1)} - \frac{x+1}{x-1} \\ \text{LCM} &= (x-1)(x+1) \\ &= \frac{4x - (x+1)(x+1)}{(x-1)(x+1)} \\ &= \frac{4x - (x^2 + 2x + 1)}{x^2 - 1} \\ &= \frac{-x^2 + 2x - 1}{x^2 - 1} = \frac{-(x^2 - 2x + 1)}{x^2 - 1} \\ &= \frac{-(x-1)^2}{(x-1)(x+1)} = \frac{-(x-1)}{x+1} \\ &= \frac{1-x}{x+1} \end{aligned}$$

$$15. \text{ Pari needs 4 hours to complete a work. His friend Yuvan needs 6 hours to complete the same work. How long will it take to complete if they work together?} \quad \text{MDL-19}$$

solution

$$\begin{aligned} \text{Pari's 1 hour work} &= \frac{1}{4} \\ \text{Yuvan's 1 hour work} &= \frac{1}{6} \end{aligned}$$

$$\begin{aligned}\text{Together in 1 hour} &= \frac{1}{4} + \frac{1}{6} \\ &= \frac{3}{12} + \frac{2}{12} = \frac{5}{12} \\ \text{Total time taken} &= \frac{1}{\text{Together's 1 hr work}} \\ &= \frac{12}{5} \text{ hours} \\ &= 2 \text{ hours } 24 \text{ minutes } \quad (2.4 \text{ hrs})\end{aligned}$$

16. Find the square root of the following expressions

(i) $256(x-a)^8(x-b)^4(x-c)^{16}(x-d)^{20}$ (ii) $\frac{144a^8b^{12}c^{16}}{81f^{12}g^4h^{14}}$ **PTA-5**

solution

$$\begin{aligned}\text{(i)} \quad & \sqrt{256(x-a)^8(x-b)^4(x-c)^{16}(x-d)^{20}} \\ &= \sqrt{256} \cdot |(x-a)^4(x-b)^2(x-c)^8(x-d)^{10}| \\ &= 16|(x-a)^4(x-b)^2(x-c)^8(x-d)^{10}| \\ \text{(ii)} \quad & \sqrt{\frac{144a^8b^{12}c^{16}}{81f^{12}g^4h^{14}}} \\ &= \frac{\sqrt{144}}{\sqrt{81}} \left| \frac{a^4b^6c^8}{f^6g^2h^7} \right| \\ &= \frac{12}{9} \left| \frac{a^4b^6c^8}{f^6g^2h^7} \right| = \frac{4}{3} \left| \frac{a^4b^6c^8}{f^6g^2h^7} \right|\end{aligned}$$

17. Find the square root of the following rational expressions. (i) $\frac{400x^4y^{12}z^{16}}{100x^8y^4z^4}$

JUL-22 (ii) $\frac{121(a+b)^8(x+y)^8(b-c)^8}{81(b-c)^4(a-b)^{12}(b-c)^4}$

solution

$$\begin{aligned}\text{(i)} \quad & \sqrt{\frac{400x^4y^{12}z^{16}}{100x^8y^4z^4}} = \frac{2y^4z^6}{x^2} \\ &= 2 \left| \frac{y^4z^6}{x^2} \right| \\ \text{(ii)} \quad & \sqrt{\frac{121(a+b)^8(x+y)^8(b-c)^8}{81(b-c)^4(a-b)^{12}(b-c)^4}} \\ &= \sqrt{\frac{121(a+b)^8(x+y)^8}{81(a-b)^{12}}} \\ &= \frac{11}{9} \left| \frac{(a+b)^4(x+y)^4}{(a-b)^6} \right|\end{aligned}$$

18. Find the sum and product of the roots (i) $x^2 + 8x - 65 = 0$ **APR-2**

(ii) $kx^2 - k^2x - 2k^3 = 0$

solution

Quadratic Eqn Form $ax^2 + bx + c = 0$

Sum of roots $= -\frac{b}{a}$, Product of roots $= \frac{c}{a}$

(i) $a = 1, b = 8, c = -65$

Sum $= -\frac{8}{1} = -8$

Product $= \frac{-65}{1} = -65$

(ii) $a = k, b = -k^2, c = -2k^3$

Sum $= -\frac{-k^2}{k} = k$

Product $= \frac{-2k^3}{k} = -2k^2$

19. Determine the quadratic equations, whose sum and product of roots are (i) -9, 20 **JUL-21** (ii) $\frac{-3}{2}$, -1 **PTA-4**

solution

Quadratic Equation Formula:

$x^2 - (\text{Sum of roots})x + (\text{Product of roots}) = 0$

(i) Sum = -9, Product = 20

$x^2 - (-9)x + 20 = 0$

$x^2 + 9x + 20 = 0$

(ii) Sum $= -\frac{3}{2}$, Product $= -1$

$x^2 - \left(-\frac{3}{2}\right)x + (-1) = 0$

$x^2 + \frac{3}{2}x - 1 = 0$

Multiply by 2: $2x^2 + 3x - 2 = 0$

20. Find the sum and product of the roots

(i) $x^2 + 3x - 28 = 0$ (ii) $3 + \frac{1}{a} = \frac{10}{a^2}$ (iii) $3y^2 - y - 4 = 0$

solution

(i) $a = 1, b = 3, c = -28$

Sum $= -\frac{b}{a} = -3$,

Product $= \frac{c}{a} = -28$

(ii) Multiply Eqn by a^2 :

$3a^2 + a = 10 \implies 3a^2 + a - 10 = 0$

Sum $= -\frac{b}{a} = -\frac{1}{3}$

Product $= \frac{c}{a} = -\frac{10}{3}$

(iii) $a = 3, b = -1, c = -4$

Sum $= -\frac{-1}{3} = \frac{1}{3}$

Product $= -\frac{4}{3}$

21. Solve $2x^2 - 2\sqrt{6}x + 3 = 0$ **PTA-6**

solution

Given Quadratic Equation $2x^2 - 2\sqrt{6}x + 3 = 0$.

Write Perfect Square Form like $a^2 + 2ab + b^2 = 0$

$(\sqrt{2}x)^2 - 2(\sqrt{2}x)(\sqrt{3}) + (\sqrt{3})^2 = 0$

$(\sqrt{2}x - \sqrt{3})^2 = 0$

$\sqrt{2}x - \sqrt{3} = 0 \implies \sqrt{2}x = \sqrt{3}$

$x = \frac{\sqrt{3}}{\sqrt{2}} = \sqrt{\frac{3}{2}}$

It's perfect square, Roots are real and equal.

$x = \sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}}$

22. Solve $2m^2 + 19m + 30 = 0$

solution

Given Quadratic Equation $2m^2 + 19m + 30 = 0$. Using Factorisation Method Solve

$2m^2 + 4m + 15m + 30 = 0$

$2m(m+2) + 15(m+2) = 0$

$(m+2)(2m+15) = 0$

$m+2 = 0 \implies \boxed{m = -2}$

$2m+15 = 0 \implies \boxed{m = -\frac{15}{2}}$

23. Solve the following quadratic equations by factorization method

(i) $3(p^2 - 6) = p(p+5)$ (ii) $\sqrt{a(a-7)} = 3\sqrt{2}$

(iii) $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$

solution

(i) $3(p^2 - 6) = p(p+5) \implies 3p^2 - 18 = p^2 + 5p$

$2p^2 - 5p - 18 = 0$

$2p^2 - 9p + 4p - 18 = 0$

$p(2p-9) + 2(2p-9) = 0$

$(p+2)(2p-9) = 0$

$\boxed{p = -2, \frac{9}{2}}$

$$(ii) \sqrt{a(a-7)} = 3\sqrt{2} \implies a(a-7) = (3\sqrt{2})^2$$

$$a^2 - 7a = 9 \times 2 = 18$$

$$a^2 - 7a - 18 = 0$$

$$(a-9)(a+2) = 0$$

$$\boxed{a = 9, -2}$$

$$(iii) \sqrt{2}x^2 + 7x + 5\sqrt{2} = 0 \implies \sqrt{2}x^2 + 2x + 5x + 5\sqrt{2} = 0$$

$$\sqrt{2}x(x + \sqrt{2}) + 5(x + \sqrt{2}) = 0$$

$$(x + \sqrt{2})(\sqrt{2}x + 5) = 0$$

$$\boxed{x = -\sqrt{2}}, \quad \boxed{x = -\frac{5}{\sqrt{2}}}$$

24. Solve $x^2 + 2x - 2 = 0$ by formula method **JUL-22**

solution

Given Quadratic Equation $x^2 + 2x - 2 = 0 \implies ax^2 + bx + c = 0$.

$$a = 1, b = 2, c = -2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(1)(-2)}}{2(1)}$$

$$x = \frac{-2 \pm \sqrt{4+8}}{2}$$

$$x = \frac{-2 \pm \sqrt{12}}{2} = \frac{-2 \pm 2\sqrt{3}}{2}$$

$$x = -1 \pm \sqrt{3}$$

25. Solve $2x^2 - 3x - 3 = 0$ by formula method.

solution

Given Quadratic Equation $2x^2 - 3x - 3 = 0 \implies ax^2 + bx + c = 0$.

$$a = 2, b = -3, c = -3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(2)(-3)}}{2(2)}$$

$$x = \frac{3 \pm \sqrt{9+24}}{4}$$

$$x = \frac{3 \pm \sqrt{33}}{4}$$

$$x = \frac{3 + \sqrt{33}}{4}, x = \frac{3 - \sqrt{33}}{4}$$

26. Solve the following quadratic equations by formula method

(i) $\sqrt{2}f^2 - 6f + 3\sqrt{2} = 0$ (ii) $3y^2 - 20y - 23 = 0$

solution

(i) $a = \sqrt{2}, b = -6, c = 3\sqrt{2}$

$$f = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$f = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(\sqrt{2})(3\sqrt{2})}}{2(\sqrt{2})}$$

$$f = \frac{6 \pm \sqrt{36-24}}{2\sqrt{2}} = \frac{6 \pm \sqrt{12}}{2\sqrt{2}}$$

$$f = \frac{6 \pm 2\sqrt{3}}{2\sqrt{2}} = \frac{3 \pm \sqrt{3}}{\sqrt{2}}$$

(ii) $a = 3, b = -20, c = -23$

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$y = \frac{-(-20) \pm \sqrt{(-20)^2 - 4(3)(-23)}}{2(3)}$$

$$y = \frac{20 \pm \sqrt{400+276}}{6} = \frac{20 \pm \sqrt{676}}{6}$$

$$y = \frac{20 \pm 26}{6}$$

$$y = \frac{46}{6} = \frac{23}{3} \quad \text{or} \quad y = \frac{-6}{6} = -1$$

27. The product of Kumaran's age (in years) two years ago and his age four years from now is one more than twice his present age. What is his present age? **PTA-1**

solution

Let Kumaran present age = x

Kumaran Age Afer 2 yrs ago = $x - 2$

Kumaran Age 4 yrs from now = $x + 4$

$$(x-2)(x+4) = 2x + 1$$

$$x^2 + 4x - 2x - 8 = 2x + 1$$

$$x^2 + 2x - 8 = 2x + 1$$

$$x^2 = 9$$

$$x = \pm 3$$

Age cannot be negative $x = 3$ years.

Kumaran Present Age is 3 years.

28. If the difference between a number and its reciprocal is $\frac{24}{5}$ find the number. **PTA-6,MAR-26**

solution

Let the number be x . Its reciprocal is $\frac{1}{x}$.

$$x - \frac{1}{x} = \frac{24}{5}$$

$$\frac{x^2 - 1}{x} = \frac{24}{5}$$

$$5(x^2 - 1) = 24x$$

$$5x^2 - 24x - 5 = 0$$

$$5x^2 - 25x + x - 5 = 0$$

$$5x(x-5) + 1(x-5) = 0$$

$$(5x+1)(x-5) = 0$$

$$\boxed{x = 5} \quad \text{or} \quad \boxed{x = -\frac{1}{5}}$$

29. Determine the nature of roots for the following quadratic equations

(i) $x^2 - x - 20 = 0$ (ii) $9x^2 - 24x + 16 = 0$ (iii) $2x^2 - 2x + 9 = 0$

solution

(i) $\Delta = b^2 - 4ac$

$$= (-1)^2 - 4(1)(-20)$$

$$= 1 + 80 = 81 > 0$$

Roots are real and distinct.

(ii) $\Delta = b^2 - 4ac$

$$= (-24)^2 - 4(9)(16)$$

$$= 576 - 576 = 0$$

Roots are real and equal.

(iii) $\Delta = b^2 - 4ac$

$$= (-2)^2 - 4(2)(9)$$

$$= 4 - 72 = -68 < 0$$

No real roots.

30. Determine the nature of the roots for the following quadratic equations

(i) $15x^2 + 11x + 2 = 0$ **JUL-21, 23, 24** (ii) $x^2 - x - 1 = 0$ (iii) $\sqrt{2}t^2 - 3t + 3\sqrt{2} = 0$ (iv) $9a^2b^2x^2 - 24abcdx + 16c^2d^2 = 0, a \neq 0, b \neq 0$

solution

Discriminant $\Delta = b^2 - 4ac$

(i) $\Delta = 11^2 - 4(15)(2) = 121 - 120 = 1 > 0$

Roots are real and distinct.

(ii) $\Delta = (-1)^2 - 4(1)(-1) = 1 + 4 = 5 > 0$

Roots are real and distinct.

(iii) $\Delta = (-3)^2 - 4(\sqrt{2})(3\sqrt{2})$

$$= 9 - 24 = -15 < 0$$

No real roots.

(iv) $\Delta = (-24abcd)^2 - 4(9a^2b^2)(16c^2d^2)$

$$= 576a^2b^2c^2d^2 - 576a^2b^2c^2d^2 = 0$$

Roots are real and equal.

31. If the difference between the roots of the equation $x^2 - 13x + k = 0$ is 17 find k.

solution

Let roots be α, β . $\alpha - \beta = 17$

$$\text{Sum of roots: } \alpha + \beta = -\left(\frac{-13}{1}\right) = 13$$

$$\text{Product of roots: } \alpha\beta = \frac{k}{1} = k$$

$$\text{WKT } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$17^2 = 13^2 - 4k$$

$$289 = 169 - 4k$$

$$4k = 169 - 289$$

$$4k = -120 \implies \boxed{k = -30}$$

32. Write each of the following expression in terms of $\alpha + \beta$ and $\alpha\beta$

(i) $\frac{\alpha}{3\beta} + \frac{\beta}{3\alpha}$ (ii) $\frac{1}{\alpha^2\beta} + \frac{1}{\beta^2\alpha}$ (iii) $(3\alpha - 1)(3\beta - 1)$ (iv) $\frac{\alpha+3}{\beta} + \frac{\beta+3}{\alpha}$

solution

$$(i) \quad \frac{\alpha}{3\beta} + \frac{\beta}{3\alpha} = \frac{\alpha^2 + \beta^2}{3\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{3\alpha\beta}$$

$$(ii) \quad \frac{1}{\alpha^2\beta} + \frac{1}{\beta^2\alpha} = \frac{\beta + \alpha}{\alpha^2\beta^2} = \frac{\alpha + \beta}{(\alpha\beta)^2}$$

$$(iii) \quad (3\alpha - 1)(3\beta - 1) = 9\alpha\beta - 3\alpha - 3\beta + 1$$

$$= 9\alpha\beta - 3(\alpha + \beta) + 1$$

$$(iv) \quad \frac{\alpha + 3}{\beta} + \frac{\beta + 3}{\alpha} = \frac{\alpha(\alpha + 3) + \beta(\beta + 3)}{\alpha\beta}$$

$$= \frac{\alpha^2 + 3\alpha + \beta^2 + 3\beta}{\alpha\beta}$$

$$= \frac{(\alpha^2 + \beta^2) + 3(\alpha + \beta)}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta + 3(\alpha + \beta)}{\alpha\beta}$$

33. If a matrix has 16 elements, what are the possible orders it can have?

solution

$$\text{Total elements} = \text{rows} \times \text{columns} = 16$$

$$\text{Pairs } (m, n) \text{ such that } m \times n = 16$$

$$\text{Possible orders: } 1 \times 16, 16 \times 1, 2 \times 8, 8 \times 2, 4 \times 4$$

34. Construct a 3×3 matrix whose elements are $a_{ij} = i^2 j^2$

solution

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$a_{ij} = i^2 j^2$$

$$a_{11} = 1^2 \cdot 1^2 = 1, a_{12} = 1^2 \cdot 2^2 = 4, a_{13} = 1^2 \cdot 3^2 = 9$$

$$a_{21} = 2^2 \cdot 1^2 = 4, a_{22} = 2^2 \cdot 2^2 = 16, a_{23} = 2^2 \cdot 3^2 = 36$$

$$a_{31} = 3^2 \cdot 1^2 = 9, a_{32} = 3^2 \cdot 2^2 = 36, a_{33} = 3^2 \cdot 3^2 = 81$$

$$A = \begin{pmatrix} 1 & 4 & 9 \\ 4 & 16 & 36 \\ 9 & 36 & 81 \end{pmatrix}$$

35. In the matrix $A = \begin{pmatrix} 8 & 9 & 4 & 3 \\ -1 & \sqrt{7} & \frac{\sqrt{3}}{2} & 5 \\ 1 & 4 & 3 & 0 \\ 6 & 8 & -11 & 1 \end{pmatrix}$, write (i) The number of

elements (ii) The order of the matrix (iii) Write the elements $a_{22}, a_{23}, a_{24}, a_{34}, a_{43}, a_{44}$

solution

(i) Number of elements = 4 rows $\times$ 4 cols = 16

(ii) Order = 4×4

(iii) a_{22} = Element in Row 2, Col 2 = $\sqrt{7}$

$$a_{23} = \frac{\sqrt{3}}{2}, \quad a_{24} = 5$$

$$a_{34} = 0, \quad a_{43} = -11, \quad a_{44} = 1$$

36. Construct a 3×3 matrix whose elements are given by (i) $a_{ij} = |i - 2j|$ **MAR-26**

solution

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$a_{ij} = |i - 2j|$$

$$\text{Row 1: } a_{11} = |1 - 2(1)| = |-1| = 1$$

$$a_{12} = |1 - 2(2)| = |-3| = 3$$

$$a_{13} = |1 - 2(3)| = |-5| = 5$$

$$\text{Row 2: } a_{21} = |2 - 2(1)| = 0$$

$$a_{22} = |2 - 2(2)| = |-2| = 2$$

$$a_{23} = |2 - 2(3)| = |-4| = 4$$

$$\text{Row 3: } a_{31} = |3 - 2(1)| = 1$$

$$a_{32} = |3 - 2(2)| = |-1| = 1$$

$$a_{33} = |3 - 2(3)| = |-3| = 3$$

$$A = \begin{pmatrix} 1 & 3 & 5 \\ 0 & 2 & 4 \\ 1 & 1 & 3 \end{pmatrix}$$

37. If $A = \begin{pmatrix} \sqrt{7} & -3 \\ -\sqrt{5} & 2 \\ \sqrt{3} & -5 \end{pmatrix}$ then find the transpose of $-A$. **JUL-20, PTA-2**

solution

$-A$ (multiply by -1):

$$-A = \begin{pmatrix} -\sqrt{7} & 3 \\ \sqrt{5} & -2 \\ -\sqrt{3} & 5 \end{pmatrix}$$

Transpose of $-A, (-A)^T$:

$$(-A)^T = \begin{pmatrix} -\sqrt{7} & \sqrt{5} & -\sqrt{3} \\ 3 & -2 & 5 \end{pmatrix}$$

38. If $A = \begin{pmatrix} 5 & 2 & 2 \\ -\sqrt{17} & 0.7 & \frac{5}{2} \\ 8 & 3 & 1 \end{pmatrix}$ then verify $(A^T)^T = A$ **JUL-2**

solution

$$A^T = \begin{pmatrix} 5 & -\sqrt{17} & 8 \\ 2 & 0.7 & 3 \\ 2 & \frac{5}{2} & 1 \end{pmatrix}$$

Transpose $A^T \implies (A^T)^T$:

$$(A^T)^T = \begin{pmatrix} 5 & 2 & 2 \\ -\sqrt{17} & 0.7 & \frac{5}{2} \\ 8 & 3 & 1 \end{pmatrix}$$

$$\boxed{(A^T)^T = A}$$

39. Find the values of x, y and z from the following equations

(i) $\begin{pmatrix} 12 & 3 \\ x & 5 \end{pmatrix} = \begin{pmatrix} y & z \\ 3 & 5 \end{pmatrix}$ **MDL-19** (ii) $\begin{pmatrix} x+y & 2 \\ 5+z & xy \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 5 & 8 \end{pmatrix}$

solution

(i) Equating corresponding elements:

$$12 = y \implies \boxed{y = 12}$$

$$3 = z \implies \boxed{z = 3}$$

$$x = 3 \implies \boxed{x = 3}$$

(ii) Equating corresponding elements:

$$5 + z = 5 \implies \boxed{z = 0} \dots (1)$$

$$x + y = 6 \implies y = 6 - x \dots (2)$$

$$xy = 8 \dots (3)$$

$$\text{Sub } y : x(6 - x) = 8$$

$$6x - x^2 = 8 \implies x^2 - 6x + 8 = 0$$

$$(x - 4)(x - 2) = 0 \implies x = 4 \text{ or } x = 2$$

$$\text{If } \boxed{x = 4, y = 2}. \quad \text{If } x = 2, y = 4.$$

40. If $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 7 & 0 \\ 1 & 3 & 1 \\ 2 & 4 & 0 \end{pmatrix}$, find $A + B$.

solution

$$A + B = \begin{pmatrix} 1+1 & 2+7 & 3+0 \\ 4+1 & 5+3 & 6+1 \\ 7+2 & 8+4 & 9+0 \end{pmatrix}$$

$$A + B = \begin{pmatrix} 2 & 9 & 3 \\ 5 & 8 & 7 \\ 9 & 12 & 9 \end{pmatrix}$$

41. If $A = \begin{pmatrix} 1 & 3 & -2 \\ 5 & -4 & 6 \\ -3 & 2 & 9 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 8 \\ 3 & 4 \\ 9 & 6 \end{pmatrix}$, find $A + B$

solution

$$\text{Order of } A = 3 \times 3$$

$$\text{Order of } B = 3 \times 2$$

Matrix Addition only both matrices have same order

$A + B$ is not defined.

42. If $A = \begin{pmatrix} 7 & 8 & 6 \\ 1 & 3 & 9 \\ -4 & 3 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 4 & 11 & -3 \\ -1 & 2 & 4 \\ 7 & 5 & 0 \end{pmatrix}$ then Find $2A + B$. **PTA-3**

solution

$$2A = 2 \begin{pmatrix} 7 & 8 & 6 \\ 1 & 3 & 9 \\ -4 & 3 & -1 \end{pmatrix} = \begin{pmatrix} 14 & 16 & 12 \\ 2 & 6 & 18 \\ -8 & 6 & -2 \end{pmatrix}$$

$$2A + B = \begin{pmatrix} 14 & 16 & 12 \\ 2 & 6 & 18 \\ -8 & 6 & -2 \end{pmatrix} + \begin{pmatrix} 4 & 11 & -3 \\ -1 & 2 & 4 \\ 7 & 5 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 14+4 & 16+11 & 12-3 \\ 2-1 & 6+2 & 18+4 \\ -8+7 & 6+5 & -2+0 \end{pmatrix}$$

$$2A + B = \begin{pmatrix} 18 & 27 & 9 \\ 1 & 8 & 22 \\ -1 & 11 & -2 \end{pmatrix}$$

43. If $A = \begin{pmatrix} 1 & 9 \\ 3 & 4 \\ 8 & -3 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 7 \\ 3 & 3 \\ 1 & 0 \end{pmatrix}$ then verify that (i) $A + B = B + A$ (ii) $A + (-A) = (-A) + A = 0$

solution

$$(i) \quad A + B = \begin{pmatrix} 1+5 & 9+7 \\ 3+3 & 4+3 \\ 8+1 & -3+0 \end{pmatrix} = \begin{pmatrix} 6 & 16 \\ 6 & 7 \\ 9 & -3 \end{pmatrix}$$

$$B + A = \begin{pmatrix} 5+1 & 7+9 \\ 3+3 & 3+4 \\ 1+8 & 0-3 \end{pmatrix} = \begin{pmatrix} 6 & 16 \\ 6 & 7 \\ 9 & -3 \end{pmatrix}$$

$$\therefore A + B = B + A$$

$$(ii) \quad -A = \begin{pmatrix} -1 & -9 \\ -3 & -4 \\ -8 & 3 \end{pmatrix}$$

$$A + (-A) = \begin{pmatrix} 1-1 & 9-9 \\ 3-3 & 4-4 \\ 8-8 & -3+3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} = 0$$

$$(-A) + A = \begin{pmatrix} -1+1 & -9+9 \\ -3+3 & -4+4 \\ -8+8 & 3-3 \end{pmatrix} = 0$$

$$\therefore A + (-A) = (-A) + A = 0$$

44. If $A = \begin{pmatrix} 0 & 4 & 9 \\ 8 & 3 & 7 \end{pmatrix}$, $B = \begin{pmatrix} 7 & 3 & 8 \\ 1 & 4 & 9 \end{pmatrix}$ find the value of
(i) $B - 5A$ (ii) $3A - 9B$

solution

$$(i) \quad 5A = 5 \begin{pmatrix} 0 & 4 & 9 \\ 8 & 3 & 7 \end{pmatrix} = \begin{pmatrix} 0 & 20 & 45 \\ 40 & 15 & 35 \end{pmatrix}$$

$$B - 5A = \begin{pmatrix} 7 & 3 & 8 \\ 1 & 4 & 9 \end{pmatrix} - \begin{pmatrix} 0 & 20 & 45 \\ 40 & 15 & 35 \end{pmatrix}$$

$$= \begin{pmatrix} 7-0 & 3-20 & 8-45 \\ 1-40 & 4-15 & 9-35 \end{pmatrix}$$

$$B - 5A = \begin{pmatrix} 7 & -17 & -37 \\ -39 & -11 & -26 \end{pmatrix}$$

$$(ii) \quad 3A = 3 \begin{pmatrix} 0 & 4 & 9 \\ 8 & 3 & 7 \end{pmatrix} = \begin{pmatrix} 0 & 12 & 27 \\ 24 & 9 & 21 \end{pmatrix},$$

$$9B = 9 \begin{pmatrix} 7 & 3 & 8 \\ 1 & 4 & 9 \end{pmatrix} = \begin{pmatrix} 63 & 27 & 72 \\ 9 & 36 & 81 \end{pmatrix}$$

$$3A - 9B = \begin{pmatrix} 0-63 & 12-27 & 27-72 \\ 24-9 & 9-36 & 21-81 \end{pmatrix}$$

$$3A - 9B = \begin{pmatrix} -63 & -15 & -45 \\ 15 & -27 & -60 \end{pmatrix}$$

45. If $A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 8 & 3 & 1 \\ 2 & 4 & 1 \\ 5 & 3 & 1 \end{pmatrix}$, find AB .

solution

$$\text{Order of } A: 2 \times 3. \quad \text{Order of } B: 3 \times 3.$$

$$\text{Result order: } 2 \times 3.$$

$$AB = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 5 \end{pmatrix} \times \begin{pmatrix} 8 & 3 & 1 \\ 2 & 4 & 1 \\ 5 & 3 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1(8) + 2(2) + 0(5) & 1(3) + 2(4) + 0(3) & 1(1) + 2(1) + 0(1) \\ 3(8) + 1(2) + 5(5) & 3(3) + 1(4) + 5(3) & 3(1) + 1(1) + 5(1) \end{pmatrix}$$

$$= \begin{pmatrix} 8+4+0 & 3+8+0 & 1+2+0 \\ 24+2+25 & 9+4+15 & 3+1+5 \end{pmatrix}$$

$$AB = \begin{pmatrix} 12 & 11 & 3 \\ 51 & 28 & 9 \end{pmatrix}$$

46. If $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ find AB and BA . Verify $AB = BA$?

APR-25

solution

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \times \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 2(2) + 1(1) & 2(0) + 1(3) \\ 1(2) + 3(1) & 1(0) + 3(3) \end{pmatrix}$$

$$= \begin{pmatrix} 4+1 & 0+3 \\ 2+3 & 0+9 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 5 & 9 \end{pmatrix}$$

$$BA = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix} \times \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 2(2) + 0(1) & 2(1) + 0(3) \\ 1(2) + 3(1) & 1(1) + 3(3) \end{pmatrix}$$

$$= \begin{pmatrix} 4+0 & 2+0 \\ 2+3 & 1+9 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 5 & 10 \end{pmatrix}$$

$$AB \neq BA.$$

47. If $A = \begin{pmatrix} 2 & -2\sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 2\sqrt{2} \\ -\sqrt{2} & 2 \end{pmatrix}$. Show that A and B satisfy commutative property with respect to matrix multiplication **PTA-5**

solution

$$AB = \begin{pmatrix} 2 & -2\sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix} \times \begin{pmatrix} 2 & 2\sqrt{2} \\ -\sqrt{2} & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2(2) + (-2\sqrt{2})(-\sqrt{2}) & 2(2\sqrt{2}) + (-2\sqrt{2})(2) \\ \sqrt{2}(2) + 2(-\sqrt{2}) & \sqrt{2}(2\sqrt{2}) + 2(2) \end{pmatrix}$$

$$= \begin{pmatrix} 4+4 & 4\sqrt{2}-4\sqrt{2} \\ 2\sqrt{2}-2\sqrt{2} & 4+4 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$$

$$BA = \begin{pmatrix} 2 & 2\sqrt{2} \\ -\sqrt{2} & 2 \end{pmatrix} \times \begin{pmatrix} 2 & -2\sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2(2) + (2\sqrt{2})(\sqrt{2}) & 2(-2\sqrt{2}) + (2\sqrt{2})(2) \\ -\sqrt{2}(2) + 2(\sqrt{2}) & -\sqrt{2}(-2\sqrt{2}) + 2(2) \end{pmatrix}$$

$$= \begin{pmatrix} 4+4 & -4\sqrt{2}+4\sqrt{2} \\ -2\sqrt{2}+2\sqrt{2} & 4+4 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$$

$$AB = BA,$$

48. A has 'a' rows and 'a + 3' columns. B has 'b' rows and '17 - b' columns, and if both products AB and BA exist, find a, b?

solution

$$\begin{aligned}\text{Order of } A &: a \times (a + 3) \\ \text{Order of } B &: b \times (17 - b) \\ \text{For AB to exist: Cols of A} &= \text{Rows of B} \\ a + 3 &= b \quad \dots (1) \\ \text{For BA to exist: Cols of B} &= \text{Rows of A} \\ 17 - b &= a \quad \dots (2) \\ \text{Sub Eq 1 into Eq 2:} \\ 17 - (a + 3) &= a \\ 14 - a &= a \implies 2a = 14 \\ a &= 7 \\ b &= 7 + 3 \\ b &= 10\end{aligned}$$

49. If A is of order $p \times q$ and B is of order $q \times r$ what is the order of AB and BA? **PTA-1**

solution

$$\begin{aligned}\text{Order of A} &= p \times q \\ \text{Order of B} &= q \times r \\ \text{Order of AB: Since cols of A (q) = rows of B (q),} \\ &\text{product AB exists with order } p \times r. \\ \text{For BA: B is } q \times r, \text{ A is } p \times q. \\ \text{If } r &= p, \text{ the order of BA will be } q \times q.\end{aligned}$$

50. If $A = \begin{pmatrix} 2 & 5 \\ 4 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -3 \\ 2 & 5 \end{pmatrix}$ find AB, BA and verify $AB = BA$?

solution

$$\begin{aligned}AB &= \begin{pmatrix} 2 & 5 \\ 4 & 3 \end{pmatrix} \times \begin{pmatrix} 1 & -3 \\ 2 & 5 \end{pmatrix} \\ &= \begin{pmatrix} 2(1) + 5(2) & 2(-3) + 5(5) \\ 4(1) + 3(2) & 4(-3) + 3(5) \end{pmatrix} \\ &= \begin{pmatrix} 2 + 10 & -6 + 25 \\ 4 + 6 & -12 + 15 \end{pmatrix} = \begin{pmatrix} 12 & 19 \\ 10 & 3 \end{pmatrix} \\ BA &= \begin{pmatrix} 1 & -3 \\ 2 & 5 \end{pmatrix} \times \begin{pmatrix} 2 & 5 \\ 4 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 1(2) + (-3)(4) & 1(5) + (-3)(3) \\ 2(2) + 5(4) & 2(5) + 5(3) \end{pmatrix} \\ &= \begin{pmatrix} 2 - 12 & 5 - 9 \\ 4 + 20 & 10 + 15 \end{pmatrix} = \begin{pmatrix} -10 & -4 \\ 24 & 25 \end{pmatrix} \\ AB &\neq BA.\end{aligned}$$

51. If $A = \begin{pmatrix} \cos \theta & 0 \\ 0 & \cos \theta \end{pmatrix}$, $B = \begin{pmatrix} \sin \theta & 0 \\ 0 & \sin \theta \end{pmatrix}$ then show that $A^2 + B^2 = I$

PTA-2

solution

$$\begin{aligned}A^2 &= \begin{pmatrix} \cos \theta & 0 \\ 0 & \cos \theta \end{pmatrix} \times \begin{pmatrix} \cos \theta & 0 \\ 0 & \cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos^2 \theta & 0 \\ 0 & \cos^2 \theta \end{pmatrix} \\ B^2 &= \begin{pmatrix} \sin \theta & 0 \\ 0 & \sin \theta \end{pmatrix} \times \begin{pmatrix} \sin \theta & 0 \\ 0 & \sin \theta \end{pmatrix} \\ &= \begin{pmatrix} \sin^2 \theta & 0 \\ 0 & \sin^2 \theta \end{pmatrix} \\ A^2 + B^2 &= \begin{pmatrix} \cos^2 \theta + \sin^2 \theta & 0 + 0 \\ 0 + 0 & \cos^2 \theta + \sin^2 \theta \end{pmatrix} \\ \cos^2 \theta + \sin^2 \theta &= 1 : \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I\end{aligned}$$

Hence $A^2 + B^2 = I$ Verified.

52. Verify that $A^2 = I$ when $A = \begin{pmatrix} 5 & -4 \\ 6 & -5 \end{pmatrix}$.

solution

$$\begin{aligned}A^2 &= \begin{pmatrix} 5 & -4 \\ 6 & -5 \end{pmatrix} \times \begin{pmatrix} 5 & -4 \\ 6 & -5 \end{pmatrix} \\ &= \begin{pmatrix} 5(5) + (-4)(6) & 5(-4) + (-4)(-5) \\ 6(5) + (-5)(6) & 6(-4) + (-5)(-5) \end{pmatrix} \\ &= \begin{pmatrix} 25 - 24 & -20 + 20 \\ 30 - 30 & -24 + 25 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ A^2 &= I \quad \text{Verified.}\end{aligned}$$

53. If -4 is a root of the equation $x^2 + px - 4 = 0$ and if the equation $x^2 + px + q = 0$ has equal roots, find the values of p and q.

solution

$$\begin{aligned}\text{Since } -4 \text{ is a root of } x^2 + px - 4 = 0, \text{ sub } x = -4 : \\ (-4)^2 + p(-4) - 4 = 0 \\ 16 - 4p - 4 = 0 \\ 12 - 4p = 0 \implies 4p = 12 \implies p = 3 \\ x^2 + px + q = 0 \text{ with } p = 3 : \\ x^2 + 3x + q = 0 \\ \text{Since it has equal roots, } \Delta = b^2 - 4ac = 0 \\ (3)^2 - 4(1)(q) = 0 \\ 9 - 4q = 0 \implies 4q = 9 \implies q = \frac{9}{4}\end{aligned}$$

54. If $\cos \theta \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} + \sin \theta \begin{pmatrix} x & -\cos \theta \\ \cos \theta & x \end{pmatrix} = I_2$, find x.

solution

$$\begin{aligned}\cos \theta \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} + \sin \theta \begin{pmatrix} x & -\cos \theta \\ \cos \theta & x \end{pmatrix} \\ = \begin{pmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ -\sin \theta \cos \theta & \cos^2 \theta \end{pmatrix} + \begin{pmatrix} x \sin \theta & -\sin \theta \cos \theta \\ \sin \theta \cos \theta & x \sin \theta \end{pmatrix} \\ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \begin{pmatrix} \cos^2 \theta + x \sin \theta & \cos \theta \sin \theta - \sin \theta \cos \theta \\ -\sin \theta \cos \theta + \sin \theta \cos \theta & \cos^2 \theta + x \sin \theta \end{pmatrix} \\ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\end{aligned}$$

Equate top-left elements:

$$\cos^2 \theta + x \sin \theta = 1$$

$$x \sin \theta = 1 - \cos^2 \theta$$

$$x \sin \theta = \sin^2 \theta \implies x = \sin \theta$$

CHAPTER - 3. ALGEBRA

5 MARK IMPORTANT QUESTIONS

5 MARK IMPORTANT QUESTIONS

1. Solve the following system of linear equations in three variables

JUL-23

$$3x - 2y + z = 2, 2x + 3y - z = 5, x + y + z = 6$$

solution

Given system of equations:

$$3x - 2y + z = 2 \quad \dots (1)$$

$$2x + 3y - z = 5 \quad \dots (2)$$

$$x + y + z = 6 \quad \dots (3)$$

Add (1) & (2) to eliminate z :

$$\begin{array}{r} 3x - 2y + z = 2 \\ + 2x + 3y - z = 5 \\ \hline 5x + y = 7 \end{array} \quad \dots (4)$$

Add (2) & (3) to eliminate z :

$$\begin{array}{r} 2x + 3y - z = 5 \\ + x + y + z = 6 \\ \hline 3x + 4y = 11 \end{array} \quad \dots (5)$$

Multiply (4) by 4 and sub (5) to eliminate y :

$$\begin{array}{r} 20x + 4y = 28 \\ - 3x + 4y = 11 \\ \hline 17x = 17 \Rightarrow x = 1 \end{array}$$

Sub $x = 1$ in (4):

$$5(1) + y = 7 \Rightarrow y = 2$$

Substitute $x = 1, y = 2$ in (3):

$$1 + 2 + z = 6 \Rightarrow z = 3$$

2. Solve $x + 2y - z = 5; x - y + z = -2; -5x - 4y + z = -11$

solution

Given system of equations:

$$x + 2y - z = 5 \quad \dots (1)$$

$$x - y + z = -2 \quad \dots (2)$$

$$-5x - 4y + z = -11 \quad \dots (3)$$

Add (1) & (2) to eliminate z :

$$\begin{array}{r} x + 2y - z = 5 \\ + x - y + z = -2 \\ \hline 2x + y = 3 \end{array} \quad \dots (4)$$

Add (1) & (3) to eliminate z :

$$\begin{array}{r} x + 2y - z = 5 \\ + -5x - 4y + z = -11 \\ \hline -4x - 2y = -6 \Rightarrow 2x + y = 3 \end{array} \quad \dots (5)$$

Solve (4) & (5) :

$$\begin{array}{r} 2x + y = 3 \\ - 2x + y = 3 \\ \hline 0 = 0 \end{array}$$

The system has infinitely many solutions:

3. Solve $3x + y - 3z = 1; -2x - y + 2z = 1; -x - y + z = 2$

solution

Given system of equations:

$$3x + y - 3z = 1 \quad \dots (1)$$

$$-2x - y + 2z = 1 \quad \dots (2)$$

$$-x - y + z = 2 \quad \dots (3)$$

Add (1) & (2) to eliminate y :

$$\begin{array}{r} 3x + y - 3z = 1 \\ + -2x - y + 2z = 1 \\ \hline x - z = 2 \end{array} \quad \dots (4)$$

Add (1) & (3) to eliminate y :

$$\begin{array}{r} 3x + y - 3z = 1 \\ (+) -x - y + z = 2 \\ \hline 2x - 2z = 3 \end{array} \quad \dots (5)$$

Add (1) & (3) to eliminate y :

$$\begin{array}{r} (4) \times 2 \Rightarrow 2x - 2z = 4 \\ (-) 2x - 2z = 3 \\ \hline 0 = 3 \end{array}$$

The system has no solution.

4. Solve the following system of linear equations in three variables

$$(i) x + y + z = 5; 2x - y + z = 9; x - 2y + 3z = 16$$

$$(ii) x + 20 = \frac{3y}{2} + 10 = 2z + 5 = 110 - (y + z) \quad \text{JUL-21, PTA-5, APR-23}$$

solution

(i) Given system of equations:

$$x + y + z = 5 \quad \dots (1)$$

$$2x - y + z = 9 \quad \dots (2)$$

$$x - 2y + 3z = 16 \quad \dots (3)$$

Add (1) & (2) to eliminate y :

$$\begin{array}{r} x + y + z = 5 \\ + 2x - y + z = 9 \\ \hline 3x + 2z = 14 \end{array} \quad \dots (4)$$

Solve (2) & (3) to eliminate y :

$$\begin{array}{r} (2) \times 2 \Rightarrow 4x - 2y + 2z = 18 \\ + -x + 2y - 3z = -16 \\ \hline 3x - z = 2 \end{array} \quad \dots (5)$$

Solve (4) & (5) to eliminate x :

$$\begin{array}{r} 3x + 2z = 14 \\ (-) 3x - z = 2 \\ \hline 3z = 12 \Rightarrow z = 4 \end{array}$$

Sub $z = 4$ in (5):

$$3x - 4 = 2 \Rightarrow 3x = 6 \Rightarrow x = 2$$

Sub $x = 2, z = 4$ in (1):

$$2 + y + 4 = 5 \Rightarrow y + 6 = 5 \Rightarrow y = -1$$

(ii) Given equations simplified:

$$x + 20 = \frac{3y}{2} + 10 \Rightarrow 2x - 3y = -20 \quad \dots (1)$$

$$\frac{3y}{2} + 10 = 2z + 5 \Rightarrow 3y - 4z = -10 \quad \dots (2)$$

$$2z + 5 = 110 - y - z \Rightarrow y + 3z = 105 \quad \dots (3)$$

Add (1) & (2) to eliminate y :

$$\begin{array}{r} 2x - 3y = -20 \\ + 3y - 4z = -10 \\ \hline 2x - 4z = -30 \Rightarrow x - 2z = -15 \end{array} \quad \dots (4)$$

Solve (2) & (3) to eliminate y :

$$\begin{array}{r} (3) \times 3 \Rightarrow 3y + 9z = 315 \\ (-) 3y - 4z = -10 \\ \hline -13z = -325 \Rightarrow z = 25 \end{array}$$

Sub $z = 25$ in (3):

$$y + 3(25) = 105 \Rightarrow y + 75 = 105 \Rightarrow y = 30$$

Sub $y = 30$ in (1):

$$2x - 3(30) = -20 \Rightarrow 2x - 90 = -20 \Rightarrow 2x = 70 \Rightarrow x = 35$$

5. Discuss the nature of solutions of the following system of equations

$$\frac{y+z}{4} = \frac{z+x}{3} = \frac{x+y}{2}, x + y + z = 27$$

solution

Given equations:

$$\frac{y+z}{4} = \frac{z+x}{3} = \frac{x+y}{2} \text{ and } x + y + z = 27$$

$$\text{Let } \frac{y+z}{4} = \frac{z+x}{3} = \frac{x+y}{2} = k$$

$$y + z = 4k, \quad z + x = 3k, \quad x + y = 2k$$

Adding all three expressions:

$$2(x + y + z) = 9k$$

Sub $x + y + z = 27$ into the equation:

$$2(27) = 9k \Rightarrow 54 = 9k \Rightarrow k = 6$$

Sub $k = 6$ to form a system of equations:

$$y + z = 24 \quad \dots (1)$$

$$x + z = 18 \quad \dots (2)$$

$$x + y = 12 \quad \dots (3)$$

Subtract (1) from (2) to eliminate z :

$$\begin{array}{rrrrr} x & + & 0y & + & z & = & 18 \\ + & 0x & - & y & - & z & = & -24 \\ \hline x & - & y & & & = & -6 & \dots (4) \end{array}$$

Add (3) & (4) to eliminate y :

$$\begin{array}{rrrrr} x & + & y & = & 12 \\ + & x & - & y & = & -6 \\ \hline 2x & & & = & 6 & \implies & x = 3 \end{array}$$

Sub $x = 3$ in (3):

$$3 + y = 12 \implies y = 9$$

Sub $y = 9$ in (1):

$$9 + z = 24 \implies z = 15$$

The system has a unique solution:

6. Vani, her father and her grand father have an average age of 53. One-half of her grand father's age plus one-third of her father's age plus one fourth of Vani's age is 65. Four years ago if Vani's grandfather was four times as old as Vani then how old are they all now? **PTA-2**

solution

Let V, F, G be the ages of Vani, her Father, and her Grandfather.

Given equations:

$$V + F + G = 53 \times 3 \implies V + F + G = 159 \quad \dots (1)$$

$$\frac{G}{2} + \frac{F}{3} + \frac{V}{4} = 65 \implies 3V + 4F + 6G = 780 \quad \dots (2)$$

$$G - 4 = 4(V - 4) \implies 4V - G = 12 \quad \dots (3)$$

Multiply (1) by 4 and subtract from (2) to eliminate F :

$$\begin{array}{rrrrr} 3V & + & 4F & + & 6G & = & 780 \\ + & -4V & - & 4F & - & 4G & = & -636 \\ \hline -V & & & + & 2G & = & 144 & \dots (4) \end{array}$$

Multiply (3) by 2 and add to (4) to eliminate G :

$$\begin{array}{rrrrr} 8V & - & 2G & = & 24 \\ + & -V & + & 2G & = & 144 \\ \hline 7V & & & = & 168 & \implies & V = 24 \end{array}$$

Sub $V = 24$ in (3):

$$4(24) - G = 12 \implies 96 - G = 12 \implies G = 84$$

Sub $V = 24, G = 84$ in (1):

$$24 + F + 84 = 159 \implies F + 108 = 159 \implies F = 51$$

Ages: Vani = 24, Father = 51, Grandfather = 84

7. Find the GCD of the polynomials $x^3 + x^2 - x + 2$ and $2x^3 - 5x^2 + 5x - 3$.

solution

Divide $2x^3 - 5x^2 + 5x - 3$ by $x^3 + x^2 - x + 2$:

$$\begin{array}{r} x^3 + x^2 - x + 2 \overline{) 2x^3 - 5x^2 + 5x - 3} \\ \underline{2x^3 + 2x^2 - 2x + 4} \quad (-) \\ -7x^2 + 7x - 7 \\ \underline{-7x^2 + 7x - 7} \\ 0 \end{array}$$

$$\begin{array}{r} x^2 - x + 1 \overline{) x^3 + x^2 - x + 2} \\ \underline{x^3 - x^2 + x} \quad (-) \\ 2x^2 - 2x + 2 \\ \underline{2x^2 - 2x + 2} \quad (-) \\ 0 \end{array}$$

GCD Of $(x^3 + x^2 - x + 2, 2x^3 - 5x^2 + 5x - 3) = x^2 - x + 1$.

8. Find the GCD of $6x^3 - 30x^2 + 60x - 48$ and $3x^3 - 12x^2 + 21x - 18$.

solution

$$P(x) = 6(x^3 - 5x^2 + 10x - 8), Q(x) = 3(x^3 - 4x^2 + 7x - 6)$$

GCD of $(6, 3) = 3$.

$$\begin{array}{r} 1 \overline{) x^3 - 4x^2 + 7x - 6} \\ \underline{x^3 - 5x^2 + 10x - 8} \quad (-) \\ -x^2 + 3x - 2 \\ \underline{-(x^2 - 3x + 2)} \\ 0 \end{array}$$

$$\begin{array}{r} x - 1 \overline{) x^3 - 4x^2 + 7x - 6} \\ \underline{x^3 - 3x^2 + 2x} \quad (-) \\ -x^2 + 5x - 6 \\ \underline{-x^2 + 3x - 2} \quad (-) \\ 2x - 4 \\ \underline{2(x - 2)} \\ 0 \end{array}$$

$$\begin{array}{r} x - 1 \overline{) x^2 - 3x + 2} \\ \underline{x^2 - 2x} \quad (-) \\ -x + 2 \\ \underline{-x + 2} \quad (-) \\ 0 \end{array}$$

GCD of $(6x^3 - 30x^2 + 60x - 48, 3x^3 - 12x^2 + 21x - 18) = 3(x - 2)$

9. Find the GCD of the given polynomials

(i) $x^4 + 3x^3 - x - 3, x^3 + x^2 - 5x + 3$ **JUL-20**

(ii) $3x^4 + 6x^3 - 12x^2 - 24x, 4x^4 + 14x^3 + 8x^2 - 8x$

(iii) $3x^3 + 3x^2 + 3x + 3, 6x^3 + 12x^2 + 6x + 12$

solution

(i) Divide $x^4 + 3x^3 + 0x^2 - x - 3$ by $x^3 + x^2 - 5x + 3$:

$$\begin{array}{r} x + 2 \overline{) x^4 + 3x^3 + 0x^2 - x - 3} \\ \underline{x^4 + x^3 - 5x^2 + 3x} \quad (-) \\ 2x^3 + 5x^2 - 4x - 3 \\ \underline{2x^3 + 2x^2 - 10x + 6} \quad (-) \\ 3x^2 + 6x - 9 \\ \underline{3(x^2 + 2x - 3)} \\ 0 \end{array}$$

$$\begin{array}{r} x - 1 \overline{) x^3 + x^2 - 5x + 3} \\ \underline{x^3 + 2x^2 - 3x} \quad (-) \\ -x^2 - 2x + 3 \\ \underline{-x^2 - 2x + 3} \quad (-) \\ 0 \end{array}$$

GCD Of $(x^4 + 3x^3 - x - 3, x^3 + x^2 - 5x + 3) = x^2 + 2x - 3$. **(ii)**

$f(x) = 3x(x^3 + 2x^2 - 4x - 8)$ and $g(x) = 2x(2x^3 + 7x^2 + 4x - 4)$. The common factor is x .

Divide $2x^3 + 7x^2 + 4x - 4$ by $x^3 + 2x^2 - 4x - 8$:

$$\begin{array}{r} 2 \overline{) 2x^3 + 7x^2 + 4x - 4} \\ \underline{2x^3 + 4x^2 - 8x - 16} \quad (-) \\ 3x^2 + 12x + 12 \\ \underline{3(x^2 + 4x + 4)} \\ 0 \end{array}$$

$$\begin{array}{r} x - 2 \overline{) x^3 + 2x^2 - 4x - 8} \\ \underline{x^3 + 4x^2 + 4x} \quad (-) \\ -2x^2 - 8x - 8 \\ \underline{-2x^2 - 8x - 8} \quad (-) \\ 0 \end{array}$$

GCD Of $(3x^4 + 6x^3 - 12x^2 - 24x, 4x^4 + 14x^3 + 8x^2 - 8x) = x(x^2 + 4x + 4)$.

(iii) $f(x) = 3(x^3 + x^2 + x + 1)$ and $g(x) = 6(x^3 + 2x^2 + x + 2)$. The

common factor is $(3, 6) = 3$.

Divide $x^3 + 2x^2 + x + 2$ by $x^3 + x^2 + x + 1$:

$$\begin{array}{r} 1 \overline{) x^3 + 2x^2 + x + 2} \\ \underline{x^3 + x^2 + x + 1} \quad (-) \\ x^2 + 1 \end{array}$$

$$\begin{array}{r}
 x+1 \\
 \hline
 x^2+1 \quad x^3+x^2+x+1 \\
 x^3+0x^2+x \quad (-) \\
 \hline
 x^2+0x+1 \\
 x^2+0x+1 \quad (-) \\
 \hline
 0
 \end{array}$$

GCD Of $(3x^3 + 3x^2 + 3x + 3, 6x^3 + 12x^2 + 6x + 12) = 3(x^2 + 1)$.

10. Find $\frac{16x^2-2x-3}{3x^2-2x-1} \div \frac{8x^2+11x+3}{3x^2-11x-4}$

solution

$$\begin{array}{r}
 16x^2 - 2x - 3 \\
 \swarrow \quad \searrow \\
 16x^2 - 8x \quad +6x - 3 \\
 | \quad \quad | \\
 8x(2x - 1) \quad +3(2x - 1) \\
 \swarrow \quad \searrow \\
 (8x + 3)(2x - 1)
 \end{array}$$

$$\begin{array}{r}
 3x^2 - 2x - 1 \\
 \swarrow \quad \searrow \\
 3x^2 - 3x \quad +x - 1 \\
 | \quad \quad | \\
 3x(x - 1) \quad +1(x - 1) \\
 \swarrow \quad \searrow \\
 (3x + 1)(x - 1)
 \end{array}$$

$$\begin{array}{r}
 8x^2 + 11x + 3 \\
 \swarrow \quad \searrow \\
 8x^2 + 8x \quad +3x + 3 \\
 | \quad \quad | \\
 8x(x + 1) \quad +3(x + 1) \\
 \swarrow \quad \searrow \\
 (8x + 3)(x + 1)
 \end{array}$$

$$\begin{array}{r}
 3x^2 - 11x - 4 \\
 \swarrow \quad \searrow \\
 3x^2 - 12x \quad +x - 4 \\
 | \quad \quad | \\
 3x(x - 4) \quad +1(x - 4) \\
 \swarrow \quad \searrow \\
 (3x + 1)(x - 4)
 \end{array}$$

$$16x^2 - 2x - 3 = 16x^2 - 8x + 6x - 3 = (8x + 3)(2x - 1)$$

$$3x^2 - 2x - 1 = 3x^2 - 3x + x - 1 = (3x + 1)(x - 1)$$

$$8x^2 + 11x + 3 = 8x^2 + 8x + 3x + 3 = (8x + 3)(x + 1)$$

$$3x^2 - 11x - 4 = 3x^2 - 12x + x - 4 = (3x + 1)(x - 4)$$

$$\begin{aligned}
 \frac{16x^2 - 2x - 3}{3x^2 - 2x - 1} \div \frac{8x^2 + 11x + 3}{3x^2 - 11x - 4} &= \frac{(8x + 3)(2x - 1)}{(3x + 1)(x - 1)} \times \frac{(3x + 1)(x - 4)}{(8x + 3)(x + 1)} \\
 &= \frac{(2x - 1)(x - 4)}{(x - 1)(x + 1)}
 \end{aligned}$$

11. Simplify: $\frac{b^2+3b-28}{b^2+4b+4} \div \frac{b^2-49}{b^2-5b-14}$ **JUL-22**

solution

$$\begin{aligned}
 \frac{b^2 + 3b - 28}{b^2 + 4b + 4} \div \frac{b^2 - 49}{b^2 - 5b - 14} &= \frac{(b + 7)(b - 4)}{(b + 2)^2} \times \frac{b^2 - 5b - 14}{b^2 - 49} \\
 &= \frac{(b + 7)(b - 4)}{(b + 2)^2} \times \frac{(b - 7)(b + 2)}{(b - 7)(b + 7)} \\
 &= \frac{b - 4}{b + 2}
 \end{aligned}$$

12. If $x = \frac{a^2+3a-4}{3a^2-3}$ and $y = \frac{a^2+2a-8}{2a^2-2a-4}$ find the value of x^2y^{-2} . **PTA-3**

solution

$$x = \frac{(a + 4)(a - 1)}{3(a^2 - 1)} = \frac{(a + 4)(a - 1)}{3(a - 1)(a + 1)} = \frac{a + 4}{3(a + 1)}$$

$$y = \frac{(a + 4)(a - 2)}{2(a^2 - a - 2)} = \frac{(a + 4)(a - 2)}{2(a - 2)(a + 1)} = \frac{a + 4}{2(a + 1)}$$

$$\frac{x}{y} = \frac{a + 4}{3(a + 1)} \times \frac{2(a + 1)}{a + 4} = \frac{2}{3}$$

$$x^2y^{-2} = \left(\frac{x}{y}\right)^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

13. Simplify $\frac{1}{x^2-5x+6} + \frac{1}{x^2-3x+2} - \frac{1}{x^2-8x+15}$

solution

$$\begin{aligned}
 &\frac{1}{x^2 - 5x + 6} + \frac{1}{x^2 - 3x + 2} - \frac{1}{x^2 - 8x + 15} \\
 &= \frac{1}{(x - 2)(x - 3)} + \frac{1}{(x - 1)(x - 2)} - \frac{1}{(x - 3)(x - 5)} \\
 &= \frac{(x - 1) + (x - 3)}{(x - 1)(x - 2)(x - 3)} - \frac{1}{(x - 3)(x - 5)} \\
 &= \frac{2(x - 2)}{(x - 1)(x - 2)(x - 3)} - \frac{1}{(x - 3)(x - 5)} \\
 &= \frac{2}{(x - 1)(x - 3)} - \frac{1}{(x - 3)(x - 5)} \\
 &= \frac{2(x - 5) - (x - 1)}{(x - 1)(x - 3)(x - 5)} = \frac{x - 9}{(x - 1)(x - 3)(x - 5)}
 \end{aligned}$$

14. If $A = \frac{2x+1}{2x-1}$, $B = \frac{2x-1}{2x+1}$ find $\frac{1}{A-B} - \frac{2B}{A^2-B^2}$

solution

$$\begin{aligned}
 \frac{1}{A - B} - \frac{2B}{A^2 - B^2} &= \frac{A + B}{(A - B)(A + B)} - \frac{2B}{(A + B)(A - B)} \\
 &= \frac{A + B - 2B}{(A + B)(A - B)} = \frac{A - B}{(A + B)(A - B)} = \frac{1}{A + B} \\
 A + B &= \frac{2x + 1}{2x - 1} + \frac{2x - 1}{2x + 1} = \frac{(2x + 1)^2 + (2x - 1)^2}{4x^2 - 1} \\
 &= \frac{4x^2 + 4x + 1 + 4x^2 - 4x + 1}{4x^2 - 1} = \frac{8x^2 + 2}{4x^2 - 1} \\
 \frac{1}{A + B} &= \frac{4x^2 - 1}{2(4x^2 + 1)}
 \end{aligned}$$

15. If $A = \frac{x}{x+1}$, $B = \frac{1}{x+1}$, prove that $\frac{(A+B)^2 + (A-B)^2}{A \div B} = \frac{2(x^2+1)}{x(x+1)^2}$

solution

$$\begin{aligned}
 \frac{(A + B)^2 + (A - B)^2}{A \div B} &= \frac{2(A^2 + B^2)}{A \div B} \\
 A^2 + B^2 &= \left(\frac{x}{x+1}\right)^2 + \left(\frac{1}{x+1}\right)^2 = \frac{x^2 + 1}{(x+1)^2} \\
 A \div B &= \frac{x}{x+1} \times \frac{x+1}{1} = x \\
 \frac{2(A^2 + B^2)}{A \div B} &= \frac{2\left(\frac{x^2+1}{(x+1)^2}\right)}{x} = \frac{2(x^2+1)}{x(x+1)^2} = \text{RHS}
 \end{aligned}$$

16. Iniya bought 50 kg of fruits consisting of apples and bananas. She paid twice as much per kg for the apple as she did for the banana. If Iniya bought 1800 worth of apples and 600 worth bananas, then how many kgs of each fruit did she buy?

solution

Let kg of apples = x , bananas = y . $x + y = 50$

Let price of banana = a . Price of apple = $2a$

Cost of apples: $x(2a) = 1800 \implies xa = 900$

Cost of bananas: $y(a) = 600$

$$\text{Divide: } \frac{xa}{ya} = \frac{900}{600} \implies \frac{x}{y} = \frac{3}{2}$$

$$x = \frac{3}{2}y$$

$$\frac{3}{2}y + y = 50 \implies \frac{5}{2}y = 50 \implies y = 20$$

$$x = 50 - 20 = 30$$

Iniya bought 30 kg apples and 20 kg bananas.

17. Find the square root of $64x^4 - 16x^3 + 17x^2 - 2x + 1$ **APR-24, JUL-21, 24**

solution

$$\begin{array}{r}
 8x^2 - 1x + 1 \\
 8x^2 \overline{) 64x^4 - 16x^3 + 17x^2 - 2x + 1} \\
 \underline{64x^4} \\
 (-) \\
 2 \times 8x^2 - 1x \\
 \underline{16x^2 - 2x + 1} \\
 16x^2 - 2x + 1 \\
 \underline{16x^2 - 2x + 1} \\
 (-) \\
 0
 \end{array}
 \quad
 \begin{array}{l}
 \sqrt{64x^4} = 8x^2 \\
 -\frac{16x^3}{16x^2} = -1x \\
 \frac{16x^2}{16x^2} = +1
 \end{array}$$

$$\sqrt{64x^4 - 16x^3 + 17x^2 - 2x + 1} = |8x^2 - x + 1|.$$

18. If $9x^4 + 12x^3 + 28x^2 + ax + b$ is a perfect square, find the values of a and b. **PTA-5**

solution

$$\begin{array}{r}
 3x^2 + 2x + 4 \\
 3x^2 \overline{) 9x^4 + 12x^3 + 28x^2 + ax + b} \\
 \underline{9x^4} \\
 (-) \\
 6x^2 + 2x \\
 \underline{12x^3 + 28x^2} \\
 12x^3 + 4x^2 \\
 \underline{12x^3} \\
 (-) \\
 6x^2 + 4x + 4 \\
 \underline{24x^2 + ax + b} \\
 24x^2 + 16x + 16 \\
 \underline{24x^2 + 16x + 16} \\
 (-) \\
 (a - 16)x + (b - 16)
 \end{array}
 \quad
 \begin{array}{l}
 \sqrt{9x^4} = 3x^2 \\
 \frac{12x^3}{6x^2} = +2x \\
 \frac{24x^2}{6x^2} = +4
 \end{array}$$

$$(a - 16) = 0 \Rightarrow a = 16, \quad (b - 16) = 0 \Rightarrow b = 16.$$

19. Find the square root of the following polynomials by division method

(i) $x^4 - 12x^3 + 42x^2 - 36x + 9$ **JUL-22**

(ii) $37x^2 - 28x^3 + 4x^4 + 42x + 9$

(iii) $16x^4 + 8x^2 + 1$

(iv) $121x^4 - 198x^3 - 183x^2 + 216x + 144$

solution

(i)

$$\begin{array}{r}
 x^2 - 6x + 3 \\
 x^2 \overline{) x^4 - 12x^3 + 42x^2 - 36x + 9} \\
 \underline{x^4} \\
 (-) \\
 2x^2 - 6x \\
 \underline{12x^3 + 42x^2} \\
 12x^3 + 36x^2 \\
 \underline{12x^3} \\
 (+) \\
 6x^2 - 36x + 9 \\
 \underline{6x^2 - 36x + 9} \\
 (-) \\
 0
 \end{array}
 \quad
 \begin{array}{l}
 \sqrt{x^4} = x^2 \\
 -\frac{12x^3}{2x^2} = -6x \\
 \frac{6x^2}{2x^2} = +3
 \end{array}$$

$$\sqrt{x^4 - 12x^3 + 42x^2 - 36x + 9} = |x^2 - 6x + 3|$$

(ii) Arrange Ascending Order: $4x^4 - 28x^3 + 37x^2 + 42x + 9$

$$\begin{array}{r}
 2x^2 - 7x - 3 \\
 2x^2 \overline{) 4x^4 - 28x^3 + 37x^2 + 42x + 9} \\
 \underline{4x^4} \\
 (-) \\
 4x^2 - 7x \\
 \underline{28x^3 + 37x^2} \\
 28x^3 + 49x^2 \\
 \underline{28x^3} \\
 (+) \\
 -12x^2 + 42x + 9 \\
 \underline{-12x^2 + 42x + 9} \\
 (-) \\
 0
 \end{array}
 \quad
 \begin{array}{l}
 \sqrt{4x^4} = 2x^2 \\
 -\frac{28x^3}{4x^2} = -7x \\
 -\frac{12x^2}{4x^2} = -3
 \end{array}$$

$$\sqrt{37x^2 - 28x^3 + 4x^4 + 42x + 9} = |2x^2 - 7x - 3|.$$

(iii) Rewrite: $16x^4 + 0x^3 + 8x^2 + 0x + 1$

$$\begin{array}{r}
 4x^2 + 1 \\
 4x^2 \overline{) 16x^4 + 8x^2 + 1} \\
 \underline{16x^4} \\
 (-) \\
 8x^2 + 1 \\
 \underline{8x^2 + 1} \\
 (-) \\
 0
 \end{array}
 \quad
 \begin{array}{l}
 \sqrt{16x^4} = 4x^2 \\
 \frac{8x^2}{8x^2} = +1
 \end{array}$$

$$\sqrt{16x^4 + 8x^2 + 1} = |4x^2 + 1|$$

(iv)

$$\begin{array}{r}
 11x^2 - 9x - 12 \\
 11x^2 \overline{) 121x^4 - 198x^3 - 183x^2 + 216x + 144} \\
 \underline{121x^4} \\
 (-) \\
 22x^2 - 9x \\
 \underline{-198x^3 - 183x^2} \\
 -198x^3 + 81x^2 \\
 \underline{-198x^3} \\
 (+) \\
 -264x^2 + 216x + 144 \\
 \underline{-264x^2 + 216x + 144} \\
 (+) \\
 0
 \end{array}
 \quad
 \begin{array}{l}
 \sqrt{121x^4} = 11x^2 \\
 -\frac{198x^3}{22x^2} = -9x \\
 -\frac{264x^2}{22x^2} = -12
 \end{array}$$

$$\sqrt{121x^4 - 198x^3 - 183x^2 + 216x + 144} = |11x^2 - 9x - 12|$$

20. Find the values of a and b if the following polynomials are perfect squares

(i) $4x^4 - 12x^3 + 37x^2 + bx + a$ **PTA-4**

(ii) $ax^4 + bx^3 + 361x^2 + 220x + 100$

solution

(i)

$$\begin{array}{r}
 2x^2 - 3x + 7 \\
 2x^2 \overline{) 4x^4 - 12x^3 + 37x^2 + bx + a} \\
 \underline{4x^4} \\
 (-) \\
 4x^2 - 3x \\
 \underline{12x^3 + 37x^2} \\
 12x^3 + 9x^2 \\
 \underline{12x^3} \\
 (+) \\
 28x^2 + bx + a \\
 \underline{28x^2 - 42x + 49} \\
 (-) \\
 0
 \end{array}
 \quad
 \begin{array}{l}
 \sqrt{4x^4} = 2x^2 \\
 -\frac{12x^3}{4x^2} = -3x \\
 \frac{28x^2}{4x^2} = +7
 \end{array}$$

$$(b + 42) = 0 \Rightarrow b = -42, \quad (a - 49) = 0 \Rightarrow a = 49$$

(ii) Rewrite Ascending Order: $100 + 220x + 361x^2 + bx^3 + ax^4$

$$\begin{array}{r}
 10 + 11x + 12x^2 \\
 10 \overline{) 100 + 220x + 361x^2 + bx^3 + ax^4} \\
 \underline{100} \\
 (-) \\
 20 + 11x \\
 \underline{220x + 361x^2} \\
 220x + 121x^2 \\
 \underline{220x} \\
 (-) \\
 240x^2 + bx^3 + ax^4 \\
 \underline{240x^2 + 264x^3 + 144x^4} \\
 (-) \\
 0
 \end{array}
 \quad
 \begin{array}{l}
 \sqrt{100} = 10 \\
 \frac{220x}{20} = +11x \\
 \frac{240x^2}{20} = +12x^2
 \end{array}$$

$$(b - 264) = 0 \Rightarrow b = 264, \quad (a - 144) = 0 \Rightarrow a = 144$$

21. Find the values of m and n if the following polynomials are perfect squares

(i) $36x^4 - 60x^3 + 61x^2 - mx + n$ **APR-22, JUL-25**

(ii) $x^4 - 8x^3 + mx^2 + nx + 16$

solution

(i)

$$\begin{array}{r}
 6x^2 - 5x + 3 \\
 6x^2 \overline{) 36x^4 - 60x^3 + 61x^2 - mx + n} \\
 \underline{36x^4} \\
 (-) \\
 \underline{-60x^3 + 61x^2} \\
 (-) \\
 \underline{-60x^3 + 25x^2} \\
 (+) \\
 \underline{36x^2 - mx + n} \\
 36x^2 - 30x + 9 \\
 (-) \\
 \underline{36x^2 - 30x + 9} \\
 0
 \end{array}
 \quad
 \begin{array}{l}
 \sqrt{36x^4} = 6x^2 \\
 -\frac{60x^3}{12x^2} = -5x \\
 \frac{36x^2}{12x^2} = +3
 \end{array}$$

$$(-m + 30) = 0 \Rightarrow m = 30, \quad (n - 9) = 0 \Rightarrow n = 9.$$

(ii)

$$\begin{array}{r}
 x^2 - 4x + 4 \\
 x^2 \overline{) x^4 - 8x^3 + mx^2 + nx + 16} \\
 \underline{x^4} \\
 (-) \\
 \underline{-8x^3 + mx^2} \\
 (-) \\
 \underline{-8x^3 + 16x^2} \\
 (+) \\
 \underline{(m-16)x^2 + nx + 16} \\
 8x^2 - 32x + 16 \\
 (-) \\
 \underline{8x^2 - 32x + 16} \\
 0
 \end{array}
 \quad
 \begin{array}{l}
 \sqrt{x^4} = x^2 \\
 -\frac{8x^3}{2x^2} = -4x \\
 \sqrt{16} = +4
 \end{array}$$

$$(m - 16 - 8) = 0 \Rightarrow m = 24, \quad (n + 32) = 0 \Rightarrow n = -32.$$

22. Solve $pqx^2 - (p+q)^2x + (p+q)^2 = 0$ **APR-22****solution**

$$\begin{aligned}
 pqx^2 - (p+q)^2x + (p+q)^2 &= 0 \\
 pqx^2 - (p^2 + 2pq + q^2)x + (p+q)^2 &= 0 \\
 pqx^2 - (p^2 + pq + pq + q^2)x + (p+q)^2 &= 0 \\
 pqx^2 - p(p+q)x - q(p+q)x + (p+q)^2 &= 0
 \end{aligned}$$

Group the terms:

$$\begin{aligned}
 px(qx - (p+q)) - (p+q)(qx - (p+q)) &= 0 \\
 (px - (p+q))(qx - (p+q)) &= 0
 \end{aligned}$$

$$\boxed{x = \frac{p+q}{p}}, \quad \boxed{x = \frac{p+q}{q}}$$

23. A bus covers a distance of 90 km at a uniform speed. Had the speed been 15 km/ hour more it would have taken 30 minutes less for the journey. Find the original speed of the bus.

solutionThe original speed of Bus = v km/h.

$$\text{Time taken } t_1 = \frac{90}{v}$$

$$\text{New time } t_2 = \frac{90}{v+15}$$

$$t_1 - t_2 = 30 \text{ mins} = \frac{1}{2} \text{ hour}$$

$$\begin{aligned}
 \frac{90}{v} - \frac{90}{v+15} &= \frac{1}{2} \\
 \frac{90(v+15) - 90v}{v(v+15)} &= \frac{1}{2}
 \end{aligned}$$

$$\frac{1350}{v^2 + 15v} = \frac{1}{2}$$

$$v^2 + 15v = 2700$$

$$v^2 + 15v - 2700 = 0$$

$$v^2 + 60v - 45v - 2700 = 0$$

$$(v+60)(v-45) = 0$$

$$v = 45 \text{ km/h (speed can't be -ve)}$$

Hence the Original Speed of Bus is $\boxed{45 \text{ km/hr}}$.

24. A girl is twice as old as her sister. Five years hence, the product of their ages (in years) will be 375. Find their present ages. **PTA-4**

solutionLet sister's present age = x Girl's present age = $2x$ Ages after 5 years = $(x+5)$ and $(2x+5)$

$$(x+5)(2x+5) = 375$$

$$2x^2 + 5x + 10x + 25 = 375$$

$$2x^2 + 15x - 350 = 0$$

$$2x^2 + 35x - 20x - 350 = 0$$

$$x(2x+35) - 10(2x+35) = 0$$

$$(x-10)(2x+35) = 0 \Rightarrow \boxed{x = 10}$$

Sister's age = 10 yrs, Girl's age = 20 yrs.

25. From a group of $2x^2$ black bees, square root of half of the group went to a tree. Again eight-ninth of the bees went to the same tree. The remaining two got caught up in a fragrant lotus. How many bees were there in total?

solution

$$\text{Total bees} = 2x^2$$

$$\text{First group} = \sqrt{\frac{1}{2}(2x^2)} = \sqrt{x^2} = x$$

$$\text{Second group} = \frac{8}{9}(2x^2) = \frac{16x^2}{9}$$

$$\text{Remaining} = 2$$

$$x + \frac{16x^2}{9} + 2 = 2x^2$$

$$\text{Multiply by 9: } 9x + 16x^2 + 18 = 18x^2$$

$$2x^2 - 9x - 18 = 0$$

$$2x^2 - 12x + 3x - 18 = 0$$

$$2x(x-6) + 3(x-6) = 0$$

$$(2x+3)(x-6) = 0 \Rightarrow \boxed{x = 6}$$

$$\text{Total bees} = 2(6)^2 = 2(36) = \boxed{72 \text{ bees.}}$$

26. The hypotenuse of a right angled triangle is 25 cm and its perimeter 56 cm. Find the length of the smallest side.

solutionLet the two legs be a and b .

$$\text{Hypotenuse } c = 25$$

$$a^2 + b^2 = 25^2 = 625 \quad \dots (1)$$

$$\text{Perimeter: } a + b + 25 = 56 \Rightarrow a + b = 31$$

$$b = 31 - a \quad \dots (2)$$

$$\text{Sub (2) into (1): } a^2 + (31-a)^2 = 625$$

$$a^2 + 961 - 62a + a^2 = 625$$

$$2a^2 - 62a + 336 = 0$$

$$a^2 - 31a + 168 = 0$$

$$a^2 - 7a - 24a + 168 = 0$$

$$a(a-7) - 24(a-7) = 0$$

$$(a-7)(a-24) = 0 \Rightarrow a = 7 \text{ or } 24$$

The sides are 7 and 24. Smallest side = 7 cm.

27. If the roots of $(a-b)x^2 + (b-c)x + (c-a) = 0$ are real and equal, then prove that b, a, c are in arithmetic progression.

solution

$$\text{Sum of the coefficients: } (a-b) + (b-c) + (c-a) = 0$$

This means $x = 1$ is always a root.

$$\text{Product of roots } (\alpha \times \beta) = \frac{c}{a} = \frac{c-a}{a-b}$$

$$1 \times 1 = \frac{c-a}{a-b}$$

$$a-b = c-a$$

$$2a = b + c$$

This is the condition for b, a, c to be in A.P.

28. If a, b are real then show that the roots of the equation $(a-b)x^2 - 6(a+b)x - 9(a-b) = 0$ are real and unequal.

solution

Given Equation $(a-b)x^2 - 6(a+b)x - 9(a-b) = 0$.

$$\begin{aligned} a &= (a-b), b = -6(a+b), c = -9(a-b) \\ \Delta &= b^2 - 4ac \\ &= [-6(a+b)]^2 - 4(a-b)[-9(a-b)] \\ &= 36(a+b)^2 + 36(a-b)^2 \\ &= 36[(a^2 + 2ab + b^2) + (a^2 - 2ab + b^2)] \\ &= 36[2a^2 + 2b^2] = 72(a^2 + b^2) \end{aligned}$$

Since a, b are real, $a^2 + b^2 > 0$

$$\Rightarrow \Delta > 0$$

$\therefore$ The roots are real and unequal.

29. If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + b^2 - ac = 0$ are real and equal prove that either $a = 0$ (or) $a^3 + b^3 + c^3 = 3abc$

PTA-6

solution

Given Equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + b^2 - ac = 0$

$$\begin{aligned} a &= (c^2 - ab), b = -2(a^2 - bc), c = (b^2 - ac) \\ \Delta &= b^2 - 4ac = 0 \\ [-2(a^2 - bc)]^2 - 4(c^2 - ab)(b^2 - ac) &= 0 \\ 4(a^4 - 2a^2bc + b^2c^2) - 4(b^2c^2 - ac^3 - ab^3 + a^2bc) &= 0 \\ \text{Divide by 4:} \\ a^4 - 2a^2bc + b^2c^2 - b^2c^2 + ac^3 + ab^3 - a^2bc &= 0 \\ a^4 - 3a^2bc + ab^3 + ac^3 &= 0 \\ a(a^3 + b^3 + c^3 - 3abc) &= 0 \\ a &= 0 \\ \text{or } a^3 + b^3 + c^3 &= 3abc \end{aligned}$$

30. If α, β are the roots of the equation $3x^2 + 7x - 2 = 0$ find the values of

(i) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (ii) $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$

solution

$$\begin{aligned} \alpha + \beta &= -\frac{7}{3}, \quad \alpha\beta = -\frac{2}{3} \\ \text{(i)} \quad \frac{\alpha}{\beta} + \frac{\beta}{\alpha} &= \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} \\ &= \frac{(-\frac{7}{3})^2 - 2(-\frac{2}{3})}{-\frac{2}{3}} = \frac{\frac{49}{9} + \frac{4}{3}}{-\frac{2}{3}} \\ &= \frac{\frac{61}{9}}{-\frac{2}{3}} = -\frac{61}{6} \\ \text{(ii)} \quad \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} &= \frac{\alpha^3 + \beta^3}{\alpha\beta} = \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{\alpha\beta} \\ &= \frac{(-\frac{7}{3})^3 - 3(-\frac{2}{3})(-\frac{7}{3})}{-\frac{2}{3}} \\ &= \frac{-\frac{343}{27} - \frac{14}{3}}{-\frac{2}{3}} = \frac{-\frac{343}{27} - \frac{126}{27}}{-\frac{18}{27}} \\ &= \frac{-469}{-18} = \frac{469}{18} \end{aligned}$$

31. If α, β are the roots of the equation $2x^2 - x - 1 = 0$ then form the equation whose roots are

(i) $\frac{1}{\alpha}, \frac{1}{\beta}$ (ii) $\alpha^2\beta, \beta^2\alpha$ **MDL-19** (iii) $2\alpha + \beta, 2\beta + \alpha$

solution

$$\begin{aligned} \alpha + \beta &= \frac{1}{2}, \quad \alpha\beta = -\frac{1}{2} \\ \text{(i)} \quad \text{Sum} &= \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{\frac{1}{2}}{-\frac{1}{2}} = -1 \\ \text{Product} &= \frac{1}{\alpha\beta} = \frac{1}{-\frac{1}{2}} = -2 \\ x^2 - (\alpha + \beta)x + \alpha\beta &= 0 \\ x^2 - (-1)x + (-2) &= 0 \Rightarrow x^2 + x - 2 = 0 \end{aligned}$$

$$\text{(ii)} \quad \text{Sum} = \alpha^2\beta + \alpha\beta^2 = \alpha\beta(\alpha + \beta) = -\frac{1}{2} \left(\frac{1}{2} \right) = -\frac{1}{4}$$

$$\text{Product} = (\alpha^2\beta)(\alpha\beta^2) = (\alpha\beta)^3 = \left(-\frac{1}{2} \right)^3 = -\frac{1}{8}$$

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$x^2 - \left(-\frac{1}{4} \right)x - \frac{1}{8} = 0 \Rightarrow 8x^2 + 2x - 1 = 0$$

$$\text{(iii)} \quad \text{Sum} = 2\alpha + \beta + 2\beta + \alpha = 3(\alpha + \beta) = \frac{3}{2}$$

$$\begin{aligned} \text{Product} &= (2\alpha + \beta)(\alpha + 2\beta) = 2(\alpha^2 + \beta^2) + 5\alpha\beta \\ &= 2[(\alpha + \beta)^2 - 2\alpha\beta] + 5\alpha\beta = 2(\alpha + \beta)^2 + \alpha\beta \\ &= 2 \left(\frac{1}{4} \right) - \frac{1}{2} = 0 \end{aligned}$$

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$x^2 - \frac{3}{2}x + 0 = 0 \Rightarrow 2x^2 - 3x = 0$$

32. The roots of the equation $2x^2 - 7x + 5 = 0$ are α and β . Without solving for the roots, find

(i) $\frac{1}{\alpha} + \frac{1}{\beta}$ (ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (iii) $\frac{\alpha+2}{\beta+2} + \frac{\beta+2}{\alpha+2}$

solution

Given quadratic Equation $2x^2 - 7x + 5 = 0$.

$$\alpha + \beta = -\frac{b}{a} = \frac{7}{2}, \quad \alpha\beta = \frac{c}{a} = \frac{5}{2}$$

$$\text{(i)} \quad \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{\frac{7}{2}}{\frac{5}{2}} = \frac{7}{5}$$

$$\begin{aligned} \text{(ii)} \quad \frac{\alpha}{\beta} + \frac{\beta}{\alpha} &= \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} \\ &= \frac{\frac{49}{4} - 5}{\frac{5}{2}} = \frac{\frac{29}{4}}{\frac{5}{2}} = \frac{29}{10} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \frac{\alpha + 2}{\beta + 2} + \frac{\beta + 2}{\alpha + 2} &= \frac{(\alpha + 2)^2 + (\beta + 2)^2}{(\alpha + 2)(\beta + 2)} \\ &= \frac{(\alpha^2 + \beta^2) + 4(\alpha + \beta) + 8}{\alpha\beta + 2(\alpha + \beta) + 4} \\ \text{Num} &= \left(\frac{29}{4} \right) + 4 \left(\frac{7}{2} \right) + 8 = \frac{29}{4} + 22 = \frac{117}{4} \\ \text{Den} &= \frac{5}{2} + 2 \left(\frac{7}{2} \right) + 4 = \frac{5}{2} + 11 = \frac{27}{2} = \frac{54}{4} \\ &= \frac{\frac{117}{4}}{\frac{54}{4}} = \frac{117}{54} = \frac{13}{6} \end{aligned}$$

33. If α and β are the roots of $x^2 + 7x + 10 = 0$ find the values of

(i) $(\alpha - \beta)$ (ii) $\alpha^2 + \beta^2$ (iii) $\alpha^3 - \beta^3$ (iv) $\alpha^4 + \beta^4$ (v) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

(vi) $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ **APR-22**

solution

Given Quadratic Equation $x^2 + 7x + 10 = 0$.

$$\alpha + \beta = -\frac{b}{a} = -7, \quad \alpha\beta = \frac{c}{a} = 10$$

$$\begin{aligned} \text{(i)} \quad \alpha - \beta &= \pm \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} \\ &= \pm \sqrt{49 - 40} = \pm \sqrt{9} = \pm 3 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= 49 - 20 = 29 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \alpha^3 - \beta^3 &= (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2) \\ &= (\pm 3)(29 + 10) = \pm 3(39) = \pm 117 \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \alpha^4 + \beta^4 &= (\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 \\ &= (29)^2 - 2(10)^2 = 841 - 200 = 641 \end{aligned}$$

$$\text{(v)} \quad \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{29}{10}$$

$$\begin{aligned} \text{(vi)} \quad \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} &= \frac{\alpha^3 + \beta^3}{\alpha\beta} = \frac{(\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)}{\alpha\beta} \\ &= \frac{-7(29 - 10)}{10} = \frac{-7(19)}{10} = -\frac{133}{10} \end{aligned}$$

34. If α, β are the roots of $7x^2 + ax + 2 = 0$ and if $\beta - \alpha = \frac{-13}{7}$. Find the values of a. **PTA-6**

solution

Given Quadratic Equation $7x^2 + ax + 2 = 0$.

$$\alpha + \beta = -\frac{b}{a} = -\frac{a}{7}, \quad \alpha\beta = \frac{c}{a} = \frac{2}{7}$$

$$\beta - \alpha = -\frac{13}{7} \implies \alpha - \beta = \frac{13}{7}$$

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$\left(\frac{13}{7}\right)^2 = \left(-\frac{a}{7}\right)^2 - 4\left(\frac{2}{7}\right)$$

$$\frac{169}{49} = \frac{a^2}{49} - \frac{8}{7}$$

$$\frac{169}{49} = \frac{a^2}{49} - \frac{56}{49}$$

$$a^2 = 169 + 56 = 225$$

$$a = \pm 15$$

35. If one root of the equation $2y^2 - ay + 64 = 0$ is twice the other then find the values of a **PTA-6**

solution

Let roots be α and 2α

$$\text{Product } (\alpha)(2\alpha) = \frac{64}{2} = 32$$

$$2\alpha^2 = 32 \implies \alpha^2 = 16 \implies \alpha = \pm 4$$

$$\text{Sum } \alpha + 2\alpha = \frac{-(-a)}{2} \implies 3\alpha = \frac{a}{2}$$

$$a = 6\alpha$$

$$\text{If } \alpha = 4, a = 24$$

$$\text{If } \alpha = -4, a = -24$$

$$\therefore a = \pm 24$$

36. If one root of the equation $3x^2 + kx + 81 = 0$ (having real roots) is the square of the other then find k.

solution

Let roots be α and α^2

$$\text{Product } (\alpha)(\alpha^2) = \frac{81}{3} = 27$$

$$\alpha^3 = 27 \implies \alpha = 3 \text{ (since roots are real)}$$

$$\text{Sum } \alpha + \alpha^2 = -\frac{k}{3}$$

$$3 + (3)^2 = -\frac{k}{3}$$

$$3 + 9 = -\frac{k}{3} \implies 12 = -\frac{k}{3}$$

$$k = -36$$

37. Find the value of a, b, c, d from the equation

$$\begin{pmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{pmatrix} = \begin{pmatrix} 1 & 5 \\ 0 & 2 \end{pmatrix}$$

solution

$$a - b = 1 \quad \dots (1)$$

$$2a - b = 0 \implies b = 2a \quad \dots (2)$$

$$2a + c = 5 \quad \dots (3)$$

$$3c + d = 2 \quad \dots (4)$$

$$\text{Sub (2) in (1): } a - 2a = 1 \implies -a = 1 \implies a = -1$$

$$\text{From (2): } b = 2(-1) \implies b = -2$$

$$\text{Sub } a \text{ in (3): } 2(-1) + c = 5 \implies -2 + c = 5 \implies c = 7$$

$$\text{Sub } c \text{ in (4): } 3(7) + d = 2 \implies 21 + d = 2 \implies d = -19$$

$$a = -1, b = -2, c = 7, d = -19$$

38. If $A = \begin{pmatrix} 4 & 3 & 1 \\ 2 & 3 & -8 \\ 1 & 0 & -4 \end{pmatrix}, B = \begin{pmatrix} 2 & 3 & 4 \\ 1 & 9 & 2 \\ -7 & 1 & -1 \end{pmatrix}, C = \begin{pmatrix} 8 & 3 & 4 \\ 1 & -2 & 3 \\ 2 & 4 & -1 \end{pmatrix}$ then verify that $A + (B + C) = (A + B) + C$

solution

$$\text{LHS: } B + C = \begin{pmatrix} 2 & 3 & 4 \\ 1 & 9 & 2 \\ -7 & 1 & -1 \end{pmatrix} + \begin{pmatrix} 8 & 3 & 4 \\ 1 & -2 & 3 \\ 2 & 4 & -1 \end{pmatrix}$$

$$B + C = \begin{pmatrix} 2+8 & 3+3 & 4+4 \\ 1+1 & 9-2 & 2+3 \\ -7+2 & 1+4 & -1-1 \end{pmatrix}$$

$$= \begin{pmatrix} 10 & 6 & 8 \\ 2 & 7 & 5 \\ -5 & 5 & -2 \end{pmatrix}$$

$$A + (B + C) = \begin{pmatrix} 4 & 3 & 1 \\ 2 & 3 & -8 \\ 1 & 0 & -4 \end{pmatrix} + \begin{pmatrix} 10 & 6 & 8 \\ 2 & 7 & 5 \\ -5 & 5 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 4+10 & 3+6 & 1+8 \\ 2+2 & 3+7 & -8+5 \\ 1-5 & 0+5 & -4-2 \end{pmatrix}$$

$$A + (B + C) = \begin{pmatrix} 14 & 9 & 9 \\ 4 & 10 & -3 \\ -4 & 5 & -6 \end{pmatrix}$$

$$\text{RHS: } A + B = \begin{pmatrix} 4 & 3 & 1 \\ 2 & 3 & -8 \\ 1 & 0 & -4 \end{pmatrix} + \begin{pmatrix} 2 & 3 & 4 \\ 1 & 9 & 2 \\ -7 & 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 6 & 5 \\ 3 & 12 & -6 \\ -6 & 1 & -5 \end{pmatrix}$$

$$(A + B) + C = \begin{pmatrix} 6 & 6 & 5 \\ 3 & 12 & -6 \\ -6 & 1 & -5 \end{pmatrix} + \begin{pmatrix} 8 & 3 & 4 \\ 1 & -2 & 3 \\ 2 & 4 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 6+8 & 6+3 & 5+4 \\ 3+1 & 12-2 & -6+3 \\ -6+2 & 1+4 & -5-1 \end{pmatrix}$$

$$(A + B) + C = \begin{pmatrix} 14 & 9 & 9 \\ 4 & 10 & -3 \\ -4 & 5 & -6 \end{pmatrix}$$

LHS = RHS, Associative property verified.

39. Find X and Y if $X + Y = \begin{pmatrix} 7 & 0 \\ 3 & 5 \end{pmatrix}$ and $X - Y = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$.

solution

$$X + Y = \begin{pmatrix} 7 & 0 \\ 3 & 5 \end{pmatrix} \quad \dots (1)$$

$$X - Y = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix} \quad \dots (2)$$

$$\text{Add (1) and (2): } 2X = \begin{pmatrix} 10 & 0 \\ 3 & 9 \end{pmatrix}$$

$$X = \frac{1}{2} \begin{pmatrix} 10 & 0 \\ 3 & 9 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ \frac{3}{2} & \frac{9}{2} \end{pmatrix}$$

$$\text{Subtract (2) from (1): } 2Y = \begin{pmatrix} 4 & 0 \\ 3 & 1 \end{pmatrix}$$

$$Y = \frac{1}{2} \begin{pmatrix} 4 & 0 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ \frac{3}{2} & \frac{1}{2} \end{pmatrix}$$

40. Find the non-zero values of x satisfying the matrix equation **PTA-4**

$$x \begin{pmatrix} 2x & 2 \\ 3 & x \end{pmatrix} + 2 \begin{pmatrix} 8 & 5x \\ 4 & 4x \end{pmatrix} = 2 \begin{pmatrix} x^2 + 8 & 24 \\ 10 & 6x \end{pmatrix}$$

solution

$$\begin{pmatrix} 2x^2 & 2x \\ 3x & x^2 \end{pmatrix} + \begin{pmatrix} 16 & 10x \\ 8 & 8x \end{pmatrix} = \begin{pmatrix} 2x^2 + 16 & 48 \\ 20 & 12x \end{pmatrix}$$

$$\begin{pmatrix} 2x^2 + 16 & 12x \\ 3x + 8 & x^2 + 8x \end{pmatrix} = \begin{pmatrix} 2x^2 + 16 & 48 \\ 20 & 12x \end{pmatrix}$$

$$\text{Equate } a_{12}: 12x = 48 \implies x = 4$$

$$\text{Equate } a_{21}: 3x + 8 = 20 \implies 3x = 12 \implies x = 4$$

$$\text{Equate } a_{22}: x^2 + 8x = 12x$$

$$x^2 - 4x = 0 \implies x(x - 4) = 0$$

$$x = 0 \text{ or } x = 4$$

41. Solve for x, y : $\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + 2 \begin{pmatrix} -2x \\ -y \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \end{pmatrix}$

solution

$$\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \begin{pmatrix} -4x \\ -2y \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \end{pmatrix}$$

$$\begin{pmatrix} x^2 - 4x \\ y^2 - 2y \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \end{pmatrix}$$

For x : $x^2 - 4x = 5 \implies x^2 - 4x - 5 = 0$

$$(x - 5)(x + 1) = 0 \implies \boxed{x = 5, -1}$$

For y : $y^2 - 2y = 8 \implies y^2 - 2y - 8 = 0$

$$(y - 4)(y + 2) = 0 \implies \boxed{y = 4, -2}$$

42. If $A = \begin{pmatrix} 1 & -1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 \\ 2 & 1 \\ 1 & 3 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ show that

$$(AB)C = A(BC).$$

solution

$$\text{LHS: } AB = \begin{pmatrix} 1 & -1 & 2 \end{pmatrix} \times \begin{pmatrix} 1 & -1 \\ 2 & 1 \\ 1 & 3 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 - 2 + 2 & -1 - 1 + 6 \end{pmatrix} = \begin{pmatrix} 1 & 4 \end{pmatrix}$$

$$(AB)C = \begin{pmatrix} 1 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 + 8 & 2 - 4 \end{pmatrix}$$

$$(AB)C = \boxed{\begin{pmatrix} 9 & -2 \end{pmatrix}}$$

$$\text{RHS: } BC = \begin{pmatrix} 1 & -1 \\ 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 - 2 & 2 + 1 \\ 2 + 2 & 4 - 1 \\ 1 + 6 & 2 - 3 \end{pmatrix} = \begin{pmatrix} -1 & 3 \\ 4 & 3 \\ 7 & -1 \end{pmatrix}$$

$$A(BC) = \begin{pmatrix} 1 & -1 & 2 \end{pmatrix} \begin{pmatrix} -1 & 3 \\ 4 & 3 \\ 7 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 - 4 + 14 & 3 - 3 - 2 \end{pmatrix}$$

$$A(BC) = \boxed{\begin{pmatrix} 9 & -2 \end{pmatrix}}$$

$$\text{LHS} = \text{RHS, Hence verified.}$$

43. If $A = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ -4 & 2 \end{pmatrix}$, $C = \begin{pmatrix} -7 & 6 \\ 3 & 2 \end{pmatrix}$ verify that $A(B + C) = AB + AC$. **PTA-1**

solution

$$\text{LHS: } B + C = \begin{pmatrix} 1 & 2 \\ -4 & 2 \end{pmatrix} + \begin{pmatrix} -7 & 6 \\ 3 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} -6 & 8 \\ -1 & 4 \end{pmatrix}$$

$$A(B + C) = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} -6 & 8 \\ -1 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} -6 - 1 & 8 + 4 \\ 6 - 3 & -8 + 12 \end{pmatrix}$$

$$A(B + C) = \boxed{\begin{pmatrix} -7 & 12 \\ 3 & 4 \end{pmatrix}}$$

$$\text{RHS: } AB = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -4 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 - 4 & 2 + 2 \\ -1 - 12 & -2 + 6 \end{pmatrix} = \begin{pmatrix} -3 & 4 \\ -13 & 4 \end{pmatrix}$$

$$AC = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} -7 & 6 \\ 3 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} -7 + 3 & 6 + 2 \\ 7 + 9 & -6 + 6 \end{pmatrix} = \begin{pmatrix} -4 & 8 \\ 16 & 0 \end{pmatrix}$$

$$AB + AC = \begin{pmatrix} -3 & 4 \\ -13 & 4 \end{pmatrix} + \begin{pmatrix} -4 & 8 \\ 16 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} -3 - 4 & 4 + 8 \\ -13 + 16 & 4 + 0 \end{pmatrix}$$

$$AB + AC = \boxed{\begin{pmatrix} -7 & 12 \\ 3 & 4 \end{pmatrix}}$$

$$\text{LHS} = \text{RHS, Hence verified.}$$

44. If $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ -1 & 4 \\ 0 & 2 \end{pmatrix}$ show that $(AB)^T = B^T A^T$

APR-25, JUL-20

solution

$$\text{LHS: } AB = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 4 \\ 0 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2 - 2 + 0 & -1 + 8 + 2 \\ 4 + 1 + 0 & -2 - 4 + 2 \end{pmatrix}$$

$$= \boxed{\begin{pmatrix} 0 & 9 \\ 5 & -4 \end{pmatrix}}$$

$$(AB)^T = \begin{pmatrix} 0 & 5 \\ 9 & -4 \end{pmatrix}$$

$$\text{RHS: } B^T = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 4 & 2 \end{pmatrix}, \quad A^T = \begin{pmatrix} 1 & 2 \\ 2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$B^T A^T = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2(1) - 1(2) + 0 & 2(2) - 1(-1) + 0 \\ -1(1) + 4(2) + 2(1) & -1(2) + 4(-1) + 2(1) \end{pmatrix}$$

$$B^T A^T = \boxed{\begin{pmatrix} 0 & 5 \\ 9 & -4 \end{pmatrix}}$$

$$\text{LHS} = \text{RHS, Hence verified.}$$

45. Given that $A = \begin{pmatrix} 1 & 3 \\ 5 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 & 2 \\ 3 & 5 & 2 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 3 & 2 \\ -4 & 1 & 3 \end{pmatrix}$ verify that $A(B + C) = AB + AC$

solution

$$\text{LHS: } B + C = \begin{pmatrix} 1 & -1 & 2 \\ 3 & 5 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 3 & 2 \\ -4 & 1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 2 & 4 \\ -1 & 6 & 5 \end{pmatrix}$$

$$A(B + C) = \begin{pmatrix} 1 & 3 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 2 & 2 & 4 \\ -1 & 6 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 2 - 3 & 2 + 18 & 4 + 15 \\ 10 + 1 & 10 - 6 & 20 - 5 \end{pmatrix}$$

$$A(B + C) = \boxed{\begin{pmatrix} -1 & 20 & 19 \\ 11 & 4 & 15 \end{pmatrix}}$$

$$\text{RHS: } AB = \begin{pmatrix} 1 & 3 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 \\ 3 & 5 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 + 9 & -1 + 15 & 2 + 6 \\ 5 - 3 & -5 - 5 & 10 - 2 \end{pmatrix}$$

$$AB = \begin{pmatrix} 10 & 14 & 8 \\ 2 & -10 & 8 \end{pmatrix}$$

$$AC = \begin{pmatrix} 1 & 3 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 & 2 \\ -4 & 1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 - 12 & 3 + 3 & 2 + 9 \\ 5 + 4 & 15 - 1 & 10 - 3 \end{pmatrix}$$

$$AC = \begin{pmatrix} -11 & 6 & 11 \\ 9 & 14 & 7 \end{pmatrix}$$

$$AB + AC = \begin{pmatrix} 10 & 14 & 8 \\ 2 & -10 & 8 \end{pmatrix} + \begin{pmatrix} -11 & 6 & 11 \\ 9 & 14 & 7 \end{pmatrix}$$

$$= \begin{pmatrix} 10 - 11 & 14 + 6 & 8 + 11 \\ 2 + 9 & -10 + 14 & 8 + 7 \end{pmatrix}$$

$$= \boxed{\begin{pmatrix} -1 & 20 & 19 \\ 11 & 4 & 15 \end{pmatrix}}$$

$$\text{LHS} = \text{RHS, Hence verified.}$$

46. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ show that $A^2 - (a+d)A = (bc-ad)I_2$.

solution

$$\begin{aligned} \text{LHS: } A^2 &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{pmatrix} \\ (a+d)A &= (a+d) \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + ad & ab + bd \\ ac + cd & ad + d^2 \end{pmatrix} \\ A^2 - (a+d)A &= \begin{pmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{pmatrix} - \begin{pmatrix} a^2 + ad & ab + bd \\ ac + cd & ad + d^2 \end{pmatrix} \\ &= \begin{pmatrix} a^2 + bc - (a^2 + ad) & ab + bd - (ab + bd) \\ ac + cd - (ac + cd) & bc + d^2 - (ad + d^2) \end{pmatrix} \\ &= \begin{pmatrix} bc - ad & 0 \\ 0 & bc - ad \end{pmatrix} \\ &= (bc - ad) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = (bc - ad)I_2 \implies \text{RHS} \end{aligned}$$

47. If $A = \begin{pmatrix} 5 & 2 & 9 \\ 1 & 2 & 8 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 7 \\ 1 & 2 \\ 5 & -1 \end{pmatrix}$ verify that $(AB)^T = B^T A^T$

JUL-23, APR-24, JUL-25

solution

$$\begin{aligned} \text{LHS: } AB &= \begin{pmatrix} 5 & 2 & 9 \\ 1 & 2 & 8 \end{pmatrix} \begin{pmatrix} 1 & 7 \\ 1 & 2 \\ 5 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 5(1) + 2(1) + 9(5) & 5(7) + 2(2) + 9(-1) \\ 1(1) + 2(1) + 8(5) & 1(7) + 2(2) + 8(-1) \end{pmatrix} \\ &= \begin{pmatrix} 5 + 2 + 45 & 35 + 4 - 9 \\ 1 + 2 + 40 & 7 + 4 - 8 \end{pmatrix} = \begin{pmatrix} 52 & 30 \\ 43 & 3 \end{pmatrix} \\ (AB)^T &= \begin{pmatrix} 52 & 43 \\ 30 & 3 \end{pmatrix} \end{aligned}$$

$$\text{RHS: } B^T = \begin{pmatrix} 1 & 1 & 5 \\ 7 & 2 & -1 \end{pmatrix}, \quad A^T = \begin{pmatrix} 5 & 1 \\ 2 & 2 \\ 9 & 8 \end{pmatrix}$$

$$\begin{aligned} B^T A^T &= \begin{pmatrix} 1 & 1 & 5 \\ 7 & 2 & -1 \end{pmatrix} \begin{pmatrix} 5 & 1 \\ 2 & 2 \\ 9 & 8 \end{pmatrix} \\ &= \begin{pmatrix} 1(5) + 1(2) + 5(9) & 1(1) + 1(2) + 5(8) \\ 7(5) + 2(2) - 1(9) & 7(1) + 2(2) - 1(8) \end{pmatrix} \\ &= \begin{pmatrix} 5 + 2 + 45 & 1 + 2 + 40 \\ 35 + 4 - 9 & 7 + 4 - 8 \end{pmatrix} \end{aligned}$$

$$B^T A^T = \begin{pmatrix} 52 & 43 \\ 30 & 3 \end{pmatrix}$$

LHS = RHS, Hence verified.

48. If $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$ show that $A^2 - 5A + 7I_2 = 0$

PTA-3, JUL-24, APR-23, MAR-26

solution

$$\begin{aligned} A^2 &= \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 9 - 1 & 3 + 2 \\ -3 - 2 & -1 + 4 \end{pmatrix} = \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix} \end{aligned}$$

$$5A = 5 \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 15 & 5 \\ -5 & 10 \end{pmatrix}$$

$$7I_2 = 7 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$$

$$\text{LHS} = A^2 - 5A + 7I_2$$

$$= \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix} - \begin{pmatrix} 15 & 5 \\ -5 & 10 \end{pmatrix} + \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$$

$$= \begin{pmatrix} 8 - 15 + 7 & 5 - 5 + 0 \\ -5 - (-5) + 0 & 3 - 10 + 7 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0 = \text{RHS}$$

49. Solve $\frac{1}{3}(x + y - 5) = y - z = 2x - 11 = 9 - (x + 2z)$.

solution

$$\text{Eq 1: } \frac{1}{3}(x + y - 5) = 2x - 11$$

$$x + y - 5 = 6x - 33 \implies 5x - y = 28 \quad \dots (1)$$

$$\text{Eq 2: } y - z = 2x - 11$$

$$2x - y + z = 11 \quad \dots (2)$$

$$\text{Eq 3: } 2x - 11 = 9 - x - 2z$$

$$3x + 2z = 20 \quad \dots (3)$$

$$\text{From (1) } y = 5x - 28.$$

$$\text{Substitute } y \text{ into (2):}$$

$$2x - (5x - 28) + z = 11$$

$$-3x + 28 + z = 11 \implies z = 3x - 17$$

$$\text{Substitute } z \text{ into (3):}$$

$$3x + 2(3x - 17) = 20$$

$$3x + 6x - 34 = 20$$

$$9x = 54 \implies \boxed{x = 6}$$

$$\text{Find } y : y = 5(6) - 28 = 30 - 28 \quad \boxed{y = 2}$$

$$\text{Find } z : z = 3(6) - 17 = 18 - 17 \quad \boxed{z = 1}$$

$$\text{Solution: } \boxed{(x, y, z) = (6, 2, 1)}$$

50. Find the GCD of the following by division algorithm: $2x^4 + 13x^3 + 27x^2 + 23x + 7, x^3 + 3x^2 + 3x + 1, x^2 + 2x + 1$.

solution

$$\begin{array}{r} x^2 + 2x + 1 \overline{) \begin{array}{l} x^3 + 3x^2 + 3x + 1 \\ x^3 + 2x^2 + x \quad (-) \\ \hline x^2 + 2x + 1 \\ x^2 + 2x + 1 \quad (-) \\ \hline 0 \end{array}} \end{array}$$

The GCD of the last two is $x^2 + 2x + 1$.

$$\begin{array}{r} 2x^2 + 9x + 7 \overline{) \begin{array}{l} 2x^4 + 13x^3 + 27x^2 + 23x + 7 \\ 2x^4 + 4x^3 + 2x^2 \\ \hline 9x^3 + 25x^2 + 23x \\ 9x^3 + 18x^2 + 9x \\ \hline 7x^2 + 14x + 7 \\ 7x^2 + 14x + 7 \\ \hline 0 \end{array}} \end{array}$$

GCD of $(2x^4 + 13x^3 + 27x^2 + 23x + 7, x^3 + 3x^2 + 3x + 1, x^2 + 2x + 1) = x^2 + 2x + 1$

51. Find the square root of $289x^4 - 612x^3 + 970x^2 - 684x + 361$.

solution

$$\begin{array}{r} 17x^2 - 18x + 19 \overline{) \begin{array}{l} 289x^4 - 612x^3 + 970x^2 - 684x + 361 \\ 289x^4 \quad (-) \\ \hline -612x^3 + 970x^2 \\ -612x^3 + 324x^2 \quad (-) \\ \hline 34x^2 - 646x^2 - 684x + 361 \\ 34x^2 - 36x + 19 \quad (-) \\ \hline 646x^2 - 684x + 361 \\ 646x^2 - 684x + 361 \quad (-) \\ \hline 0 \end{array}} \end{array}$$

$$\sqrt{289x^4 - 612x^3 + 970x^2 - 684x + 361} = |17x^2 - 18x + 19|$$

52. A boat takes 1.6 hours longer to go 36 kms up a river than down the river. If the speed of the water current is 4 km per hr, what is the speed of the boat in still water?

solution

Let boat speed in still water = v km/hr.

Upstream speed = $v - 4$

Downstream speed = $v + 4$

$$\text{Time Up} - \text{Time Down} = 1.6 \text{ hours}$$

$$\frac{36}{v-4} - \frac{36}{v+4} = 1.6 = \frac{16}{10} = \frac{8}{5}$$

$$36 \left[\frac{(v+4) - (v-4)}{(v-4)(v+4)} \right] = \frac{8}{5}$$

$$36 \left[\frac{8}{v^2 - 16} \right] = \frac{8}{5}$$

$$\frac{36}{v^2 - 16} = \frac{1}{5}$$

$$v^2 - 16 = 180 \implies v^2 = 196$$

$$v = 14 \text{ km/hr}$$

(Since speed cannot be negative)

The Speed of the Boat in Water is 14 km/hr

53. If α and β are the roots of the polynomial $f(x) = x^2 - 2x + 3$, find the polynomial whose roots are

(i) $\alpha + 2, \beta + 2$

(ii) $\frac{\alpha-1}{\alpha+1}, \frac{\beta-1}{\beta+1}$

solution

Given Quadratic Equation $f(x) = x^2 - 2x + 3$.

$$\alpha + \beta = -\frac{b}{a} = -\left(\frac{-2}{1}\right) = 2, \quad \alpha\beta = \frac{c}{a} = \frac{3}{1} = 3$$

$$\begin{aligned} \text{(i) Sum} &= (\alpha + 2) + (\beta + 2) = \alpha + \beta + 4 \\ &= 2 + 4 = 6 \end{aligned}$$

$$\begin{aligned} \text{Product} &= (\alpha + 2)(\beta + 2) = \alpha\beta + 2(\alpha + \beta) + 4 \\ &= 3 + 2(2) + 4 = 11 \end{aligned}$$

$$x^2 - (\text{Sum of roots})x + (\text{Product of roots}) = 0$$

$$\boxed{x^2 - 6x + 11 = 0}$$

$$\begin{aligned} \text{(ii) Sum} &= \frac{\alpha-1}{\alpha+1} + \frac{\beta-1}{\beta+1} \\ &= \frac{(\alpha-1)(\beta+1) + (\beta-1)(\alpha+1)}{(\alpha+1)(\beta+1)} \\ &= \frac{2\alpha\beta - 2}{\alpha\beta + \alpha + \beta + 1} = \frac{2(3) - 2}{3 + 2 + 1} = \frac{4}{6} = \frac{2}{3} \end{aligned}$$

$$\begin{aligned} \text{Product} &= \frac{(\alpha-1)(\beta-1)}{(\alpha+1)(\beta+1)} = \frac{\alpha\beta - (\alpha + \beta) + 1}{\alpha\beta + \alpha + \beta + 1} \\ &= \frac{3 - 2 + 1}{3 + 2 + 1} = \frac{2}{6} = \frac{1}{3} \end{aligned}$$

$$x^2 - (\text{Sum of roots})x + (\text{Product of roots}) = 0$$

$$x^2 - \frac{2}{3}x + \frac{1}{3} = 0$$

$$\boxed{3x^2 - 2x + 1 = 0}$$

54. Given $A = \begin{pmatrix} p & 0 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -q \\ 1 & 0 \end{pmatrix}$, $C = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix}$ and if $BA = C^2$, find p and q.

solution

$$\begin{aligned} BA &= \begin{pmatrix} 0 & -q \\ 1 & 0 \end{pmatrix} \begin{pmatrix} p & 0 \\ 0 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 0(p) - q(0) & 0(0) - q(2) \\ 1(p) + 0(0) & 1(0) + 0(2) \end{pmatrix} \end{aligned}$$

$$\boxed{BA = \begin{pmatrix} 0 & -2q \\ p & 0 \end{pmatrix}}$$

$$\begin{aligned} C^2 &= \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 4-4 & -4-4 \\ 4+4 & -4+4 \end{pmatrix} \end{aligned}$$

$$\boxed{C^2 = \begin{pmatrix} 0 & -8 \\ 8 & 0 \end{pmatrix}}$$

$$BA = C^2 :$$

$$\begin{pmatrix} 0 & -2q \\ p & 0 \end{pmatrix} = \begin{pmatrix} 0 & -8 \\ 8 & 0 \end{pmatrix}$$

$$-2q = -8 \implies \boxed{q = 4}$$

$$\boxed{p = 8}$$

55. $A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 6 & 3 \\ 8 & 5 \end{pmatrix}$, $C = \begin{pmatrix} 3 & 6 \\ 1 & 1 \end{pmatrix}$ find the matrix D, such that $CD - AB = 0$

solution

$$CD - AB = 0 \implies CD = AB$$

$$AB = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 6 & 3 \\ 8 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 18+0 & 9+0 \\ 24+40 & 12+25 \end{pmatrix}$$

$$AB = \begin{pmatrix} 18 & 9 \\ 64 & 37 \end{pmatrix}$$

$$\text{Let } D = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$$

$$CD = \begin{pmatrix} 3 & 6 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x & y \\ z & w \end{pmatrix}$$

$$CD = \begin{pmatrix} 3x+6z & 3y+6w \\ x+z & y+w \end{pmatrix}$$

$$\text{Equating } CD = AB :$$

$$\begin{pmatrix} 3x+6z & 3y+6w \\ x+z & y+w \end{pmatrix} = \begin{pmatrix} 18 & 9 \\ 64 & 37 \end{pmatrix}$$

$$x+z = 64 \implies z = 64 - x$$

$$3x+6z = 18 \implies 3x+6(64-x) = 18$$

$$3x+384-6x = 18 \implies -3x = 18-384 = -366$$

$$x = 122 \implies z = 64 - 122 = -58$$

$$y+w = 37 \implies w = 37 - y$$

$$3y+6w = 9 \implies 3y+6(37-y) = 9$$

$$3y+222-6y = 9 \implies -3y = 9-222 = -213$$

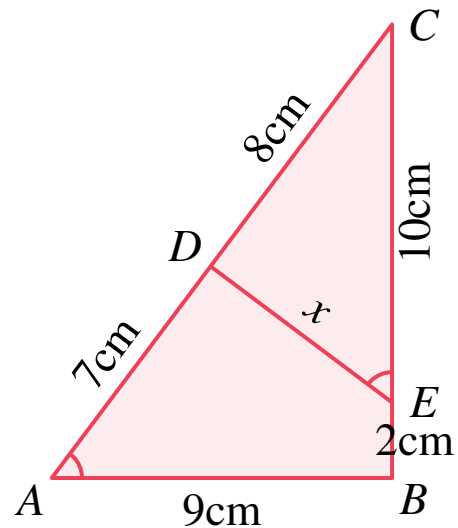
$$y = 71 \implies w = 37 - 71 = -34$$

$$\therefore D = \begin{pmatrix} 122 & 71 \\ -58 & -34 \end{pmatrix}$$

CHAPTER - 4. GEOMETRY

2 MARK IMPORTANT QUESTIONS

1. In Fig $\angle A = \angle CED$. Prove that $\angle CAB = \angle CED$. Also find the value of x .



Solution:

In $\triangle CAB$ and $\triangle CED$, $\angle C$ is common, $\angle A = \angle CED$

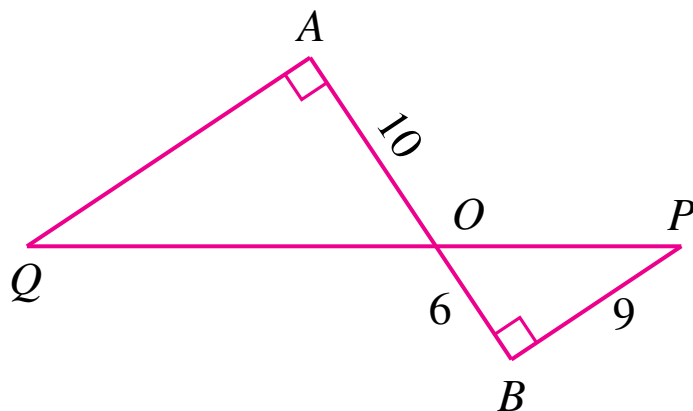
$\triangle CAB \sim \triangle CED$ (By AA similarity)

$$\frac{CA}{CE} = \frac{AB}{DE} = \frac{CB}{CD}$$

$$\frac{AB}{DE} = \frac{CB}{CD}$$

$$\frac{9}{x} = \frac{10+2}{8} \Rightarrow x = \frac{8 \times 9}{12} = 6 \text{ cm.}$$

2. In Fig.4.21, QA and PB are perpendiculars to AB. If $AO = 10$ cm, $BO = 6$ cm and $PB = 9$ cm. Find AQ. **JUL-24**



Solution:

In $\triangle AOQ$ and $\triangle BOP$, $\angle OAQ = \angle OBP = 90^\circ$

$\angle AOQ = \angle BOP$ (Vertically opposite angles)

By AA Criterion of similarity,

$\triangle AOQ \sim \triangle BOP$

$$\frac{AO}{BO} = \frac{OQ}{OP} = \frac{AQ}{BP}$$

$$\frac{10}{6} = \frac{AQ}{9} \Rightarrow AQ = \frac{10 \times 9}{6} = 15 \text{ cm}$$

3. The perimeters of two similar triangles ABC and PQR are respectively 36 cm and 24 cm. If $PQ = 10$ cm, find AB.

Solution:

Given $\triangle ABC \sim \triangle PQR$,

$$\frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle PQR} = \frac{AB}{PQ}$$

$$\frac{36}{24} = \frac{AB}{10}$$

$$\frac{3}{2} = \frac{AB}{10}$$

$$AB = \frac{3 \times 10}{2} = 15 \text{ cm.}$$

4. If $\triangle ABC$ is similar to $\triangle DEF$ such that $BC = 3$ cm, $EF = 4$ cm and area of $\triangle ABC = 54 \text{ cm}^2$. Find the area of $\triangle DEF$. **PTA-2**

Solution:

Given $\triangle ABC \sim \triangle DEF$,

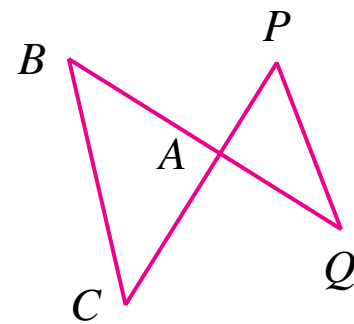
$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} = \left(\frac{BC}{EF}\right)^2$$

$$\frac{54}{\text{Area of } \triangle DEF} = \left(\frac{3}{4}\right)^2$$

$$\frac{54}{\text{Area of } \triangle DEF} = \frac{9}{16}$$

$$\text{Area of } \triangle DEF = \frac{54 \times 16}{9} = 6 \times 16 = 96 \text{ cm}^2.$$

5. In the adjacent figure, $\triangle ACB \sim \triangle APQ$. If $BC = 8$ cm, $PQ = 4$ cm, $BA = 6.5$ cm and $AP = 2.8$ cm, find CA and AQ.



Solution:

Given $\triangle ACB \sim \triangle APQ$.

$BC = 8$, $PQ = 4$, $BA = 6.5$, $AP = 2.8$

$$\frac{AC}{AP} = \frac{CB}{PQ} = \frac{AB}{AQ}$$

$$\frac{CA}{2.8} = \frac{8}{4} = \frac{6.5}{AQ}$$

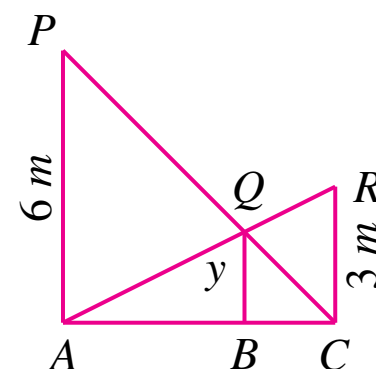
$$\frac{CA}{2.8} = \frac{8}{4} :$$

$$\frac{CA}{2.8} = 2 \Rightarrow CA = 2 \times 2.8 = \boxed{5.6 \text{ cm.}}$$

$$\frac{6.5}{AQ} = \frac{8}{4} :$$

$$\frac{6.5}{AQ} = 2 \Rightarrow AQ = \frac{6.5}{2} = \boxed{3.25 \text{ cm.}}$$

6. Two vertical poles of heights 6 m and 3 m are erected above a horizontal ground AC. Find the value of y . **PTA-5**



Solution:

Let the height of I pole, $PA = 6 \text{ m} = a$

Let the height of II pole, $RC = 3 \text{ m} = b$

Let the height of intersection, $QB = y$

Using the direct formula, $y = \frac{ab}{a+b}$

$$y = \frac{6 \times 3}{6+3}$$

$$y = \frac{18}{9}$$

$$y = \boxed{2 \text{ m.}}$$

7. D and E are respectively the points on the sides AB and AC of a $\triangle ABC$ such that $AB = 5.6\text{ cm}$, $AD = 1.4\text{ cm}$, $AC = 7.2\text{ cm}$, $AE = 1.8\text{ cm}$. Show that $BE \parallel BC$. **MAR-26**

Solution:

$$AB = 5.6\text{ cm}, AD = 1.4\text{ cm}, AC = 7.2\text{ cm}, AE = 1.8\text{ cm}.$$

$$BD = AB - AD = 5.6 - 1.4 = 4.2\text{ cm}$$

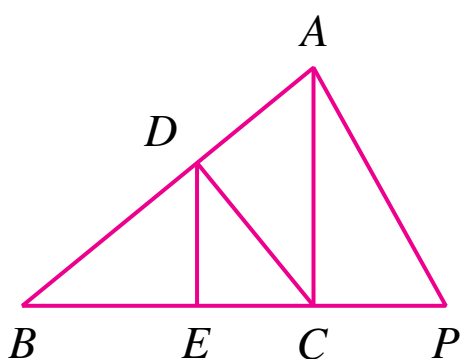
$$EC = AC - AE = 7.2 - 1.8 = 5.4\text{ cm}.$$

$$\frac{AD}{DB} = \frac{1.4}{4.2} = \frac{1}{3} \text{ and } \frac{AE}{EC} = \frac{1.8}{5.4} = \frac{1}{3}$$

$$\frac{AD}{DB} = \frac{AE}{EC}$$

By converse of Basic Proportionality Theorem DE is parallel to BC.
Hence proved.

8. In the Fig.4.38, $DE \parallel AC$ and $DC \parallel AP$. Prove that $\frac{BE}{EC} = \frac{BC}{CP}$. **PTA-4.**



Solution:

In $\triangle BCA$, $DE \parallel CA$.

By Basic Proportionality Theorem (Thales Theorem):

$$\frac{BE}{EC} = \frac{BD}{DA} \dots\dots(1)$$

In $\triangle BPA$, $DC \parallel PA$.

By Basic Proportionality Theorem:

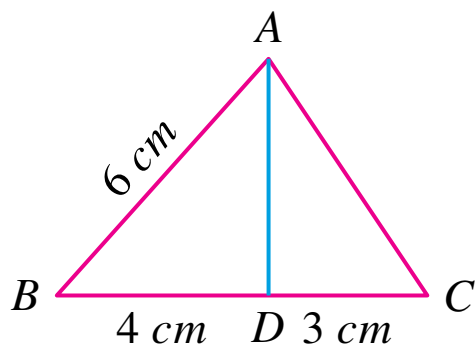
$$\frac{BC}{CP} = \frac{BD}{DA} \dots\dots(2)$$

From (1) and (2):

$$\frac{BE}{EC} = \frac{BC}{CP}$$

(Hence Proved)

9. In the Fig.4.39, AD is the bisector of $\angle A$. If $BD = 4\text{ cm}$, $DC = 3\text{ cm}$ and $AB = 6\text{ cm}$, find AC. **PTA-5, APR-22**



Solution:

In $\triangle ABC$, AD is the bisector of $\angle A$.

By Angle Bisector Theorem

$$\frac{AB}{AC} = \frac{BD}{DC}$$

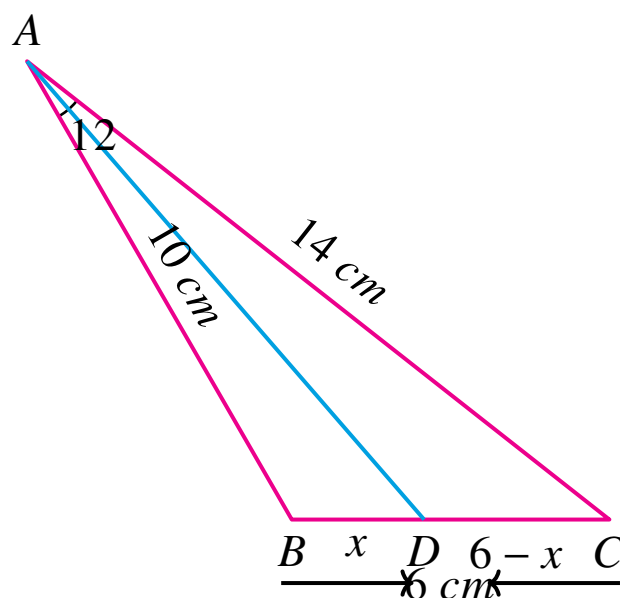
Sub($BD = 4$, $DC = 3$, $AB = 6$):

$$\frac{4}{3} = \frac{6}{AC}$$

$$4 \times AC = 6 \times 3 \implies 4 \times AC = 18$$

$$AC = \frac{18}{4} = 4.5\text{ cm}.$$

10. In the Fig. 4.40, AD is the bisector of $\angle BAC$, if $AB = 10\text{ cm}$, $AC = 14\text{ cm}$ and $BC = 6\text{ cm}$. Find BD and DC. **PTA-3, APR-23.**



Solution:

Since AD is the bisector of $\angle BAC$

By Angle Bisector Theorem:

$$\frac{BD}{DC} = \frac{AB}{AC}$$

Let $BD = x\text{ cm}$. Since $BC = 6\text{ cm}$, $DC = (6 - x)\text{ cm}$.

$$\frac{x}{6 - x} = \frac{10}{14} = \frac{5}{7}$$

$$7x = 5(6 - x)$$

$$7x = 30 - 5x \implies 12x = 30$$

$$x = \frac{30}{12} = 2.5\text{ cm}$$

$$BD = 2.5\text{ cm and } DC = 6 - 2.5 = 3.5\text{ cm}.$$

11. In $\triangle ABC$, D and E are points on the sides AB and AC respectively such that $DE \parallel BC$ (i) If $\frac{AD}{DB} = \frac{3}{4}$ and $AC = 15\text{ cm}$ find AE. **JUL-21**

Solution:

Given $DE \parallel BC$ in $\triangle ABC$.

By Basic Proportionality Theorem (Thales Theorem):

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$\frac{AD}{DB} = \frac{3}{4}, \frac{AE}{EC} = \frac{3}{4}.$$

$AE : EC = 3 : 4$. Let $AE = 3k$ and $EC = 4k$.

Since $AC = AE + EC = 15\text{ cm}$:

$$3k + 4k = 15 \implies 7k = 15 \implies k = \frac{15}{7}$$

$$AE = 3k = 3 \left(\frac{15}{7} \right) = \frac{45}{7} \approx 6.43\text{ cm}.$$

12. Check whether AD is bisector of $\angle A$ of $\triangle ABC$ in each of the following
(i) $AB = 5\text{ cm}$, $AC = 10\text{ cm}$, $BD = 1.5\text{ cm}$ and $CD = 3.5\text{ cm}$. **JUL-20**

Solution:

- (ii) $AB = 4\text{ cm}$, $AC = 6\text{ cm}$, $BD = 1.6\text{ cm}$ and $CD = 2.4\text{ cm}$. **JUL-23**

By the converse of the Angle Bisector Theorem

$$AD \text{ is the bisector of } \angle A \text{ if } \frac{AB}{AC} = \frac{BD}{CD}.$$

(i):

$$\frac{AB}{AC} = \frac{5}{10} = \frac{1}{2}$$

$$\frac{BD}{CD} = \frac{1.5}{3.5} = \frac{15}{35} = \frac{3}{7}$$

$\frac{1}{2} \neq \frac{3}{7}$, AD is **not** the angle bisector $\angle A$.

(ii):

$$\frac{AB}{AC} = \frac{4}{6} = \frac{2}{3}$$

$$\frac{BD}{CD} = \frac{1.6}{2.4} = \frac{16}{24} = \frac{2}{3}$$

$\frac{AB}{AC} = \frac{BD}{CD}$, AD is the angle bisector $\angle A$.

13. In $\triangle ABC$ D and E are points on the sides AB and AC respectively. Show that $DE \parallel BC$ if $AB = 12$ cm, $AD = 8$ cm, $AE = 12$ cm and $AC = 18$ cm.

Solution:

To prove $DE \parallel BC$

By Converse of Basic Proportionality Theorem:

Find DB : $DB = AB - AD = 12 - 8 = 4$ cm.

Find EC : $EC = AC - AE = 18 - 12 = 6$ cm.

$$\frac{AD}{DB} = \frac{8}{4} = 2$$

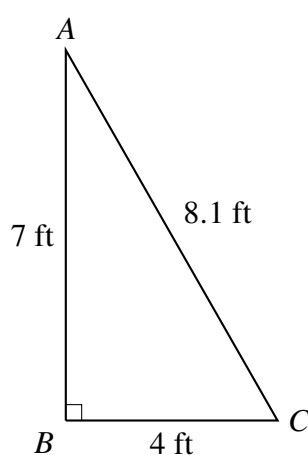
$$\frac{AE}{EC} = \frac{12}{6} = 2$$

$$\frac{AD}{DB} = \frac{AE}{EC}$$

By the converse of Thales Theorem, $DE \parallel BC$.

14. What length of ladder is needed to reach a height of 7 ft along the wall when the base of the ladder is 4 ft from the wall? Round off your answer to the next tenth place. **MDL-19**

Solution:



Let the length of the ladder be AC . Height $AB = 7$ ft, base $BC = 4$ ft.

By Pythagoras theorem:

$$AC^2 = AB^2 + BC^2$$

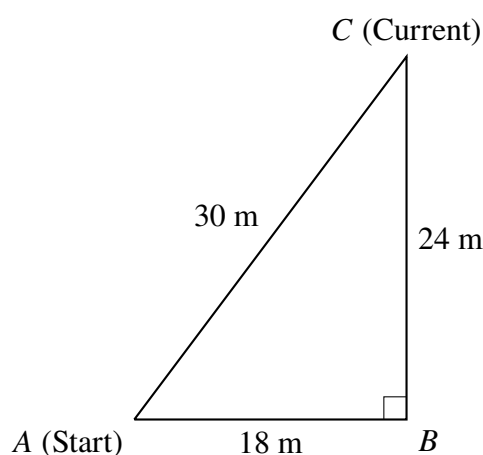
$$AC^2 = 7^2 + 4^2 = 49 + 16 = 65$$

$$AC = \sqrt{65} \approx 8.062 \text{ ft}$$

The length of the ladder needed is 8.1 ft.

15. A man goes 18 m due east and then 24 m due north. Find the distance of his current position from the starting point? **APR-24, JUL-22**

Solution:



Given $AB = 18$ m, $BC = 24$ m.

$$AC^2 = 18^2 + 24^2$$

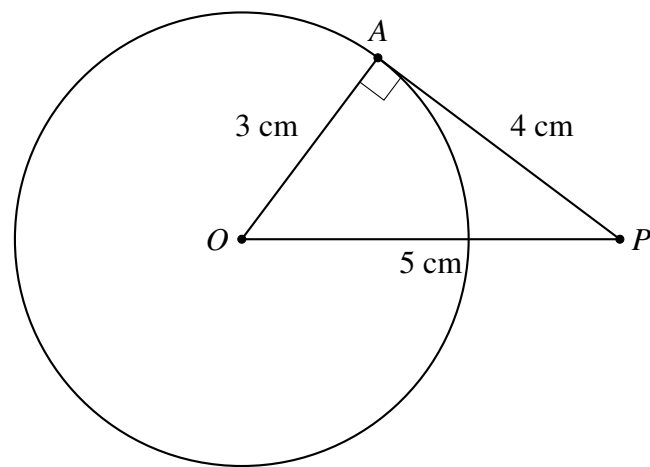
$$AC^2 = 324 + 576 = 900$$

$$AC = \sqrt{900} = 30 \text{ m.}$$

The distance between starting to Current position is 30 m.

16. Find the length of the tangent drawn from a point whose distance from the centre of a circle is 5 cm and radius of the circle is 3 cm.

Solution:



In $\triangle OAP$ is a right-angled triangle at P .

Given: $OP = 5$ cm (hypotenuse) and $OA = 3$ cm (radius).

By Pythagoras theorem:

$$OP^2 = OA^2 + PA^2$$

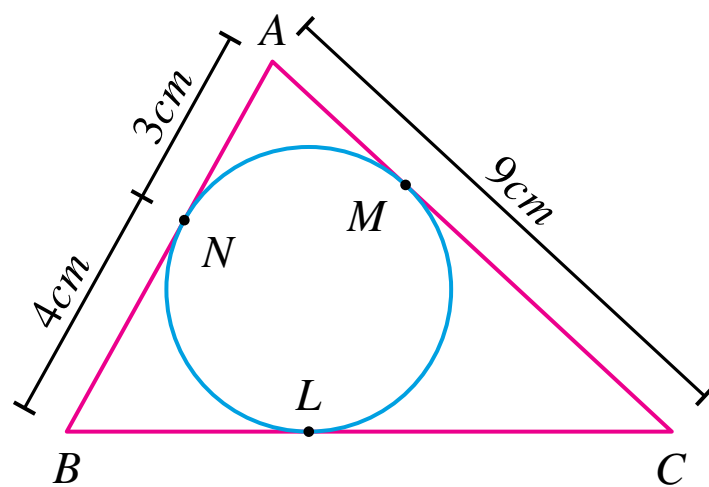
$$5^2 = 3^2 + PA^2$$

$$25 = 9 + PA^2 \Rightarrow PA^2 = 16 \Rightarrow \boxed{PT = 4 \text{ cm.}}$$

The length of the tangent is 4 cm.

17. In Fig.4.65, $\triangle ABC$ is circumscribing a circle. Find the length of BC .

JUL-25



Solution:

Tangents drawn from an external point to a circle are equal in length.

The tangents from A are AM and AN . Thus, $AM = AN = 3$ cm.

$BL = BN = 4$ cm.

$AC = 9$ cm. $AC = AM + MC$.

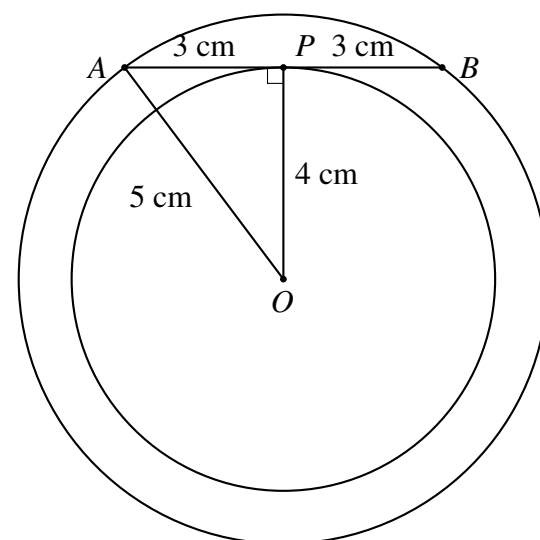
$$9 = 3 + MC \Rightarrow \boxed{MC = 6 \text{ cm.}}$$

$CL = MC = 6$ cm.

The length of $BC = BL + CL = 4 + 6 = \boxed{10 \text{ cm.}}$

18. If radii of two concentric circles are 4 cm and 5 cm then find the length of the chord of one circle which is a tangent to the other circle.

Solution:



Radius of the larger circle $OA = 5$ cm.

Radius of the smaller circle $OP = 4$ cm.

Since AB is tangent to the smaller circle at P , $OP \perp AB$.

In the right-angled $\triangle OPA$:

$$OA^2 = OP^2 + AP^2$$

$$5^2 = 4^2 + AP^2$$

$$25 = 16 + AP^2$$

$$AP^2 = 9$$

$$AP = 3 \text{ cm.}$$

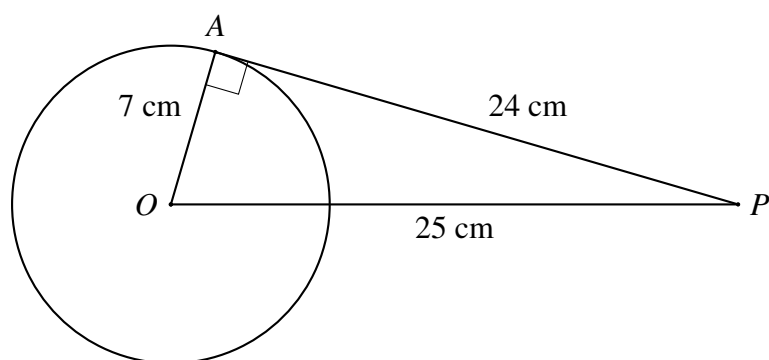
A perpendicular drawn from the center to a chord bisects the chord.

$$AB = 2 \times AP = 2 \times 3 = 6 \text{ cm.}$$

The length of the chord is 6 cm.

19. The length of the tangent to a circle from a point P, which is 25 cm away from the centre is 24 cm. What is the radius of the circle?

Solution:



Given: $OP = 25$ cm, $PA = 24$ cm.

Radius is perpendicular to the tangent, $\angle OAP = 90^\circ$.

In $\triangle OAP$, By Pythagoras theorem:

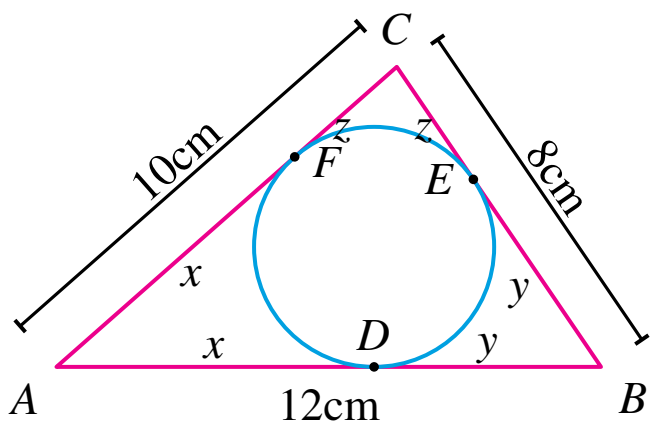
$$OP^2 = OA^2 + PA^2$$

$$25^2 = OA^2 + 24^2 \implies 625 = OA^2 + 576$$

$$OA^2 = 625 - 576 = 49 \implies OA = 7 \text{ cm.}$$

The radius of the circle is 7 cm.

20. A circle is inscribed in $\triangle ABC$ having sides 8 cm, 10 cm and 12 cm as shown in figure, Find AD, BE and CF.



Solution:

Let $AD = AF = x$, $BD = BE = y$, and $CE = CF = z$.

Given $AB = 12$, $BC = 8$, and $AC = 10$:

$$x + y = 12 \dots (1)$$

$$y + z = 8 \dots (2)$$

$$x + z = 10 \dots (3)$$

$$2(x + y + z) = 12 + 8 + 10 = 30 \implies x + y + z = 15 \dots (4)$$

To find z : subtract (1) from (4): $15 - 12 = 3$

$$z = 3 \text{ cm. } \boxed{CF = 3 \text{ cm.}}$$

To find x : subtract (2) from (4): $15 - 8 = 7$

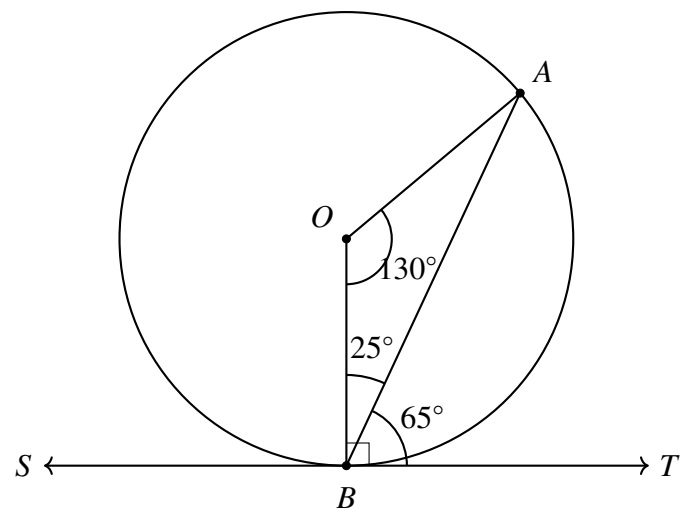
$$x = 7 \text{ cm. } \boxed{AD = 7 \text{ cm.}}$$

To find y : subtract (3) from (4): $15 - 10 = 5$

$$y = 5 \text{ cm. } \boxed{BE = 5 \text{ cm.}}$$

21. A tangent ST to a circle touches it at B . AB is a chord such that $\angle ABT = 65^\circ$. Find $\angle AOB$, where "O" is the centre of the circle.

Solution:



Since ST is tangent at B , $OB \perp ST$, so $\angle OBT = 90^\circ$.

The angle between the chord AB and the radius OB is:

$$\angle OBA = \angle OBT - \angle ABT = 90^\circ - 65^\circ = 25^\circ.$$

In $\triangle AOB$, $OA = OB$ (radii of the same circle).

Hence, it is an isosceles triangle.

$$\angle OAB = \angle OBA = 25^\circ.$$

Sum of angles in $\triangle AOB$:

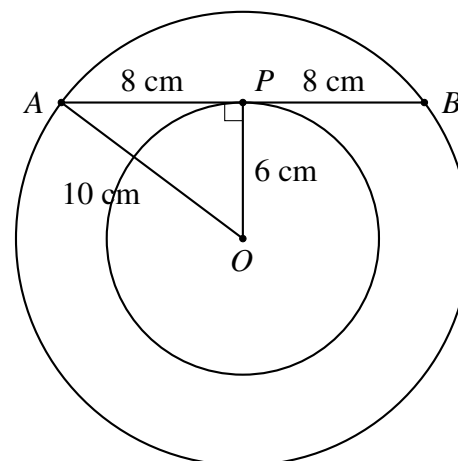
$$\angle AOB + \angle OAB + \angle OBA = 180^\circ$$

$$\angle AOB + 25^\circ + 25^\circ = 180^\circ$$

$$\angle AOB + 50^\circ = 180^\circ \implies \boxed{\angle AOB = 130^\circ}.$$

22. In two concentric circles, a chord of length 16 cm of larger circle becomes a tangent to the smaller circle whose radius is 6 cm. Find the radius of the larger circle.

Solution:



Let the center of the concentric circles be O .

AB chord of the larger circle tangent to the smaller circle at P .

Since AB is tangent to the smaller circle at P , radius $OP \perp AB$.

Given: length of chord $AB = 16$ cm,

Radius of smaller circle $OP = 6$ cm.

A perpendicular from the center to a chord bisects

$$AP = PB = \frac{16}{2} = 8 \text{ cm.}$$

In right-angled $\triangle OPA$,

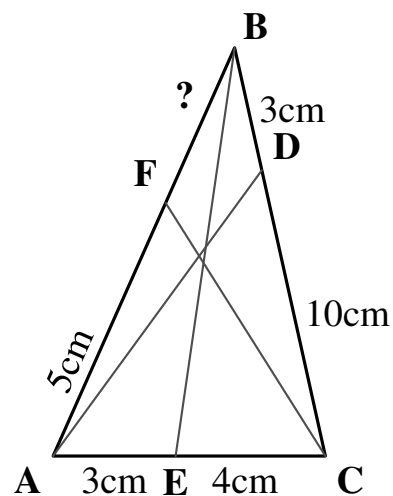
By Pythagoras theorem:

$$\boxed{OA^2 = OP^2 + AP^2}$$

$$OA^2 = 6^2 + 8^2 = 36 + 64 = 100$$

$$OA = \sqrt{100} = \boxed{10 \text{ cm.}}$$

23. An artist has created a triangular stained glass window and has one strip of small length left before completing the window. She needs to figure out the length of left out portion based on the lengths of the other sides as shown in the figure.



Solution:

By Ceva's Theorem

$$\frac{AF}{FB} \times \frac{BD}{DC} \times \frac{CE}{EA} = 1$$

$$AF = 5 \text{ cm}, BD = 3 \text{ cm},$$

$$DC = 10 \text{ cm}, CE = 4 \text{ cm}, EA = 3 \text{ cm}.$$

$$\frac{5}{FB} \times \frac{3}{10} \times \frac{4}{3} = 1$$

$$\frac{5}{FB} \times \frac{12}{30} = 1$$

$$\frac{5}{FB} \times \frac{2}{5} = 1$$

$$FB = 2 \text{ cm}$$

The length of the left out portion BF is 2 cm.

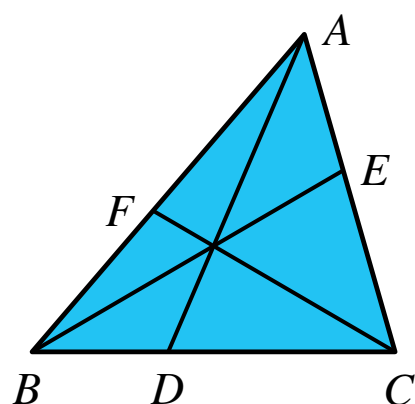
24. State Ceva's Theorem.**(Also Cevian)

Solution:

Ceva's Theorem: Let ABC be a triangle and let D, E, F be points on the lines BC, CA, AB respectively. The cevians AD, BE , and CF are concurrent (intersect at a single point) if and only if:

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1$$

Cevian: A cevian is a line segment that joins a vertex of a triangle with a point on the opposite side (or its extension). For example, medians, altitudes, and angle bisectors are cevians.

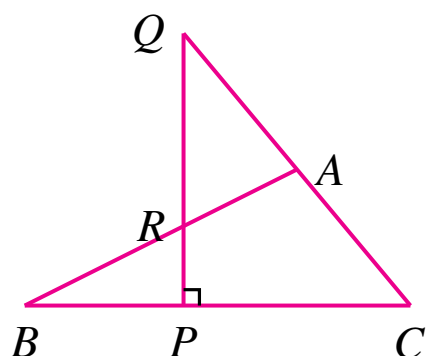


25. State Menelaus's Theorem.**

Solution:

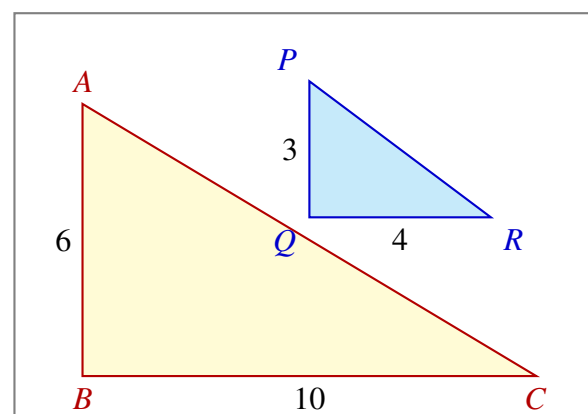
Menelaus's Theorem: Let ABC be a triangle and let P, Q, R be points on the lines BC, CA, AB respectively (or their extensions). The points P, Q , and R are collinear if and only if the directed segments satisfy the relation:

$$\frac{BP}{PC} \times \frac{CQ}{QA} \times \frac{AR}{RB} = -1$$



26. Is $\triangle ABC \sim \triangle PQR$?

Solution:



In $\triangle ABC$ and $\triangle PQR$,

$$\frac{PQ}{AB} = \frac{3}{6} = \frac{1}{2}; \quad \frac{QR}{BC} = \frac{4}{10} = \frac{2}{5}$$

$$\frac{1}{2} \neq \frac{2}{5}, \quad \frac{PQ}{AB} \neq \frac{QR}{BC}.$$

The corresponding sides are not proportional.

$\triangle ABC$ is not similar to $\triangle PQR$.

Chapter-4: GEOMETRY

5 MARK IMPORTANT QUESTIONS

1. STATE AND PROVE THALES THEOREM OR BPT. **APR-22, 24**

Statement

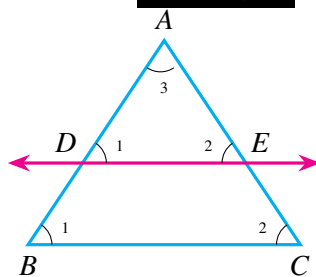
A straight line drawn parallel to a side of triangle intersecting the other two sides, divides the sides in the same ratio.

Proof

Given: In $\triangle ABC$, D is a point on AB and E is a point on AC .

To prove: $\frac{AD}{DB} = \frac{AE}{EC}$

Construction: Draw a line $DE \parallel BC$



No.	Statement	Reason
1.	$\angle ABC = \angle ADE = \angle 1$	Corresponding angles equal $DE \parallel BC$
2.	$\angle ACB = \angle AED = \angle 2$	Corresponding angles equal $DE \parallel BC$
3.	$\angle DAE = \angle BAC = \angle 3$	Both $\triangle$ common angle
4.	$\triangle ABC \sim \triangle ADE$ $\frac{AB}{AD} = \frac{AC}{AE}$ $\frac{AD + DB}{AD} = \frac{AE + EC}{AE}$ $1 + \frac{DB}{AD} = 1 + \frac{EC}{AE}$ $\frac{DB}{AD} = \frac{EC}{AE}$ $\frac{AD}{DB} = \frac{AE}{EC}$	By AAA similarity Corresponding sides proportional Split AB and AC D and E . On simplify Cancel 1 on both sides Take reciprocal
Hence proved		

2. STATE AND PROVE ANGLE BISECTOR THEOREM OR ABT. **APR-23, 25, MDL-19, JUL-22**

Statement

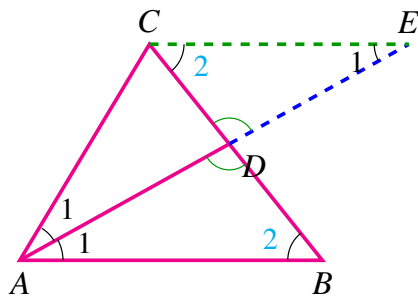
The internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the corresponding sides containing the angle.

Proof

Given: In $\triangle ABC$, AD is the internal bisector

To prove: $\frac{AB}{AC} = \frac{BD}{CD}$

Construction: Draw a line through C parallel to AB . Extend AD to meet line through C at E



No	Statement	Reason
1.	$\angle AEC = \angle BAE = \angle 1$ $\angle ABD = \angle ECD = \angle 2$	Two parallel lines cut by a transversal make alternate angles equal.
2.	$\triangle ACE$ is isosceles $AC = CE \dots (1)$	In $\triangle ACE$, $\angle CAE = \angle CEA$
3.	$\triangle ABD \sim \triangle ECD$ $\frac{AB}{CE} = \frac{BD}{CD}$	By AA Similarity
4.	$\frac{AB}{AC} = \frac{BD}{CD}$	From (1) $AC = CE$. Hence proved.

3. STATE AND PROVE PYTHAGORAS THEOREM OR BAUDHAYANA THEOREM. **JUL-23, 24, 25, SEP-21**

Statement

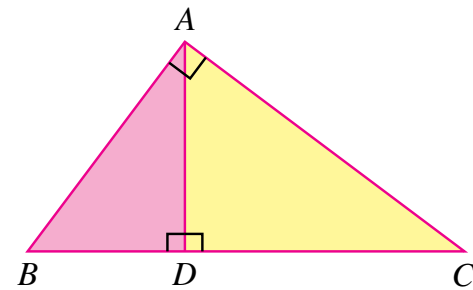
In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

Proof

Given: In $\triangle ABC$, $\angle A = 90^\circ$

To prove: $AB^2 + AC^2 = BC^2$

Construction: Draw $AD \perp BC$



No.	Statement	Reason
1.	Compare $\triangle ABC$ and $\triangle DBA$ $\angle B$ is common $\angle BAC = \angle BDA = 90^\circ$ $\triangle ABC \sim \triangle DBA$ $\frac{AB}{BD} = \frac{BC}{AB}$ $AB^2 = BC \times BD \dots (1)$	Given $\angle BAC = 90^\circ$ and $\angle BDA = 90^\circ$ By AA similarity
2.	Compare $\triangle ABC$ and $\triangle DAC$ $\angle C$ is common $\angle BAC = \angle ADC = 90^\circ$ $\triangle ABC \sim \triangle DAC$ $\frac{AC}{DC} = \frac{BC}{AC}$ $AC^2 = BC \times DC \dots (2)$	Given $\angle BAC = 90^\circ$ and $\angle ADC = 90^\circ$ By AA similarity

Add (1) and (2)

$$AB^2 + AC^2 = BC \times BD + BC \times DC$$

$$= BC(BD + DC) = BC \times BC$$

$$AB^2 + AC^2 = BC^2.$$

Hence the theorem is proved.

4. STATE AND PROVE ALTERNATE SEGMENT OR CHORD TANGENT THEOREM. **PTA-4**

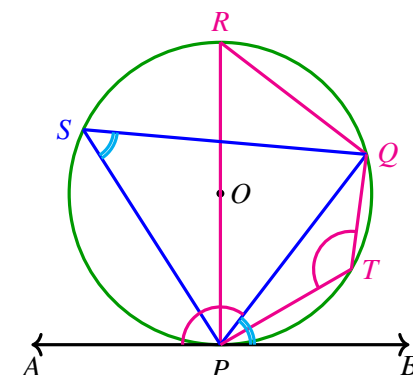
Statement

If a line touches a circle and from the point of contact a chord is drawn, the angles between the tangent and the chord are respectively equal to the angles in the corresponding alternate segments.

Given: A circle with centre at O , tangent AB touches the circle at P and PQ is a chord. S and T are two points on the circle in the opposite sides of chord PQ .

To prove: (i) $\angle QPB = \angle PSQ$
(ii) $\angle QPA = \angle PTQ$

Construction: Draw the diameter POR . Draw QR , QS and PS .



No.	Statement	Reason
1.	$\angle RPB = 90^\circ$ Now, $\angle RPQ + \angle QPB = 90^\circ \dots (1)$	Diameter RP is perpendicular to tangent AB .
2.	In $\triangle RPQ$, $\angle PQR = 90^\circ \dots (2)$	Angle in a semicircle is 90° .
3.	$\angle QRP + \angle RPQ = 90^\circ \dots (3)$	In a right angled triangle, sum of the two acute angles is 90° .
4.	$\angle RPQ + \angle QPB = \angle QRP + \angle RPQ$ $\angle QPB = \angle QRP \dots (4)$	From (1) and (3).
5.	$\angle QRP = \angle PSQ \dots (5)$	Angles in the same segment are equal.
6.	$\angle QPB = \angle PSQ \dots (6)$	From (4) and (5); Hence (i) is proved.
7.	$\angle QPB + \angle QPA = 180^\circ \dots (7)$	Linear pair of angles.
8.	$\angle PSQ + \angle PTQ = 180^\circ \dots (8)$	Sum of opposite angles of a cyclic quadrilateral is 180° .
9.	$\angle QPB + \angle QPA = \angle PSQ + \angle PTQ$	From (7) and (8).
10.	$\angle QPB + \angle QPA = \angle QPB + \angle PTQ$ $\angle QPA = \angle PTQ$	$\angle QPB = \angle PSQ$ from (6)
11.	$\angle QPA = \angle PTQ$	Hence (ii) is proved.

5. Two poles of height 'a' metres and 'b' metres are 'p' metres apart. Prove that the height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is given by $\frac{ab}{a+b}$ metres. **APR-23**

Solution:

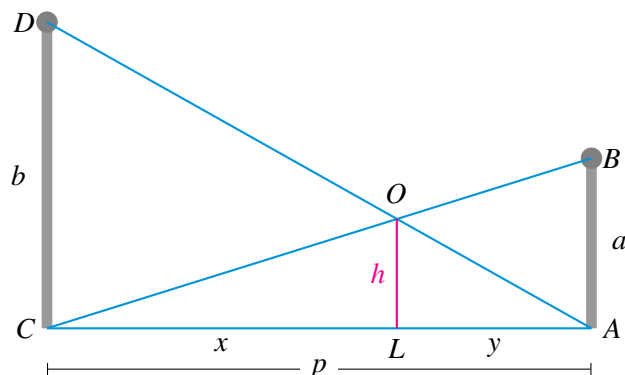


Fig. 4.23

Let AB and CD be two poles of height 'a' metres and 'b' metres the poles are 'p' metres apart. $AC = p$ metres. Let $OL = h$ metres.

Let $CL = x$ and $LA = y$. $x + y = p$

In $\triangle ABC$ and $\triangle LOC$,

$$\angle CAB = \angle CLO \quad [\text{each equal to } 90^\circ]$$

$$\angle C = \angle C \quad [C \text{ is common}]$$

$$\triangle CAB \sim \triangle CLO \quad [\text{By AA similarity}]$$

$$\frac{CA}{CL} = \frac{AB}{LO} \Rightarrow \frac{p}{x} = \frac{a}{h}$$

$$x = \frac{ph}{a} \quad \dots (1)$$

In $\triangle ALO$ and $\triangle ACD$,

$$\angle ALO = \angle ACD \quad [\text{each equal to } 90^\circ]$$

$$\angle A = \angle A \quad [A \text{ is common}]$$

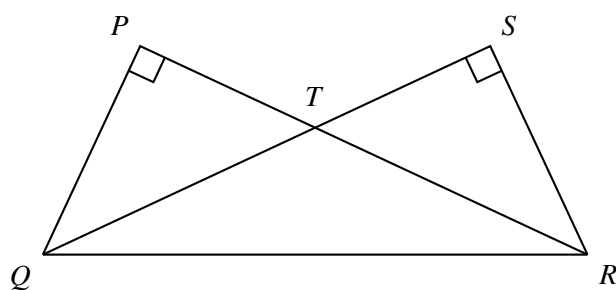
$$\triangle ALO \sim \triangle ACD \quad [\text{by AA similarity}]$$

$$\frac{AL}{AC} = \frac{OL}{DC} \Rightarrow \frac{y}{p} = \frac{h}{b} \Rightarrow y = \frac{ph}{b} \quad \dots (2)$$

$$\begin{aligned} (1) + (2) \Rightarrow x + y &= \frac{ph}{a} + \frac{ph}{b} \\ p &= ph \left(\frac{1}{a} + \frac{1}{b} \right) \quad (x + y = p) \\ 1 &= h \left(\frac{a+b}{ab} \right) \\ h &= \frac{ab}{a+b} \end{aligned}$$

6. Two triangles QPR and QSR , right angled at P and S respectively are drawn on the same base QR and on the same side of QR . If PR and SQ intersect at T , prove that $PT \times TR = ST \times TQ$ **PTA-6**

Solution:



Let $\triangle PQT$ and $\triangle SRT$. Given $\angle QPR = 90^\circ$ and $\angle QSR = 90^\circ$. $\angle PQT = 90^\circ$ and $\angle RST = 90^\circ$.

In $\triangle PQT$ and $\triangle SRT$:

$$\angle QPT = \angle RST = 90^\circ \quad (\text{Given})$$

$$\angle PTQ = \angle STR \quad (\text{Vertically opposite angles})$$

By AA similarity criterion,

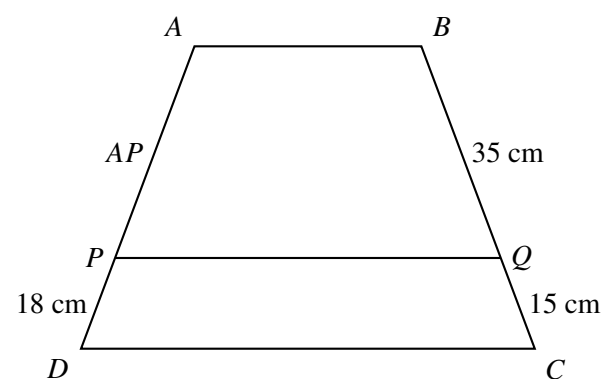
$$\triangle PQT \sim \triangle SRT$$

Since ratio of their corresponding sides is equal:

$$\begin{aligned} \frac{PT}{ST} &= \frac{TQ}{TR} \\ PT \times TR &= ST \times TQ \end{aligned}$$

7. $ABCD$ is a trapezium in which $AB \parallel DC$ and P, Q are points on AD and BC respectively, such that $PQ \parallel DC$ if $PD = 18$ cm, $BQ = 35$ cm and $QC = 15$ cm, find AD .

Solution:



In trapezium $ABCD$, $AB \parallel DC$. Given $PQ \parallel DC$. $PQ \parallel AB \parallel DC$.

$$\frac{AP}{PD} = \frac{BQ}{QC}$$

$$\frac{AP}{18} = \frac{35}{15}$$

$$\frac{AP}{18} = \frac{7}{3}$$

$$AP = 18 \times \frac{7}{3}$$

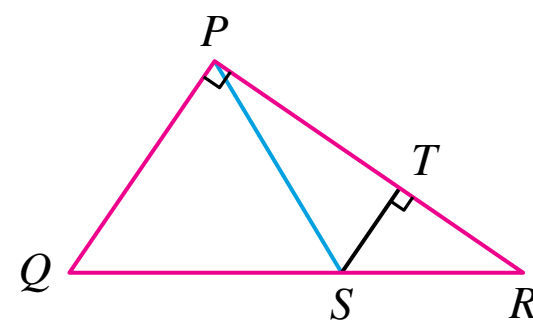
$$AP = 6 \times 7 = 42 \text{ cm}$$

$$AD = AP + PD$$

$$AD = 42 + 18$$

$$AD = 60 \text{ cm}$$

8. In figure $\angle QPR = 90^\circ$, PS is its bisector. If $ST \perp PR$, Prove that $ST \times (PQ + PR) = PQ \times PR$. **PTA-2**



Solution:

Given $\angle QPR = 90^\circ$ and PS is the bisector, so $\angle QPS = \angle SPR = 45^\circ$.

Draw a perpendicular SU from S to PQ . Since $\angle QPR = 90^\circ$, $ST \perp PR$, and $SU \perp PQ$, quadrilateral $PTSU$ is a rectangle.

The diagonal PS bisects the 90° angle, $PTSU$ is a square $ST = SU$.

$$\text{Area of } \triangle PQR = \frac{1}{2} \times PQ \times PR$$

$$\text{Area of } \triangle PQS = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times PQ \times SU$$

$$= \frac{1}{2} \times PQ \times ST$$

$$\text{Area of } \triangle PSR = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times PR \times ST$$

$$\text{Area of } \triangle PQR = \text{Area of } \triangle PQS + \text{Area of } \triangle PSR:$$

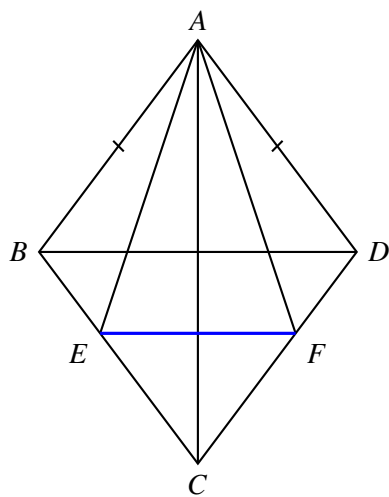
$$\frac{1}{2} \times PQ \times PR = \left(\frac{1}{2} \times PQ \times ST \right) + \left(\frac{1}{2} \times PR \times ST \right)$$

$$PQ \times PR = ST(PQ) + ST(PR)$$

$$PQ \times PR = ST(PQ + PR)$$

9. $ABCD$ is a quadrilateral in which $AB = AD$, the bisector of $\angle BAC$ and $\angle CAD$ intersect the sides BC and CD at the points E and F respectively. Prove that $EF \parallel BD$. **JUL-22**

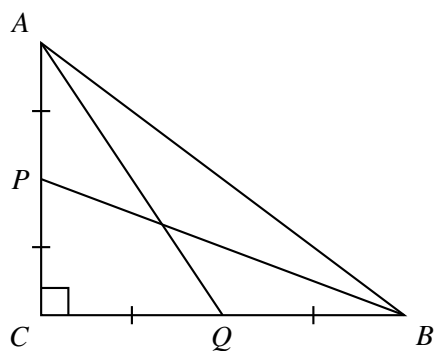
Solution:



$$\begin{aligned} \frac{BE}{EC} &= \frac{AB}{AC} & (1) \text{ In } \triangle ABC \text{ By the ABT Theorem} \\ \frac{DF}{FC} &= \frac{AD}{AC} & (2) \text{ In } \triangle ADC \text{ By the ABT Theorem} \\ \frac{BE}{EC} &= \frac{AD}{AC} & (3) \text{ Sub given } AB = AD \text{ in (1)} \\ \frac{BE}{EC} &= \frac{DF}{FC} & \text{Compare (2) and (3)} \\ \frac{CE}{EB} &= \frac{CF}{FD} & \text{Taking reciprocals} \\ EF &\parallel BD & \text{By the converse of BPT in } \triangle CBD \end{aligned}$$

10. P and Q are the mid-points of the sides CA and CB respectively of a $\triangle ABC$, right angled at C. Prove that $4(AQ^2 + BP^2) = 5AB^2$. **APR-22**

Solution:



In $\triangle ABC$, $\angle C = 90^\circ$. P and Q are the midpoints of sides CA and CB respectively.
 $CP = \frac{AC}{2}$ and $CQ = \frac{BC}{2}$.

In right-angled $\triangle ACQ$ By Pythagoras theorem

$$AQ^2 = AC^2 + CQ^2$$

$$AQ^2 = AC^2 + \left(\frac{BC}{2}\right)^2$$

$$AQ^2 = AC^2 + \frac{BC^2}{4} \quad (1)$$

In right-angled $\triangle BCP$ By Pythagoras theorem

$$BP^2 = BC^2 + CP^2$$

$$BP^2 = BC^2 + \left(\frac{AC}{2}\right)^2$$

$$BP^2 = BC^2 + \frac{AC^2}{4} \quad (2)$$

$$AQ^2 + BP^2 = AC^2 + \frac{BC^2}{4} + BC^2 + \frac{AC^2}{4}$$

$$AQ^2 + BP^2 = \frac{5}{4}AC^2 + \frac{5}{4}BC^2$$

$$AQ^2 + BP^2 = \frac{5}{4}(AC^2 + BC^2)$$

$$AC^2 + BC^2 = AB^2$$

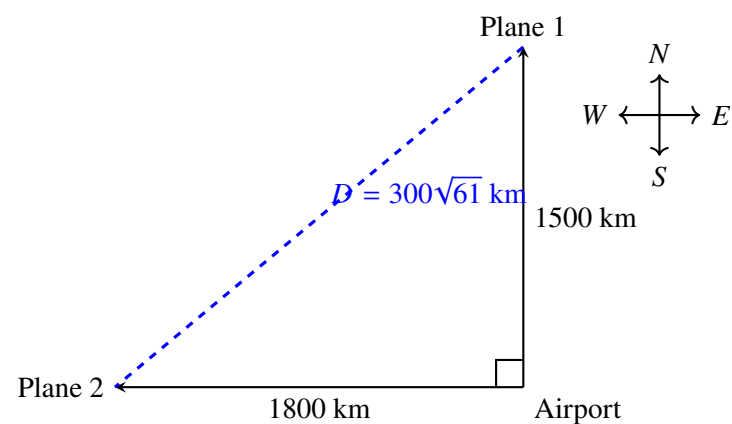
$$AQ^2 + BP^2 = \frac{5}{4}AB^2$$

$$4(AQ^2 + BP^2) = 5AB^2$$

Hence Proved

11. An Aeroplane after take off from an airport and flies due north at a speed of 1000 km/hr. At the same time, another aeroplane take off from the same airport and flies due west at a speed of 1200 km/hr. How far apart will be the two planes after $1\frac{1}{2}$ hours? **APR-22**

Solution:



$$\text{Time} = 1\frac{1}{2} \text{ hours} = 1.5 \text{ hours}$$

Distance I plane due North (y-axis)

$$\text{Distance} = \text{Speed} \times \text{time}$$

$$d_1 = 1000 \times 1.5 = 1500 \text{ km}$$

Distance II plane due West (x-axis)

$$\text{Distance} = \text{Speed} \times \text{time}$$

$$d_2 = 1200 \times 1.5 = 1800 \text{ km}$$

In Right Angle, By Pythagoras Thm $D^2 = d_1^2 + d_2^2$

$$D^2 = (1500)^2 + (1800)^2$$

$$D^2 = 2250000 + 3240000$$

$$D^2 = 5490000$$

$$D = \sqrt{5490000}$$

$$D = \sqrt{90000 \times 61}$$

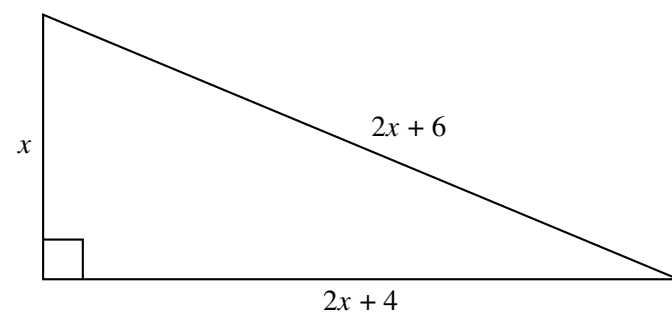
$$D = 300\sqrt{61} \text{ km}$$

The planes are $300\sqrt{61}$ km apart (approx. 2343 km).

12. The hypotenuse of a right triangle is 6 m more than twice of the shortest side. If the third side is 2 m less than the hypotenuse, find the sides of the triangle.

PTA-3

Solution:



Let the shortest side of the right triangle be x metres.

$$\text{Hypotenuse (h)} = 2x + 6$$

$$\text{Third side (y)} = (2x + 6) - 2$$

$$= 2x + 4$$

$$\text{By Pythagoras Thm } AB^2 + AC^2 = BC^2$$

$$x^2 + (2x + 4)^2 = (2x + 6)^2$$

$$x^2 + 4x^2 + 16x + 16 = 4x^2 + 24x + 36$$

$$x^2 + 16x + 16 = 24x + 36$$

$$x^2 - 8x - 20 = 0$$

$$(x - 10)(x + 2) = 0$$

$$x = 10 \text{ or } -2$$

$$x = 10$$

$$\text{Shortest side} = 10 \text{ m}$$

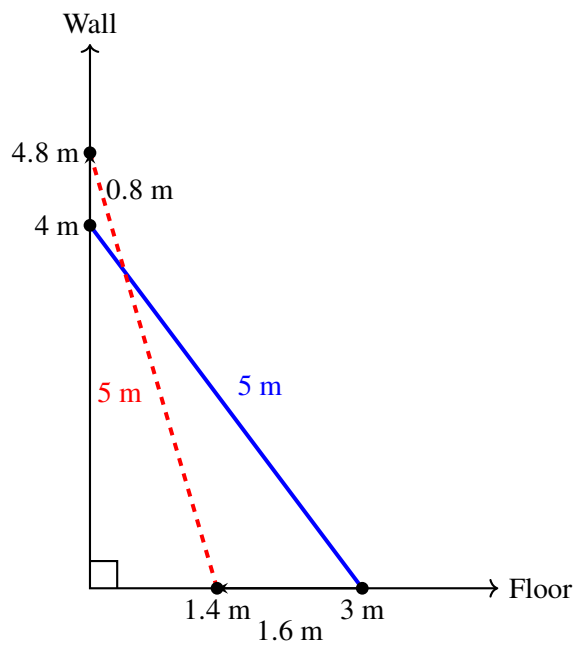
$$\text{Third side} = 2(10) + 4 = 24 \text{ m}$$

$$\text{Hypotenuse} = 2(10) + 6 = 26 \text{ m}$$

The sides of the triangle are 10 m, 24 m, and 26 m.

13. 5 m long ladder is placed leaning towards a vertical wall such that it reaches the wall at a point 4 m high. If the foot of the ladder is moved 1.6 m towards the wall, then find the distance by which the top of the ladder would slide upwards on the wall.

Solution:



Let the initial distance of the foot of the ladder be b_1 and the initial height be $h_1 = 4$ m. Length of the ladder $L = 5$ m.

$$L^2 = h_1^2 + b_1^2 \quad (\text{By Pythagoras theorem})$$

$$5^2 = 4^2 + b_1^2$$

$$25 = 16 + b_1^2$$

$$b_1^2 = 9$$

$$b_1 = 3 \text{ m}$$

$$b_2 = b_1 - 1.6 \quad (\text{Foot moved } 1.6 \text{ m})$$

$$b_2 = 3 - 1.6$$

$$b_2 = 1.4 \text{ m}$$

$$L^2 = h_2^2 + b_2^2$$

$$5^2 = h_2^2 + 1.4^2$$

$$25 = h_2^2 + 1.96$$

$$h_2^2 = 25 - 1.96$$

$$h_2^2 = 23.04$$

$$h_2 = \sqrt{23.04}$$

$$h_2 = 4.8 \text{ m}$$

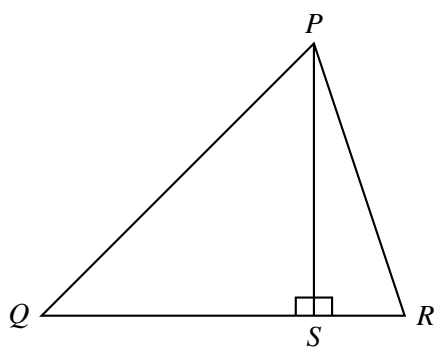
$$\text{Distance slid upwards} = h_2 - h_1$$

$$= 4.8 - 4$$

$$= 0.8 \text{ m}$$

14. The perpendicular PS on the base QR of a $\triangle PQR$ intersects QR at S , such that $QS = 3SR$. Prove that $2PQ^2 = 2PR^2 + QR^2$ **PTA-6**

Solution:



Given in $\triangle PQR$, $PS \perp QR$ and $QS = 3SR$.

$$QR = QS + SR$$

$$QR = 3SR + SR \quad (\text{Sub } QS = 3SR)$$

$$QR = 4SR$$

$$SR = \frac{QR}{4}$$

$$QS = 3\left(\frac{QR}{4}\right) = \frac{3QR}{4}$$

(In right-angled $\triangle PSQ$ by Pythagoras theorem)

$$PQ^2 = PS^2 + QS^2$$

$$PQ^2 = PS^2 + \left(\frac{3QR}{4}\right)^2$$

$$PQ^2 = PS^2 + \frac{9QR^2}{16} \quad \text{--- (1)}$$

(In right-angled $\triangle PSR$ by Pythagoras theorem)

$$PR^2 = PS^2 + SR^2$$

$$PR^2 = PS^2 + \left(\frac{QR}{4}\right)^2$$

$$PR^2 = PS^2 + \frac{QR^2}{16} \quad \text{--- (2)}$$

$$PQ^2 - PR^2 = \left(PS^2 + \frac{9QR^2}{16}\right) - \left(PS^2 + \frac{QR^2}{16}\right)$$

$$PQ^2 - PR^2 = \frac{8QR^2}{16}$$

$$PQ^2 - PR^2 = \frac{QR^2}{2}$$

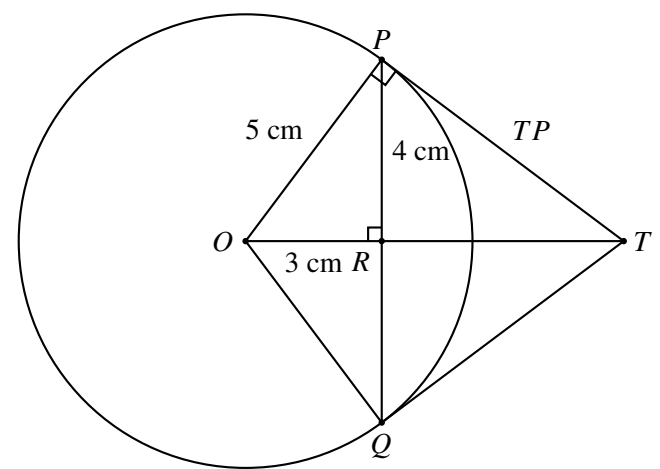
$$2(PQ^2 - PR^2) = QR^2$$

$$2PQ^2 - 2PR^2 = QR^2$$

$$2PQ^2 = 2PR^2 + QR^2 \quad (\text{Hence Proved})$$

15. PQ is a chord of length 8 cm to a circle of radius 5 cm. The tangents at P and Q intersect at a point T . Find the length of the tangent TP . **MDL-19**

Solution:



Let O be the center of the circle. Join OT . OT is the perpendicular bisector of chord PQ . Let OT intersect PQ at R . Then $PR = RQ = \frac{8}{2} = 4$ cm.

$$OP^2 = OR^2 + PR^2 \quad (\text{In right-angled } \triangle POR \text{ By Pythagoras theorem})$$

$$5^2 = OR^2 + 4^2$$

$$25 = OR^2 + 16$$

$$OR^2 = 9$$

$$OR = 3 \text{ cm}$$

$$\angle OPT = 90^\circ \quad TP \text{ is a tangent, } OP \perp TP$$

$$\tan \theta = \frac{PR}{OR} \quad (\text{Let } \angle POR = \theta \text{ in } \triangle PRO)$$

$$\tan \theta = \frac{4}{3}$$

$$\tan \theta = \frac{TP}{OP} \quad (\text{In right-angled } \triangle OPT)$$

$$\frac{TP}{OP} = \frac{4}{3}$$

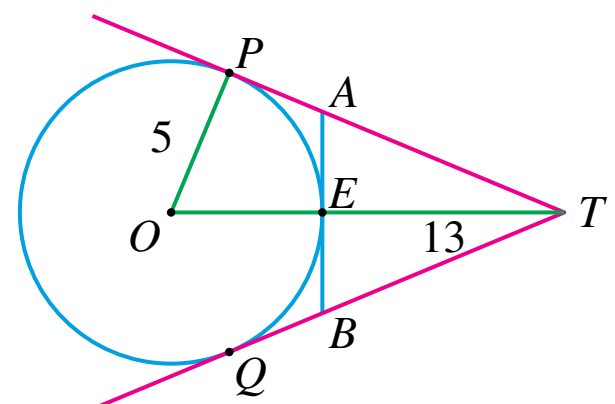
$$\frac{TP}{5} = \frac{4}{3} \quad (\text{Substituting } OP = 5)$$

$$TP = \frac{20}{3}$$

$$TP = 6.67 \text{ cm}$$

The length of the Tangent is 6.67 cm.

16. In figure, O is the centre of the circle with radius 5 cm. T is a point such that $OT = 13$ cm and OT intersects the circle E , if AB is the tangent to the circle at E , find the length of AB . **PTA-4**



Solution:

Since PT is a tangent to the circle at P , $OP \perp PT$.

In right-angled $\triangle OPT$, $OP = 5$ cm and $OT = 13$ cm.

$$PT^2 = OT^2 - OP^2 \quad (\text{By Pythagoras theorem})$$

$$PT^2 = 13^2 - 5^2$$

$$PT^2 = 169 - 25$$

$$PT^2 = 144$$

$$PT = 12 \text{ cm}$$

$$ET = OT - OE \quad OE = 5 \text{ cm}$$

$$ET = 13 - 5$$

$$ET = 8 \text{ cm}$$

$$\angle AET = \angle OPT = 90^\circ \quad (\text{Since } AB \perp OT \text{ } PT \perp OP)$$

$$\angle T = \angle T \quad \text{Common angle}$$

$$\triangle AET \sim \triangle OPT \quad (\text{By AA similarity})$$

$$\frac{AE}{OP} = \frac{ET}{PT}$$

$$\frac{AE}{5} = \frac{8}{12}$$

$$\frac{AE}{5} = \frac{2}{3}$$

$$AE = \frac{10}{3} \text{ cm}$$

$$AB = 2AE \quad \text{Since E is a Midpoint}$$

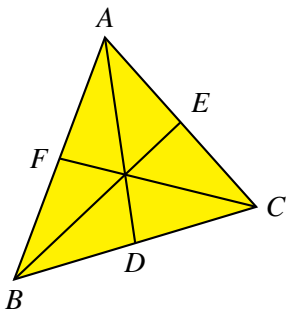
$$AB = 2 \left(\frac{10}{3} \right)$$

$$AB = \frac{20}{3} \text{ cm} = 6.67 \text{ cm}$$

17. show that in a triangle, the Medians are concurrent. **JUL-21**

Solution:

The Medians are the cevians where D , E , F are midpoints of BC , CA and AB respectively.



$$\text{Since } D \text{ is a midpoint of } BC, BD = DC \text{ so } \frac{BD}{DC} = 1 \quad \dots (1)$$

$$\text{Since } E \text{ is a midpoint of } CA, CE = EA \text{ so } \frac{CE}{EA} = 1 \quad \dots (2)$$

$$\text{Since } F \text{ is a midpoint of } AB, AF = FB \text{ so } \frac{AF}{FB} = 1 \quad \dots (3)$$

Multiply (1), (2) and (3)

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1 \times 1 \times 1 = 1$$

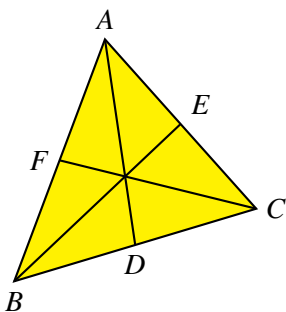
And so, Ceva's theorem is satisfied.

Hence the Medians are concurrent.

18. Show that the angle bisectors of a triangle are concurrent. **PTA-4,MAR-26**

Solution:

The Medians are the cevians where D , E , F are midpoints of BC , CA and AB respectively.



$$\text{Since } AD \text{ is a angle bisector of } \angle A, \frac{AB}{AC} = \frac{BD}{DC} \quad \dots (1)$$

$$\text{Since } BE \text{ is a angle bisector of } \angle B, \frac{BC}{BA} = \frac{CE}{EA} \quad \dots (2)$$

$$\text{Since } CF \text{ is a angle bisector of } \angle C, \frac{CA}{CB} = \frac{AF}{FB} \quad \dots (3)$$

Multiply (1), (2) and (3)

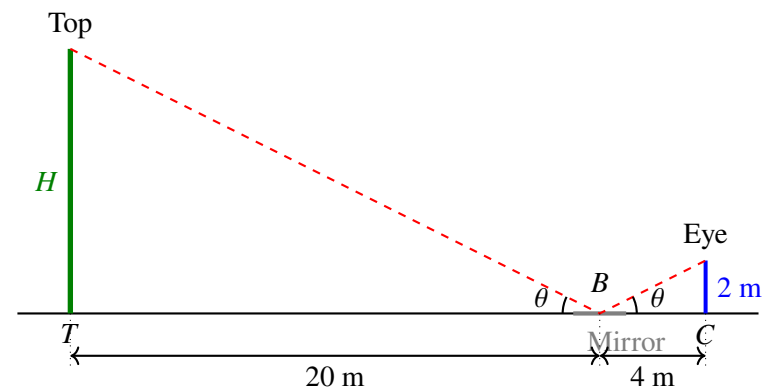
$$\frac{AB}{AC} \times \frac{BC}{BA} \times \frac{CA}{CB} = \frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1$$

And so, Ceva's theorem is satisfied.

Hence the angle bisectors of a triangle are concurrent.

19. A man whose eye-level is 2 m above the ground wishes to find the height of a tree. He places a mirror horizontally on the ground 20 m from the tree and finds that if he stands at a point C which is 4 m from the mirror B, he can see the reflection of the top of the tree. How height is the tree?

Solution:



Let the tree height be H and the tree base be T . Let the mirror be at B .

Distance from tree to mirror: $TB = 20$ m.

Distance from man to mirror: $CB = 4$ m.

Man's eye level: $h = 2$ m.

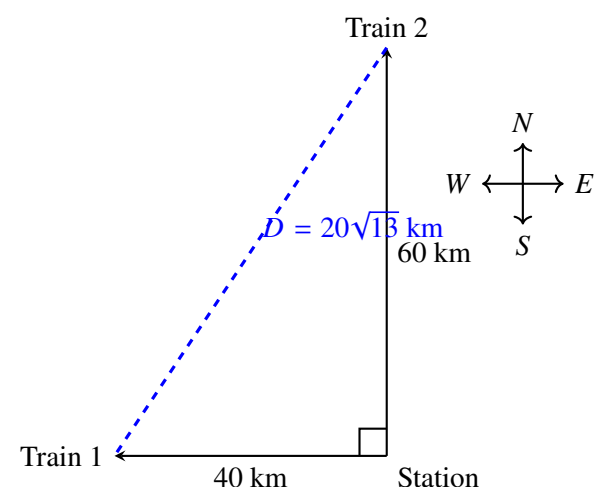
By the laws of reflection, the angle of incidence equal.

$$\begin{aligned} \frac{\text{Height of Tree}}{\text{Distance from Tree to Mirror}} &= \frac{\text{Height of Man's Eye}}{\text{Distance from Man to Mirror}} \\ \frac{H}{20} &= \frac{2}{4} \\ \frac{H}{20} &= \frac{1}{2} \\ H &= \frac{20}{2} \\ H &= 10 \text{ m} \end{aligned}$$

The height of the tree is 10 m.

20. Two trains leave a railway station at the same time. The first train travels due west and the second train due north. The first train travels at a speed of 20 km/hr and the second train travels at 30 km/hr. After 2 hours, what is the distance between them?

Solution:



$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{Time} = 2 \text{ hours}$$

$$d_W = 20 \times 2 = 40 \text{ km} \quad (\text{I train due West})$$

$$d_N = 30 \times 2 = 60 \text{ km} \quad (\text{II train due North})$$

$$D^2 = d_W^2 + d_N^2 \quad (\text{By Pythagoras theorem})$$

$$D^2 = 40^2 + 60^2$$

$$D^2 = 1600 + 3600$$

$$D^2 = 5200$$

$$D = \sqrt{5200}$$

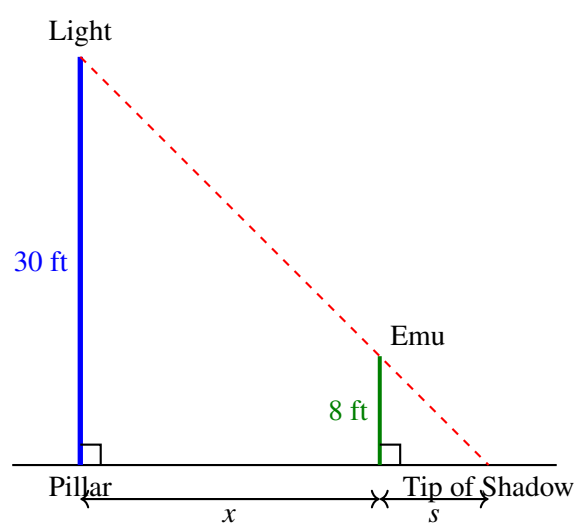
$$D = \sqrt{400 \times 13}$$

$$D = 20\sqrt{13} \text{ km}$$

The distance between the two trains after 2 hours is $20\sqrt{13}$ km.

21. An Emu which is 8 feet tall is standing at the foot of a pillar which is 30 feet high. It walks away from the pillar. The shadow of the Emu falls beyond Emu. What is the relation between the length of the shadow and the distance from the Emu to the pillar?

Solution:



Let the height of the pillar be $P = 30$ ft .

The height of the Emu be $E = 8$ ft.

Let the distance walked by the Emu from the pillar be x .

The length of its shadow be s .

Base of large triangle = $x + s$, Height of large triangle = 30.

Base of small triangle = s , Height of small triangle = 8.

$$\frac{s}{8} = \frac{x + s}{30} \quad (\text{By AA Similarity})$$

$$30s = 8(x + s)$$

$$30s = 8x + 8s$$

$$30s - 8s = 8x$$

$$22s = 8x$$

$$11s = 4x$$

$$s = \frac{4}{11}x$$

The relation is $s = \frac{4}{11}x$.

CHAPTER - 5. CO-ORDINATE GEOMETRY

HALF YEARLY EXAM IMPORTANT QUESTIONS

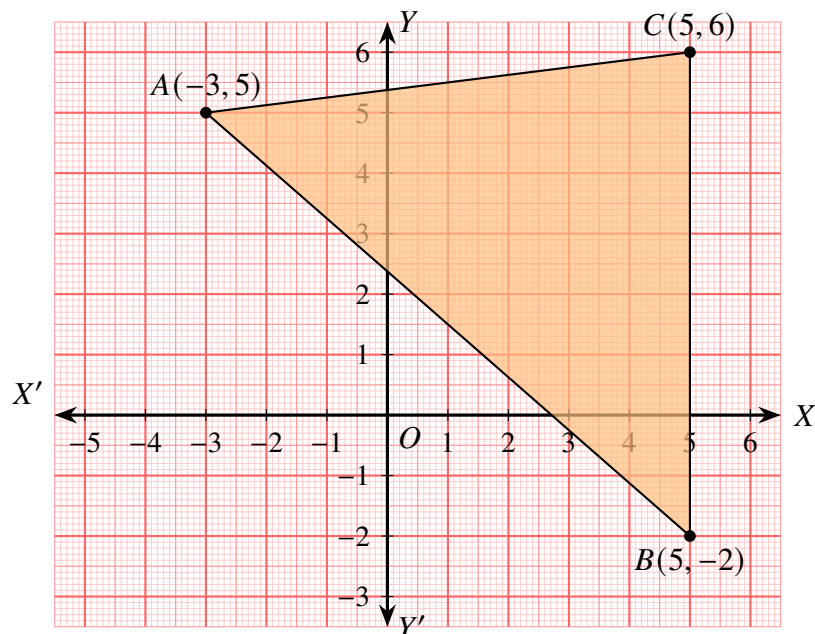
2 MARK IMPORTANT QUESTIONS

CHAPTER - 5. COORDINATE GEOMETRY

2 MARK IMPORTANT QUESTIONS

1. Find the area of the triangle whose vertices are $(-3, 5)$, $(5, 6)$ and $(5, -2)$

solution



Plot the points take counter-clockwise order.

Let the vertices be

$A(-3, 5)$	$B(5, -2)$	$C(5, 6)$
$\downarrow$	$\downarrow$	$\downarrow$
(x_1, y_1)	(x_2, y_2)	(x_3, y_3)

$$\begin{aligned}
 \text{The Area of } \triangle ABC &= \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix} \\
 &= \frac{1}{2} \begin{vmatrix} -3 & 5 & 5 & -3 \\ 5 & -2 & 6 & 5 \end{vmatrix} \\
 &= \frac{1}{2} \{ (6 + 30 + 25) - (25 - 10 - 18) \} \\
 &= \frac{1}{2} \{ 61 + 3 \} \\
 &= \frac{1}{2} (64) = \boxed{32 \text{ sq. units}}
 \end{aligned}$$

2. Show that the points $P(-1.5, 3)$, $Q(6, -2)$, $R(-3, 4)$ are collinear.

APR-22, PTA-4

solution

The points are $P(-1.5, 3)$, $Q(6, -2)$, $R(-3, 4)$

$$\begin{aligned}
 \text{The Area of } \triangle PQR &= \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix} \\
 &= \frac{1}{2} \begin{vmatrix} -1.5 & 6 & -3 & -1.5 \\ 3 & -2 & 4 & 3 \end{vmatrix} \\
 &= \frac{1}{2} \{ (3 + 24 - 9) - (18 + 6 - 6) \} \\
 &= \frac{1}{2} \{ 18 - 18 \} = 0 \text{ sq. units}
 \end{aligned}$$

The given points are collinear.

3. If the area of the triangle formed by the vertices $A(-1, 2)$, $B(k, -2)$ and $C(7, 4)$ (taken in order) is 22 sq. units, find the value of k. **JUL-22**

solution

The vertices are $A(-1, 2)$, $B(k, -2)$ and $C(7, 4)$

$$\begin{aligned}
 \text{The Area of } \triangle ABC &= \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix} \\
 &= \frac{1}{2} \begin{vmatrix} -1 & k & 7 & -1 \\ 2 & -2 & 4 & 2 \end{vmatrix} \\
 &= \frac{1}{2} \{ (2 + 4k + 14) - (2k - 14 - 4) \} = 22 \\
 2k + 34 &= 44 \Rightarrow 2k = 10 \Rightarrow \boxed{k = 5}
 \end{aligned}$$

4. If the points $P(-1, -4)$, $Q(b, c)$ and $R(5, -1)$ are collinear and if $2b + c = 4$, then find the values of b and c. **JUL-21**

solution

$$\text{The Area of } \triangle PQR = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}$$

Since the three points $P(-1, -4)$, $Q(b, c)$ and $R(5, -1)$ are collinear,

Area of triangle PQR = 0

$$\begin{aligned}
 \frac{1}{2} \begin{vmatrix} -1 & b & 5 & -1 \\ -4 & c & -1 & -4 \end{vmatrix} &= 0 \\
 \frac{1}{2} \{ (-c - b - 20) - (-4b + 5c + 1) \} &= 0 \\
 -c - b - 20 + 4b - 5c - 1 &= 0 \\
 \boxed{b - 2c = 7} \quad \dots(1)
 \end{aligned}$$

$$b - 2c = 7 \quad \text{--- ①}$$

$$2b + c = 4 \quad \text{--- ②}$$

Solve ① & ② .

$$\text{②} \times 2 \Rightarrow 4b + 2c = 8$$

$$\text{①} \Rightarrow b - 2c = 7$$

$$5b = 15$$

$$b = \frac{15}{5}$$

$$\boxed{b = 3}$$

$$\text{Sub } b = 3 \text{ in } \text{②} . \quad 2 \times 3 + c = 4$$

$$6 + c = 4$$

$$c = 4 - 6$$

$$\boxed{c = -2}$$

5. Find the area of the triangle formed by the points

(i) $(1, -1)$, $(-4, 6)$ and $(-3, -5)$

(ii) $(-10, -4)$, $(-8, -1)$ and $(-3, -5)$.

solution

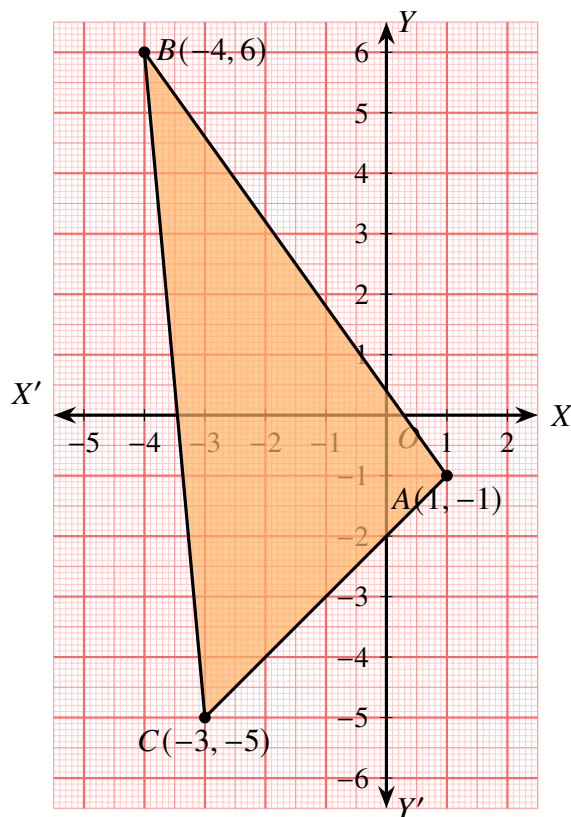
(i) Plot the points take counter-clockwise order.

Let the vertices be

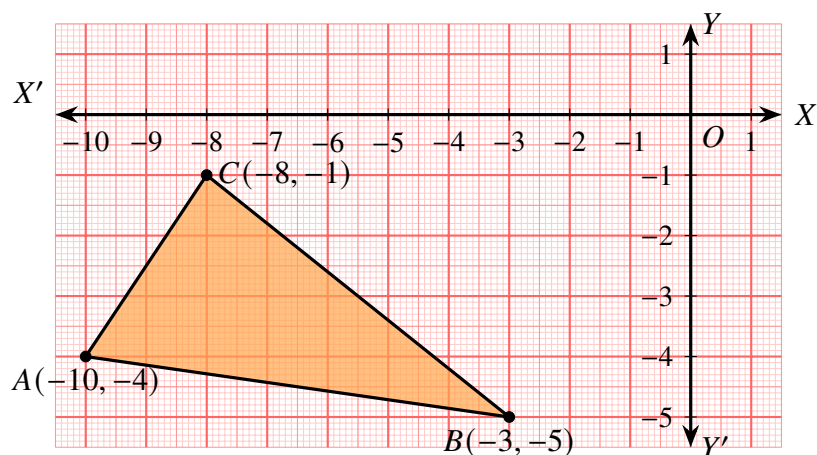
$A(1, -1)$	$B(-4, 6)$	$C(-3, -5)$
$\downarrow$	$\downarrow$	$\downarrow$
(x_1, y_1)	(x_2, y_2)	(x_3, y_3)

$$\text{The Area of } \triangle ABC = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}$$

$$\begin{aligned}
 &= \frac{1}{2} \begin{vmatrix} 1 & -4 & -3 & 1 \\ -1 & 6 & -5 & -1 \end{vmatrix} \\
 &= \frac{1}{2} \{(6 + 20 + 3) - (4 - 18 - 5)\} \\
 &= \frac{1}{2} \{29 + 19\} \\
 &= \frac{1}{2} (48) = \boxed{24 \text{ sq. units}}
 \end{aligned}$$



(ii)



Plot the points take counter-clockwise order.

Let the vertices be

A(-10, -4)	B(-3, -5)	C(-8, -1)
↓	↓	↓
(x ₁ , y ₁)	(x ₂ , y ₂)	(x ₃ , y ₃)

$$\text{The Area of } \triangle ABC = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}$$

$$\begin{aligned}
 &= \frac{1}{2} \begin{vmatrix} -10 & -3 & -8 & -10 \\ -4 & -5 & -1 & -4 \end{vmatrix} \\
 &= \frac{1}{2} \{(50 + 3 + 32) - (12 + 40 + 10)\} \\
 &= \frac{1}{2} \{85 - 62\} \\
 &= \frac{1}{2} (23) = \boxed{11.5 \text{ sq. units}}
 \end{aligned}$$

6. Determine whether the sets of points are collinear? (i) $(-\frac{1}{2}, 3)$, $(-5, 6)$ and $(-8, 8)$.

solutionThe points are $A(-\frac{1}{2}, 3)$, $B(-5, 6)$, $C(-8, 8)$

$$\text{The Area of } \triangle ABC = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}$$

$$\begin{aligned}
 &= \frac{1}{2} \begin{vmatrix} -\frac{1}{2} & -5 & -8 & -\frac{1}{2} \\ 3 & 6 & 8 & 3 \end{vmatrix} \\
 &= \frac{1}{2} \{(-3 - 40 - 24) - (-15 - 48 - 4)\} \\
 &= \frac{1}{2} \{-67 + 67\} = 0 \text{ sq. units}
 \end{aligned}$$

The given points are collinear.

7. Find the value of 'a' for which the given points are collinear. (i) $(2, 3)$, $(4, a)$ and $(6, -3)$.

solutionThe points are $A((2, 3)$, $B(4, a)$, $C(6, -3)$

$$\text{The Area of } \triangle ABC = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}$$

$$\begin{aligned}
 &= \frac{1}{2} \begin{vmatrix} 2 & 4 & 6 & 2 \\ 3 & a & -3 & 3 \end{vmatrix} \\
 &\{(2a - 12 + 18) - (12 + 6a - 6)\} = 0 \\
 &\{2a + 6 - 6 - 6a\} = 0 \Rightarrow \boxed{a = 0}
 \end{aligned}$$

8. If the points $A(-3, 9)$, $B(a, b)$ and $C(4, -5)$ are collinear and if $a + b = 1$ then find a and b. **APR-24**.

solution

$$\text{The Area of } \triangle ABC = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}$$

Since the three points $A(-3, 9)$, $B(a, b)$ and $C(4, -5)$ are collinear,

Area of triangle ABC = 0

$$\begin{aligned}
 &\frac{1}{2} \begin{vmatrix} -3 & a & 4 & -3 \\ 9 & b & -5 & 9 \end{vmatrix} = 0 \\
 &\frac{1}{2} \{(-3b - 5a + 36) - (9a + 4b + 15)\} = 0 \\
 &-5a - 9a - 3b - 4b + 21 = 0 \\
 &\boxed{2a + b = 3} \dots(1)
 \end{aligned}$$

$$2a + b = 3 \text{ --- ①}$$

$$a + b = 1 \text{ --- ②}$$

Solve ① & ② .

$$\text{①} \Rightarrow 2a + b = 3$$

$$\text{②} \Rightarrow a + b = 1(-)$$

$$\boxed{a = 2}$$

$$\text{Sub } a = 2 \text{ in ②. } 2 \times 2 + b = 1$$

$$b = 1 - 2$$

$$\boxed{b = -1}$$

9. (i) What is the slope of a line whose inclination is 30° ?
What is the inclination of a line whose slope is $\sqrt{3}$?

solution

- (i) Given $\theta = 30^\circ$

$$\begin{aligned}\text{Slope } m &= \tan \theta \\ m &= \tan 30^\circ\end{aligned}$$

$$m = \frac{1}{\sqrt{3}}$$

- (ii) Given $m = \sqrt{3}$

$$\begin{aligned}\text{Slope } m &= \tan \theta \\ \sqrt{3} &= \tan \theta\end{aligned}$$

$$\theta = 60^\circ$$

10. Find the slope of a line joining the given points

- (i) $(-6, 1)$ and $(-3, 2)$ **JUL-25**

- (ii) $(-\frac{1}{3}, \frac{1}{2})$ and $(\frac{2}{7}, \frac{3}{7})$

- (iii) $(14, 10)$ and $(14, -6)$ **JUL-20, MAR-26**

solution

- (i) **Points** $(-6, 1)$ and $(-3, 2)$

$$\begin{aligned}\text{The slope} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{2 - 1}{-3 - (-6)} \\ &= \frac{1}{3}\end{aligned}$$

- (ii) **Points** $(-\frac{1}{3}, \frac{1}{2})$ and $(\frac{2}{7}, \frac{3}{7})$

$$\begin{aligned}\text{The slope} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{\frac{3}{7} - \frac{1}{2}}{\frac{2}{7} - (-\frac{1}{3})} \\ &= \frac{\frac{6-7}{14}}{\frac{6+7}{21}}\end{aligned}$$

$$= -\frac{1}{14^2} \times \frac{21^3}{13} = -\frac{3}{26}$$

- (iii) **Points** $(14, 10)$ and $(14, -6)$

$$\begin{aligned}\text{The slope} &= \frac{y_2 - y_1}{x_2 - x_1} \\ \text{The slope} &= \frac{-6 - 10}{14 - 14} \\ &= \frac{-16}{0}\end{aligned}$$

The slope is undefined.

11. The line r passes through the points $(-2, 2)$ and $(5, 8)$ and the line s passes through the points $(-8, 7)$ and $(-2, 0)$. Is the line r perpendicular to s ?

solution

$$\text{The slope of line } r \text{ is } m_1 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{8 - 2}{5 - (-2)} = \frac{6}{7}$$

$$\text{The slope of line } s \text{ is } m_2 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - 7}{-2 - (-8)} = \frac{-7}{6}$$

$$\text{The product of slopes} = \frac{6}{7} \times \frac{-7}{6} = -1$$

$$m_1 m_2 = -1$$

The line r is perpendicular to line s .

12. The line p passes through the points $(3, -2)$, $(12, 4)$ and the line q passes through the points $(6, -2)$ and $(12, 2)$. Is p parallel to q ?

JUL-22, 24, APR-22

solution

$$\text{The slope of line P is } m_1 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 2}{12 - 3} = \frac{2}{9} = \frac{2}{9}$$

$$\text{The slope of line Q is } m_2 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - (-2)}{12 - 6} = \frac{4}{6} = \frac{2}{3}$$

Thus slope of line P = slope of line Q

The line P is parallel to the line Q.

13. Show that the points $(-2, 5)$, $(6, -1)$ and $(2, 2)$ are collinear.

solution

The vertices are $A(-2, 5)$, $B(6, -1)$ and $C(2, 2)$.

$$\begin{aligned}\text{Slope of AB} &= \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1 - 5}{6 - (-2)} \\ &= \frac{-6}{8} = \frac{-3}{4}\end{aligned}$$

$$\begin{aligned}\text{Slope of BC} &= \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - 1}{2 - 6} \\ &= \frac{1}{-4} = \frac{-1}{4}\end{aligned}$$

Slope of AB = Slope of BC

Hence the points A, B, and C are collinear.

14. What is the inclination of a line whose slope is 1? **PTA-3**

solution

$$\begin{aligned}\text{Slope } m &= 1 \\ \tan \theta &= 1 \\ &= \tan 45^\circ \\ \therefore \theta &= 45^\circ\end{aligned}$$

15. Find the slope of a line joining the points

- (i) $(5, \sqrt{5})$ with the origin **JUL-22, 23**

- (ii) $(\sin \theta, -\cos \theta)$ and $(-\sin \theta, \cos \theta)$ **PTA-2**

solution

- (i) Given points $(5, \sqrt{5})$ and $(0, 0)$

$$\begin{aligned}\text{Slope of a line} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{\sqrt{5} - 0}{5 - 0} = \frac{\sqrt{5}}{5} \\ &= \frac{1}{\sqrt{5}}\end{aligned}$$

- (ii) Given points $(\sin \theta, -\cos \theta)$ and $(-\sin \theta, \cos \theta)$

$$\begin{aligned}\text{Slope of a line} &= \frac{y_1 - y_2}{x_1 - x_2} \\ &= \frac{-\cos \theta - \cos \theta}{\sin \theta + \sin \theta} \\ &= \frac{-2 \cos \theta}{2 \sin \theta} = -\cot \theta\end{aligned}$$

16. Show that the given points are collinear: $(-3, -4)$, $(7, 2)$ and $(12, 5)$

JUL-21

solution Given Points $A(-3, -4)$, $B(7, 2)$ and $C(12, 5)$

$$\text{Slope of AB} = \frac{y_1 - y_2}{x_1 - x_2} = \frac{-4 - 2}{-3 - 7} = \frac{-6}{-10} = \frac{3}{5}$$

$$\text{Slope of BC} = \frac{y_1 - y_2}{x_1 - x_2} = \frac{2 - 5}{7 - 12} = \frac{-3}{-5} = \frac{3}{5}$$

Slope of AB = Slope of BC

$\therefore$ The points A, B and C are collinear.

17. If the three points $(3, -1)$, $(a, 3)$ and $(1, -3)$ are collinear, find the value of a .

solution

Given Points $A(3, -1)$, $B(a, 3)$ and $C(1, -3)$

$$\text{Slope of AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1 - 3}{3 - a} = \frac{-4}{3 - a}$$

$$\text{Slope of BC} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 + 3}{a - 1} = \frac{6}{a - 1}$$

$$\text{Slope of AB} = \text{Slope of BC}$$

$$\frac{-4}{3 - a} = \frac{6}{a - 1}$$

$$-2(a - 1) = 3(3 - a)$$

$$-2a + 2 = 9 - 3a$$

$$3a - 2a = 9 - 2$$

$$\boxed{a = 7.}$$

18. The line through the points $(-2, a)$ and $(9, 3)$ has slope $-\frac{1}{2}$. Find the value of a . **APR-25**

solution

Given Point $(-2, a)$ and $(9, 3)$ and slope $-\frac{1}{2}$

$$\text{Slope } m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$-\frac{1}{2} = \frac{a - 3}{-2 - 9}$$

$$-\frac{1}{2} = \frac{a - 3}{-11}$$

$$11 = 2(a - 3)$$

$$11 + 6 = 2a$$

$$\boxed{a = \frac{17}{2}}$$

19. The line through the points $(-2, 6)$ and $(4, 8)$ is perpendicular to the line through the points $(8, 12)$ and $(x, 24)$. Find the value of x . **PTA-6**

solution

Slope of the line joining the points $(-2, 6)$ and $(4, 8)$

$$m_1 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 - 8}{-2 - 4} = \frac{-2}{-6} = \frac{1}{3}$$

Slope of the line joining the points $(8, 12)$ and $(x, 24)$

$$m_2 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{12 - 24}{8 - x} = -\frac{12}{8 - x}$$

The lines are perpendicular

$$\therefore m_1 \times m_2 = -1.$$

$$\frac{1}{3} \times \frac{-12}{8 - x} = -1$$

$$4 = 8 - x$$

$$x = 8 - 4$$

$$\boxed{x = 4}$$

20. Find the equation of a straight line whose

(i) Slope is 5 and y intercept is -9

(ii) Inclination is 45° and y intercept is 11.

solution

(i) Given, Slope = 5, Y- intercept $c = -9$

Equation of a straight line is

$$y = mx + c$$

$$y = 5x - 9 \Rightarrow \boxed{5x - y - 9 = 0}$$

(ii) Given, $\theta = 45^\circ$, Y-intercept $c = 11$

$$\text{Slope } m = \tan \theta = \tan 45^\circ = 1$$

Equation of a straight line is $y = mx + c$

$$y = x + 11 \Rightarrow \boxed{x - y + 11 = 0}$$

21. Calculate the slope and y intercept of the straight line $8x - 7y + 6 = 0$

JUL-21

solution

Equation of the given straight line is

$$8x - 7y + 6 = 0$$

$$\Rightarrow 7y = 8x + 6$$

$$y = \frac{8}{7}x + \frac{6}{7} \quad \dots(1)$$

Comparing (1) with $y = mx + c$

$$\text{Slope } m = \frac{8}{7} \quad \text{Y-intercept } c = \frac{6}{7}$$

22. Find the equation of a line passing through the point $(3, -4)$ and having slope $\frac{-5}{7}$.

solution

Given point $(3, -4)$ and slope $\frac{-5}{7}$

Point-Slope Form $y - y_1 = m(x - x_1)$

$$y + 4 = -\frac{5}{7}(x - 3)$$

$$7y + 28 = -5x + 15$$

$$5x + 7y + 28 - 15 = 0$$

$$\boxed{5x + 7y + 13 = 0}$$

23. Find the equation of a line passing through the point $A(1, 4)$ and perpendicular to the line joining points $(2, 5)$ and $(4, 7)$.

solution

Given Points $A(1, 4)$ $B(2, 5)$ and $C(4, 7)$

$$\text{Slope of line BC} = \frac{7 - 5}{4 - 2} = \frac{2}{2} = 1$$

Let m be the slope of the required line

$$\Rightarrow m \times 1 = -1 \Rightarrow m = -1$$

The required line passes through the point $A(1, 4)$.

The equation of the required S.L

$$y - y_1 = m(x - x_1)$$

$$y - 4 = -1(x - 1)$$

$$y - 4 = -x + 1$$

$$\boxed{x + y - 5 = 0}$$

24. Find the equation of a straight line passing through $(5, -3)$ and $(7, -4)$.

solution

Given Points $(5, -3)$ and $(7, -4)$.

Two Point Form

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\Rightarrow \frac{y + 3}{-4 + 3} = \frac{x - 5}{7 - 5}$$

$$\Rightarrow 2y + 6 = -x + 5$$

$$x + 2y + 1 = 0$$

25. Find the equation of a line which passes through $(5, 7)$ and makes intercepts on the axes equal in magnitude but opposite in sign.

solution

The X-intercept be ' a ' and Y-intercept be ' $-a$ '.

Intercept form

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\Rightarrow \frac{x}{a} + \frac{y}{-a} = 1$$

$$x - y = a \quad \dots(1)$$

Since (1) passes through (5, 7)

$$5 - 7 = a \Rightarrow a = -2$$

The required equation of the S.L

$$x - y = -2; \text{ or } \boxed{x - y + 2 = 0}$$

- 26.** Find the intercepts made by the line $4x - 9y + 36 = 0$ on the coordinate axes.

solution

Equation of the given line is

$$4x - 9y + 36 = 0 \Rightarrow 4x - 9y = -36 \quad (\div -36)$$

$$\frac{4x}{-36} - \frac{9y}{-36} = \frac{-36}{-36}$$

$$\frac{x}{-9} + \frac{y}{4} = 1 \quad \dots(1)$$

X-intercept $a = -9$; Y-intercept $b = 4$.

- 27.** The equation of a straight line is $2(x - y) + 5 = 0$ Find its slope, inclination and intercept on the Y axis.

solution

$2x - 2y + 5 = 0$ is in the form $ax + by + c = 0$

$$\text{Slope} = -\frac{a}{b} = \frac{-2}{-2} = 1$$

$$\text{Slope } \boxed{m = 1}$$

$$\tan \theta = 1 = \tan 45^\circ$$

$$\boxed{\theta = 45^\circ}$$

$$2x - 2y + 5 = 0$$

$$2x - 2y = -5 \quad (\div -2)$$

$$\frac{x}{(-\frac{5}{2})} + \frac{y}{(\frac{5}{2})} = 1$$

$$\text{y intercept is } \boxed{\frac{5}{2}}$$

- 28.** Find the equation of a line whose inclination is 30° and making an intercept -3 on the Y axis.

solution

Given inclination $\theta = 30^\circ$

$$\therefore \text{Slope } m = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

y intercept $c = -3$

slope and 'y' intercept Form

$$y = mx + c$$

$$\Rightarrow y = \frac{1}{\sqrt{3}}x - 3$$

$$\sqrt{3}y = x - 3\sqrt{3}$$

$$\boxed{x - \sqrt{3}y - 3\sqrt{3} = 0}$$

- 29.** Find the slope and y intercept of $\sqrt{3}x + (1 - \sqrt{3})y = 3$.

solution

$$\sqrt{3}x + (1 - \sqrt{3})y = 3$$

Change in to $y = mx + c$

$$(1 - \sqrt{3})y = -\sqrt{3}x + 3$$

$$= -\frac{\sqrt{3}}{1 - \sqrt{3}} + \frac{3}{1 - \sqrt{3}}$$

$$= -\frac{\sqrt{3}}{1 - \sqrt{3}} \times \frac{1 + \sqrt{3}}{1 + \sqrt{3}}$$

$$\boxed{m = \frac{\sqrt{3} + 3}{2}}$$

$$\text{y intercept} = \frac{3}{1 - \sqrt{3}} \times \frac{1 + \sqrt{3}}{1 + \sqrt{3}}$$

$$= \frac{3 + 3\sqrt{3}}{1 - 3}$$

$$\boxed{c = \frac{3 + 3\sqrt{3}}{-2}}$$

- 30.** Find the value of 'a', if the line through $(-2, 3)$ and $(8, 5)$ is perpendicular to $y = ax + 2$.

solution

Slope of a line passing through (x_1, y_1) and (x_2, y_2)

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 5}{-2 - 8} = \frac{-2}{-10} = \frac{1}{5} = m_1$$

Slope of the line $y = ax + 2$ is 'a' $= m_2$

The lines are perpendicular

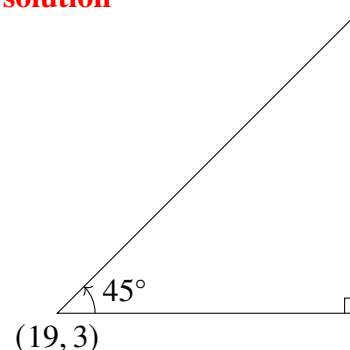
$$\therefore m_1 \times m_2 = -1$$

$$\frac{1}{5} \times a = -1$$

$$a = -5.$$

- 31.** The hill in the form of a right triangle has its foot at $(19, 3)$. The inclination of the hill to the ground is 45° . Find the equation of the hill joining the foot and top. **PTA-2.**

solution



Foot of the hill is at $(19, 3)$

Angle of inclination $\theta = 45^\circ$

$$\text{Slope } m = \tan 45^\circ = 1$$

Equation of line passing through (x_1, y_1)

$$\text{Slope 'm' is } y - y_1 = m(x - x_1)$$

$$\Rightarrow y - 3 = 1(x - 19)$$

$$y - 3 = x - 19 \Rightarrow \boxed{x - y - 16 = 0}$$

- 32.** A cat is located at the point $(-6, -4)$ in xy plane. A bottle of milk is kept at $(5, 11)$. The cat wish to consume the milk travelling through shortest possible distance. Find the equation of the path it needs to take its milk. **APR-23.**

solution

Two Points A (x_1, y_1) B (x_2, y_2)

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y + 4}{11 + 4} = \frac{x + 6}{5 + 6}$$

$$11(y + 4) = 15(x + 6)$$

$$11y + 44 = 15x + 90$$

$$\boxed{15x - 11y + 46 = 0}$$

33. Find the equation of a straight line which has slope $\frac{-5}{4}$ and passing through the point $(-1, 2)$. **APR-22,24.**

solution

Given Point (x_1, y_1) and Slope $\frac{-5}{4}$

Point-Slope Form

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -\frac{5}{4}(x + 1)$$

$$4y - 8 = -5x - 5$$

$$\boxed{5x + 4y - 3 = 0}$$

34. Find the equation of a line through the given pair of points

(i) $(2, \frac{2}{3})$ and $(\frac{-1}{2}, -2)$

(ii) $(2, 3)$ and $(-7, -1)$.

solution

(i) Given points (x_1, y_1) and (x_2, y_2)
 $(2, \frac{2}{3})$ and $(\frac{-1}{2}, -2)$

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y - \frac{2}{3}}{-2 - \frac{2}{3}} = \frac{x - 2}{-\frac{1}{2} - 2}$$

$$\frac{\frac{3y-2}{3}}{\frac{-6-2}{3}} = \frac{\frac{2x-4}{2}}{\frac{-1-4}{2}}$$

$$-5(3y - 2) = -8(2x - 4)$$

$$-15y + 10 = -16x + 32$$

$$\boxed{16x - 15y - 22 = 0}$$

(ii) Two points (x_1, y_1) and (x_2, y_2)
 $(2, 3)$ and $(-7, -1)$

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y - 3}{-1 - 3} = \frac{x - 2}{-7 - 2}$$

$$9(y - 3) = 4(x - 2)$$

$$9y - 27 = 4x - 8$$

$$\boxed{4x - 9y + 19 = 0}$$

35. Find the Equation of a line whose intercepts on the x and y axes (i) $4, -6$ **MAR-26** (ii) $-5, \frac{3}{4}$.

36. Find the intercepts made by the following lines on the coordinate axes.

(i) $3x - 2y - 6 = 0$ **JUL-21**

(ii) $4x + 3y + 12 = 0$ **JUL-25.**

solution

(i) Given line $3x - 2y - 6 = 0$

$$3x - 2y = 6 \quad (\div 6)$$

$$\frac{3x}{6} - \frac{2y}{6} = 1$$

$$\frac{x}{2} + \frac{y}{(-3)} = 1$$

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\text{'x' intercept} = a = 2$$

$$\text{y intercept} = b = -3$$

(ii) $4x + 3y + 12 = 0$

$$4x + 3y = -12 \quad (\div -12)$$

$$\frac{4x}{-12} + \frac{3y}{-12} = 1$$

$$\frac{x}{(-3)} + \frac{y}{(-4)} = 1$$

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\therefore \text{x intercept} = a = -3$$

$$\text{y intercept} = b = -4.$$

37. Find the slope of the straight line $6x + 8y + 7 = 0$ **JUL-24.**

solution

$$\text{Given } 6x + 8y + 7 = 0$$

$$\text{slope } m = \frac{-\text{Co-efficient of } x}{\text{Co-efficient of } y} = \frac{-a}{b}$$

$$= -\frac{6}{8} = -\frac{3}{4}$$

$$\text{The slope of the straight line} = -\frac{3}{4}.$$

38. Show that the straight lines $2x + 3y - 8 = 0$ and $4x + 6y + 18 = 0$ are parallel.

solution

Slope of the straight line $2x + 3y - 8 = 0$ is

$$m_1 = \frac{-a}{b}$$

$$m_1 = \frac{-2}{3}$$

Slope of the straight line $4x + 6y + 18 = 0$ is

$$m_2 = \frac{-a}{b}$$

$$m_2 = \frac{-4}{6} = \frac{-2}{3}$$

$$m_1 = m_2$$

That is, slopes are equal

The Lines are parallel.

39. Show that the straight lines $x - 2y + 3 = 0$ and $6x + 3y + 8 = 0$ are perpendicular. **APR-25, PTA-5.**

solution

Slope of the straight line is $x - 2y + 3 = 0$

$$m_1 = \frac{-a}{b} = \frac{-1}{-2} = \frac{1}{2}$$

Slope of the straight line is $6x + 3y + 8 = 0$

$$m_2 = \frac{-a}{b} = \frac{-6}{3} = -2$$

$$m_1 \times m_2 = \frac{1}{2} \times (-2) = -1$$

The two straight lines are perpendicular.

40. Find the equation of a straight line which is parallel to the line $3x - 7y = 12$ and passing through the point $(6, 4)$. **JUL-23.**

solution

Equation of the straight line, parallel to

$$3x - 7y - 12 = 0$$

$$3x - 7y + k = 0.$$

Since it passes through the point $(6, 4)$.

$$3(6) - 7(4) + k = 0$$

$$k = 28 - 18$$

$$k = 10$$

Equation of the required S.L is

$$3x - 7y + 10 = 0.$$

41. Find the slope of the line which is

(i) parallel to $y = 0.7x - 11$

(ii) perpendicular to the line $x = -11$.

solution

(i) Given line is $y = 0.7x - 11$ is in the form

$$y = mx + c$$

$$\text{Slope } m = 0.7$$

Two lines are parallel

$$m = 0.7.$$

(ii) Given line is $x = -11$

$$\text{Slope } m = -\frac{a}{b} = -\frac{1}{0} = \text{undefined.}$$

$\therefore$ Slope of the line perpendicular to

$$x = -11 \text{ is } 0.$$

42. Check whether the given lines are parallel or perpendicular

(i) $\frac{x}{3} + \frac{y}{4} + \frac{1}{7} = 0$ and $\frac{2x}{3} + \frac{y}{2} + \frac{1}{10} = 0$

(ii) $5x + 23y + 14 = 0$ and $23x - 5y + 9 = 0$.

solution: Alliter Method

(i) Given lines are $\frac{x}{3} + \frac{y}{4} + \frac{1}{7} = 0$ and

$$\frac{2x}{3} + \frac{y}{2} + \frac{1}{10} = 0$$

$$a_1 = \frac{1}{3}, b_1 = \frac{1}{4} \text{ and } a_2 = \frac{2}{3}, b_2 = \frac{1}{2}$$

$$\frac{a_1}{a_2} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}, \quad \frac{b_1}{b_2} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}. \text{ The lines are parallel.}$$

(ii) Given lines $5x + 23y + 14 = 0$ and

$$23x - 5y + 9 = 0$$

$$\therefore a_1 = 5, b_1 = 23 \text{ and } a_2 = 23, b_2 = -5$$

$$a_1 a_2 + b_1 b_2 = (5)(23) + (23)(-5) = 0$$

$\therefore$ The lines are perpendicular.

43. If the straight lines $12y = -(p+3)x + 12$, $12x - 7y = 16$ are perpendicular then find 'p'. **APR-23.**

solution

Given lines are $12y = -(p+3)x + 12$ and

$$12x - 7y = 16$$

$$12y = -(p+3)x + 12$$

$$(p+3)x + 12y - 12 = 0 \dots (1)$$

$$12x - 7y - 16 = 0 \dots (2)$$

$$\therefore a_1 = (p+3), b_1 = 12 \text{ and } a_2 = 12, b_2 = -7$$

Given that the lines are perpendicular,

$$\therefore a_1 a_2 + b_1 b_2 = 0$$

$$(p+3)(12) + (12)(-7) = 0$$

$$12[p+3-7] = 0$$

$$p - 4 = 0$$

$$p = 4$$

44. Find the equations of the lines, whose sum and product of intercepts are 1 and -6 respectively.

solution

Given, sum of the intercepts $a + b = 1 \dots (1)$

Product of the intercepts $ab = -6 \dots (2)$

$$b = \frac{-6}{a}$$

Sub in (1)

$$a - \frac{6}{a} = 1$$

$$a^2 - a - 6 = 0$$

$$(a-3)(a+2) = 0$$

$$a = -2, 3$$

When $a = -2$, $b = \frac{-6}{-2} = 3$

When $a = 3$, $b = \frac{-6}{3} = -2$

Equation of the line in intercepts form is

$$\frac{x}{a} + \frac{y}{b} = 1$$

(i) $a = -2, b = 3 \Rightarrow \frac{x}{-2} + \frac{y}{3} = 1$

$$3x - 2y + 6 = 0$$

(ii) $a = 3, b = -2 \Rightarrow \frac{x}{3} + \frac{y}{-2} = 1$

$$-2x + 3y + 6 = 0$$

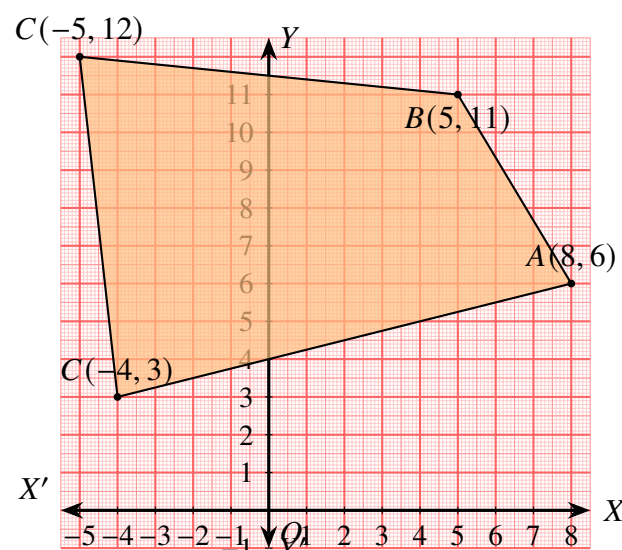
$$\Rightarrow 2x - 3y - 6 = 0$$

CHAPTER - 5. COORDINATE GEOMETRY (Continued)

5 MARK IMPORTANT QUESTIONS

1. Find the area of the quadrilateral formed by the points (8, 6), (5, 11), (-5, 12) and (-4, 3) **JUL-22, APR-23, MAR-26**

solution



Plot the points take counter-clockwise order.

$$A(8, 6), B(5, 11), C(-5, 12), D(-4, 3)$$

Let the vertices be

$$(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$$

$$\text{The Area of Quadrilateral} = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_4 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_1 \end{vmatrix}$$

$$= \frac{1}{2} \begin{vmatrix} 8 & 5 & -5 & -4 & 8 \\ 6 & 11 & 12 & 3 & 5 \end{vmatrix}$$

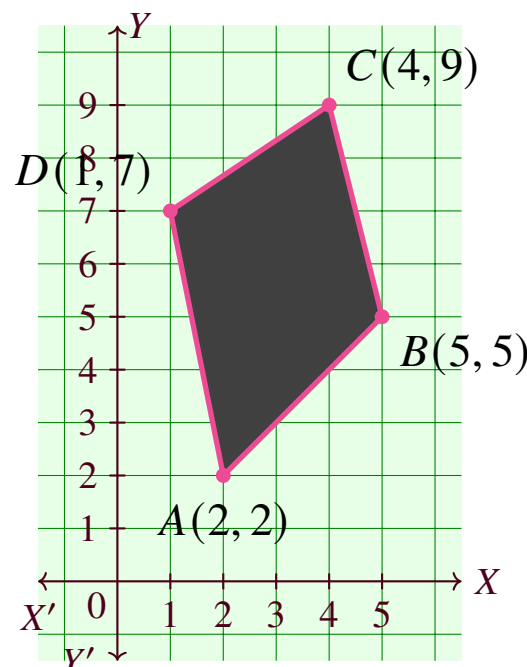
$$= \frac{1}{2} \{(88 + 60 - 15 - 24) - (30 - 55 - 48 + 24)\}$$

$$= \frac{1}{2} \{109 + 49\}$$

$$= \frac{1}{2} (158) = 79 \text{ sq. units}$$

2. The given diagram shows a plan for constructing a new parking lot at a campus. It is estimated that such construction would cost 1300 per square feet. What will be the total cost for making the parking lot?

solution



Let the vertices be

$$A(2, 2), B(5, 5), C(4, 9), D(1, 7)$$

$$(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$$

$$\text{The Area of Quadrilateral} = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_4 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_1 \end{vmatrix}$$

$$\begin{aligned}
 &= \frac{1}{2} \begin{vmatrix} 2 & 5 & 4 & 1 & 2 \\ 2 & 5 & 9 & 7 & 2 \end{vmatrix} \\
 &= \frac{1}{2} \{ (10 + 45 + 28 + 2) - (10 + 20 + 9 + 14) \} \\
 &= \frac{1}{2} \{ 85 - 53 \} \\
 &= \frac{1}{2} (32) = 16 \text{ sq. units.}
 \end{aligned}$$

Area of parking lot = 16 sq. ft

Construction rate per Sq.ft = ₹1300

Total cost of Parking Lot = 16×1300

= ₹20800.

3. Find the area of the quadrilateral whose vertices are at

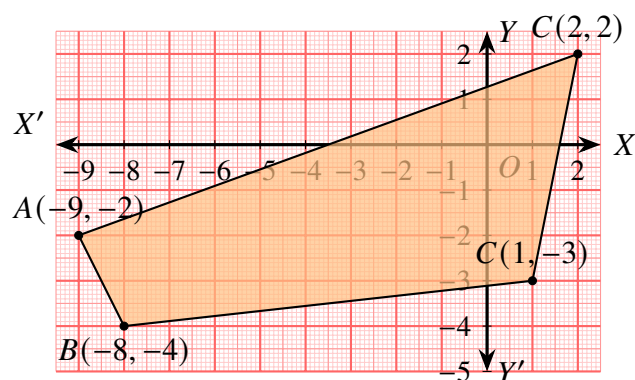
APR-24, JUL-23, 24, 25

(i) $(-9, -2)$, $(-8, -4)$, $(2, 2)$ and $(1, -3)$

(ii) $(-9, 0)$, $(-8, 6)$, $(-1, -2)$ and $(-6, -3)$.

solution

(i)



Plot the points take counter-clockwise order.

A $(-9, -2)$, B $(-8, -4)$, C $(1, -3)$, D $(2, 2)$

Let the vertices be

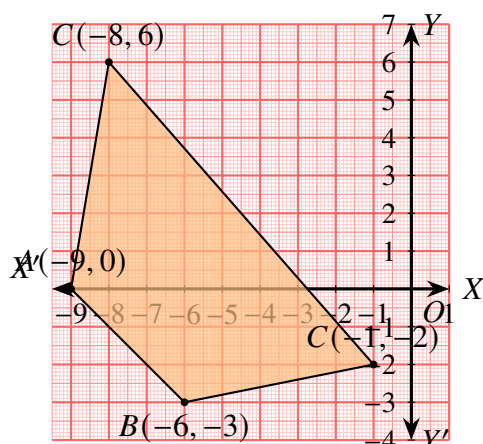
(x_1, y_1) (x_2, y_2) (x_3, y_3) (x_4, y_4)

$$\text{The Area of Quadrilateral} = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_4 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_1 \end{vmatrix}$$

$$\begin{aligned}
 &= \frac{1}{2} \begin{vmatrix} -9 & -8 & 1 & 2 & -9 \\ -2 & -4 & -3 & 2 & -2 \end{vmatrix} \\
 &= \frac{1}{2} [(36 + 24 + 2 - 4) - (16 - 4 - 6 - 18)]
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} [58 - (-12)] \\
 &= \frac{1}{2} [70] = 35 \text{ sq. units}
 \end{aligned}$$

(ii)



Plot the points take counter-clockwise order.

A $(-9, 0)$, B $(-6, -3)$, C $(-1, -2)$, D $(-8, 6)$

Let the vertices be

(x_1, y_1) (x_2, y_2) (x_3, y_3) (x_4, y_4)

$$\text{The Area of Quadrilateral} = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_4 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_1 \end{vmatrix}$$

$$\begin{aligned}
 &= \frac{1}{2} \begin{vmatrix} -9 & -6 & -1 & -8 & -9 \\ 0 & -3 & -2 & 6 & 0 \end{vmatrix} \\
 &= \frac{1}{2} \{ (27 + 12 - 6 - 0) - (0 + 3 + 16 - 54) \} \\
 &= \frac{1}{2} \{ 33 + 35 \} \\
 &= \frac{1}{2} (68) \\
 &= 34 \text{ sq. units}
 \end{aligned}$$

4. Find the value of k, if the area of a quadrilateral is 28 sq.units, whose vertices are taken in the order $(-4, -2)$, $(-3, k)$, $(3, -2)$ and $(2, 3)$

APR-25, JUL-20, PTA-5

solution

A $(-4, -2)$, B $(-3, k)$, C $(3, -2)$, D $(2, 3)$

Let the vertices be

(x_1, y_1) (x_2, y_2) (x_3, y_3) (x_4, y_4)

$$\text{The Area of Quadrilateral} = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_4 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_1 \end{vmatrix}$$

$$\begin{aligned}
 28 &= \frac{1}{2} \begin{vmatrix} -4 & -3 & 3 & 2 & -4 \\ -2 & k & -2 & 3 & -2 \end{vmatrix} \\
 28 &= \frac{1}{2} \{ (-4k + 6 + 9 - 4) - (6 + 3k - 4 - 12) \} \\
 56 &= \{-7k + 21\} \\
 7k &= 21 - 56 = -35 \\
 k &= \frac{-35}{7}
 \end{aligned}$$

$$k = -5$$

Alliter Method

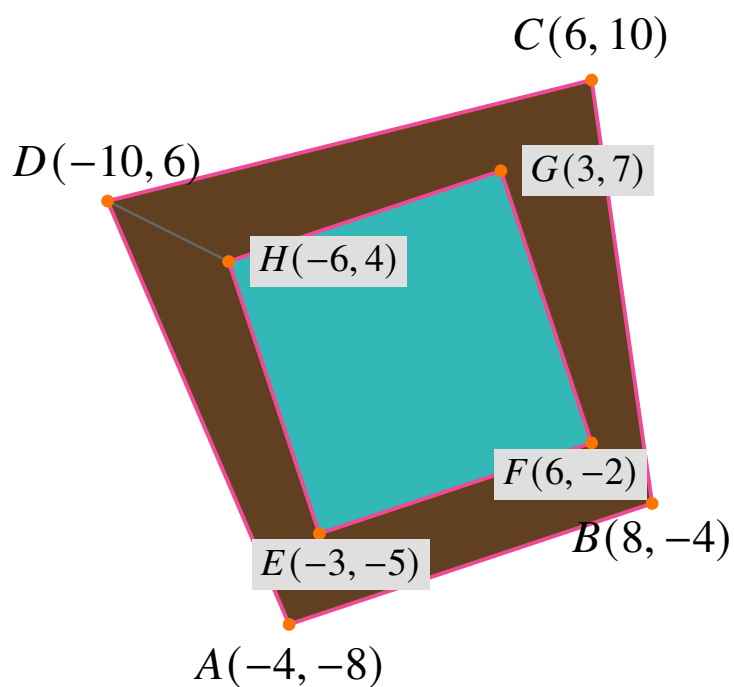
$$\text{The Area of Quadrilateral} = \frac{1}{2} \begin{vmatrix} x_1 - x_3 & x_2 - x_4 \\ y_1 - y_3 & y_2 - y_4 \end{vmatrix}$$

$$= \frac{1}{2} \{ (x_1 - x_3)(y_2 - y_4) - (x_2 - x_4)(y_1 - y_3) \}$$

$$\text{The Area of Triangle} = \frac{1}{2} \begin{vmatrix} x_1 - x_2 & x_1 - x_3 \\ y_1 - y_2 & y_1 - y_3 \end{vmatrix}$$

$$= \frac{1}{2} \{ (x_1 - x_2)(y_1 - y_3) - (x_1 - x_3)(y_1 - y_2) \}$$

5. In the figure, the quadrilateral swimming pool shown is surrounded by concrete patio. Find the area of the patio. **PTA-2**

**solution****Area of Quad ABCD**

A(-4, -8), B(8, -4), C(6, 10) and D(-10, 6)

$$\begin{aligned}
 \text{Area} &= \frac{1}{2} [(x_1 - x_3)(y_2 - y_4) - (x_2 - x_4)(y_1 - y_3)] \\
 &= \frac{1}{2} [(-4 - 6)(-4 - 6) - (8 + 10)(-8 - 10)] \\
 &= \frac{1}{2} [100 + 324] \\
 &= \frac{424}{2} = 212 \text{ sq. units}
 \end{aligned}$$

Area of Quad EFGH

E(-3, -5), F(6, -2), G(3, 7) and H(-6, 4)

$$\begin{aligned}
 \text{Area} &= \frac{1}{2} [(-3 - 3)(-2 - 4) - (6 + 6)(-5 - 7)] \\
 &= \frac{1}{2} [36 + 144] \\
 &= \frac{180}{2} = 90 \text{ sq. units}
 \end{aligned}$$

Area of patio

$$\begin{aligned}
 &= \text{Area}(ABCD) - \text{Area}(EFGH) \\
 &= 212 - 90 \\
 &= 122 \text{ sq. units}
 \end{aligned}$$

6. Without using Pythagoras theorem, show that the points (1, -4), (2, -3) and (4, -7) form a right angled triangle. **PTA-4**

solution

The vertices A (1, -4), B(2, -3) and C(4, -7).

$$\begin{aligned}
 \text{The slope of AB} &= \frac{y_2 - y_1}{x_2 - x_1} = \frac{-3 + 4}{2 - 1} \\
 &= \frac{1}{1} = 1
 \end{aligned}$$

$$\begin{aligned}
 \text{The slope of BC} &= \frac{y_2 - y_1}{x_2 - x_1} = \frac{-7 + 3}{4 - 2} \\
 &= \frac{-4}{2} = -2
 \end{aligned}$$

$$\begin{aligned}
 \text{The slope of AC} &= \frac{y_2 - y_1}{x_2 - x_1} = \frac{-7 + 4}{4 - 1} \\
 &= \frac{-3}{3} = -1
 \end{aligned}$$

$$\text{Slope of AB} \times \text{Slope of AC} = (1)(-1) = -1$$

$$m_1 \times m_2 = -1$$

AB is perpendicular to AC. $\angle A = 90^\circ$ ΔABC is a right angled triangle.

7. Show that the given points form a right angled triangle

(i) A(1, -4), B(2, -3) and C(4, -7) (ii) L(0, 5), M(9, 12) and N(3, 14).

solution

(i) Given Points A(1, -4), B(2, -3) C(4, -7)

$$\text{Slope of the line} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope of AB} = \frac{-4 + 3}{1 - 2} = \frac{-1}{-1} = 1$$

$$\text{Slope of BC} = \frac{-3 + 7}{2 - 4} = \frac{4}{-2} = -2$$

$$\text{Slope of AC} = \frac{-4 + 7}{1 - 4} = \frac{3}{-3} = -1$$

$$(\text{Slope of AB}) \times (\text{Slope of AC}) = -1$$

 $\therefore$ AB is perpendicular to AC.

The given vertices form a right angled triangle

(ii) L(0, 5), M(9, 12) and N(3, 14)

$$\text{Slope of a line} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope of LM} = \frac{5 - 12}{0 - 9} = \frac{-7}{-9} = \frac{7}{9}$$

$$\text{Slope of MN} = \frac{12 - 14}{9 - 3} = \frac{-2}{6} = \frac{-1}{3}$$

$$\text{Slope of LN} = \frac{5 - 14}{0 - 3} = \frac{-9}{-3} = 3$$

$$(\text{Slope of MN}) \times (\text{Slope of LN}) = \left(-\frac{1}{3}\right) \times 3 = -1.$$

 $\therefore$ MN is perpendicular to LN.

The given vertices form a right angled triangle

8. Show that the given points form a parallelogram A(2.5, 3.5), B(10, -4), C(2.5, -2.5) and D(-5, 5).

solution

Given points A (2.5, 3.5), B (10, -4), C (2.5, -2.5) and D (-5, 5)

$$\text{Slope of a line} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope of AB} = \frac{3.5 + 4}{2.5 - 10} = \frac{7.5}{-7.5} = -1$$

$$\text{Slope of BC} = \frac{-4 + 2.5}{10 - 2.5} = \frac{-1.5}{7.5} = -\frac{1}{5}$$

$$\text{Slope of CD} = \frac{-2.5 - 5}{2.5 + 5} = \frac{-7.5}{7.5} = -1$$

$$\text{Slope of AD} = \frac{3.5 - 5}{2.5 + 5} = \frac{-1.5}{7.5} = -\frac{1}{5}$$

$$\text{Slope of AB} = \text{Slope of CD} = -1$$

$$\text{Slope of BC} = \text{Slope of AD} = -\frac{1}{5}$$

 $\therefore$ AB is parallel to CD and BC is parallel to AD.

The given points form a parallelogram.

9. If the points A(2, 2), B(-2, -3), C(1, -3) and D(x, y) form a parallelogram then find the value of x and y.

solution

Given Points A(2, 2), B(-2, -3), C(1, -3) and D(x, y)

$$\text{Slope of a line} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope of AB} = \frac{2 + 3}{2 + 2} = \frac{5}{4}$$

$$\text{Slope of BC} = \frac{-3 + 3}{-2 - 1} = 0$$

$$\text{Slope of CD} = \frac{-3 - y}{1 - x}$$

$$\text{Slope of AD} = \frac{2 - y}{2 - x}$$

The points form a parallelogram

 $\therefore$ Slope of AB = Slope of CD

$$\frac{5}{4} = \frac{-3 - y}{1 - x}$$

$$5(1 - x) = 4(-3 - y)$$

$$5 - 5x = -12 - 4y$$

$$5x - 4y = 17 \quad \dots (1)$$

$$\text{Slope of BC} = \text{Slope of AD}$$

$$0 = \frac{2-y}{2-x}$$

$$\Rightarrow 2 - y = 0$$

$$y = 2$$

Sub in (1)

$$5x - 4(2) = 17$$

$$5x = 17 + 8 = 25$$

$$x = \frac{25}{5} = 5$$

$$\therefore x = 5, y = 2.$$

10. Let $A(3, -4)$, $B(9, -4)$, $C(5, -7)$ and $D(7, -7)$. Show that ABCD is a trapezium.

solution

Given Points $A(3, -4)$, $B(9, -4)$, $C(5, -7)$ and $D(7, -7)$.

$$\text{Slope of a line} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope of AB} = \frac{-4 + 4}{3 - 9} = 0$$

$$\text{Slope of BD} = \frac{-4 + 7}{9 - 7} = \frac{3}{2}$$

$$\text{Slope of CD} = \frac{-7 + 7}{5 - 7} = 0$$

$$\text{Slope of AC} = \frac{-4 + 7}{3 - 5} = \frac{3}{-2} = -\frac{3}{2}$$

$$\text{Slope of AB} = \text{Slope of CD}$$

$\therefore$ AB is parallel to CD

BD is not parallel to AC

The given points form a Trapezium.

11. A quadrilateral has vertices at $A(-4, -2)$, $B(5, -1)$, $C(6, 5)$ and $D(-7, 6)$. Show that the mid-points of its sides form a parallelogram. **APR-22**

solution

Given Points $A(-4, -2)$, $B(5, -1)$, $C(6, 5)$ and $D(-7, 6)$

Mid point of AB

$$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= P \left(\frac{-4 + 5}{2}, \frac{-2 - 1}{2} \right)$$

$$= P \left(\frac{1}{2}, -\frac{3}{2} \right)$$

Mid point of BC

$$= Q \left(\frac{5 + 6}{2}, \frac{-1 + 5}{2} \right) = Q \left(\frac{11}{2}, 2 \right)$$

Mid point of CD

$$= R \left(\frac{6 - 7}{2}, \frac{5 + 6}{2} \right) = R \left(-\frac{1}{2}, \frac{11}{2} \right)$$

Mid point of AD

$$= S \left(\frac{-4 - 7}{2}, \frac{-2 + 6}{2} \right) = S \left(-\frac{11}{2}, 2 \right)$$

$$\text{Slope of PQ} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-\frac{3}{2} - 2}{\frac{1}{2} - \frac{11}{2}} = \frac{-\frac{7}{2}}{-\frac{10}{2}} = \frac{7}{10}$$

$$\text{Slope of QR} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - \frac{11}{2}}{\frac{11}{2} + \frac{1}{2}} = \frac{-\frac{7}{2}}{\frac{12}{2}} = -\frac{7}{12}$$

$$\text{Slope of RS} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\frac{11}{2} - 2}{-\frac{1}{2} + \frac{11}{2}} = \frac{\frac{7}{2}}{\frac{10}{2}} = \frac{7}{10}$$

$$\text{Slope of PS} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-\frac{3}{2} - 2}{\frac{1}{2} + \frac{11}{2}} = \frac{-\frac{7}{2}}{\frac{12}{2}} = -\frac{7}{12}$$

$$\text{Slope of PQ} = \text{Slope of RS} \Rightarrow PQ \parallel RS$$

$$\text{Slope of QR} = \text{Slope of PS} \Rightarrow QR \parallel PS$$

The mid points form a parallelogram.

12. A line makes positive intercepts on coordinate axes whose sum is 7 and it passes through $(-3, 8)$. Find its equation. **APR-25**

solution

If a and b are the intercepts

$$a + b = 7 \text{ or } b = 7 - a$$

$$\text{Intercept form } \frac{x}{a} + \frac{y}{b} = 1 \quad \dots (1)$$

$$\frac{x}{a} + \frac{y}{7-a} = 1$$

As this line passes through the point $(-3, 8)$,

$$\frac{-3}{a} + \frac{8}{7-a} = 1$$

$$\Rightarrow -3(7-a) + 8a = a(7-a)$$

$$-21 + 3a + 8a = 7a - a^2$$

$$a^2 + 4a - 21 = 0$$

$$\text{Solve } (a-3)(a+7) = 0$$

$$a = 3 \text{ or } a = -7$$

Since a is positive,

$$a = 3 \text{ and } b = 7 - a = 7 - 3 = 4.$$

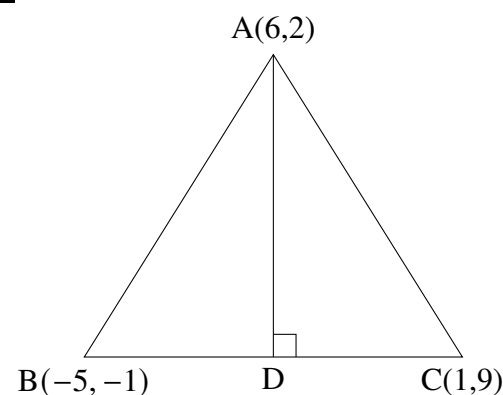
$$\frac{x}{3} + \frac{y}{4} = 1$$

$$4x + 3y - 12 = 0$$

13. If the vertices of a $\triangle ABC$ are $A(6, 2)$, $B(-5, -1)$ and $C(1, 9)$ through the vertex A **JUL-21, PTA-6**

(i) find the equation of median (ii) find the equation of altitude

MDL-19



solution

Given vertices are A $(6, 2)$, B $(-5, -1)$ and C $(1, 9)$

Median through A:

$$\text{Mid point of BC} = D \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= D \left(\frac{-5 + 1}{2}, \frac{-1 + 9}{2} \right)$$

$$= D(-2, 4)$$

AD is the median.

$$\text{Equation of AD} \Rightarrow \frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y - 2}{4 - 2} = \frac{x - 6}{-2 - 6}$$

$$\frac{y - 2}{2} = \frac{x - 6}{-8}$$

$$-4y + 8 = x - 6$$

$$x + 4y - 14 = 0$$

Altitude through A:

Altitude is through A Perpendicular to BC

$$\text{Slope of BC} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-1 - 9}{-5 - 1} = \frac{-10}{-6} = \frac{5}{3}$$

$$\therefore \text{Slope of Altitude} = -\frac{3}{5}$$

Point- Slope Form

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -\frac{3}{5}(x - 6)$$

$$5y - 10 = -3x + 18$$

$$3x + 5y - 28 = 0$$

14. Find the equation of a straight line

- (i) passing through $(1, -4)$ and has intercepts which are in the ratio 2:5
 (ii) passing through $(-8, 4)$ and making equal intercepts on the coordinate axes

solution

(i) Given that intercepts are in the ratio 2 : 5

$$\frac{a}{b} = \frac{2}{5}$$

$$a = \frac{2b}{5}$$

Eqn of Intercepts form $\frac{x}{a} + \frac{y}{b} = 1$

$$\Rightarrow \frac{x}{\left(\frac{2b}{5}\right)} + \frac{y}{b} = 1$$

$$\Rightarrow \frac{5x}{2b} + \frac{y}{b} = 1$$

$$\Rightarrow 5x + 2y = 2b$$

This passes through $(1, -4)$

$$\therefore 5(1) + 2(-4) = 2b$$

$$5 - 8 = 2b \Rightarrow b = -\frac{3}{2}$$

$$a = \frac{2b}{5} = \frac{2\left(-\frac{3}{2}\right)}{5} = -\frac{3}{5}$$

$\therefore$ Equation of a straight line is

$$\frac{x}{a} + \frac{y}{b} = 1 \Rightarrow \frac{x}{\left(-\frac{3}{5}\right)} + \frac{y}{\left(-\frac{3}{2}\right)} = 1$$

$$\frac{5x}{-3} + \frac{2y}{-3} = 1$$

$$5x + 2y + 3 = 0.$$

(ii) Given that intercepts are equal.

$$\therefore a = b$$

Equation of the line in intercepts form is

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x}{a} + \frac{y}{a} = 1$$

$$x + y = a$$

This passes through $(-8, 4)$

$$\therefore -8 + 4 = a$$

$$a = -4 \quad [\because a = b]$$

$$b = -4$$

$\therefore$ Equation of the straight line is

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x}{-4} + \frac{y}{-4} = 1$$

$$x + y + 4 = 0.$$

15. Find the equation of a straight line parallel to Y axis and passing through the point of intersection of the lines $4x + 5y = 13$ and $x - 8y + 9 = 0$.

solution

$$4x + 5y - 13 = 0 \quad \dots (1)$$

$$x - 8y + 9 = 0 \quad \dots (2)$$

Multiply equation (2) by 4:

$$4(x - 8y + 9) = 4(0)$$

$$4x - 32y + 36 = 0 \quad \dots (3)$$

Subtract equation (3) from equation (1):

$$(4x + 5y - 13) - (4x - 32y + 36) = 0$$

$$4x + 5y - 13 - 4x + 32y - 36 = 0$$

$$37y - 49 = 0$$

$$37y = 49$$

$$y = \frac{49}{37}$$

Sub the value of y in equation (2):

$$x - 8\left(\frac{49}{37}\right) + 9 = 0$$

$$x - \frac{392}{37} + 9 = 0$$

$$x = \frac{392}{37} - 9$$

$$x = \frac{392 - 333}{37}$$

$$x = \frac{59}{37}$$

The point of intersection

$$(x, y) = \left(\frac{59}{37}, \frac{49}{37}\right)$$

The equ of line parallel to Y-axis $x = c$.

$$\text{It passes through } (x, y) = \left(\frac{59}{37}, \frac{49}{37}\right).$$

$$c = \frac{59}{37}.$$

$$\text{The equ of the line } x = \frac{59}{37}$$

$$\Rightarrow 37x - 59 = 0.$$

16. Find the equation of a line passing through $(6, -2)$ and perpendicular to the line joining the points $(6, 7)$ and $(2, -3)$.

solution

Slope of the points $(6, 7)$ and $(2, -3)$ is

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{7 + 3}{6 - 2}$$

$$m_1 = \frac{10}{4} = \frac{5}{2}$$

Slope of the perpendicular line

$$m_1 \times m_2 = -\frac{1}{(5/2)} = -\frac{2}{5}$$

$$\text{Point}(6, -2) \quad \text{slope } -\frac{2}{5}$$

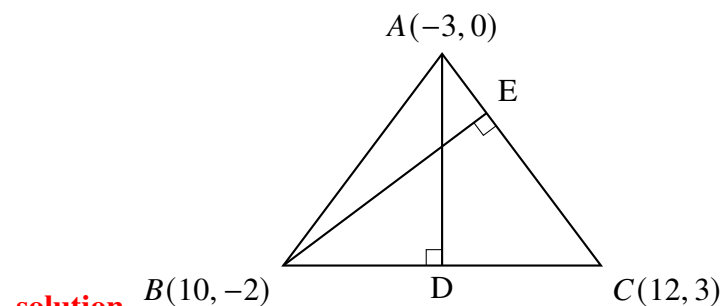
$$y - y_1 = m(x - x_1)$$

$$y + 2 = -\frac{2}{5}(x - 6)$$

$$5y + 10 = -2x + 12$$

$$2x + 5y - 2 = 0$$

17. $A(-3, 0)$, $B(10, -2)$ and $C(12, 3)$ are the vertices of $\triangle ABC$. Find the equation of the altitude through A and B. **JUL-23, APR-24, MAR-26**



solution

$$\text{Slope of BC} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-2 - 3}{10 - 12} = \frac{-5}{-2} = \frac{5}{2}$$

Altitude AD is perp to BC through $A(-3, 0)$

$$\therefore \text{Slope of AD} = -\frac{2}{5}$$

$$\text{Equ of AD } y - y_1 = m(x - x_1)$$

$$y - 0 = -\frac{2}{5}(x + 3)$$

$$5y = -2x - 6$$

$$2x + 5y + 6 = 0$$

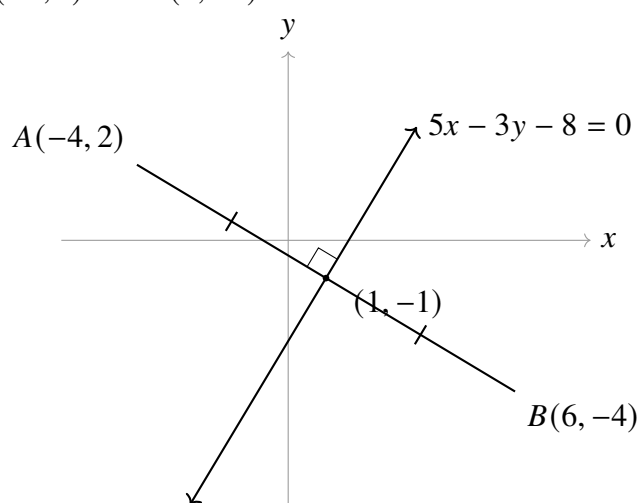
$$\begin{aligned}\text{Slope of AC} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{0 - 3}{-3 - 12} = \frac{-3}{-15} = \frac{1}{5}\end{aligned}$$

Altitude BE is perp to AC B(10, -2).

$$\text{Slope of BE} = -5$$

$$\begin{aligned}\therefore \text{Equation of BE } y - y_1 &= m(x - x_1) \\ y + 2 &= -5(x - 10) \\ y + 2 &= -5x + 50 \\ 5x + y - 48 &= 0\end{aligned}$$

18. Find the equation of the perpendicular bisector of the line joining the points A(-4, 2) and B(6, -4).



solution

points A(-4, 2) and B(6, -4)

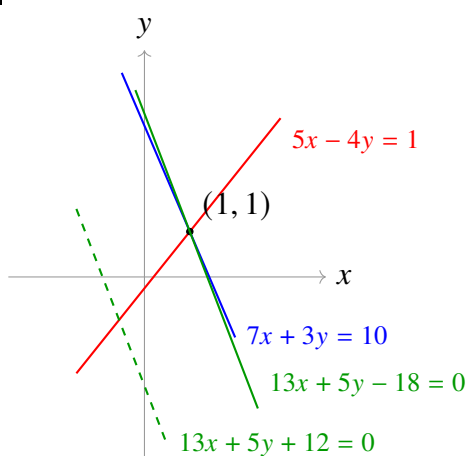
$$\begin{aligned}\text{Mid point of AB} &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left(\frac{-4 + 6}{2}, \frac{2 + (-4)}{2} \right) = (1, -1) \\ \text{Slope of AB} &= \frac{y_1 - y_2}{x_1 - x_2} = \frac{2 - (-4)}{-4 - 6} = \frac{6}{-10} = -\frac{3}{5} \\ \text{Slope of perp line} &= \frac{5}{3}\end{aligned}$$

$$\text{Perp bisector passing } (1, -1), \text{ slope } \frac{5}{3}.$$

$\therefore$ The Eqn of perp bisector

$$\begin{aligned}y - y_1 &= m(x - x_1) \\ y + 1 &= \frac{5}{3}(x - 1) \\ 3y + 3 &= 5x - 5 \\ 5x - 3y - 8 &= 0\end{aligned}$$

19. Find the equation of a straight line through the intersection of lines $7x + 3y = 10$, $5x - 4y = 1$ and parallel to the line $13x + 5y + 12 = 0$ **JUL-24**.



solution

Given lines are $7x + 3y = 10$, $5x - 4y = 1$.

Solve two Eqn

$$\begin{aligned}7x + 3y &= 10 \quad \dots (1) \\ 5x - 4y &= 1 \quad \dots (2) \\ (1) \times 4 &\Rightarrow 28x + 12y = 40 \quad \dots (3) \\ (2) \times 3 &\Rightarrow 15x - 12y = 3 \quad \dots (4) \\ (3) + (4) &\Rightarrow 43x = 43 \\ x &= \frac{43}{43} = 1\end{aligned}$$

Sub $x = 1$ in (1)

$$7(1) + 3y = 10 \Rightarrow 3y = 3 \Rightarrow y = 1$$

The point intersection (1, 1)

$$\text{Slope } 13x + 5y + 12 = 0 \Rightarrow -\frac{a}{b} = -\frac{13}{5}$$

$$\therefore \text{Slope of parallel line is } \frac{-13}{5}$$

$$\text{Equ of line passing } (1, 1), \text{ slope } \frac{-13}{5}$$

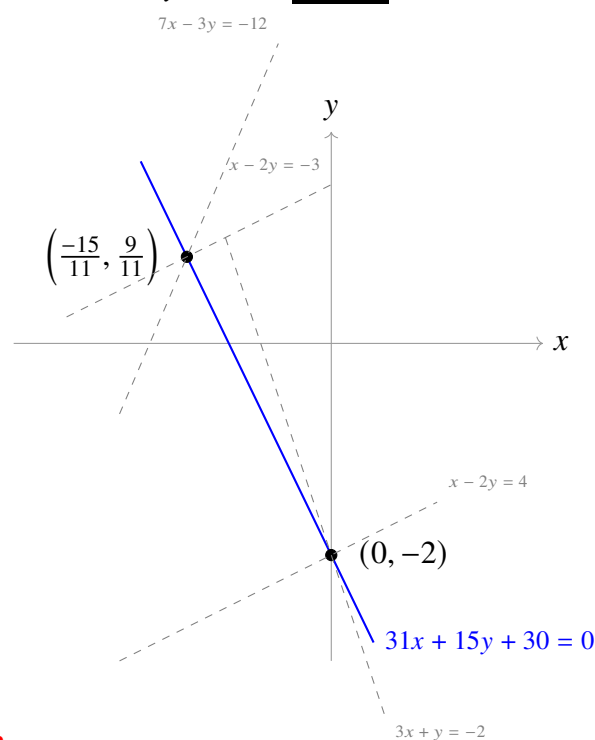
$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{-13}{5}(x - 1)$$

$$5y - 5 = -13x + 13$$

$$13x + 5y - 18 = 0$$

20. Find the equation of a straight line joining the point of intersection of $3x + y + 2 = 0$ and $x - 2y - 4 = 0$ to the point of intersection of $7x - 3y = -12$ and $2y = x + 3$ **PTA-3**



solution

$$3x + y = -2 \quad \dots (1)$$

$$x - 2y = 4 \quad \dots (2)$$

$$(1) \times 2 \Rightarrow 6x + 2y = -4 \quad \dots (3)$$

$$x - 2y = 4 \quad \dots (2)$$

$$(3) + (2) \Rightarrow 7x = 0$$

$$x = 0$$

Sub $x = 0$ in (1)

$$3(0) + y = -2$$

$$y = -2$$

The point intersection (1) (2) (0, -2)

Solve $7x - 3y = -12$ and $2y = x + 3$

$$7x - 3y = -12 \quad \dots (5)$$

$$x - 2y = -3 \quad \dots (2)$$

$$(2) \times 7 \Rightarrow 7x - 14y = -21 \quad \dots (6)$$

$$(5) \Rightarrow 7x - 3y = -12$$

$$(6) - (5) \Rightarrow -11y = -9$$

$$y = \frac{9}{11}$$

Sub $y = \frac{9}{11}$ in (2)

$$x - 2\left(\frac{9}{11}\right) = -3$$

$$x = -3 + \frac{18}{11} = \frac{-15}{11}$$

The point of intersection of

$$(5) \text{ and } (2) \text{ is } \left(\frac{-15}{11}, \frac{9}{11} \right)$$

Equation of the line (0, -2) and $\left(\frac{-15}{11}, \frac{9}{11} \right)$ is

$$\begin{aligned}\frac{y - y_1}{y_2 - y_1} &= \frac{x - x_1}{x_2 - x_1} \\ \frac{y + 2}{\frac{9}{11} + 2} &= \frac{x - 0}{-\frac{15}{11} - 0} \\ \frac{11(y + 2)}{9 + 22} &= \frac{-11x}{15} \\ 15(y + 2) &= -31x \\ 15y + 30 &= -31x \\ 31x + 15y + 30 &= 0\end{aligned}$$

21. The area of a triangle is 5 sq.units. Two of its vertices are (2, 1) and (3, -2). The third vertex is (x, y) where $y = x + 3$. Find the coordinates of the third vertex. **PTA-1**

solution Given Points (2, 1), (3, -2), (x, y)

$$\begin{aligned}\text{Area of triangle} &= \frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1) \\ &\quad + x_3(y_1 - y_2)] = 5 \\ 2(-2 - y) + 3(y - 1) + x(1 + 2) &= 10 \\ -4 - 2y + 3y - 3 + 3x &= 10 \\ 3x + y &= 17 \quad \dots (1) \\ y &= x + 3, \text{ Sub in (1)} \\ \therefore 3x + x + 3 &= 17 \\ 4x &= 14 \\ x &= 7/2 \\ \text{and } y &= 7/2 + 3 \\ &= 13/2\end{aligned}$$

Third vertex is (7/2, 13/2)

22. If vertices of a quadrilateral are at A(-5, 7), B(-4, k), C(-1, -6) and D(4, 5) and its area is 72 sq. units. Find the value of k.

solution

$$\begin{array}{cccc} A(-5, 7) & B(-4, k) & C(-1, -6) & D(4, 5) \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{Let the vertices be} & (x_1, y_1) & (x_2, y_2) & (x_3, y_3) & (x_4, y_4) \end{array}$$

$$\text{The Area of Quadrilateral} = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_4 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_1 \end{vmatrix}$$

$$72 = \frac{1}{2} \begin{vmatrix} -5 & -4 & -1 & 4 & -5 \\ 7 & k & -6 & 5 & 7 \end{vmatrix}$$

$$28 = \frac{1}{2}\{(-5k + 24 - 5 + 28) - (-28 - k - 24 - 25)\}$$

$$144 = \{-4k + 124\}$$

$$4k = 124 - 144 = -20$$

$$k = \frac{-20}{4}$$

$$\boxed{k = -5}$$

23. Without using distance formula, show that the points (-2, -1), (4, 0), (3, 3) and (-3, 2) are vertices of a parallelogram.

solution

Given Points A(-2, -1), B(4, 0), C(3, 3) and D(-3, 2)

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope of AB} = \frac{-1 - 0}{-2 - 4} = \frac{1}{6}$$

$$\text{Slope of BC} = \frac{0 - 3}{4 - 3} = \frac{-3}{1} = -3$$

$$\text{Slope of CD} = \frac{3 - 2}{-3 - 3} = \frac{1}{-6}$$

$$\text{Slope of DA} = \frac{2 + 1}{-3 + 2} = \frac{3}{-1} = -3$$

Slope of AB = Slope of CD $\Rightarrow$ AB is parallel to CD

Slope of BC = Slope of DA $\Rightarrow$ BC is parallel to DA

The given points form a parallelogram

24. Find the equation of a line passing through the point of intersection of the lines $4x + 7y - 3 = 0$ and $2x - 3y + 1 = 0$ that has equal intercepts on the axes.

solution

$$4x + 7y - 3 = 0 \Rightarrow 4x + 7y = 3 \quad \dots (1)$$

$$2x - 3y + 1 = 0 \Rightarrow 2x - 3y = -1 \quad \dots (2)$$

Solve (1) and (2)

$$4x + 7y = 3 \quad \dots (1)$$

$$(2) \times 2 \Rightarrow 4x - 6y = -2 \quad \dots (3)$$

$$(1) - (3) \Rightarrow 13y = 5$$

$$y = \frac{5}{13}$$

$$\text{Sub in (1)} \Rightarrow 4x + 7\left(\frac{5}{13}\right) = 3$$

$$\Rightarrow x = \frac{1}{13}$$

$$\text{intersection Point} \left(\frac{1}{13}, \frac{5}{13}\right)$$

The line has equal intercepts

$$a = b$$

$$\text{Intercept form } \frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x}{a} + \frac{y}{a} = 1 \Rightarrow x + y = a$$

$$\text{This passes through} \left(\frac{1}{13}, \frac{5}{13}\right)$$

$$\frac{1}{13} + \frac{5}{13} = a \Rightarrow a = \frac{6}{13}$$

$$\therefore b = \frac{6}{13}$$

$$\text{Tee equation of Line } \frac{x}{\frac{6}{13}} + \frac{y}{\frac{6}{13}} = 1$$

$$13x + 13y - 6 = 0.$$

CHAPTER - 6. TRIGONOMETRY

2 MARK IMPORTANT QUESTIONS

1. Prove that $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$ **JUL-23**

solution

$$\tan^2 \theta - \sin^2 \theta = \tan^2 \theta - \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \cos^2 \theta$$

$$= \tan^2 \theta (1 - \cos^2 \theta)$$

$$= \tan^2 \theta \sin^2 \theta$$

Hence Proved

2. Prove that $\sec \theta - \cos \theta = \tan \theta \sin \theta$

solution

$$\sec \theta - \cos \theta = \frac{1}{\cos \theta} - \cos \theta = \frac{1 - \cos^2 \theta}{\cos \theta}$$

$$= \frac{\sin^2 \theta}{\cos \theta} \quad [\text{since } 1 - \cos^2 \theta = \sin^2 \theta]$$

$$= \frac{\sin \theta}{\cos \theta} \times \sin \theta$$

$$= \tan \theta \sin \theta$$

Hence Proved

3. Prove that $\frac{\sec \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} = \cot \theta$ **APR-23**

solution

$$\frac{\sec \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} = \frac{\frac{1}{\cos \theta}}{\sin \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1}{\sin \theta \cos \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 - \sin^2 \theta}{\sin \theta \cos \theta}$$

$$= \frac{\cos^2 \theta}{\sin \theta \cos \theta} = \cot \theta$$

Hence Proved

4. Prove that $\sqrt{\frac{1+\cos \theta}{1-\cos \theta}} = \operatorname{cosec} \theta + \cot \theta$ **APR-24, MDL-19, MAR-26**

solution

$$\begin{aligned}\sqrt{\frac{1+\cos \theta}{1-\cos \theta}} &= \sqrt{\frac{1+\cos \theta}{1-\cos \theta} \times \frac{1+\cos \theta}{1+\cos \theta}} \\ &\text{[multiply numerator and denominator} \\ &\text{conjugate of } 1-\cos \theta] \\ &= \sqrt{\frac{(1+\cos \theta)^2}{1-\cos^2 \theta}} = \frac{1+\cos \theta}{\sqrt{\sin^2 \theta}} \\ &\text{[since } \sin^2 \theta + \cos^2 \theta = 1] \\ &= \frac{1+\cos \theta}{\sin \theta} \\ &= \operatorname{cosec} \theta + \cot \theta \\ &\text{Hence Proved}\end{aligned}$$

5. Prove the following identities. $1 + \frac{\cot^2 \theta}{1+\operatorname{cosec} \theta} = \operatorname{cosec} \theta$

solution

$$\begin{aligned}1 + \frac{\cot^2 \theta}{1+\operatorname{cosec} \theta} &= 1 + \frac{\cot^2 \theta}{1+\operatorname{cosec} \theta} \\ &= 1 + \frac{\operatorname{cosec}^2 \theta - 1}{\operatorname{cosec} \theta + 1} \quad \text{[since } \operatorname{cosec}^2 \theta - 1 = \cot^2 \theta] \\ &= 1 + \frac{(\operatorname{cosec} \theta + 1)(\operatorname{cosec} \theta - 1)}{\operatorname{cosec} \theta + 1} \\ &= 1 + (\operatorname{cosec} \theta - 1) \\ &= \operatorname{cosec} \theta \\ &\text{Hence Proved}\end{aligned}$$

6. Prove the following identities.

(i) $\cot \theta + \tan \theta = \sec \theta \operatorname{cosec} \theta$ (ii) $\tan^4 \theta + \tan^2 \theta = \sec^4 \theta - \sec^2 \theta$

solution

$$\begin{aligned}\text{(i) } \cot \theta + \tan \theta &= \frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} \\ &= \frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cdot \cos \theta} \\ &= \frac{1}{\sin \theta \cdot \cos \theta} \\ &= \frac{1}{\cos \theta} \times \frac{1}{\sin \theta} \\ &= \sec \theta \cdot \operatorname{cosec} \theta \\ &\text{Hence Proved}\end{aligned}$$

$$\begin{aligned}\text{(ii) } \tan^4 \theta + \tan^2 \theta &= \tan^2 \theta (\tan^2 \theta + 1) \\ &= (\sec^2 \theta - 1) \cdot (\sec^2 \theta) \\ &= \sec^4 \theta - \sec^2 \theta \\ &\text{Hence Proved}\end{aligned}$$

7. Prove the following identities. (i) $\frac{1-\tan^2 \theta}{\cot^2 \theta - 1} = \tan^2 \theta$ **JUL-24**

(ii) $\frac{\cos \theta}{1+\sin \theta} = \sec \theta - \tan \theta$ **JUL-25**

solution

$$\begin{aligned}\text{(i) } \frac{1-\tan^2 \theta}{\cot^2 \theta - 1} &= \frac{1-\tan^2 \theta}{\frac{1}{\tan^2 \theta} - 1} \\ &= \frac{1-\tan^2 \theta}{\frac{1-\tan^2 \theta}{\tan^2 \theta}} \\ &= \tan^2 \theta \\ &= \text{Hence Proved}\end{aligned}$$

$$\begin{aligned}\text{(ii) } \frac{\cos \theta}{1+\sin \theta} &= \frac{\cos \theta}{1+\sin \theta} \times \frac{1-\sin \theta}{1-\sin \theta} \\ &= \frac{\cos \theta (1-\sin \theta)}{\cos^2 \theta} \\ &= \frac{1-\sin \theta}{\cos \theta} \\ &= \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} \\ &= \sec \theta - \tan \theta\end{aligned}$$

8. Prove the following identities.

(i) $\sqrt{\frac{1+\sin \theta}{1-\sin \theta}} = \sec \theta + \tan \theta$

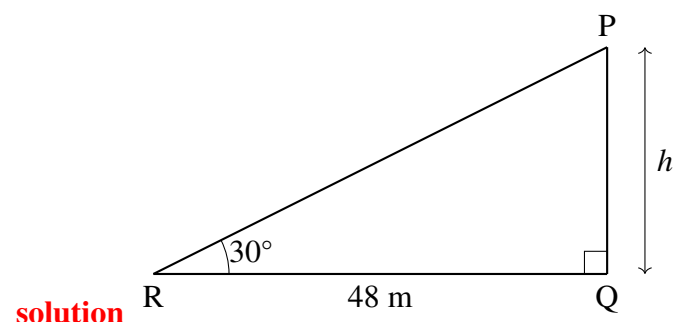
(ii) $\sqrt{\frac{1+\sin \theta}{1-\sin \theta}} + \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = 2 \sec \theta$ **JUL-23**

solution

$$\begin{aligned}\text{(i) } \sqrt{\frac{1+\sin \theta}{1-\sin \theta}} &= \sqrt{\frac{1+\sin \theta}{1-\sin \theta} \times \frac{1+\sin \theta}{1+\sin \theta}} \\ &= \sqrt{\frac{(1+\sin \theta)^2}{\cos^2 \theta}} \\ &= \frac{1+\sin \theta}{\cos \theta} \\ &= \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} \\ &= \sec \theta + \tan \theta \\ &= \text{Hence Proved}\end{aligned}$$

$$\begin{aligned}\text{(ii) } &= \sqrt{\frac{1+\sin \theta}{1-\sin \theta} \times \frac{1+\sin \theta}{1+\sin \theta}} + \sqrt{\frac{1-\sin \theta}{1+\sin \theta} \times \frac{1-\sin \theta}{1-\sin \theta}} \\ &= \sqrt{\frac{(1+\sin \theta)^2}{1^2 - \sin^2 \theta}} + \sqrt{\frac{(1-\sin \theta)^2}{1^2 - \sin^2 \theta}} \\ &= \sqrt{\frac{(1+\sin \theta)^2}{\cos^2 \theta}} + \sqrt{\frac{(1-\sin \theta)^2}{\cos^2 \theta}} \\ &= \frac{1+\sin \theta}{\cos \theta} + \frac{1-\sin \theta}{\cos \theta} \\ &= \frac{1+\sin \theta + 1-\sin \theta}{\cos \theta} \\ &= 2 \times \frac{1}{\cos \theta} \\ &= 2 \sec \theta \\ &= \text{Hence Proved}\end{aligned}$$

9. A tower stands vertically on the ground. From a point on the ground, which is 48 m away from the foot of the tower, the angle of elevation of the top of the tower is 30° . Find the height of the tower. **PTA-1**



solution

PQ = height of the tower. $PQ = h$

In right angle triangle PQR ,

$$\angle PRQ = 30^\circ$$

$$\tan \theta = \frac{PQ}{QR}$$

$$\tan 30^\circ = \frac{h}{48} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{48} \Rightarrow h = 16\sqrt{3}$$

The height of the tower is $16\sqrt{3}$ m.

10. A kite is flying at a height of 75 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° . Find the length of the string, assuming that there is no slack in the string. **JUL-21, 22**

solution

AB = height of the kite. $AB = 75$.

AC = the length of the string.

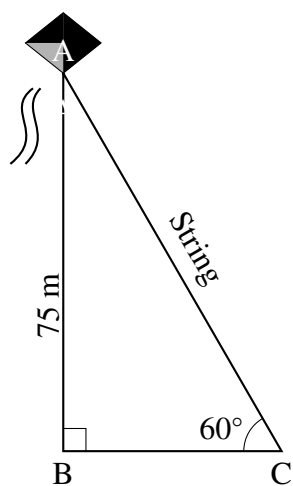
In right angle triangle ABC, $\angle ACB = 60^\circ$

$$\sin \theta = \frac{AB}{AC} \Rightarrow \sin 60^\circ = \frac{75}{AC}$$

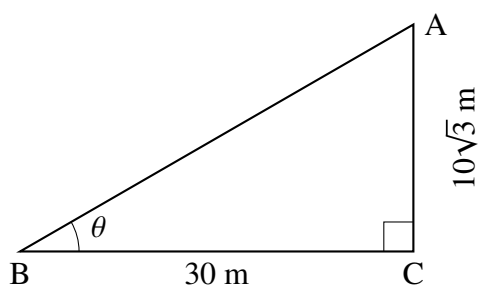
$$\frac{\sqrt{3}}{2} = \frac{75}{AC}$$

$$AC = \frac{150}{\sqrt{3}} \times \frac{\sqrt{3}\sqrt{3}}{3} = 50\sqrt{3}$$

The length of the string is $50\sqrt{3}$ m.



11. Find the angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of a tower of height $10\sqrt{3}$ m. **JUL-21, 22, PTA-2**



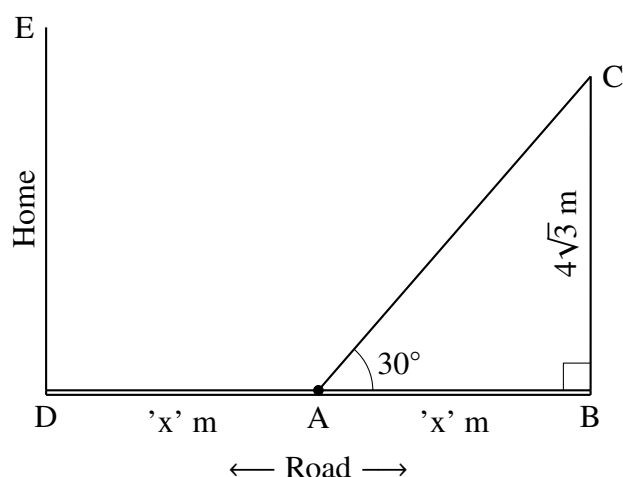
solution Given Tower Height is $10\sqrt{3}$ m. Distance = 30 m.

From the right angle $\triangle ABC$

$$\begin{aligned}\tan \theta &= \frac{AC}{BC} \\ &= \frac{10\sqrt{3} \text{ m}}{30 \text{ m}} = \frac{\sqrt{3}}{3} \\ &= \frac{\sqrt{3}}{\sqrt{3}\sqrt{3}} = \frac{1}{\sqrt{3}} \\ \tan \theta &= \frac{1}{\sqrt{3}} \\ \theta &= \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) \\ \theta &= 30^\circ\end{aligned}$$

$\therefore$ Angle of elevation is 30°

12. A road is flanked on either side by continuous rows of houses of height $4\sqrt{3}$ m with no space in between them. A pedestrian is standing on the median of the road facing a row house. The angle of elevation from the pedestrian to the top of the house is 30° . Find the width of the road.



solution

AB = x distance between foot of the House road.

$\therefore$ DB = 2x is the width of the road.

Height of the house BC = $4\sqrt{3}$ m

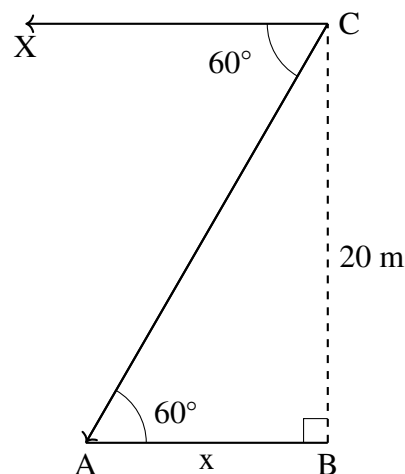
From the right angle triangle $\triangle ABC$

$$\begin{aligned}\therefore \tan 30^\circ &= \frac{BC}{AB} \\ \frac{1}{\sqrt{3}} &= \frac{4\sqrt{3}}{x} \\ x &= 4\sqrt{3} \times \sqrt{3} = 4 \times 3 = 12 \text{ m}\end{aligned}$$

Width of the road = $2 \times x = 2 \times 12 = 24$ m

$\therefore$ The Width of the road is = 24 m.

13. A player sitting on the top of a tower of height 20 m observes the angle of depression of a ball lying on the ground as 60° . Find the distance between the foot of the tower and the ball. ($\sqrt{3} = 1.732$) **PTA-3**



solution

BC = height of the tower

BC = 20 m and $\angle XCA = 60^\circ = \angle CAB$

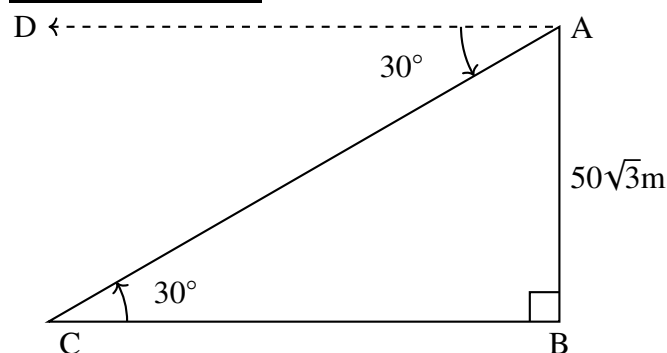
Let AB = x metres.

In right triangle ABC,

$$\begin{aligned}\tan 60^\circ &= \frac{BC}{AB} \\ \sqrt{3} &= \frac{20}{x} \\ x &= \frac{20 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = \frac{20 \times 1.732}{3} = 11.54 \text{ m}\end{aligned}$$

The distance between the foot of the tower is 11.54 m.

14. From the top of a rock $50\sqrt{3}$ m high, the angle of depression of a car on the ground is observed to be 30° . Find the distance of the car from the rock. **PTA-6, APR-22, 25**



solution

AB = The rock. C = The position of the car

$\angle DAC = \angle ACB = 30^\circ$

In right angle triangle $\triangle ABC$

$$\begin{aligned}\tan 30^\circ &= \frac{AB}{BC} \\ \frac{1}{\sqrt{3}} &= \frac{50\sqrt{3}}{BC} \\ BC &= 50\sqrt{3} \times \sqrt{3} \\ &= 50 \times 3 = 150 \text{ m}\end{aligned}$$

$\therefore$ Distance of the car from the rock = 150 m.

5 MARK IMPORTANT QUESTIONS

1. If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, then prove that $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$

solution

$$\cos \theta + \sin \theta = \sqrt{2} \cos \theta$$

Squaring both sides,

$$(\cos \theta + \sin \theta)^2 = (\sqrt{2} \cos \theta)^2$$

$$\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta = 2 \cos^2 \theta$$

$$2 \cos^2 \theta - \cos^2 \theta - \sin^2 \theta = 2 \sin \theta \cos \theta$$

$$\cos^2 \theta - \sin^2 \theta = 2 \sin \theta \cos \theta$$

$$(\cos \theta + \sin \theta)(\cos \theta - \sin \theta) = 2 \sin \theta \cos \theta$$

$$\cos \theta - \sin \theta = \frac{2 \sin \theta \cos \theta}{\cos \theta + \sin \theta} = \frac{2 \sin \theta \cos \theta}{\sqrt{2} \cos \theta}$$

$$= \sqrt{2} \sin \theta$$

$$[\text{since } \cos \theta + \sin \theta = \sqrt{2} \cos \theta]$$

$$\cos \theta - \sin \theta = \sqrt{2} \sin \theta$$

Hence Proved

2. If $\operatorname{cosec} \theta + \cot \theta = P$, then prove that $\cos \theta = \frac{P^2-1}{P^2+1}$

solution

$$\operatorname{cosec} \theta + \cot \theta = P \quad \dots(1)$$

$$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1 \text{ (identity)}$$

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{\operatorname{cosec} \theta + \cot \theta}$$

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{P} \quad \dots\dots\dots(2)$$

Add (1) and (2)

$$2 \operatorname{cosec} \theta = P + \frac{1}{P}$$

$$2 \operatorname{cosec} \theta = \frac{P^2 + 1}{P} \quad \dots\dots\dots(3)$$

Subtract (2) from (1)

$$2 \cot \theta = P - \frac{1}{P}$$

$$2 \cot \theta = \frac{P^2 - 1}{P} \quad \dots\dots\dots(4)$$

Divide (4) by (3)

$$\frac{2 \cot \theta}{2 \operatorname{cosec} \theta} = \frac{P^2 - 1}{P} \times \frac{P}{P^2 + 1}$$

$$\cos \theta = \frac{P^2 - 1}{P^2 + 1}$$

Hence Proved

3. Prove that $\left(\frac{\cos^3 A - \sin^3 A}{\cos A - \sin A} \right) - \left(\frac{\cos^3 A + \sin^3 A}{\cos A + \sin A} \right) = 2 \sin A \cos A$

solution

$$= \left(\frac{\cos^3 A - \sin^3 A}{\cos A - \sin A} \right) - \left(\frac{\cos^3 A + \sin^3 A}{\cos A + \sin A} \right)$$

$$\left[\begin{array}{l} \text{Formula } a^3 - b^3 = (a - b)(a^2 + b^2 + ab) \\ a^3 + b^3 = (a + b)(a^2 + b^2 - ab) \\ \cos^2 \theta + \sin^2 \theta = 1 \end{array} \right]$$

$$= \left(\frac{(\cos A - \sin A)(\cos^2 A + \sin^2 A + \cos A \sin A)}{\cos A - \sin A} \right) - \left(\frac{(\cos A + \sin A)(\cos^2 A + \sin^2 A - \cos A \sin A)}{\cos A + \sin A} \right)$$

$$= (1 + \cos A \sin A) - (1 - \cos A \sin A)$$

$$= 2 \cos A \sin A$$

Hence Proved

4. If $\frac{\cos^2 \theta}{\sin \theta} = p$ and $\frac{\sin^2 \theta}{\cos \theta} = q$ then prove that $p^2 q^2 (p^2 + q^2 + 3) = 1$

solution

$$\frac{\cos^2 \theta}{\sin \theta} = p \quad \dots (1)$$

$$\frac{\sin^2 \theta}{\cos \theta} = q \quad \dots (2)$$

$$p^2 q^2 (p^2 + q^2 + 3) = \left(\frac{\cos^2 \theta}{\sin \theta} \right)^2 \left(\frac{\sin^2 \theta}{\cos \theta} \right)^2$$

$$\times \left[\left(\frac{\cos^2 \theta}{\sin \theta} \right)^2 + \left(\frac{\sin^2 \theta}{\cos \theta} \right)^2 + 3 \right]$$

$$= \left(\frac{\cos^4 \theta}{\sin^2 \theta} \right) \left(\frac{\sin^4 \theta}{\cos^2 \theta} \right) \times \left[\frac{\cos^4 \theta}{\sin^2 \theta} + \frac{\sin^4 \theta}{\cos^2 \theta} + 3 \right]$$

$$= (\cos^2 \theta \times \sin^2 \theta)$$

$$\times \left[\left(\frac{\cos^6 \theta + \sin^6 \theta + 3 \sin^2 \theta \cos^2 \theta}{\sin^2 \theta \cos^2 \theta} \right) \right]$$

$$= \cos^6 \theta + \sin^6 \theta + 3 \sin^2 \theta \cos^2 \theta$$

$$\left[\text{Formula } a^3 + b^3 = (a + b)^3 - 3ab(a + b) \right]$$

$$= (\cos^2 \theta)^3 + (\sin^2 \theta)^3 + 3 \sin^2 \theta \cos^2 \theta$$

$$= [(\cos^2 \theta + \sin^2 \theta)^3 - 3 \cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta)]$$

$$+ 3 \sin^2 \theta \cos^2 \theta$$

$$= 1 - 3 \cos^2 \theta \sin^2 \theta (1) + 3 \cos^2 \theta \sin^2 \theta$$

$$= 1.$$

5. Prove the following identities. $\frac{\sin^3 A + \cos^3 A}{\sin A + \cos A} + \frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} = 2$

solution

$$= \frac{\sin^3 A - \cos^3 A}{\sin A + \cos A} + \frac{\sin^3 A + \cos^3 A}{\sin A - \cos A}$$

$$\left[\begin{array}{l} \text{Formula } a^3 - b^3 = (a - b)(a^2 + b^2 + ab) \\ a^3 + b^3 = (a + b)(a^2 + b^2 - ab) \\ \cos^2 \theta + \sin^2 \theta = 1 \end{array} \right]$$

$$= \frac{(\sin A + \cos A) \cdot (\sin^2 A - \sin A \cos A + \cos^2 A)}{\sin A + \cos A}$$

$$+ \frac{(\sin A - \cos A) \cdot (\sin^2 A + \sin A \cos A + \cos^2 A)}{\sin A - \cos A}$$

$$= (1 - \sin A \cos A) + (1 + \sin A \cos A)$$

$$= 2$$

Hence Proved

6. If $\sqrt{3} \sin \theta - \cos \theta = 0$, then show that $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$ **PTA-3**

solution

$$\sqrt{3} \sin \theta - \cos \theta = 0$$

$$\sqrt{3} \sin \theta = \cos \theta$$

$$\frac{\sqrt{3} \sin \theta}{\cos \theta} = 1$$

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{\sqrt{3}}$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\therefore \theta = 30^\circ$$

$$\text{LHS} = \tan 3\theta$$

$$= \tan 3(30^\circ)$$

$$= \tan 90^\circ$$

$$= \text{undefined} \quad \dots (1)$$

$$\text{RHS} = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

$$= \frac{3 \tan 30^\circ - \tan^3 30^\circ}{1 - 3 \tan^2 30^\circ}$$

$$= \frac{3 \cdot \frac{1}{\sqrt{3}} - \left(\frac{1}{\sqrt{3}} \right)^3}{1 - 3 \left(\frac{1}{\sqrt{3}} \right)^2}$$

$$= \frac{\frac{3}{\sqrt{3}} - \frac{1}{3\sqrt{3}}}{1 - 3 \times \frac{1}{3}}$$

$$= \frac{\frac{3 \times 3 - 1}{3\sqrt{3}}}{1 - 1}$$

$$= \frac{9 - 1}{3\sqrt{3}(0)} = \frac{8}{0}$$

$$= \text{undefined} \quad \dots (2)$$

From (1) and (2), LHS = RHS.

Hence Proved

7. If $\frac{\cos \theta}{1 + \sin \theta} = \frac{1}{a}$, then prove that $\frac{a^2 - 1}{a^2 + 1} = \sin \theta$

solution

$$\frac{\cos \theta}{1 + \sin \theta} = \frac{1}{a}$$

To Prove : $\frac{a^2 - 1}{a^2 + 1} = \sin \theta$

$$\therefore a = \frac{1 + \sin \theta}{\cos \theta}$$

$$= (1 + \sin \theta) \sec \theta$$

$$\boxed{a = \sec \theta + \tan \theta}$$

$$\frac{a^2 - 1}{a^2 + 1} = \frac{(\sec \theta + \tan \theta)^2 - 1}{(\sec \theta + \tan \theta)^2 + 1}$$

$$\begin{aligned}
&= \frac{\sec^2 \theta + \tan^2 \theta + 2 \sec \theta \tan \theta - 1}{\sec^2 \theta + \tan^2 \theta + 2 \sec \theta \tan \theta + 1} \\
&= \frac{2 \tan^2 \theta + 2 \sec \theta \tan \theta}{2 \sec^2 \theta + 2 \sec \theta \tan \theta} \\
&= \frac{2 \tan \theta (\tan \theta + \sec \theta)}{2 \sec \theta (\sec \theta + \tan \theta)} \\
&= \frac{\frac{\sin \theta}{\cos \theta}}{\frac{1}{\cos \theta}} \\
&= \sin \theta \\
&= \text{Hence Proved}
\end{aligned}$$

8. If $\cot \theta + \tan \theta = x$ and $\sec \theta - \cos \theta = y$ then prove that $(x^2 y)^{\frac{2}{3}} - (xy^2)^{\frac{2}{3}} = 1$.

solution

$$\begin{aligned}
x &= \cot \theta + \tan \theta & y &= \sec \theta - \cos \theta \\
&= \frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} & &= \frac{1}{\cos \theta} - \cos \theta \\
&= \frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cos \theta} & &= \frac{1 - \cos^2 \theta}{\cos \theta} \\
\boxed{x} &= \frac{1}{\sin \theta \cdot \cos \theta} & \boxed{y} &= \frac{\sin^2 \theta}{\cos \theta}
\end{aligned}$$

To Prove : $(x^2 y)^{\frac{2}{3}} - (xy^2)^{\frac{2}{3}} = 1$

$$\begin{aligned}
\text{LHS} &= \left(\frac{1}{\sin^2 \theta \cos^2 \theta} \times \frac{\sin^2 \theta}{\cos \theta} \right)^{\frac{2}{3}} \\
&\quad - \left(\frac{1}{\sin \theta \cdot \cos \theta} \times \frac{\sin^4 \theta}{\cos^2 \theta} \right)^{\frac{2}{3}} \\
&= \left(\frac{1}{\cos^3 \theta} \right)^{\frac{2}{3}} - \left(\frac{\sin^3 \theta}{\cos^3 \theta} \right)^{\frac{2}{3}} \\
&= \left(\sec^3 \theta \right)^{\frac{2}{3}} - \left(\tan^3 \theta \right)^{\frac{2}{3}} \\
&= \sec^2 \theta - \tan^2 \theta \\
&= 1 \\
&= \text{Hence Proved}
\end{aligned}$$

9. If $\sin \theta + \cos \theta = p$ and $\sec \theta + \operatorname{cosec} \theta = q$, then prove that $q(p^2 - 1) = 2p$.

solution

$$\begin{aligned}
\text{Given } p &= \sin \theta + \cos \theta, \\
q &= \sec \theta + \operatorname{cosec} \theta \\
\text{To Prove : } q(p^2 - 1) &= 2p \\
&= (\sec \theta + \operatorname{cosec} \theta) [(\sin \theta + \cos \theta)^2 - 1] \\
&= \left(\frac{1}{\cos \theta} + \frac{1}{\sin \theta} \right) [\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta - 1] \\
&= \frac{\sin \theta + \cos \theta}{\cos \theta \cdot \sin \theta} [1 + 2 \sin \theta \cos \theta - 1] \\
&= \frac{\sin \theta + \cos \theta}{\cos \theta \cdot \sin \theta} \times 2 \sin \theta \cos \theta \\
&= 2(\sin \theta + \cos \theta) \\
&= 2p \\
&= \text{Hence Proved}
\end{aligned}$$

10. Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are 30° and 45° respectively. If the lighthouse is 200 m high, find the distance between the two ships. ($\sqrt{3} = 1.732$)

PTA-5, APR-24, MAR-26

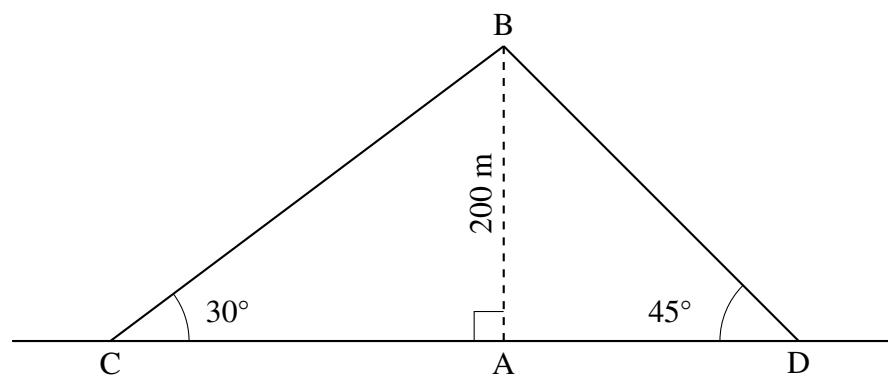
solution

AB = Height of lighthouse

C and D = positions of two ships.

$$AB = 200 \text{ m},$$

$$\angle ACB = 30^\circ, \angle ADB = 45^\circ$$



In right triangle BAC, $\tan 30^\circ = \frac{AB}{AC}$

$$\frac{1}{\sqrt{3}} = \frac{200}{AC} \Rightarrow \boxed{AC = 200\sqrt{3}} \quad \dots (1)$$

In right triangle BAD, $\tan 45^\circ = \frac{AB}{AD}$

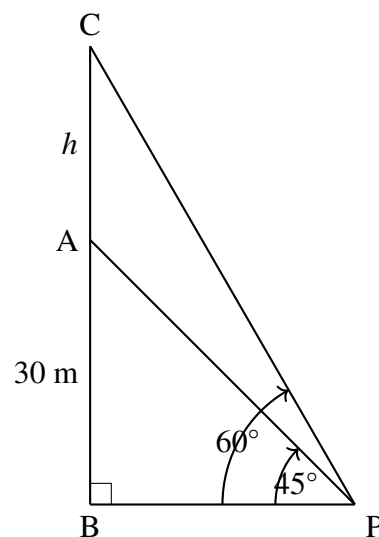
$$1 = \frac{200}{AD} \Rightarrow \boxed{AD = 200} \quad \dots (2)$$

$$\boxed{CD = AC + AD} = 200\sqrt{3} + 200 \text{ [by (1) and (2)]}$$

$$CD = 200(\sqrt{3} + 1) = 200 \times 2.732 = 546.4$$

Distance between two ships is 546.4 m.

11. From a point on the ground, the angles of elevation of the bottom and top of a tower fixed at the top of a 30 m high building are 45° and 60° respectively. Find the height of the tower. ($\sqrt{3} = 1.732$) **APR-22**



solution

Let AC be the height of the tower.

Let AB be the height of the building.

$$AC = h \text{ metres, } AB = 30 \text{ m}$$

In right triangle CBP,

$$\angle CPB = 60^\circ$$

$$\tan \theta = \frac{BC}{BP}$$

$$\tan 60^\circ = \frac{AB + AC}{BP}$$

$$\boxed{\sqrt{3} = \frac{30 + h}{BP}} \quad \dots\dots (1)$$

In right triangle ABP,

$$\angle APB = 45^\circ$$

$$\tan \theta = \frac{AB}{BP}$$

$$\tan 45^\circ = \frac{30}{BP}$$

$$\boxed{BP = 30} \quad \dots\dots (2)$$

Sub (2) in (1)

$$\sqrt{3} = \frac{30 + h}{30}$$

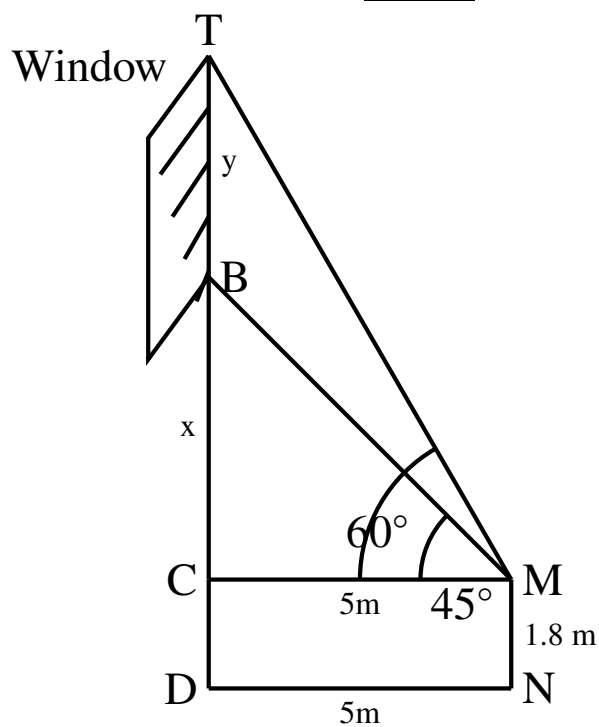
$$h = 30(\sqrt{3} - 1)$$

$$= 30(1.732 - 1)$$

$$= 30(0.732) = 21.96$$

The height of the tower is 21.96 m.

12. To a man standing outside his house, the angles of elevation of the top and bottom of a window are 60° and 45° respectively. If the height of the man is 180 cm and if he is 5 m away from the wall, what is the

height of the window? ($\sqrt{3} = 1.732$) **JUL-22****solution**

$MN = 180 \text{ cm} = 1.8 \text{ m}$
 $= \text{height of the man}$
 $DN = 5 \text{ m} = \text{Dist. between}$
 $\text{Man \& Wall}$
 $T, B \rightarrow \text{Top \& Bottom of Window}$
 $BC = x, TB = y \text{ (Height of window)}$

$$\text{In } \triangle CMB, \tan 45^\circ = \frac{x}{5}$$

$$1 = \frac{x}{5}, \boxed{x = 5}$$

$$\text{In } \triangle CMT, \tan 60^\circ = \frac{x+y}{5}$$

$$\sqrt{3} = \frac{5+y}{5}$$

$$5+y = 5\sqrt{3}$$

$$y = 5\sqrt{3} - 5$$

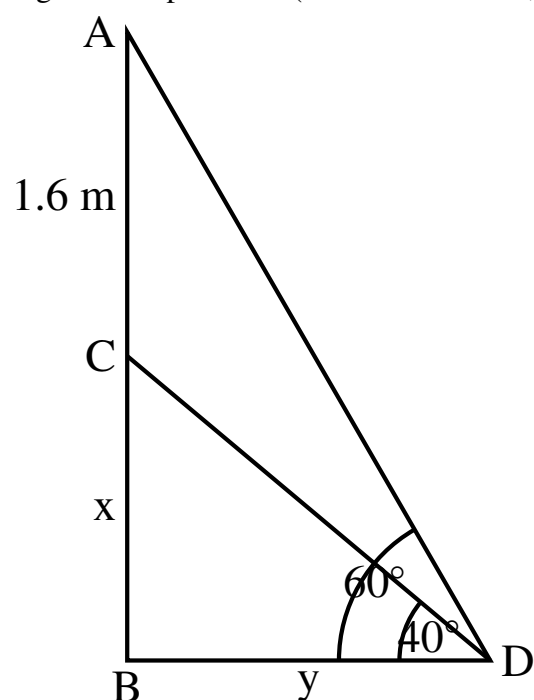
$$= 5(\sqrt{3} - 1)$$

$$= 5(0.732)$$

$$\boxed{y = 3.66 \text{ m}}$$

$\therefore$ Height of the window = 3.66 m

13. A statue 1.6 m tall stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the top of the pedestal is 40° . Find the height of the pedestal. ($\tan 40^\circ = 0.8391$, $\sqrt{3} = 1.732$)

**solution**

$D \rightarrow \text{Point of observation on the ground}$
 $BC = x \text{ m} = \text{height of Pedestal}$
 $AC = 1.6 \text{ m} = \text{height of statue}$
 $\angle BDC = 40^\circ, \angle BDA = 60^\circ, BD = y \text{ m}$

$$\text{In } \triangle BCD, \tan 40^\circ = \frac{x}{y}$$

$$y = \frac{x}{\tan 40^\circ}$$

$$\boxed{y = \frac{x}{0.8391}} \quad \dots\dots\dots(1)$$

$$\text{In } \triangle BAD, \tan 60^\circ = \frac{AB}{BD}$$

$$\sqrt{3} = \frac{x+1.6}{y}$$

$$\boxed{y = \frac{x+1.6}{\sqrt{3}}} \quad \dots\dots\dots(2)$$

$\therefore$ From (1) & (2)

$$\frac{x}{0.8391} = \frac{x+1.6}{1.732}$$

$$1.732x = 0.8391x + 1.6(0.8391)$$

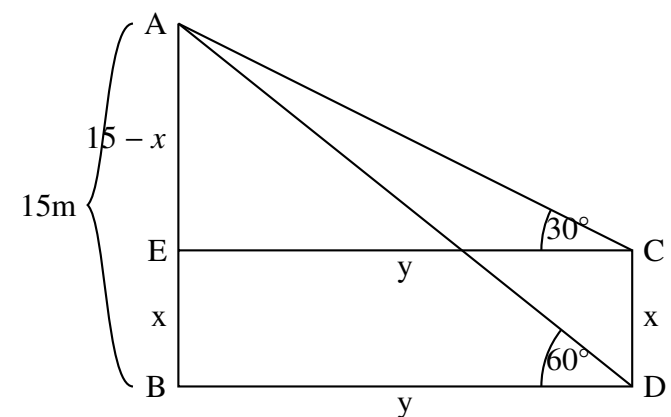
$$0.8929x = 1.343$$

$$x = \frac{1.343}{0.8929}$$

$$\boxed{x = 1.5 \text{ m}}$$

Height of the statue is = 1.5 m

14. The top of a 15 m high tower makes an angle of elevation of 60° with the bottom of an electronic pole and angle of elevation of 30° with the top of the pole. What is the height of the electric pole?

**solution**

$AB = 15 \text{ m} = \text{Height of the tower}$
 $CD = x \text{ m} = \text{Height of the pole} = BE$
 $\therefore AE = 15 - x$

$BD = EC = y$

$$\text{In } \triangle ADB, \tan 60^\circ = \frac{AB}{BD}$$

$$\sqrt{3} = \frac{15}{y}$$

$$\boxed{y = \frac{15}{\sqrt{3}} = 5\sqrt{3}} \quad \dots\dots\dots(1)$$

$$\text{In } \triangle ACE, \tan 30^\circ = \frac{AE}{EC}$$

$$\frac{1}{\sqrt{3}} = \frac{15-x}{y}$$

$$5\sqrt{3} = (15-x)\sqrt{3} \quad (\text{From (1)})$$

$$5 = 15 - x$$

$$\boxed{x = 10}$$

$\therefore$ Height of the pole is = 10m

15. From the top of a tower 50 m high, the angles of depression of the top and bottom of a tree are observed to be 30° and 45° respectively. Find the height of the tree. ($\sqrt{3} = 1.732$)

solution

The height of the tower $AB = 50 \text{ m}$

The height of the tree

$CD = y$ and $BD = x$

From the diagram,

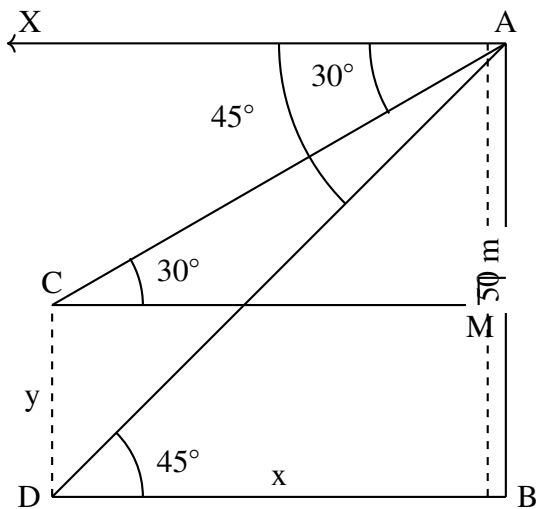
$\angle XAC = 30^\circ = \angle ACM$

$$\angle XAD = 45^\circ = \angle ADB$$

In right triangle ABD,

$$\tan 45^\circ = \frac{AB}{BD}$$

$$1 = \frac{50}{x} \rightarrow x = 50 \text{ m}$$



In right triangle AMC,

$$\tan 30^\circ = \frac{AM}{CM}$$

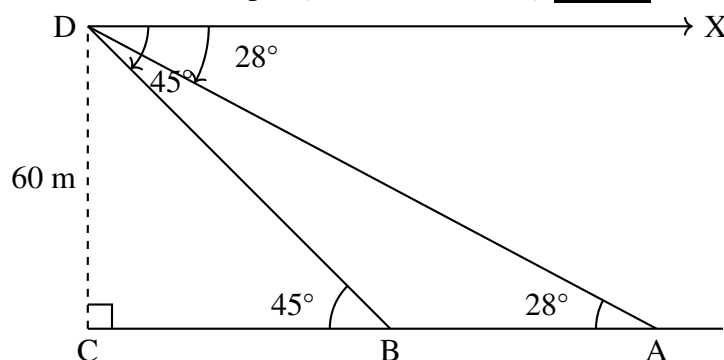
$$\frac{1}{\sqrt{3}} = \frac{AM}{50} \quad [\text{since } DB = CM]$$

$$AM = \frac{50}{\sqrt{3}} = \frac{50\sqrt{3}}{3} = \frac{50 \times 1.732}{3} = 28.85 \text{ m}$$

The Height of the tree is =

$$CD = MB = AB - AM = 50 - 28.85 = 21.15 \text{ m}$$

16. As observed from the top of a 60 m high lighthouse from the sea level, the angles of depression of two ships are 28° and 45° . If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. ($\tan 28^\circ = 0.5317$) **PTA-1**



solution

The observer on the lighthouse CD be at D.

Height of the lighthouse CD = 60 m

From the diagram,

$$\angle XDA = 28^\circ = \angle DAC$$

$$\angle XDB = 45^\circ = \angle DBC$$

In right triangle DCB,

$$\tan 45^\circ = \frac{DC}{BC}$$

$$1 = \frac{60}{BC} \rightarrow BC = 60 \text{ m}$$

In right triangle DCA,

$$\tan 28^\circ = \frac{DC}{AC}$$

$$0.5317 = \frac{60}{AC}$$

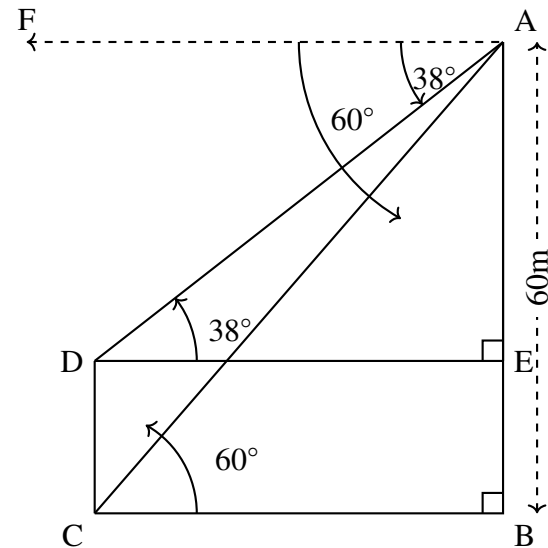
$$\text{gives } AC = \frac{60}{0.5317}$$

$$AC = 112.85$$

Distance between the two ships

$$AB = AC - BC = 52.85 \text{ m}$$

17. From the top of the tower 60 m high the angles of depression of the top and bottom of a vertical lamp post are observed to be 38° and 60° respectively. Find the height of the lamp post. ($\tan 38^\circ = 0.7813$, $\sqrt{3} = 1.732$) **JUL-20, MDL-19**



solution

Let AB be the building of height 60 m.

DC be the lamp post.

$$DC = BE$$

$$\angle FAD = \angle ADE = 38^\circ$$

$$\angle FAC = \angle ACB = 60^\circ$$

In the right triangle $\triangle ADE$

$$\tan 38^\circ = \frac{AE}{DE}$$

$$0.7813 = \frac{AE}{CB}$$

$$CB = \frac{AE}{0.7813} \quad \dots (1)$$

In the right triangle $\triangle ACB$

$$\tan 60^\circ = \frac{AB}{BC}$$

$$\sqrt{3} = \frac{60}{BC}$$

$$BC = \frac{60}{\sqrt{3}} = \frac{60 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$$

$$= \frac{60\sqrt{3}}{3}$$

$$CB = 20\sqrt{3} \quad \dots (2)$$

From (1) and (2)

$$\frac{AE}{0.7813} = 20\sqrt{3}$$

$$AE = 20 \times 1.732 \times 0.7813$$

$$= 34.64 \times 0.7813 = 27.064232 = 27.06 \text{ m}$$

The Height of the lamp post

$$EB = AB - AE = 60 - 27.06 = 32.93 \text{ m}$$

$$\therefore \text{Height of the lamp post} = 32.93 \text{ m}$$

18. An aeroplane at an altitude of 1800 m finds that two boats are sailing towards it in the same direction. The angles of depression of the boats as observed from the aeroplane are 60° and 30° respectively. Find the distance between the two boats. ($\sqrt{3} = 1.732$)

solution

$$AP = 1800 \text{ m}$$

= height of the plane from the ground

B_1, B_2 = Positions of 2 boats

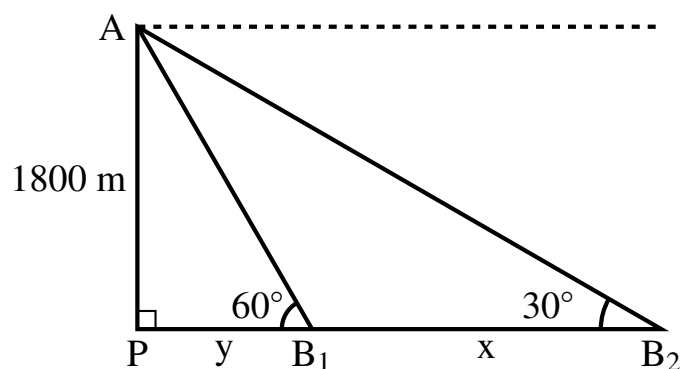
Let $B_1, B_2 = x$ m; $PB_1 = y$

In $\triangle APB_1$,

$$\tan 60^\circ = \frac{1800}{y}$$

$$\sqrt{3} = \frac{1800}{y}$$

$$y = \frac{1800}{\sqrt{3}} = 600\sqrt{3} \quad \dots (1)$$



In triangle $\triangle APB_2$,

$$\tan 30^\circ = \frac{1800}{y+x}$$

$$\frac{1}{\sqrt{3}} = \frac{1800}{y+x}$$

$$y+x = 1800\sqrt{3}$$

$$600\sqrt{3} + x = 1800\sqrt{3}$$

(From (1))

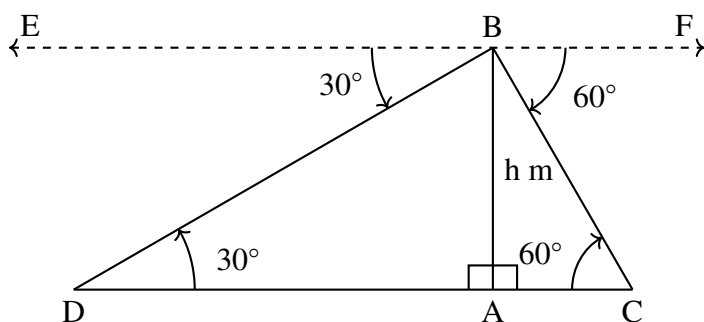
$$x = 1200\sqrt{3}$$

$$x = 1200(1.732)$$

$$\boxed{x = 2078.4 \text{ m}}$$

Distance between two bots = 2078.4 m

19. From the top of a lighthouse, the angle of depression of two ships on the opposite sides of it are observed to be 30° and 60° . If the height of the lighthouse is h meters and the line joining the ships passes through the foot of the lighthouse, show that the distance between the ships is $\frac{4h}{\sqrt{3}}$ m. **APR-23**



solution

D and C the positions of two ships

AB the light house of height 'h' m.

$$\angle EBD = \angle BDA = 30^\circ$$

$$\angle FBC = \angle BCA = 60^\circ$$

In right triangle BAC

$$\tan 60^\circ = \frac{AB}{AC}$$

$$\sqrt{3} = \frac{h}{AC}$$

$$\boxed{AC = \frac{h}{\sqrt{3}}}$$

... (1)

In right triangle $\triangle BAD$

$$\tan 30^\circ = \frac{AB}{AD}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{AD}$$

$$\boxed{AD = h\sqrt{3}}$$

... (2)

$$(1) + (2) \Rightarrow AC + AD = \frac{h}{\sqrt{3}} + h\sqrt{3}$$

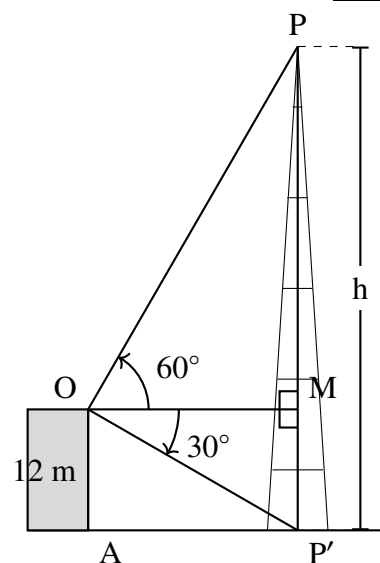
$$DC = \frac{h + h\sqrt{3}\sqrt{3}}{\sqrt{3}}$$

$$DC = \frac{h + 3h}{\sqrt{3}} = \frac{4h}{\sqrt{3}} \text{ m}$$

$\therefore$ Distance between the ships is $\frac{4h}{\sqrt{3}}$ m.

20. From the top of a 12 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 30° .

Determine the height of the tower. **JUL-25**



solution

The angle of elevation of P, $\angle MOP = 60^\circ$

The angle of depression of P', $\angle MOP' = 30^\circ$

Height of the cable tower

$$PP' = h \text{ metres.}$$

$$MP = h - OA = h - 12$$

In right triangle OMP, $\frac{MP}{OM} = \tan 60^\circ$

$$\frac{h - 12}{OM} = \sqrt{3}$$

$$OM = \frac{h - 12}{\sqrt{3}} \quad \dots (1)$$

In right triangle OMP' , $\frac{MP'}{OM} = \tan 30^\circ$

$$\frac{12}{OM} = \frac{1}{\sqrt{3}}$$

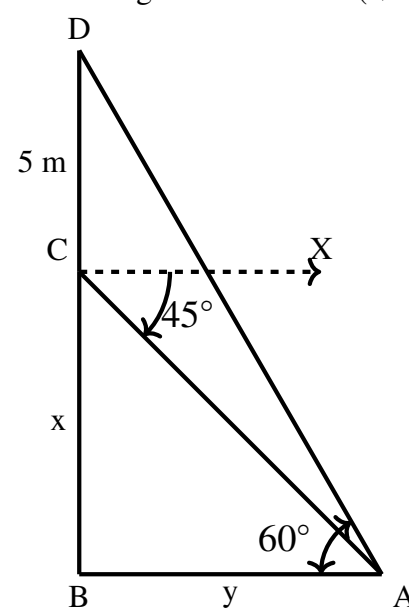
$$OM = 12\sqrt{3} \quad \dots (2)$$

$$\text{From (1) and (2)} \frac{h - 12}{\sqrt{3}} = 12\sqrt{3}$$

$$h - 12 = 12\sqrt{3} \times \sqrt{3} \Rightarrow h = 48$$

The required height of the cable tower is 48 m.

21. A pole 5 m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point 'A' on the ground is 60° and the angle of depression to the point 'A' from the top of the tower is 45° . Find the height of the tower. ($\sqrt{3} = 1.732$)



solution

BC = height of the tower

CD = height of the pole.

A = the point of observation.

BC = x and AB = y.

From the diagram,

$$\angle BAD = 60^\circ \text{ and } \angle XCA = 45^\circ = \angle BAC$$

In right triangle ABC, $\tan 45^\circ = \frac{BC}{AB}$

$$1 = \frac{x}{y} \rightarrow \boxed{x = y} \quad \dots (1)$$

In right triangle ABD,

$$\tan 60^\circ = \frac{BD}{AB} = \frac{BC + CD}{AB}$$

$$\sqrt{3} = \frac{x+5}{y} \rightarrow \sqrt{3}y = x+5$$

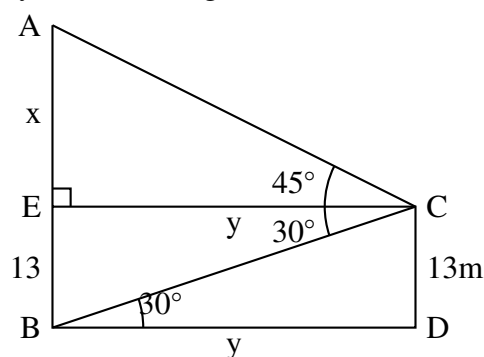
$$\sqrt{3}x = x+5 \quad [\text{From (1)}]$$

$$x = \frac{5}{\sqrt{3}-1} = \frac{5}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$$

$$= \frac{5(1.732+1)}{2} = 6.83$$

The Height of the tower is 6.83 m.

22. From the top of a tree of height 13 m the angle of elevation and depression of the top and bottom of another tree are 45° and 30° respectively. Find the height of the second tree. ($\sqrt{3} = 1.732$)



solution

$$CD = 13\text{m} = \text{height of Tree 1}$$

$$AB = x + 13 = \text{height of Tree 2}$$

$$BD = EC = y\text{m} = \text{distance 2 trees}$$

In right angle $\triangle BCD$,

$$\tan 30^\circ = \frac{CD}{BD}$$

$$\frac{1}{\sqrt{3}} = \frac{13}{y}$$

$$y = 13\sqrt{3}\text{ m}$$

In right angle $\triangle ACE$,

$$\tan 45^\circ = \frac{AE}{EC}$$

$$1 = \frac{x}{y}$$

$$y = x$$

$$x = 13\sqrt{3}\text{ m}$$

$$\therefore \text{Height of 2nd tree} = x + 13$$

$$= 13\sqrt{3} + 13$$

$$= 13(\sqrt{3} + 1)$$

$$= 13 \times 2.732$$

$$= 35.516$$

$$\boxed{x \approx 35.52\text{ m}}$$

23. The angles of elevation and depression of the top and bottom of a lamp post from the top 66 m high apartment are 60° and 30° respectively. Find

- (i) The height of the lamp post.
(ii) The difference between height of the lamp post and the apartment.
(iii) The distance between the lamp post and the apartment. ($\sqrt{3} = 1.732$)

solution

$$AT = \text{height of apartment} = 66\text{ m}$$

$$LP = \text{height of the lamp post} = x + 66$$

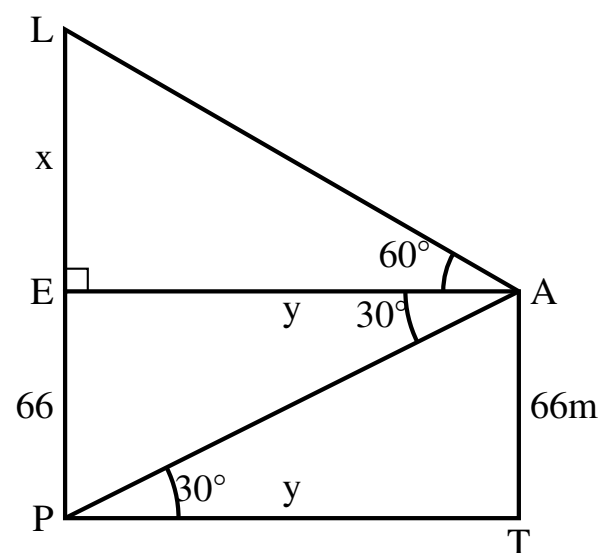
$$EA = y\text{ m} = \text{dist. between post and apartment}$$

In right angle $\triangle APT$,

$$\tan 30^\circ = \frac{AT}{y}$$

$$\frac{1}{\sqrt{3}} = \frac{66}{y}$$

$$y = 66\sqrt{3} \quad \dots\dots(1)$$



In right angle $\triangle ALE$,

$$\tan 60^\circ = \frac{LE}{EA}$$

$$\sqrt{3} = \frac{x}{y}$$

$$y = \frac{x}{\sqrt{3}} \quad \dots\dots(2)$$

$\therefore$ From (1) & (2)

$$\frac{x}{\sqrt{3}} = 66\sqrt{3}$$

$$x = 66 \times 3$$

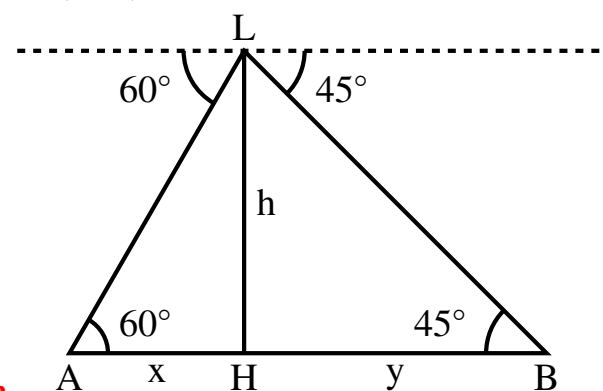
$$= 198$$

i) Height of the lamp post = $x + 66$
 $= 198 + 66$
 $= 264\text{ m}$

ii) Difference between height of lamp post
& apartment = $264 - 66 = 198\text{ m}$

iii) Dist. between lamp post & the apartment
 $\Rightarrow y = 66\sqrt{3}\text{ m}$
 $= 66(1.732)$
 $= 114.312\text{ m}$

24. Two ships are sailing in the sea on either side of the lighthouse. The angles of depression of two ships as observed from the top of the lighthouse are 60° and 45° respectively. If the distance between the ships is $200\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$ metres, find the height of the lighthouse.



solution

$$LH = h\text{ m} = \text{height of the light house}$$

$$AB = \text{Dist. between 2 ships}$$

$$x + y = 200\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$$

In right angle $\triangle LBH$,

$$\tan 45^\circ = \frac{h}{y}$$

$$1 = \frac{h}{y}$$

$$y = h \quad \dots\dots(1)$$

In right angle $\triangle LAH$,

$$\tan 60^\circ = \frac{h}{x}$$

$$\sqrt{3} = \frac{h}{x}$$

$$x = \frac{h}{\sqrt{3}} \quad \dots\dots(2)$$

$\therefore$ Add (1) & (2)

$$x + y = h + \frac{h}{\sqrt{3}}$$

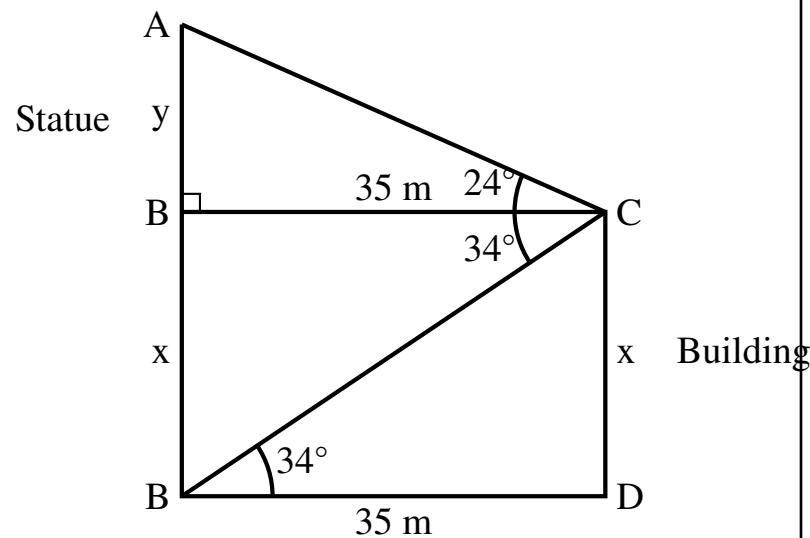
$$200 \left(\frac{\sqrt{3} + 1}{\sqrt{3}} \right) = h \left(1 + \frac{1}{\sqrt{3}} \right)$$

$$\frac{200(\sqrt{3} + 1)}{\sqrt{3}} = h \left(\frac{\sqrt{3} + 1}{\sqrt{3}} \right)$$

$$\therefore h = 200$$

$\therefore$ Height of the light house = 200 m.

25. A building and a statue are in opposite side of a street from each other 35 m apart. From a point on the roof of building the angle of elevation of the top of statue is 24° and the angle of depression of base of the statue is 34° . Find the height of the statue. ($\tan 24^\circ = 0.4452$, $\tan 34^\circ = 0.6745$) **PTA-4**



solution

$CD = x$ m = Height of the building = EB

$AB = x + y$ m = Height of the statue

In right angle $\triangle CBD$,

$$\tan 34^\circ = \frac{CD}{BD}$$

$$\tan 34^\circ = \frac{x}{35}$$

$$\boxed{x = 35 \tan 34^\circ} \quad \dots\dots\dots(1)$$

In right angle $\triangle ACE$,

$$\tan 24^\circ = \frac{AE}{EC}$$

$$\tan 24^\circ = \frac{y}{35}$$

$$\boxed{y = 35 \tan 24^\circ} \quad \dots\dots\dots(2)$$

Height of the statue = $x + y$

$$= 35(\tan 34^\circ + \tan 24^\circ)$$

$$= 35(0.6745 + 0.4452)$$

$$= 35(1.1197)$$

$$= 39.1895$$

$$\approx 39.19 \text{ m}$$

CHAPTER - 7. MENSURATION**PUBLIC EXAM IMPORTANT QUESTIONS****2 MARK IMPORTANT QUESTIONS****CHAPTER - 7.MESURATION****2 MARK IMPORTANT QUESTIONS**

1. A cylindrical drum has a height of 20 cm and base radius of 14 cm. Find its curved surface area and the total surface area. **JUL-22, 24**

solutionGiven height of the cylinder $h = 20$ cm ; radius $r = 14$ cmC.S.A. of the cylinder $= 2\pi rh$ sq. units

$$= 2 \times \frac{22}{7} \times 14 \times 20$$

$$= 2 \times 22 \times 2 \times 20 = 1760 \text{ cm}^2$$

T.S.A. of the cylinder $= 2\pi r(h + r)$ sq. units

$$= 2 \times \frac{22}{7} \times 14 \times (20 + 14)$$

$$= 2 \times \frac{22}{7} \times 14 \times 34$$

$$= 2992 \text{ cm}^2$$

$$CSA = 1760 \text{ cm}^2, TSA = 2992 \text{ cm}^2$$

2. The curved surface area of a right circular cylinder of height 14 cm is 88 cm^2 . Find the diameter of the cylinder. **JUL-23, 25**

solutionGiven Height of Cylinder $= 14$ cm, C.S.A. of the cylinder $= 88$ sq. cm

$$2\pi rh = 88$$

$$2 \times \frac{22}{7} \times r \times 14 = 88$$

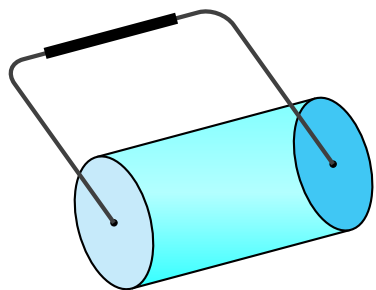
$$r = \frac{88 \times 7}{2 \times 22 \times 14}$$

$$r = 1$$

$$d = 2 \times r = 2 \times 1 = 2$$

Diameter of the cylinder 2 cm

3. A garden roller whose length is 3 m long and whose diameter is 2.8 m is rolled to level a garden. How much area will it cover in 8 revolutions? **MAR-26**

solutionCylinder diameter $d = 2.8\text{m}$ and height $= 3\text{m}$, radius $r = 1.4\text{m}$ Area covered in one revolution $=$ curved surface area of the cylinder

$$CSA \text{ of Cylinder} = 2\pi rh \text{ sq. units}$$

$$= 2 \times \frac{22}{7} \times 1.4 \times 3$$

$$= 26.4\text{m}^2$$

Area covered in 1 revolution $= 26.4 \text{ m}^2$

Area covered in 8 revolutions

$$= 8 \times 26.4 = 211.2\text{m}^2$$

Area covered 8 Revolution is 211.2 m^2

4. The radius of a conical tent is 7 m and the height is 24 m. Calculate the length of the canvas used to make the tent if the width of the rectangular canvas is 4 m? **APR-23, PTA-3**

solutionGiven r and h radius and height of the cone.radius $r = 7\text{m}$ and Height $h = 24\text{m}$

$$l = \sqrt{r^2 + h^2}$$

$$= \sqrt{49 + 576}$$

$$= \sqrt{625}$$

$$l = 25 \text{ m}$$

C.S.A. of the conical tent $= \pi rl$ sq. units.

$$\text{Area of the canvas} = \frac{22}{7} \times 7 \times 25$$

$$= 550 \text{ m}^2$$

$$\text{length of the canvas} = \frac{\text{Area of canvas}}{\text{Width}}$$

$$= \frac{550}{4} = 137.5 \text{ m}$$

The length of the canvas is 137.5 m .

5. If the total surface area of a cone of radius 7 cm is 704 cm^2 , then find its slant height. **JUL-22**

solutionGiven radius $r = 7 \text{ cm}$ TSA the cone $= \pi r(l + r)$ sq. unitsT.S.A. of cone $= 704 \text{ cm}^2$

$$704 = \frac{22}{7} \times 7(l + 7)$$

$$32 = l + 7$$

$$l = 32 - 7$$

$$= 25 \text{ cm}$$

Slant Height of the cone is 25 cm .

6. Find the diameter of a sphere whose surface area is 154 m^2 **JUL-20**

solutionGiven Surface area of sphere $= 154 \text{ m}^2$ Area of the Sphere $= 4\pi r^2$

$$4\pi r^2 = 154$$

$$4 \times \frac{22}{7} \times r^2 = 154$$

$$r^2 = 154 \times \frac{1}{4} \times \frac{7}{22}$$

$$r^2 = \frac{49}{4}$$

$$r = \sqrt{\frac{49}{4}}$$

$$\boxed{r = \frac{7}{2}}$$

$$\text{diameter} = 2 \times r = 2 \times \frac{7}{2}$$

diameter $= 7 \text{ m}$.

7. The radius of a spherical balloon increases from 12 cm to 16 cm as air being pumped into it. Find the ratio of the surface area of the balloons in the two cases. **APR-22**

solutionLet r_1 and r_2 be the radii of the balloons.

$$\frac{r_1}{r_2} = \frac{12}{16} = \frac{3}{4}$$

$$\text{Ratio of C.S.A. of balloons} = \frac{4\pi r_1^2}{4\pi r_2^2}$$

$$= \frac{r_1^2}{r_2^2} = \left(\frac{r_1}{r_2}\right)^2$$

$$= \left(\frac{3}{4}\right)^2 = \frac{9}{16}$$

Ratio of C.S.A. of balloons is 9 : 16.

8. If the base area of a hemispherical solid is 1386 sq. metres, then find its total surface area? **APR-24**

solution

Given Base area = $\pi r^2 = 1386$ sq. m

$$\text{T.S.A. Hemisphere} = 3\pi r^2 \text{ sq.m}$$

$$= 3 \times 1386$$

$$= 4158 \text{ m}^2$$

T.S.A. of hemispherical solid 4158 m^2 .

9. The slant height of a frustum of a cone is 5 cm and the radii of its ends are 4 cm and 1 cm. Find its curved surface area. **APR-25, JUL-25**

solution

Given $l = 5 \text{ cm}$, $R = 4 \text{ cm}$, $r = 1 \text{ cm}$.

$$\text{C.S.A. of the frustum} = \pi(R + r)l \text{ sq. units}$$

$$= \frac{22}{7} \times (4 + 1) \times 5$$

$$= \frac{22}{7} \times 25$$

$$= \frac{550}{7}$$

$$\text{C.S.A. of Frustum} = 78.57 \text{ cm}^2$$

10. A sphere, a cylinder and a cone are of the same height which is equal to its radius, where as cone and cylinder are of same height. Find the ratio of their curved surface areas.

solution

Required Ratio = C.S.A. of the sphere: C.S.A. of the cylinder : C.S.A. of the cone

$$= 4\pi r^2 : 2\pi rh : \pi rl,$$

$$(l = \sqrt{r^2 + h^2} = \sqrt{2r^2} = \sqrt{2}r \text{ units})$$

$$= 4\pi r^2 : 2\pi r^2 : \sqrt{2}\pi r^2$$

$$= 4 : 2 : \sqrt{2}$$

$$= 2 \times 2 : 2 : \sqrt{2}$$

$$= 2 \times \sqrt{2}\sqrt{2} : \sqrt{2}\sqrt{2} : \sqrt{2}$$

$$= 2\sqrt{2} : \sqrt{2} : 1$$

11. The ratio of the radii of two right circular cones of same height is 1:3. Find the ratio of their curved surface area when the height of each cone is 3 times the radius of the smaller cone. **PTA-2**

solution

$$\text{Given ratio of radii, } \frac{r_1}{r_2} = \frac{1}{3}$$

$$r_1 = r \text{ and } r_2 = 3r$$

$$\text{cone height } h = 3 \times (\text{radius of smaller cone})$$

$$h = 3r$$

$$\text{Slant height of cone-I, } l_1 = \sqrt{r_1^2 + h^2}$$

$$= \sqrt{r^2 + (3r)^2}$$

$$= \sqrt{r^2 + 9r^2}$$

$$= \sqrt{10r^2} = r\sqrt{10}$$

$$\text{Slant height of cone-II, } l_2 = \sqrt{r_2^2 + h^2}$$

$$= \sqrt{(3r)^2 + (3r)^2}$$

$$= \sqrt{9r^2 + 9r^2}$$

$$= \sqrt{18r^2} = 3r\sqrt{2}$$

$$\begin{aligned} \text{Ratio of their curved surface areas} &= \frac{\text{CSA}_1}{\text{CSA}_2} \\ &= \frac{\pi r_1 l_1}{\pi r_2 l_2} \\ &= \frac{\pi \cdot r \cdot r\sqrt{10}}{\pi \cdot 3r \cdot 3r\sqrt{2}} \\ &= \frac{r^2\sqrt{10}}{9r^2\sqrt{2}} \\ &= \frac{r^2\sqrt{10}}{9r^2\sqrt{2}} \\ &= \frac{\sqrt{5} \cdot \cancel{r^2}}{9\cancel{r^2}} \\ &= \frac{\sqrt{5}}{9} \end{aligned}$$

$$\therefore \text{Ratio of their CSA} = \sqrt{5} : 9$$

12. The radius and height of a cylinder are in the ratio 5:7 and its curved surface area is 5500 sq. cm. Find its radius and height. **APR-23, JUL-22**

solution

Let the Cylinder radius, r and height h be $5x$ and $7x$.

$$\text{Curved Surface Area (CSA)} = 5500 \text{ cm}^2$$

$$2\pi rh = 5500$$

$$2 \times \frac{22}{7} \times 5x \times 7x = 5500$$

$$2 \times \frac{22}{7} \times 5x \times 7x = 5500$$

$$2 \times 22 \times 5x \times x = 5500$$

$$220x^2 = 5500$$

$$x^2 = \frac{5500}{220}$$

$$x^2 = \frac{5500}{220}$$

$$x^2 = 25$$

$$x = 5$$

$$\text{Radius, } r = 5x$$

$$= 5 \times 5$$

$$= 25 \text{ cm}$$

$$\text{Height, } h = 7x$$

$$= 7 \times 5$$

$$= 35 \text{ cm}$$

13. A solid iron cylinder has total surface area of 1848 sq.cm. Its curved surface area is five-sixth of its total surface area. Find the radius and height of the iron cylinder.

solution

Let the radius and height of the cylinder be r and h respectively.

$$\text{Total Surface Area (TSA)} = 1848 \text{ cm}^2$$

$$\text{Curved Surface Area (CSA)} = \frac{5}{6} \times \text{TSA}$$

$$2\pi rh = \frac{5}{6} \times 1848$$

$$2\pi rh = 5 \times 308$$

$$2\pi rh = 1540$$

$$\text{WKT TSA} = \text{CSA} + 2\pi r^2$$

$$1848 = 1540 + 2\pi r^2$$

$$2\pi r^2 = 1848 - 1540$$

$$2\pi r^2 = 308$$

$$2 \times \frac{22}{7} \times r^2 = 308$$

$$\frac{44}{7} \times r^2 = 308$$

$$r^2 = \frac{308 \times 7}{44}$$

$$r^2 = 7 \times 7$$

$$r = 7 \text{ cm}$$

Sub $r = 7$ in CSA $(2\pi rh) = 1540$

$$2 \times \frac{22}{7} \times 7 \times h = 1540$$

$$2 \times \frac{22}{7} \times 7 \times h = 1540$$

$$44h = 1540$$

$$h = \frac{1540}{44}$$

$$h = 35 \text{ cm}$$

$\therefore$ Radius, $r = 7 \text{ cm}$

Height, $h = 35 \text{ cm}$

14. 4 persons live in a conical tent whose slant height is 19 m. If each person require 22 m^2 of the floor area, then find the height of the tent.

solution

Let the base radius and height of the conical tent be r and h respectively.

Slant height, $l = 19 \text{ m}$

Floor area required 1 person = 22 m^2

Total floor area required 4 persons = 4×22

$$\pi r^2 = 88 \text{ m}^2$$

$$\frac{22}{7} \times r^2 = 88$$

$$r^2 = \frac{88 \times 7}{22}$$

$$r^2 = \frac{88 \times 7}{22}$$

$$r^2 = 4 \times 7$$

$$r^2 = 28$$

$$\text{WKT } l^2 = r^2 + h^2$$

$$h^2 = l^2 - r^2$$

$$h^2 = 19^2 - 28$$

$$h^2 = 361 - 28$$

$$h^2 = 333$$

$$h = \sqrt{333}$$

Height of the tent, $h = \sqrt{333} \text{ m} \approx 18.25 \text{ m}$

15. Find the volume of a cylinder whose height is 2 m and whose base area is 250 m^2 . **APR-24, 25, JUL-21**

solution

Given, Height of the cylinder, $h = 2 \text{ m}$

Base area of the cylinder $(\pi r^2) = 250 \text{ m}^2$

Volume of a cylinder, $V = \text{Base area} \times \text{height}$

$$V = \pi r^2 \times h$$

$$= 250 \times 2$$

$$= 500$$

$\therefore$ Volume of the cylinder = 500 m^3

16. The volume of a solid right circular cone is 11088 cm^3 . If its height is 24 cm then find the radius of the cone. **JUL-23, PTA-2**

solution

Let the radius of the cone be r .

Given, Height, $h = 24 \text{ cm}$

Volume of the cone, $V = 11088 \text{ cm}^3$

$$\frac{1}{3} \pi r^2 h = 11088$$

$$\frac{1}{3} \times \frac{22}{7} \times r^2 \times 24 = 11088$$

$$\frac{1}{3} \times \frac{22}{7} \times r^2 \times 24 = 11088$$

$$\frac{22}{7} \times r^2 \times 8 = 11088$$

$$r^2 = \frac{11088 \times 7}{22 \times 8}$$

$$r^2 = \frac{11088 \times 7}{176}$$

$$r^2 = \frac{11088 \times 7}{176}$$

$$r^2 = 63 \times 7$$

$$r^2 = 441$$

$$r = \sqrt{441}$$

$$r = 21 \text{ cm}$$

$\therefore$ Radius of the cone, $r = 21 \text{ cm}$

17. The ratio of the volumes of two cones is 2:3. Find the ratio of their radii if the height of second cone is double the height of the first.

solution

Let the radii of the two cones be r_1 and r_2 .

Let their corresponding heights be h_1 and h_2 .

Given ratio of volumes, $\frac{V_1}{V_2} = \frac{2}{3}$

condition for heights, $h_2 = 2h_1$

volume of a cone = $\frac{1}{3} \pi r^2 h$

$$\Rightarrow \frac{\frac{1}{3} \pi r_1^2 h_1}{\frac{1}{3} \pi r_2^2 h_2} = \frac{2}{3}$$

$$\text{Sub } h_2 = 2h_1, \quad \frac{\frac{1}{3} \pi r_1^2 h_1}{\frac{1}{3} \pi r_2^2 (2h_1)} = \frac{2}{3}$$

$$\frac{\frac{1}{3} \cdot \pi \cdot r_1^2 \cdot h_1}{\frac{1}{3} \cdot \pi \cdot r_2^2 \cdot 2h_1} = \frac{2}{3}$$

$$\frac{r_1^2}{2r_2^2} = \frac{2}{3}$$

$$\frac{r_1^2}{r_2^2} = \frac{2 \times 2}{3}$$

$$\frac{r_1^2}{r_2^2} = \frac{4}{3}$$

$$\frac{r_1}{r_2} = \sqrt{\frac{4}{3}}$$

$$\frac{r_1}{r_2} = \frac{2}{\sqrt{3}}$$

$\therefore$ Ratio of their radii = $2 : \sqrt{3}$

18. The volumes of two cones of same base radius are 3600 cm^3 and 5040 cm^3 . Find the ratio of heights. **JUL-24, APR-22, PTA-4**

solution

Let the base radius of the two cones be r .

Let their heights be h_1 and h_2 .

Volume of cone-I, $V_1 = 3600 \text{ cm}^3$

Volume of cone-II, $V_2 = 5040 \text{ cm}^3$

Ratio of their volumes, $\frac{V_1}{V_2} = \frac{3600}{5040}$

$$\frac{\frac{1}{3} \pi r^2 h_1}{\frac{1}{3} \pi r^2 h_2} = \frac{3600}{5040}$$

$$\frac{\frac{1}{3} \cdot \pi \cdot r^2 \cdot h_1}{\frac{1}{3} \cdot \pi \cdot r^2 \cdot h_2} = \frac{3600}{5040}$$

$$\frac{h_1}{h_2} = \frac{3600}{5040}$$

$$\frac{h_1}{h_2} = \frac{360}{504}$$

$$\frac{h_1}{h_2} = \frac{5}{7}$$

$\therefore$ Ratio of their heights = $5 : 7$

19. If the ratio of radii of two spheres is 4:7, find the ratio of their volumes.

APR-23, 25

solution

Let the radii of the two spheres be r_1 and r_2 .

$$\text{Given ratio of radii, } \frac{r_1}{r_2} = \frac{4}{7}$$

$$\text{volume of a sphere, } V = \frac{4}{3}\pi r^3$$

$$\begin{aligned} \text{Ratio of their volumes, } \frac{V_1}{V_2} &= \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3} \\ &= \frac{\frac{4}{3} \cdot \pi \cdot r_1^3}{\frac{4}{3} \cdot \pi \cdot r_2^3} \\ &= \frac{r_1^3}{r_2^3} \\ &= \left(\frac{r_1}{r_2}\right)^3 \\ &= \left(\frac{4}{7}\right)^3 \\ &= \frac{4 \times 4 \times 4}{7 \times 7 \times 7} \\ &= \frac{64}{343} \end{aligned}$$

$$\therefore \text{Ratio of their volumes} = 64 : 343$$

20. A cone of height 24 cm is made up of modeling clay. A child reshapes it in the form of a cylinder of same radius as cone. Find the height of the cylinder. **PTA-3**

solution

Let the radius of the cone and cylinder be r .

Height of the cone $h_1 = 24$ cm. The height of the cylinder be h_2 .

Volume of cylinder = Volume of cone

$$\begin{aligned} \pi r^2 h_2 &= \frac{1}{3} \pi r^2 h_1 \\ \pi \cdot r^2 \cdot h_2 &= \frac{1}{3} \cdot \pi \cdot r^2 \cdot 24 \\ h_2 &= \frac{\frac{1}{3} \cdot \pi \cdot r^2 \cdot 24}{\pi \cdot r^2} \\ h_2 &= \frac{1}{3} \times 24 \\ h_2 &= \frac{24^1}{3^1} \\ h_2 &= 8 \text{ cm} \end{aligned}$$

$$\therefore \text{Height of the cylinder} = 8 \text{ cm}$$

21. Find the Volume of the iron used to make a hollow cylinder of height 9 cm and whose internal and external radii are 21 cm and 28 cm respectively.

solution

The external radius of Hollow Cylinder $R = 28$ cm. The internal radius $r = 21$ cm. The height $h = 9$ cm

$$\begin{aligned} \text{Volume } V &= \pi(R^2 - r^2)h \\ &= \frac{22}{7} \times (28^2 - 21^2) \times 9 \\ &= \frac{22}{7} \times (28 + 21)(28 - 21) \times 9 \\ &= \frac{22}{7} \times (49) \times (7) \times 9 \\ &= \frac{22}{7} \times 49 \times 7 \times 9 \\ &= 22 \times 49 \times 9 \\ &= 1078 \times 9 \\ &= 9702 \end{aligned}$$

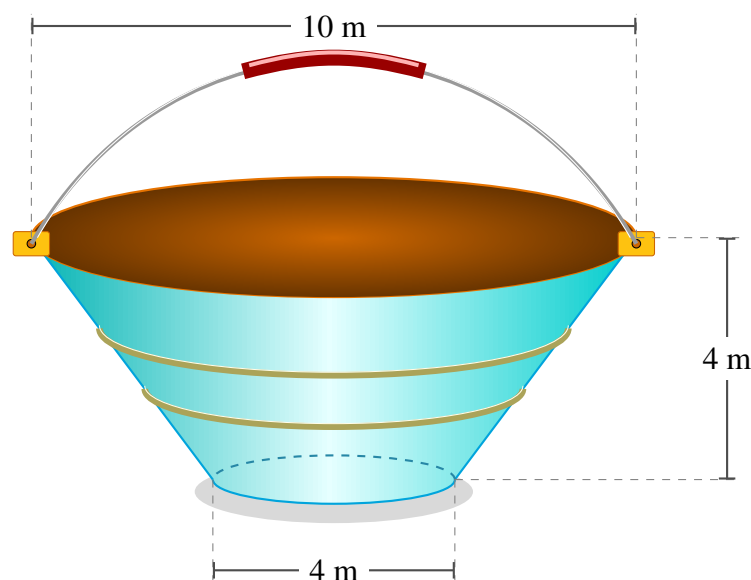
$$\therefore \text{Volume of the iron used} = 9702 \text{ cm}^3$$

CHAPTER - 7.MESURATION

5 MARK IMPORTANT QUESTIONS

1. An industrial metallic bucket is in the shape of the frustum of a right circular cone whose top and bottom diameters are 10 m and 4 m and whose height is 4 m. Find the curved and total surface area of the bucket.

solution



Given Frustum Diameter of the top = 10 m;

Radius of the top $R = \frac{d}{2} = \frac{10}{2} = 5 \text{ m}$.

Diameter of the bottom = 4 m;

radius of the bottom $r = \frac{d}{2} = \frac{4}{2} = 2 \text{ m}$, height $h = 4 \text{ m}$.

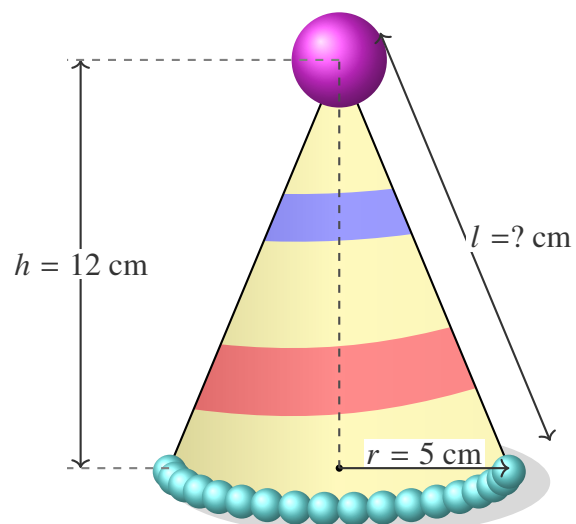
$$\begin{aligned} l &= \sqrt{h^2 + (R - r)^2} \\ &= \sqrt{4^2 + (5 - 2)^2} \\ l &= \sqrt{16 + 9} = \sqrt{25} = 5 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{C.S.A} &= \pi(R + r)l \text{ sq.units} \\ &= \frac{22}{7}(5 + 2) \times 5 = 110 \text{ m}^2 \\ \text{T.S.A.} &= \pi(R + r)l + \pi r^2 \text{ sq.units} \\ &= \frac{22}{7}[(5 + 2)5 + 4] \\ &= \frac{858}{7} = 122.7 \end{aligned}$$

$$\text{C.S.A. of Bucket} = 110 \text{ m}^2 \text{ and } \text{T.S.A. of Bucket} = 122.7 \text{ m}^2$$

2. A girl wishes to prepare birthday caps in the form of right circular cones for her birthday party, using a sheet of paper whose area is 5720 cm², how many caps can be made with radius 5 cm and height 12 cm.

solution



Festive 3D Conical Birthday Cap

Given : Cone:

$$\text{Area of paper sheet} = 5720 \text{ cm}^2$$

$$\text{Radius of one cap, } r = 5 \text{ cm}$$

$$\text{Height of one cap, } h = 12 \text{ cm}$$

To find the slant height (l) of the cone using the Pythagorean theorem:

$$l = \sqrt{r^2 + h^2}$$

$$l = \sqrt{5^2 + 12^2}$$

$$l = \sqrt{25 + 144}$$

$$l = \sqrt{169}$$

$$l = 13 \text{ cm}$$

The paper required to make one conical cap is equal to its Curved Surface Area (CSA):

$$\text{CSA of one cap} = \pi r l$$

$$= \frac{22}{7} \times 5 \times 13 \text{ cm}^2$$

The number of caps(n)

$$n = \frac{\text{Total area of the paper sheet}}{\text{CSA of one cap}}$$

$$n = \frac{5720}{\frac{22}{7} \times 5 \times 13}$$

$$n = \frac{5720 \times 7}{22 \times 5 \times 13}$$

$$n = \frac{5720 \times 7}{22 \times 5 \times 13}$$

$$n = \frac{5720 \times 7}{22 \times 5 \times 13}$$

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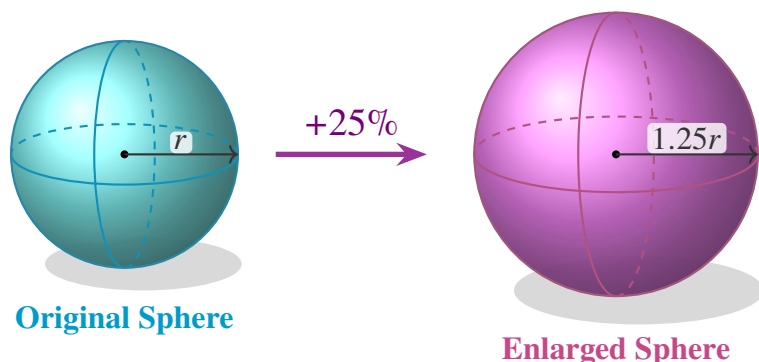
$$n = \frac{5720 \times 7}{22 \times 5 \times 13}$$

$$n = \frac{5720 \times 7}{22 \times 5 \times 13}$$

$\therefore$ 28 birthday caps can be made from the given sheet of paper.

3. The radius of a sphere increases by 25%. Find the percentage increase in its surface area. **MDL-19**

solution



The original radius of the sphere r and surface area (S_1) is:

$$S_1 = 4\pi r^2$$

The radius increases by 25%. The new radius be R .

$$R = r + (25\% \text{ of } r)$$

$$R = r + \frac{25}{100}r$$

$$R = r + \frac{1}{4}r$$

$$R = \frac{5}{4}r$$

The new surface area (S_2):

$$S_2 = 4\pi R^2$$

$$S_2 = 4\pi \left(\frac{5}{4}r\right)^2$$

$$S_2 = 4\pi \left(\frac{25}{16}r^2\right)$$

$$S_2 = \frac{25}{4}\pi r^2$$

The increase surface area (ΔS):

$$\Delta S = S_2 - S_1$$

$$\Delta S = \frac{25}{4}\pi r^2 - 4\pi r^2$$

$$\Delta S = \left(\frac{25 - 16}{4}\right)\pi r^2$$

$$\Delta S = \frac{9}{4}\pi r^2$$

$$\text{Percentage Increase} = \frac{\frac{9}{4}\pi r^2}{4\pi r^2} \times 100\%$$

$$= \frac{\frac{9}{4}r^2}{4r^2} \times 100\%$$

$$= \left(\frac{9}{4}\right) \times 100\%$$

$$= \frac{9}{16} \times 100\%$$

$$= \frac{9}{16} \times 100\%$$

$$= \frac{9 \times 25}{16} \%$$

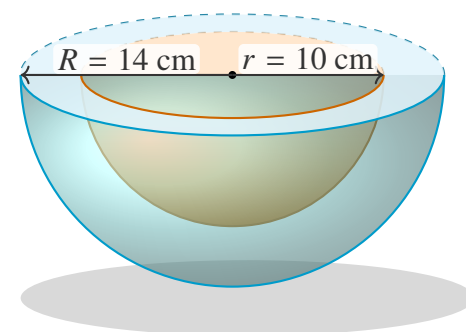
$$= \frac{225}{16} \%$$

$$= 14.0625\%$$

$\therefore$ The percentage increase in the surface area of the sphere is **56.25%**.

4. The internal and external diameters of a hollow hemispherical vessel are 20 cm and 28 cm respectively. Find the cost to paint the vessel all over at 0.14 per cm^2 . **JUL-24**

solution



Hollow Hemispherical Vessel

Given : Hollow Hemisphere:

$$\text{External diameter} = 28 \text{ cm} \implies R = \frac{28}{2} = 14 \text{ cm}$$

$$\text{Internal diameter} = 20 \text{ cm} \implies r = \frac{20}{2} = 10 \text{ cm}$$

$$\text{Cost rate to paint} = \text{Rs. } 0.14 \text{ per cm}^2$$

$$\text{TSA} = \text{Outer CSA} + \text{Inner CSA} + \text{Area of Top Ring}$$

$$= 2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2)$$

$$= \pi(2R^2 + 2r^2 + R^2 - r^2)$$

$$\text{TSA} = \pi(3R^2 + r^2)$$

$$\text{TSA} = \frac{22}{7} [3(14)^2 + (10)^2]$$

$$= \frac{22}{7} [3(196) + 100]$$

$$= \frac{22}{7} [588 + 100]$$

$$= \frac{22}{7} \times 688 \text{ cm}^2$$

$$\text{Total Cost} = \text{TSA} \times \text{Rate}$$

$$\text{Total Cost} = \left(\frac{22}{7} \times 688\right) \times 0.14$$

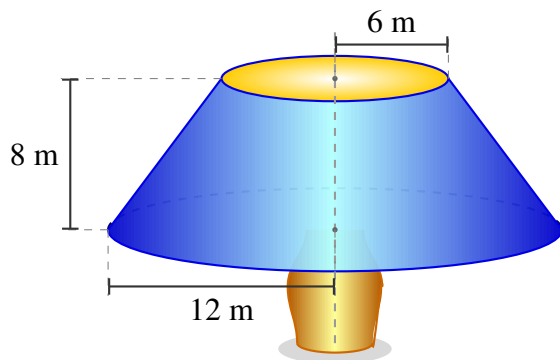
$$\text{Total Cost} = \frac{22}{7} \times 688 \times \frac{14}{100}$$

$$\begin{aligned}
 \text{Total Cost} &= 22 \times 688 \times \frac{14^2}{7 \times 100} \\
 &= \frac{22 \times 688 \times 2}{100} \\
 &= \frac{44 \times 688}{100} \\
 &= \frac{30272}{100} \\
 \text{Total Cost} &= 302.72
 \end{aligned}$$

∴ The cost to paint the hollow hemispherical vessel all over is **Rs. 302.72**.

5. The frustum shaped outer portion of the table lamp has to be painted including the top part. Find the total cost of painting the lamp if the cost of painting 1 sq. cm is 2.

solution



Given: Frustum shaped lamp $R = 12\text{ m}, r = 6\text{ m}, h = 8\text{ m}$

$$\begin{aligned}
 l &= \sqrt{(R-r)^2 + h^2} \\
 &= \sqrt{36 + 64} \\
 &= \sqrt{100} = 10\text{ cm}
 \end{aligned}$$

∴ Required portion to be painted = CSA of frustum + πr^2

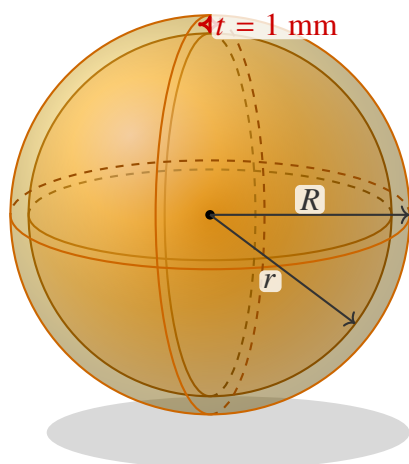
$$\begin{aligned}
 &= \pi(R+r)l + \pi r^2 \\
 &= \pi[18(10) + 36] \\
 &= \frac{22}{7} \times 216 \\
 &= \frac{4752}{7} \\
 &= 678.86\text{ m}^2
 \end{aligned}$$

Given cost of painting = Rs.2/m²

$$\begin{aligned}
 \therefore \text{Total cost} &= 678.86 \times 2 \\
 &= \text{Rs.1357.72}
 \end{aligned}$$

6. Calculate the mass of a hollow brass sphere if the inner diameter is 14 cm and thickness is 1 mm, and whose density is 17.3 g/cm³. (Hint: mass = density × volume) **MDL-19**

solution



Hollow Brass Sphere

Given : Hollow Sphere:

Inner diameter, $d = 14\text{ cm}$

Thickness, $t = 1\text{ mm}$

Density, $\rho = 17.3\text{ g/cm}^3$

$$r = \frac{d}{2} = \frac{14}{2} = 7\text{ cm}$$

$$t = 0.1\text{ cm}$$

$$R = r + t$$

$$R = 7 + 0.1 = 7.1\text{ cm}$$

$$\begin{aligned}
 \text{Volume of Hollow Sphere} &= \frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3 \\
 &= \frac{4}{3}\pi(R^3 - r^3) \\
 &= \frac{4}{3} \times \frac{22}{7} \times ((7.1)^3 - (7)^3) \\
 &= \frac{88}{21} \times (357.911 - 343) \\
 &= \frac{88}{21} \times 14.911 \\
 &\approx 4.1904 \times 14.911 \\
 V &\approx 62.48\text{ cm}^3
 \end{aligned}$$

Mass = Density × Volume

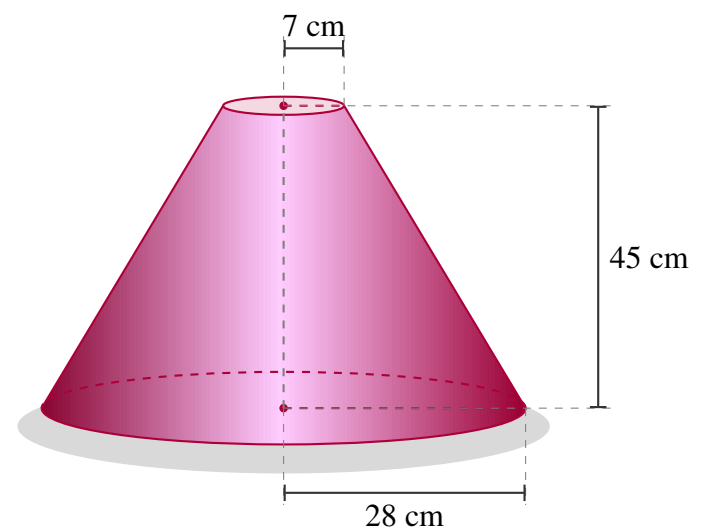
$$\text{Mass} = 17.3 \times 62.48$$

$$\text{Mass} \approx 1080.90\text{ g}$$

The mass of the hollow brass sphere is approximately **1080.9 grams** (or **1.08 kg**).

7. If the radii of the circular ends of a frustum which is 45 cm high are 28 cm and 7 cm, find the volume of the frustum. **APR-24, JUL-21, PTA-5**

solution



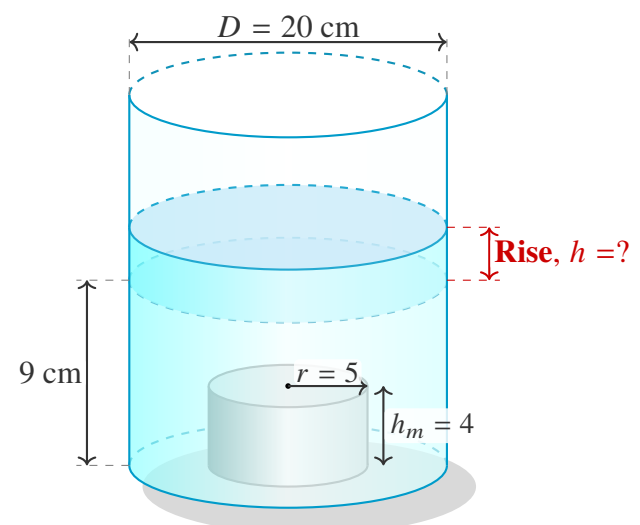
Given Frustum: $h = 45\text{ cm}, R = 28\text{ cm}, r = 7\text{ cm}$

$$\begin{aligned}
 \text{Volume of Frustum} &= \frac{1}{3}\pi[R^2 + Rr + r^2]h\text{ cu.units} \\
 &= \frac{1}{3} \times \frac{22}{7} \times [28^2 + (28 \times 7) + 7^2] \times 45 \\
 &= \frac{1}{3} \times \frac{22}{7} \times 147 \times 15 = 48510
 \end{aligned}$$

The Volume of the frustum is **48510 cm³**

8. A cylindrical glass with diameter 20 cm has water to a height of 9 cm. A small cylindrical metal of radius 5 cm and height 4 cm is immersed completely. Calculate the raise of the water in the glass? **JUL-20**

solution



Water Displacement in Glass

Given: Cylindrical Class

Diameter of the glass = 20 cm

Radius of the glass, $R = \frac{20}{2} = 10$ cmRadius of the metal, $r = 5$ cmHeight of the metal, $h_m = 4$ cm

Volume of raised water = Volume of metal cylinder

$$\pi R^2 h = \pi r^2 h_m$$

$$\pi R^2 h = \pi r^2 h_m$$

$$R^2 h = r^2 h_m$$

$$(10)^2 \times h = (5)^2 \times 4$$

$$100 \times h = 25 \times 4$$

$$100h = 100$$

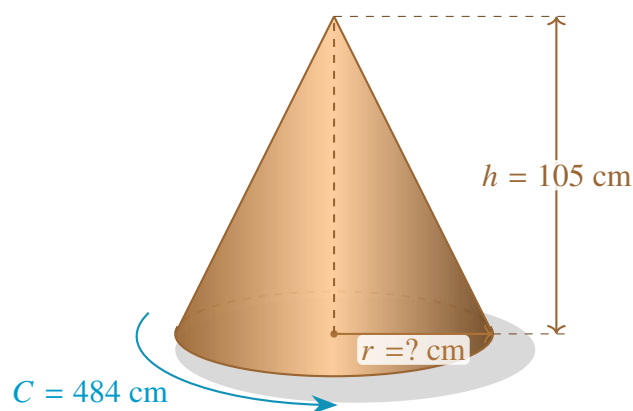
$$h = \frac{100}{100}$$

$$h = \frac{100}{100}$$

$$h = 1 \text{ cm}$$

The raise of the water in the cylindrical glass is **1 cm**.

9. If the circumference of a conical wooden piece is 484 cm then find its volume when its height is 105 cm. **JUL-22**

solution**Conical Wooden Piece****Given Conical Wood (Cone):**Circumference, $C = 484$ cmHeight, $h = 105$ cmTo find the radius (r) using the circumference formula $C = 2\pi r$.

$$2\pi r = 484$$

$$2 \times \frac{22}{7} \times r = 484$$

$$\frac{44}{7} \times r = 484$$

$$r = \frac{484 \times 7}{44}$$

$$r = \frac{484 \times 7}{44}$$

$$r = 77 \text{ cm}$$

$$\text{Volume of Cone} = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3} \times \frac{22}{7} \times r^2 \times h$$

$$= \frac{1}{3} \times \frac{22}{7} \times (77)^2 \times 105$$

$$= \frac{1}{3} \times \frac{22}{7} \times 77 \times 77 \times 105$$

$$= 22 \times 11 \times 77 \times 35$$

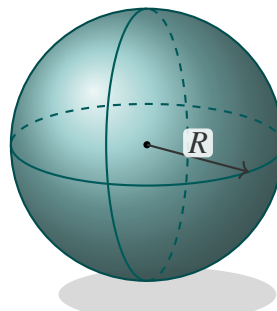
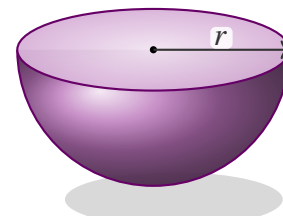
$$= (22 \times 11) \times (77 \times 35)$$

$$= 242 \times 2695$$

$$V = 652190 \text{ cm}^3$$

The volume of the conical wooden piece is **652,190 cm³**.

10. A solid sphere and a solid hemisphere have equal total surface area. Prove that the ratio of their volume is $3\sqrt{3} : 4$. **PTA-6**

solution**Solid Sphere****Solid Hemisphere****Given : Solid Sphere and Hemisphere**TSA of Solid Sphere = $4\pi R^2$ TSA of Solid Hemisphere = $3\pi r^2$ (CSA $2\pi r^2$ + Base area πr^2)

Given that their total surface areas are equal:

$$4\pi R^2 = 3\pi r^2$$

$$4\pi R^2 = 3\pi r^2$$

$$4R^2 = 3r^2$$

$$\frac{R^2}{r^2} = \frac{3}{4}$$

$$\frac{R}{r} = \sqrt{\frac{3}{4}}$$

$$\frac{R}{r} = \frac{\sqrt{3}}{2} \dots (1)$$

To find the ratio of their volumes ($\frac{V_1}{V_2}$):

$$\frac{V_1}{V_2} = \frac{\frac{4}{3}\pi R^3}{\frac{2}{3}\pi r^3}$$

$$= \frac{4 \times \frac{1}{3}\pi \times R^3}{2 \times \frac{1}{3}\pi \times r^3}$$

$$= \frac{4}{2} \times \left(\frac{R}{r}\right)^3$$

$$\frac{V_1}{V_2} = 2 \left(\frac{R}{r}\right)^3$$

$$= 2 \left(\frac{\sqrt{3}}{2}\right)^3$$

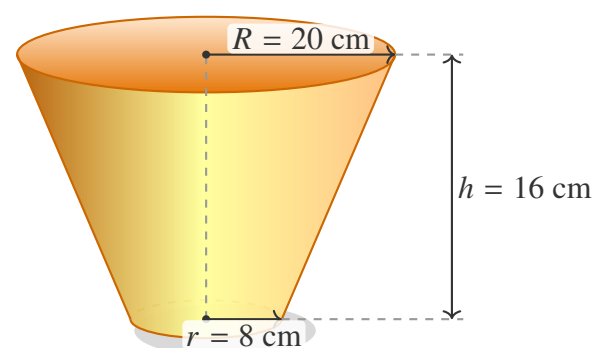
$$= 2 \left(\frac{(\sqrt{3})^3}{2^3}\right)$$

$$= 2 \times \frac{3\sqrt{3}}{8}$$

$$\frac{V_1}{V_2} = \frac{3\sqrt{3}}{4}$$

The ratio of their volumes is $3\sqrt{3} : 4$.**Hence Proved.**

11. A container open at the top is in the form of a frustum of a cone of height 16 cm with radii of its lower and upper ends are 8 cm and 20 cm respectively. Find the cost of milk which can completely fill a container at the rate of 40 per litre. **APR-22**

solution**Container (Frustum of a Cone)**

Given : Frustum: $R = 20$ cm, $r = 8$ cm, $h = 16$ cm

$$\text{Volume of Frustum } V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$$

$$\begin{aligned} V &= \frac{1}{3} \times \frac{22}{7} \times 16 \times (20^2 + 8^2 + (20 \times 8)) \\ &= \frac{1}{3} \times \frac{22}{7} \times 16 \times (400 + 64 + 160) \\ &= \frac{1}{3} \times \frac{22}{7} \times 16 \times 624 \\ &= \frac{22}{7} \times 16 \times \frac{624}{3} \\ &= \frac{22}{7} \times 16 \times 208 \\ &= \frac{22}{7} \times 3328 \\ V &= 10449.92 \text{ cm}^3 \end{aligned}$$

To convert the volume from cubic centimeters (cm^3) to litres. WKT
1 litre = 1000 cm^3 :

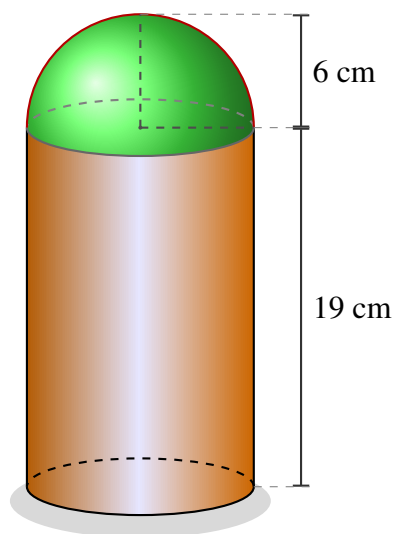
$$\begin{aligned} \text{Volume in litres} &= \frac{10449.92}{1000} \\ &= 10.44992 \text{ litres} \end{aligned}$$

$$\begin{aligned} \text{Total Cost} &= \text{Volume in litres} \times \text{Rate per litre} \\ &= 10.44992 \times 40 \\ &= 417.9968 \end{aligned}$$

The total cost of the milk is approximately **Rs. 418**.

12. A toy is in the shape of a cylinder surmounted by a hemisphere. The height of the toy is 25 cm. Find the total surface area of the toy if its common diameter is 12 cm. **JUL-21, 25**

solution



Given : Cylinder & Hemisphere: diameter $d = 12$ cm, radius $r = 6$ cm Total height of the toy is 25 cm

The Height of the cylindrical portion

$$= 25 - 6 = 19 \text{ cm}$$

T.S.A. of the toy = C.S.A. of the cylinder +

C.S.A. of the hemisphere

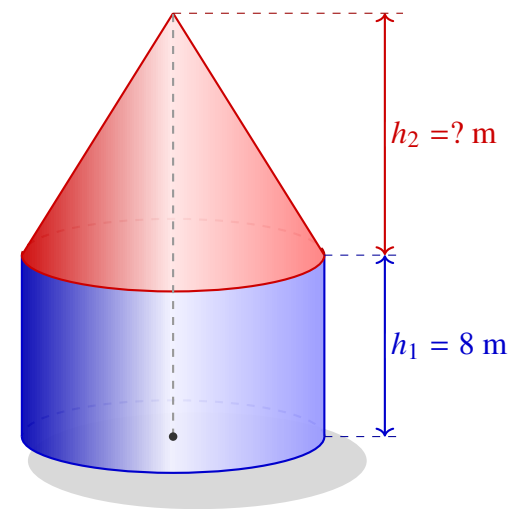
+ Base Area of the cylinder

$$\begin{aligned} &= 2\pi rh + 2\pi r^2 + \pi r^2 \\ &= \pi r(2h + 3r) \text{ sq.units} \\ &= \frac{22}{7} \times 6 \times (38 + 18) \\ &= \frac{22}{7} \times 6 \times 56 = 1056 \end{aligned}$$

T.S.A. of the toy is **1056 cm^2**

13. Arul has to make arrangements for the accommodation of 150 persons for his family function. For this purpose, he plans to build a tent which is in the shape of cylinder surmounted by a cone. Each person occupies 4 sq. m of the space on ground and 40 cu. meter of air to breathe. What should be the height of the conical part of the tent if the height of cylindrical part is 8 m? **APR-23**

solution



Ground Area = 600 m^2

Tent (Cylinder + Cone)

Given : Cylinder & Cone :

Number of persons = 150

Ground space required per person = 4 m^2

Air volume required per person = 40 m^3

Height of the cylindrical part, $h_1 = 8$ m

Total Base Area = Number of persons $\times$ Space per person

$$\pi r^2 = 150 \times 4$$

$$\pi r^2 = 600 \text{ m}^2$$

Total Volume (V) = Number of persons $\times$ Air per person

$$V = 150 \times 40$$

$$V = 6000 \text{ m}^3$$

Total Volume (V) = Volume of Cylinder + Volume of Cone

$$V = \pi r^2 h_1 + \frac{1}{3} \pi r^2 h_2$$

$$V = \pi r^2 \left(h_1 + \frac{1}{3} h_2 \right)$$

$$6000 = 600 \left(8 + \frac{1}{3} h_2 \right)$$

$$\frac{6000}{600} = 8 + \frac{1}{3} h_2$$

$$10 = 8 + \frac{1}{3} h_2$$

$$10 - 8 = \frac{1}{3} h_2$$

$$2 = \frac{1}{3} h_2$$

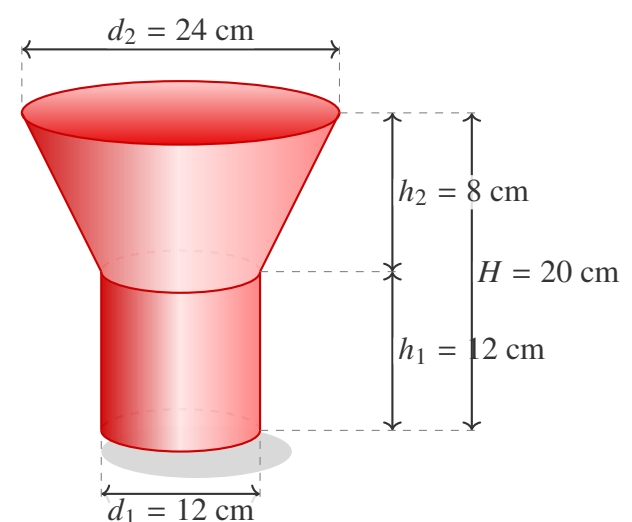
$$h_2 = 2 \times 3$$

$$h_2 = 6 \text{ m}$$

The height of the conical part of the tent should be **6 meters**.

14. A funnel consists of a frustum of a cone attached to a cylindrical portion 12 cm long attached at the bottom. If the total height be 20 cm, diameter of the cylindrical portion be 12 cm and the diameter of the top of the funnel be 24 cm. Find the outer surface area of the funnel. **PTA-1**

solution



Funnel (Frustum + Cylinder)

Given : Cylinder & FrustumTotal height, $H = 20$ cmHeight of cylinder, $h_1 = 12$ cmDiameter of cylinder, $d_1 = 12$ cm $\Rightarrow r_1 = \frac{12}{2} = 6$ cmTop diameter of funnel, $d_2 = 24$ cm $\Rightarrow r_2 = \frac{24}{2} = 12$ cmTo find the height of the frustum portion (h_2):

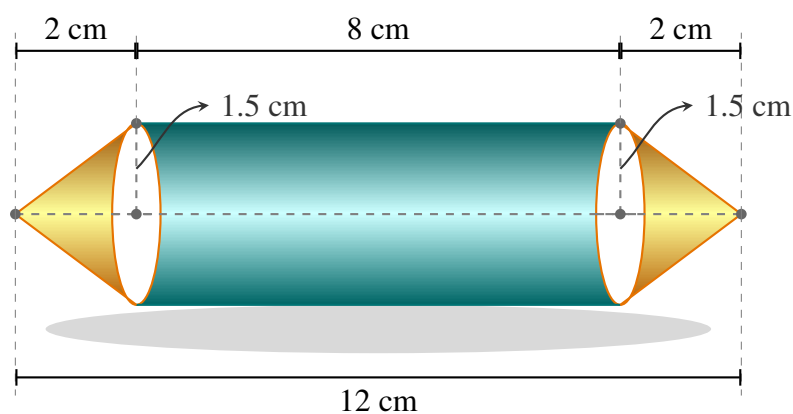
$$\begin{aligned}
 h_2 &= H - h_1 \\
 &= 20 - 12 \\
 h_2 &= 8 \text{ cm} \\
 l &= \sqrt{h_2^2 + (r_2 - r_1)^2} \\
 &= \sqrt{8^2 + (12 - 6)^2} \\
 &= \sqrt{64 + 36} \\
 &= \sqrt{100} \\
 l &= 10 \text{ cm}
 \end{aligned}$$

Total Outer Surface Area = CSA of Cylinder + CSA of Frustum

$$\begin{aligned}
 A &= 2\pi r_1 h_1 + \pi(r_1 + r_2)l \\
 &= 2\pi(6)(12) + \pi(6 + 12)(10) \\
 &= 144\pi + \pi(18)(10) \\
 &= 144\pi + 180\pi \\
 &= 324\pi \text{ cm}^2 \\
 &\approx 324 \times 3.1416 \\
 A &\approx 1017.88 \text{ cm}^2
 \end{aligned}$$

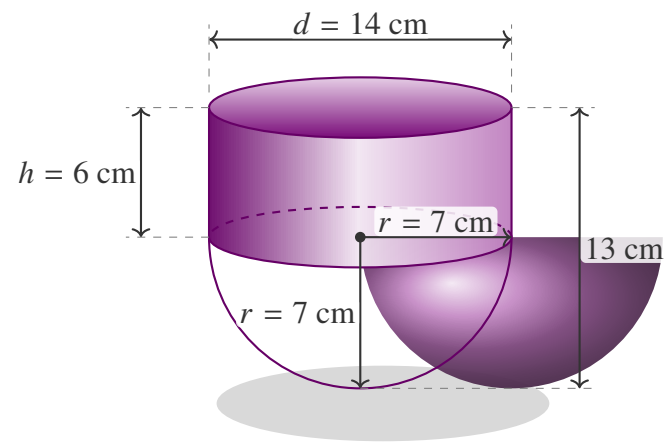
The outer surface area of the funnel is $324\pi \text{ cm}^2$ (or approximately 1017.88 cm^2).

15. Nathan, an engineering student was asked to make a model shaped like a cylinder with two cones attached at its two ends. The diameter of the model is 3 cm and its length is 12 cm. If each cone has a height of 2 cm, find the volume of the model that Nathan made. **APR-22, 25**

solution**Given : Cone :** Height $h = 2$ cm, radius $r = \frac{3}{2} = 1.5$ cm**Given : Cylinder :** Height $H = 8$ cm, radius $r = \frac{3}{2} = 1.5$ cm $\therefore$ Volume of the model = 2 (Vol. of Cone) + Vol. of Cylinder

$$\begin{aligned}
 &= \frac{2}{3}\pi r^2 h + \pi r^2 H \\
 &= \pi r^2 \left[\frac{2h}{3} + H \right] \\
 &= \frac{22}{7} \times \frac{9}{4} \left[\frac{4}{3} + 8 \right] \\
 &= \frac{11 \times 9}{7 \times 2} \left[\frac{28}{3} \right] \\
 &= \frac{11 \times 3 \times 14^2}{7} \\
 \text{Vol. of Model} &= 66 \text{ cm}^3
 \end{aligned}$$

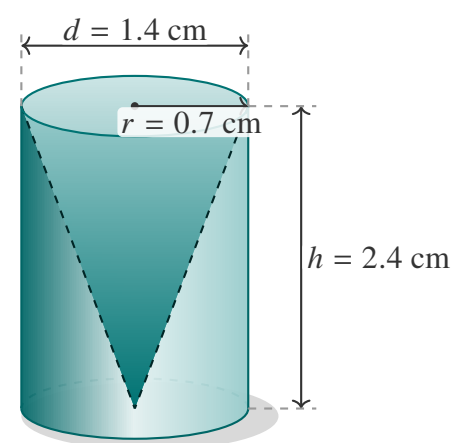
16. A vessel is in the form of a hemispherical bowl mounted by a hollow cylinder. The diameter is 14 cm and the height of the vessel is 13 cm. Find the capacity of the vessel.

solution**Hollow Vessel****Given : Cylinder & Hemisphere :**Diameter, $d = 14$ cmRadius, $r = \frac{14}{2} = 7$ cmTotal height of vessel, $H = 13$ cmThe height of the cylindrical part (h) is: $h = \text{Total height} - \text{Radius of hemisphere}$ $h = 13 \text{ cm} - 7 \text{ cm}$ $h = 6 \text{ cm}$ Total Capacity, $V = \text{Volume of Cylinder} + \text{Volume of Hemisphere}$

$$\begin{aligned}
 &= \pi r^2 h + \frac{2}{3}\pi r^3 \\
 V &= \left(\frac{22}{7} \times 7^2 \times 6 \right) + \left(\frac{2}{3} \times \frac{22}{7} \times 7^3 \right) \\
 &= \left(\frac{22}{7} \times 7 \times 7 \times 6 \right) + \left(\frac{2}{3} \times \frac{22}{7} \times 7 \times 7 \times 7 \right) \\
 &= (22 \times 7 \times 6) + \left(\frac{2}{3} \times 22 \times 49 \right) \\
 &= (154 \times 6) + \left(\frac{2}{3} \times 1078 \right) \\
 &= 924 + \frac{2156}{3} \\
 &= \frac{924 \times 3}{3} + \frac{2156}{3} \\
 &= \frac{2772 + 2156}{3} \\
 &= \frac{4928}{3} \\
 V &= 1642.66... \text{ cm}^3
 \end{aligned}$$

The capacity of the vessel is $\frac{4928}{3} \text{ cm}^3$ or approximately **1642.67 cm³**.

17. From a solid cylinder whose height is 2.4 cm and the diameter 1.4 cm, a cone of the same height and same diameter is carved out. Find the volume of the remaining solid to the nearest cm^3 .

solution**Remaining Solid**

Given : Cylinder & Cone :

Diameter, $d = 1.4$ cm

Radius, $r = \frac{1.4}{2} = 0.7 = \frac{7}{10}$ cm

Height, $h = 2.4 = \frac{24}{10}$ cm

$V_{\text{rem}} = \text{Volume of Cylinder} - \text{Volume of Cone}$

$$= \pi r^2 h - \frac{1}{3} \pi r^2 h$$

$$= \frac{2}{3} \pi r^2 h$$

$$= \frac{2}{3} \times \frac{22}{7} \times (0.7)^2 \times 2.4$$

$$= \frac{2}{3} \times \frac{22}{7} \times \left(\frac{7}{10}\right)^2 \times \frac{24}{10}$$

$$V_{\text{rem}} = \frac{2}{3} \times \frac{22}{7} \times \frac{7}{10} \times \frac{7}{10} \times \frac{24}{10}$$

$$V_{\text{rem}} = \frac{2 \times 22 \times 7 \times 8}{10 \times 10 \times 10}$$

$$= \frac{44 \times 56}{1000}$$

$$= \frac{2464}{1000}$$

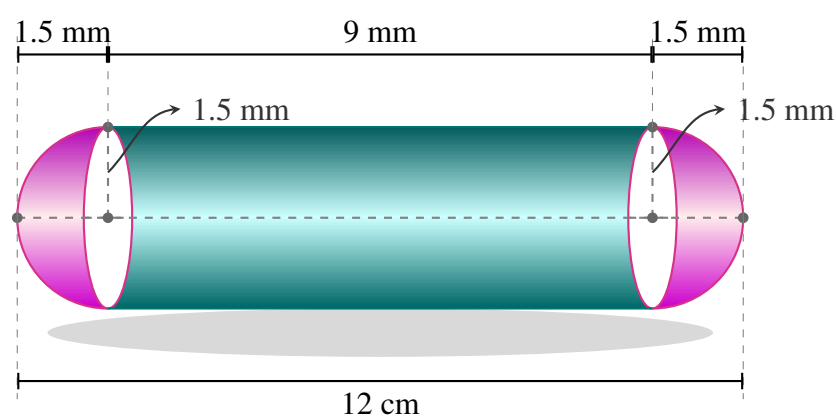
$$= \frac{2464}{1000}$$

$$= 2.464 \text{ cm}^3$$

The volume of the remaining solid to the nearest 2.464 or **2 cm³**

18. A capsule is in the shape of a cylinder with two hemisphere stuck to each of its ends. If the length of the entire capsule is 12 mm and the diameter of the capsule is 3 mm, how much medicine it can hold?

solution



Given : Cylinder: Height $H = 9$ mm, radius $r = \frac{3}{2} = 1.5$ mm.

Given : Hemisphere: radius $r = \frac{3}{2} = 1.5$ mm.

$\therefore$ Volume of the Capsule = Vol. of Cylinder + 2 (Vol. of hemisphere)

$$= \pi r^2 H + 2 \left(\frac{2}{3} \pi r^3 \right)$$

$$= \frac{22}{7} \left[\frac{9}{4} \times 9 + \frac{4}{3} \times \frac{27}{8} \right]$$

$$= \frac{22}{7} \left[\frac{81}{4} + \frac{9}{2} \right]$$

$$= \frac{22}{7} \left[\frac{81 + 18}{4} \right]$$

$$= \frac{22^{11} \times 99}{28_{14}}$$

$$= \frac{11 \times 99}{14}$$

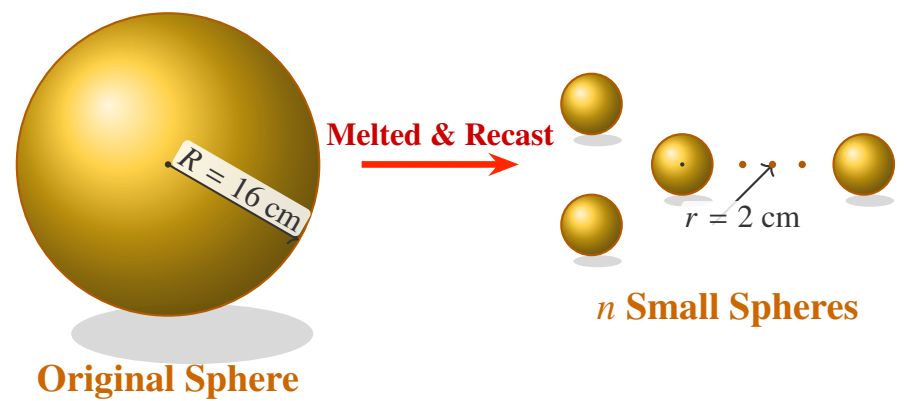
$$\text{Vol. of Capsule} = 77.78 \text{ mm}^3$$

$\therefore$ The Medicine hold on 77.78 cm^3 .

19. A metallic sphere of radius 16 cm is melted and recast into small spheres each of radius 2 cm. How many small spheres can be obtained?

APR-25, JUL-23, 24

solution



Given large metallic sphere radius R and Each small sphere radius r .
The total number of small spheres obtained be n .

$R = 16$ cm

$r = 2$ cm

Volume of Large Sphere = $n \times$ (Volume of one Small Sphere)

$$\frac{4}{3} \pi R^3 = n \times \left(\frac{4}{3} \pi r^3 \right)$$

$$R^3 = n \times r^3$$

$$= \frac{R^3}{r^3}$$

$$= \left(\frac{R}{r} \right)^3$$

$$= \left(\frac{16}{2} \right)^3$$

$$= (8)^3$$

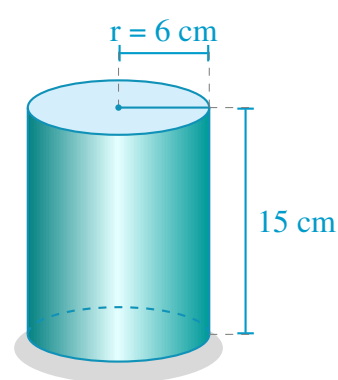
$$= 8 \times 8 \times 8$$

$$n = 512$$

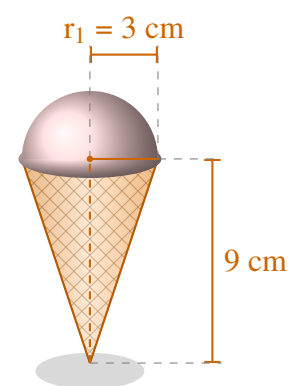
$\therefore$ **512** small spheres can be obtained.

20. A right circular cylindrical container of base radius 6 cm and height 15 cm is full of ice cream. The ice cream is to be filled in cones of height 9 cm and base radius 3 cm, having a hemispherical cap. Find the number of cones needed to empty the container. **APR-24, PTA-5, 6**

solution



Container



Ice Cream Cone

Given: Cylinder: Height $h = 15$ cm, , Radius $r = 6$ cm

Volume of the container $V = \pi r^2 h$ cubic units.

$$= \pi \times 6 \times 6 \times 15 \dots (1)$$

Given : Cone : Radius of Cone $r_1 = 3$ cm, Height of Cone $h_1 = 9$ cm

Given : Hemisphere Cap : Radius $r_1 = 3$ cm. Volume of one ice cream cone = (Volume of the cone + Volume of the hemispherical cap)

$$= \frac{1}{3} \pi r_1^2 h_1 + \frac{2}{3} \pi r_1^3$$

$$= \frac{1}{3} \times \pi \times 3 \times 3 \times 9 + \frac{2}{3} \times \pi \times 3 \times 3 \times 3$$

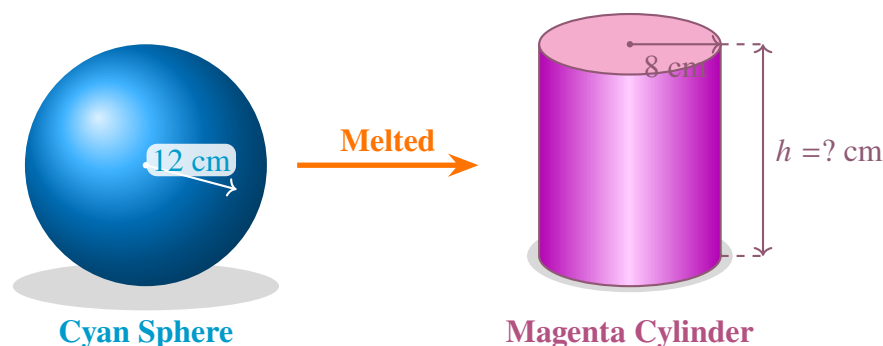
$$= \pi \times 9(3 + 2) = \pi \times 45 \dots (2)$$

$$\begin{aligned} \text{Number of cones} &= \frac{\text{Volume of the cylinder}}{\text{Volume of one ice cream cone}} \\ &= \frac{\pi \times 6^2 \times 15}{\pi \times 45} \\ &= 12 \end{aligned}$$

$\therefore$ 12 ice cream cones are required to empty the cylindrical container.

21. An aluminium sphere of radius 12 cm is melted to make a cylinder of radius 8 cm. Find the height of the cylinder. **JUL-25**

solution



Given radius of the aluminium sphere r_1 and the radius of the newly formed cylinder r_2 . The height of the cylinder h .

$$r_1 = 12 \text{ cm}$$

$$r_2 = 8 \text{ cm}$$

Volume of Cylinder = Volume of Sphere

$$\pi r_2^2 h = \frac{4}{3} \pi r_1^3$$

$$r_2^2 h = \frac{4}{3} r_1^3$$

$$(8)^2 \times h = \frac{4}{3} \times (12)^3$$

$$8 \times 8 \times h = \frac{4}{3} \times 12 \times 12 \times 12$$

$$8 \times 8 \times h = 4 \times 4 \times 12 \times 12$$

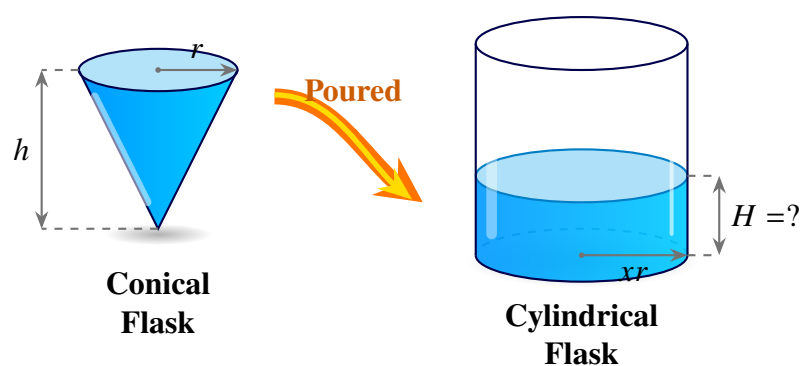
$$h = \frac{4 \times 4 \times 12 \times 12}{8 \times 8}$$

$$h = 36$$

The height of the newly formed cylinder is **36 cm**.

22. A conical flask is full of water. The flask has base radius r units and height h units, the water is poured into a cylindrical flask of base radius xr units. Find the height of water in the cylindrical flask.

solution



Given volume of the water in conical flask V_1 and the volume of the water in the cylindrical flask V_2 .

Given Conical flask: radius r , Height h .

The volume of Conical Flask:

$$V_1 = \frac{1}{3} \pi (\text{radius})^2 (\text{height})$$

$$V_1 = \frac{1}{3} \pi r^2 h$$

Given cylindrical flask: Radius $R = xr$, Height H . The volume of water in the cylindrical flask is:

$$V_2 = \pi R^2 H$$

$$V_2 = \pi (xr)^2 H$$

$$V_2 = \pi x^2 r^2 H$$

$$V_2 = V_1$$

$$\pi x^2 r^2 H = \frac{1}{3} \pi r^2 h$$

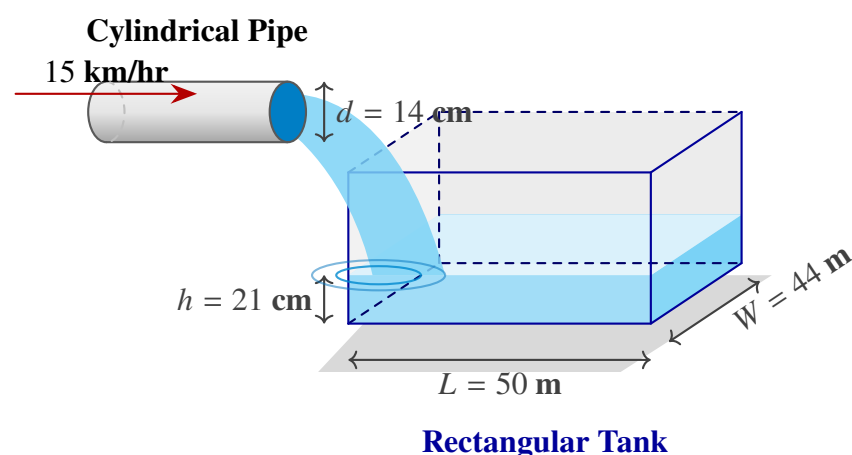
$$x^2 H = \frac{1}{3} h$$

$$H = \frac{h}{3x^2}$$

The height of the water in the cylindrical flask is $\frac{h}{3x^2}$ units.

23. Water is flowing at the rate of 15 km per hour through a pipe of diameter 14 cm into a rectangular tank which is 50 m long and 44 m wide. Find the time in which the level of water in the tanks will rise by 21 cm.

solution



Given : cylindrical pipe:

Diameter $d = 14 \text{ cm}$

$$\text{Radius, } r = \frac{14}{2} = 7 \text{ cm} = \frac{7}{100} \text{ m}$$

$$\text{Rate of flow } v = 15 \text{ km/hr} \times 1000 = 15000 \text{ m/hr}$$

$$\text{Height } h = 15000 \text{ m}$$

Given Rectangular tank:

Length, $L = 50 \text{ m}$

Width, $W = 44 \text{ m}$

$$\text{Required rise in height, } H = 21 \text{ cm} = \frac{21}{100} \text{ m}$$

$$\begin{aligned} \text{Time} &= \frac{\text{Vol. of Rectangle tank}}{\text{Vol. of Cylindrical Pipe}} \\ &= \frac{L \times W \times H}{\pi r^2 h} \\ &= \frac{50 \times 44 \times \frac{21}{100}}{\pi \times \left(\frac{7}{100}\right)^2 \times 15000} \\ &= \frac{\frac{22}{7} \times \frac{7}{100} \times \frac{7}{100} \times 15000}{22 \times 21} \\ &= \frac{22 \times 21}{11 \times 7 \times 3} \end{aligned}$$

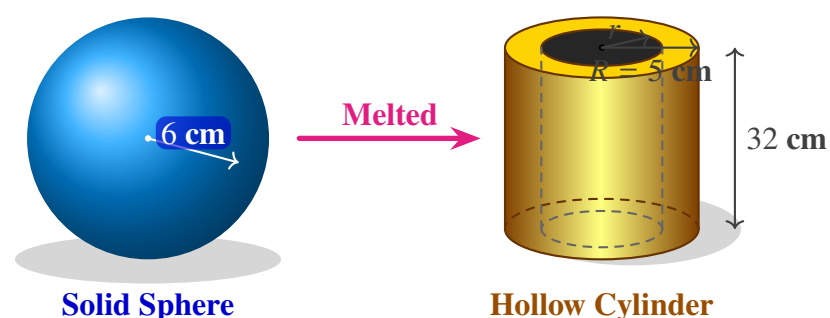
$$\text{Time Required} = 2$$

The water level in the tank will rise by 21 cm in **2 hours**.

24. A solid right circular cone of diameter 14 cm and height 8 cm is melted to form a hollow sphere. If the external diameter of the sphere is 10 cm, Find the internal diameter.

25. A solid sphere of radius 6 cm is melted into a hollow cylinder of uniform thickness. If the external radius of the base of the cylinder is 5 cm and its height is 32 cm, then find the thickness of the cylinder.

solution



Given: Solid Sphere: Radius $r_1 = 6 \text{ cm}$,

Hollow Cylinder : External Radius $R = 5 \text{ cm}$, Height $h = 32 \text{ cm}$, Internal Radius = ?..

Volume of Solid Sphere = Volume of Hollow Cylinder

$$\frac{4}{3} \pi R_s^3 = \pi h (R^2 - r^2)$$

$$\frac{4}{3} \pi (6)^3 = \pi (32) (5^2 - r^2)$$

$$\begin{aligned}\frac{4}{3} \times 216 &= 32(25 - r^2) \\ 4 \times 72 &= 32(25 - r^2) \\ 288 &= 32(25 - r^2) \\ \frac{288}{32} &= 25 - r^2 \\ 9 &= 25 - r^2 \\ r^2 &= 25 - 9 \\ r^2 &= 16\end{aligned}$$

$$r = 4 \text{ cm}$$

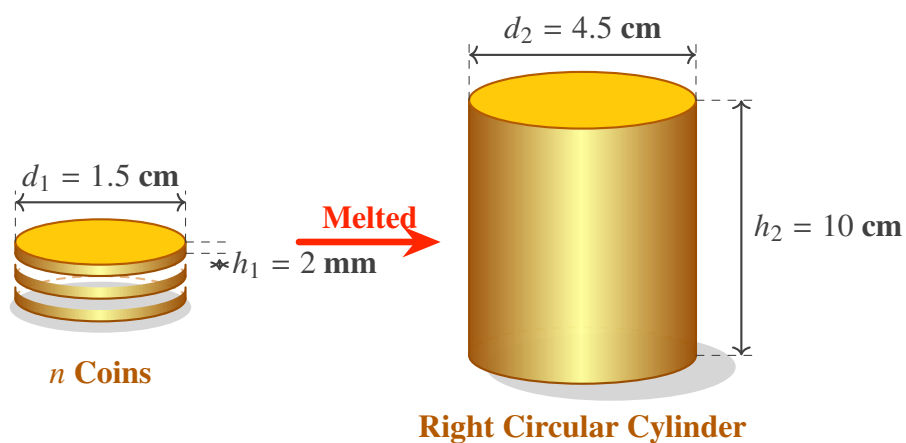
The internal radius of the cylinder is 4 cm.

$$\begin{aligned}\text{Thickness} &= R - r \\ &= 5 \text{ cm} - 4 \text{ cm} \\ T &= 1 \text{ cm}\end{aligned}$$

The thickness of the hollow cylinder is **1 cm**.

26. Find the number of coins, 1.5 cm in diameter and 2 mm thick, to be melted to form a right circular cylinder of height 10 cm and diameter 4.5 cm. **PTA-1**

solution



Given n number of coins needed to form the right circular cylinder.

For a single coin:

$$\begin{aligned}\text{Diameter, } d_1 &= 1.5 \text{ cm} \\ \text{Radius, } r_1 &= \frac{1.5}{2} \text{ cm} = \frac{15}{20} \text{ cm} = \frac{3}{4} \text{ cm} \\ \text{Thickness (Height), } h_1 &= 2 \text{ mm} = \frac{2}{10} \text{ cm} = \frac{1}{5} \text{ cm}\end{aligned}$$

Given right circular cylinder:

$$\begin{aligned}\text{Diameter, } d_2 &= 4.5 \text{ cm} \\ \text{Radius, } r_2 &= \frac{4.5}{2} \text{ cm} = \frac{45}{20} \text{ cm} = \frac{9}{4} \text{ cm} \\ \text{Height, } h_2 &= 10 \text{ cm}\end{aligned}$$

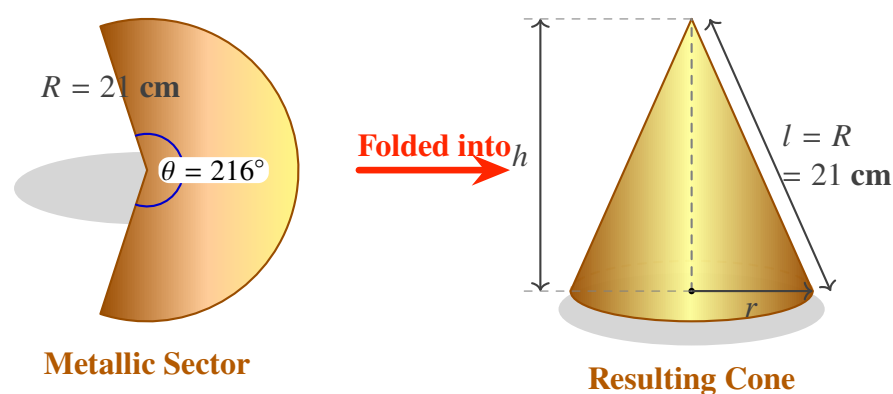
$$n \times (\text{Volume of one coin}) = \text{Volume of the large cylinder}$$

$$\begin{aligned}n \times \pi r_1^2 h_1 &= \pi r_2^2 h_2 \\ n \times \pi \left(\frac{3}{4}\right)^2 \times \frac{1}{5} &= \pi \left(\frac{9}{4}\right)^2 \times 10 \\ n \times \pi \times \frac{9}{16} \times \frac{1}{5} &= \pi \times \frac{81}{16} \times 10 \\ n &= \frac{\pi \times \frac{81}{16} \times 10}{\pi \times \frac{9}{16} \times \frac{1}{5}} \\ &= \frac{\pi \times \frac{81}{16} \times 10}{\pi \times \frac{9}{16} \times \frac{1}{5}} \\ &= \frac{81 \times 10}{9 \times \frac{1}{5}} \\ &= 810 \times \frac{5}{9} \\ &= 90 \times 5 \\ n &= 450\end{aligned}$$

The exactly **450 coins** need to be melted to form the cylinder.

27. A metallic sheet in the form of a sector of a circle of radius 21 cm has central angle of 216° . The sector is made into a cone by bringing the bounding radii together. Find the volume of the cone formed. **PTA-2**

solution



Given: Metallic sector:

Radius of the sector, $R = 21 \text{ cm} \implies$ Slant height of cone, $l = 21 \text{ cm}$
Central angle, $\theta = 216^\circ$

To Find the radius of the cone (r)

Circumference of cone base = Arc length of sector

$$\begin{aligned}2\pi r &= \frac{\theta}{360^\circ} \times 2\pi R \\ &= \frac{216}{360} \times 21 \\ &= \frac{3}{5} \times 21 \\ r &= \frac{63}{5} \text{ cm (or } 12.6 \text{ cm)}\end{aligned}$$

To Find the height of the cone (h)

$$\begin{aligned}h &= \sqrt{l^2 - r^2} \\ &= \sqrt{21^2 - \left(\frac{63}{5}\right)^2} \\ h &= \sqrt{441 - \frac{3969}{25}} \\ &= \sqrt{\frac{11025 - 3969}{25}} \\ &= \sqrt{\frac{7056}{25}} \\ h &= \frac{84}{5} \text{ cm (or } 16.8 \text{ cm)}\end{aligned}$$

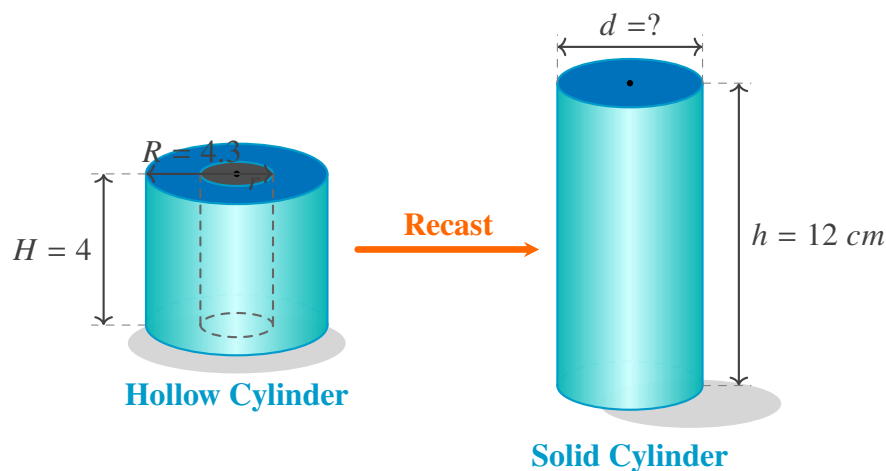
To Calculate the volume of the cone (V)

$$\begin{aligned}\text{Vol of Cone } V &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \times \frac{22}{7} \times \left(\frac{63}{5}\right)^2 \times \left(\frac{84}{5}\right) \\ V &= \frac{1}{3} \times \frac{22}{7} \times \frac{63}{5} \times \frac{63}{5} \times \frac{84}{5} \\ &= \frac{22 \times 63 \times 63 \times 84}{(3 \times 7) \times (5 \times 5 \times 5)} \\ &= \frac{22 \times 63 \times 63 \times 84}{21 \times 125} \\ &= \frac{22 \times 63 \times 63 \times 84}{21 \times 125} \\ &= \frac{22 \times 63 \times 63 \times 4}{125} \\ &= \frac{349272}{125} \\ V &= 2794.176 \text{ cm}^3\end{aligned}$$

$\therefore$ The volume of the cone formed is **2794.176 cm³**.

28. A hollow metallic cylinder whose external radius is 4.3 cm and internal radius is 1.1 cm and whole length is 4 cm is melted and recast into a solid cylinder of 12 cm long. Find the diameter of solid cylinder.

solution



Given: Hollow Cylinder

R = the external radius, r = the internal radius, and H = length of the hollow cylinder.

Given: Solid Cylinder r_s = the radius and h = length of the new solid cylinder.

$$R = 4.3 \text{ cm} \implies r = 1.1 \text{ cm}$$

$$H = 4 \text{ cm} \implies h = 12 \text{ cm}$$

Volume of Solid Cylinder = Volume of Hollow Cylinder

$$\pi r_s^2 h = \pi (R^2 - r^2) H$$

$$\pi r_s^2 h = \pi (R^2 - r^2) H$$

$$r_s^2 h = (R^2 - r^2) H$$

$$r_s^2 \times 12 = ((4.3)^2 - (1.1)^2) \times 4$$

$$r_s^2 \times 12 = (18.49 - 1.21) \times 4$$

$$r_s^2 \times 12 = 17.28 \times 4$$

$$= \frac{17.28 \times 4}{12}$$

$$= \frac{17.28 \times 4}{12}$$

$$= \frac{17.28}{3}$$

$$= 5.76$$

$$r_s = \sqrt{5.76}$$

$$r_s = 2.4 \text{ cm}$$

The diameter (d) of the solid cylinder is twice its radius:

$$d = 2 \times r_s = 2 \times 2.4$$

$$= 4.8 \text{ cm}$$

The diameter of the solid cylinder is **4.8 cm**.

CHAPTER - 8. STATISTICS & PROBABILITY

CHAPTER - 8. STATISTICS & PROBABILITY

2 MARK IMPORTANT QUESTIONS

1. Find the range and coefficient of range of the following data: 25, 67, 48, 53, 18, 39, 44. **APR-24**

solution

Given data: 25, 67, 48, 53, 18, 39, 44.

$$\text{Largest value } (L) = 67$$

$$\text{Smallest value } (S) = 18$$

$$\text{Range} = L - S$$

$$= 67 - 18$$

$$R = 49$$

$$\begin{aligned} \text{Coefficient of Range} &= \frac{L - S}{L + S} \\ &= \frac{67 - 18}{67 + 18} \\ &= \frac{49}{85} \end{aligned}$$

$$C.R \approx 0.576$$

2. Find the range of the following distribution. **APR-25, PTA-5**

Age (in years)	16-18	18-20	20-22	22-24	24-26	26-28
Number of students	0	4	6	8	2	2

solution

Given frequency of the class interval 16 – 18 is 0. We don't consider it. The lowest class interval with a non-zero frequency is 18 – 20.

- Largest value (L) = Upper limit of the highest class interval = 28
- Smallest value (S) = Lower limit of the lowest class interval (with non-zero frequency) = 18

$$\text{Range } (R) = L - S$$

$$R = 28 - 18 = 10$$

The range of the given distribution is 10 years.

3. The range of a set of data is 13.67 and the largest value is 70.08. Find the smallest value. **PTA-4**

solution

Given Data: Range (R) = 13.67, Largest value (L) = 70.08.

$$\text{Range} = L - S$$

$$13.67 = 70.08 - S$$

$$S = 70.08 - 13.67$$

$$S = 56.41$$

$$\text{The smallest value } (S) = 56.41$$

4. Find the range and coefficient of range of the following data.

(i) 63, 89, 98, 125, 79, 108, 117, 68 **APR-23, JUL-20**

(ii) 43.5, 13.6, 18.9, 38.4, 61.4, 29.8

solution

(i) Given data: 63, 89, 98, 125, 79, 108, 117, 68

Largest value (L) = 125

Smallest value (S) = 63

$$\begin{aligned}
 \text{Range } (R) &= L - S \\
 &= 125 - 63 \\
 R &= 62 \\
 \text{Coefficient of Range} &= \frac{L - S}{L + S} \\
 &= \frac{125 - 63}{125 + 63} \\
 &= \frac{62}{188} \\
 C.R &\approx 0.33 \\
 \text{(ii) Given data: } &43.5, 13.6, 18.9, 38.4, 61.4, 29.8 \\
 \text{Largest value } (L) &= 61.4 \\
 \text{Smallest value } (S) &= 13.6 \\
 \text{Range } (R) &= L - S \\
 &= 61.4 - 13.6 \\
 R &= 47.8 \\
 \text{Coefficient of Range} &= \frac{L - S}{L + S} \\
 &= \frac{61.4 - 13.6}{61.4 + 13.6} \\
 &= \frac{47.8}{75} \\
 C.R &\approx 0.637
 \end{aligned}$$

5. If the range and the smallest value of a set of data are 36.8 and 13.4 respectively, then find the largest value. **JUL-2**

solution

$$\begin{aligned}
 \text{Given data:} \\
 \text{Range } (R) &= 36.8 \\
 \text{Smallest value } (S) &= 13.4 \\
 \text{Range} &= L - S \\
 36.8 &= L - 13.4 \\
 L &= 36.8 + 13.4 \\
 L &= 50.2 \\
 \text{The largest value } (L) &= 50.2
 \end{aligned}$$

6. Calculate the range of the following data.

Income	400-450	450-500	500-550	550-600	600-650
No.of workers	12	30	8	21	6

solution

$$\begin{aligned}
 \text{Largest value } (L) &= \text{Upper limit of the highest C.I} \\
 L &= 650 \\
 \text{Smallest value } (S) &= \text{Lower limit of the lowest C.L} \\
 S &= 400 \\
 \text{Range } (R) &= L - S \\
 &= 650 - 400 \\
 R &= 250
 \end{aligned}$$

7. Find the standard deviation of first 21 natural numbers. **JUL-23, 24, PTA-6, MAR-26**

solution

$$\begin{aligned}
 \text{No.of natural numbers } (n) &= 21 \\
 \text{S.D of first } n \text{ natural numbers } (\sigma) &= \sqrt{\frac{n^2 - 1}{12}} \\
 \sigma &= \sqrt{\frac{21^2 - 1}{12}} \\
 &= \sqrt{\frac{441 - 1}{12}} \\
 &= \sqrt{\frac{440}{12}} \\
 &= \sqrt{36.66} \\
 S.D &\approx 6.05
 \end{aligned}$$

8. The mean of a data is 25.6 and its coefficient of variation is 18.75. Find the standard deviation. **PTA-3**

solution

$$\begin{aligned}
 \text{Given data:} \\
 \text{Mean } (\bar{x}) &= 25.6 \\
 \text{Coefficient of Variation } (C.V.) &= 18.75 \\
 \text{Coefficient of Variation:} \\
 C.V. &= \frac{\sigma}{\bar{x}} \times 100 \\
 18.75 &= \frac{\sigma}{25.6} \times 100 \\
 \sigma &= \frac{18.75 \times 25.6}{100} \\
 \sigma &= \frac{480}{100} \\
 \sigma &= 4.8 \\
 \text{The standard deviation } (\sigma) &= 4.8
 \end{aligned}$$

9. The standard deviation and mean of a data are 6.5 and 12.5 respectively. Find the coefficient of variation. **MDL-19**

solution

$$\begin{aligned}
 \text{Given data:} \\
 \text{Standard deviation } (\sigma) &= 6.5 \\
 \text{Mean } (\bar{x}) &= 12.5 \\
 \text{Coefficient of Variation:} \\
 C.V. &= \frac{\sigma}{\bar{x}} \times 100\% \\
 &= \frac{6.5}{12.5} \times 100\% \\
 &= \frac{65}{125} \times 100\% \\
 &= 0.52 \times 100\% \\
 C.V &= 52\%
 \end{aligned}$$

10. If the mean and coefficient of variation of a data are 15 and 48 respectively, then find the value of standard deviation.

solution

$$\begin{aligned}
 \text{Given data:} \\
 \text{Mean } (\bar{x}) &= 15 \\
 \text{Coefficient of Variation } (C.V.) &= 48 \\
 \text{Coefficient of Variation:} \\
 C.V. &= \frac{\sigma}{\bar{x}} \times 100 \\
 48 &= \frac{\sigma}{15} \times 100 \\
 \sigma &= \frac{48 \times 15}{100} \\
 \sigma &= \frac{720}{100} \\
 \sigma &= 7.2 \\
 \text{The standard deviation } (\sigma) &= 7.2
 \end{aligned}$$

11. If $n = 5$, $\bar{x} = 6$, $\Sigma x^2 = 765$, then calculate the coefficient of variation. **solution**

$$\begin{aligned}
 \text{Given data:} \\
 n &= 5 \\
 \bar{x} &= 6 \\
 \Sigma x^2 &= 765 \\
 \text{Standard Deviation } (\sigma) : \\
 \sigma &= \sqrt{\frac{\Sigma x^2}{n} - (\bar{x})^2} \\
 &= \sqrt{\frac{765}{5} - (6)^2} \\
 &= \sqrt{153 - 36} \\
 &= \sqrt{117} \\
 S.D &\approx 10.82
 \end{aligned}$$

$$\begin{aligned}\text{Coefficient of Variation (C.V.)} &= \frac{\sigma}{\bar{x}} \times 100\% \\ &= \frac{10.82}{6} \times 100\% \\ &= 1.8033 \times 100\% \\ \text{C.V.} &\approx 180.33\%\end{aligned}$$

12. Two coins are tossed together. What is the probability of getting different faces on the coins? **APR-22, JUL-25**

solution

$$\begin{aligned}\text{Sample space (S)} &= \{HH, HT, TH, TT\} \\ n(S) &= 4\end{aligned}$$

Let A be the event of getting different faces.

$$\begin{aligned}A &= \{HT, TH\} \\ n(A) &= 2 \\ \text{Probability } P(A) &= \frac{n(A)}{n(S)} \\ P(A) &= \frac{2}{4} \\ P(A) &= \frac{1}{2}\end{aligned}$$

13. What is the probability that a leap year selected at random will contain 53 Saturdays.

solution

Number of days in a leap year = 366.

366 days = 52 weeks and 2 days. Sample space for the remaining 2 days (S):

$$\begin{aligned}S &= \{(\text{Sun, Mon}), (\text{Mon, Tue}), (\text{Tue, Wed}), (\text{Wed, Thu}), \\ &\quad (\text{Thu, Fri}), (\text{Fri, Sat}), (\text{Sat, Sun})\} \\ n(S) &= 7\end{aligned}$$

Let A be the event of getting 53 Saturdays.

One of the remaining 2 days must be a Saturday.

$$\begin{aligned}A &= \{(\text{Fri, Sat}), (\text{Sat, Sun})\} \\ n(A) &= 2 \\ \text{Probability } P(A) &= \frac{n(A)}{n(S)} \\ P(A) &= \frac{2}{7}\end{aligned}$$

14. What is the probability that a non-leap year selected at random will contain 53 Saturdays.

solution

Number of days in a non-leap year = 365.

365 days = 52 weeks and 1 day. Sample space for the remaining 1 day (S):

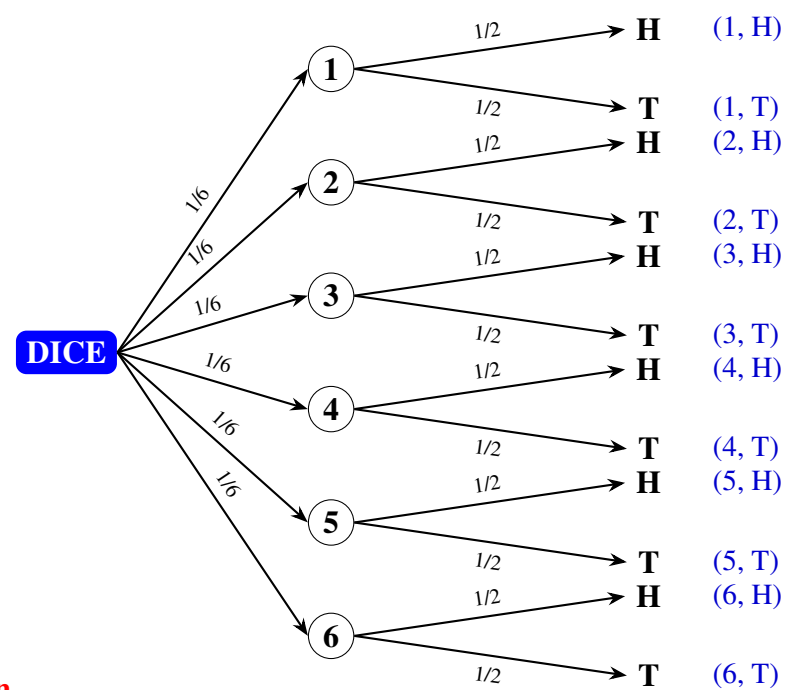
$$\begin{aligned}S &= \{\text{Sun, Mon, Tue, Wed, Thu, Fri, Sat}\} \\ n(S) &= 7\end{aligned}$$

Let A be the event of getting 53 Saturdays.

For this, the remaining 1 day must be a Saturday.

$$\begin{aligned}A &= \{\text{Sat}\} \\ n(A) &= 1 \\ \text{Probability } P(A) &= \frac{n(A)}{n(S)} \\ P(A) &= \frac{1}{7}\end{aligned}$$

15. A die is rolled and a coin is tossed simultaneously. Find the probability that the die shows an odd number and the coin shows a head. **JUL-21, 22, MAR-26**



solution

Sample space (S):

$$S = \{(1, H), (2, H), (3, H), (4, H), (5, H), (6, H), (1, T), (2, T), (3, T), (4, T), (5, T), (6, T)\}$$

$$n(S) = 12$$

Let A be the event of getting an odd number on the die and a head on the coin.

$$\begin{aligned}A &= \{(1, H), (3, H), (5, H)\} \\ n(A) &= 3\end{aligned}$$

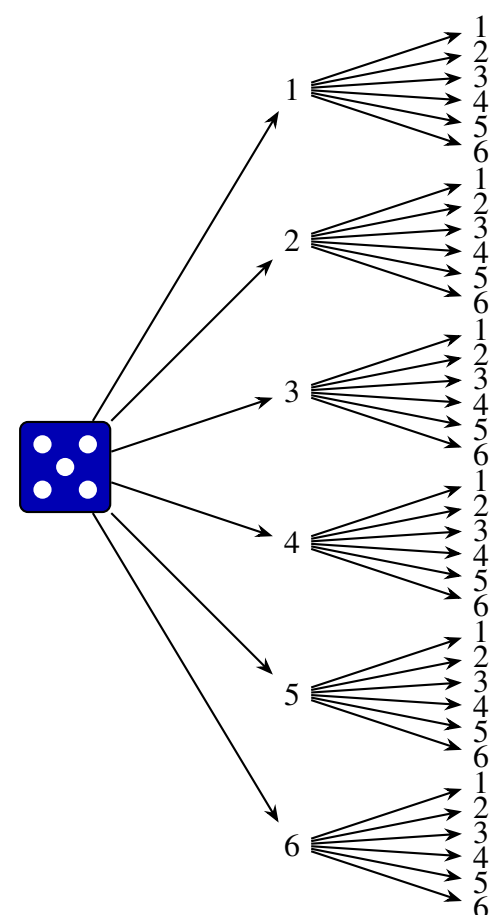
$$\text{Probability } P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{3}{12}$$

$$P(A) = \frac{1}{4}$$

16. Express the sample space for rolling two dice using tree diagram.

solution

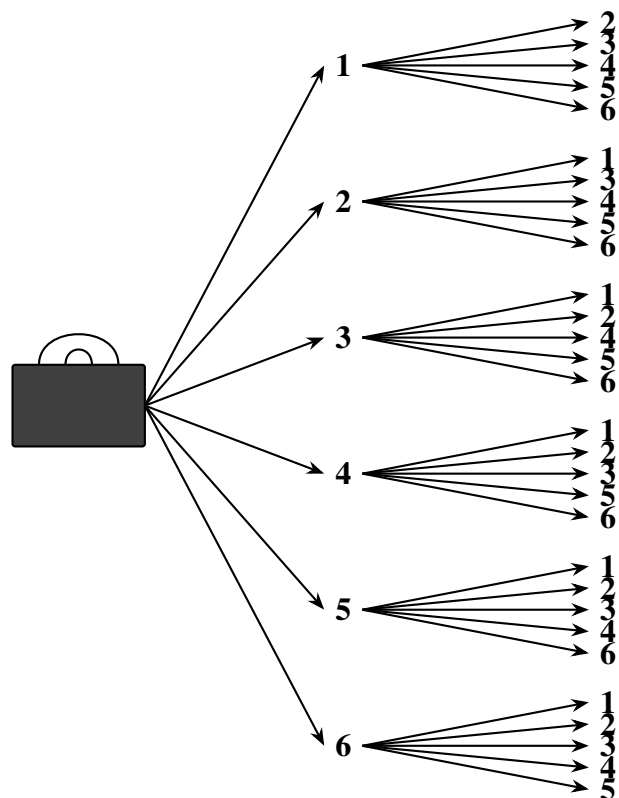


$$\begin{aligned}\text{Sample Space (S)} &= \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ &\quad (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ &\quad (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), \\ &\quad (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ &\quad (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), \\ &\quad (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}\end{aligned}$$

$$n(S) = 36$$

17. Write the sample space for selecting two balls at a time from a bag containing 6 balls numbered 1 to 6 (using tree diagram). **PTA-4**

solution



Sample Space (S) = $\{(1, 2), (1, 3), (1, 4), (1, 5), (1, 6),$
 $(2, 1), (2, 3), (2, 4), (2, 5), (2, 6),$
 $(3, 1), (3, 2), (3, 4), (3, 5), (3, 6),$
 $(4, 1), (4, 2), (4, 3), (4, 5), (4, 6),$
 $(5, 1), (5, 2), (5, 3), (5, 4), (5, 6),$
 $(6, 1), (6, 2), (6, 3), (6, 4), (6, 5)\}$
 $n(S) = 30$

18. A coin is tossed thrice. What is the probability of getting two consecutive tails?.

solution

Sample space (S):

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$
 $n(S) = 8$

Let A be the event of getting two consecutive tails.

$A = \{HTT, TTH, TTT\}$
 $n(A) = 3$

Probability $P(A) = \frac{n(A)}{n(S)}$
 $P(A) = \frac{3}{8}$

19. If $P(A) = 0.37$, $P(B) = 0.42$, $P(A \cap B) = 0.09$ then find $P(A \cup B)$.

JUL-24

solution

Given data:

$P(A) = 0.37$ $P(B) = 0.42$ $P(A \cap B) = 0.09$

WKT The Addition Theorem of Probability:

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= 0.37 + 0.42 - 0.09$
 $= 0.79 - 0.09$
 $P(A \cup B) = 0.70$

20. A flower is selected at random from a basket containing 80 yellow, 70 red and 50 white flowers. Find the probability of selecting a yellow or red flower? **APR-25**

solution

Number of yellow flowers, $n(Y) = 80$

Number of red flowers, $n(R) = 70$

Number of white flowers, $n(W) = 50$ Total number of flowers in the sample space (S):

$$n(S) = 80 + 70 + 50$$

$$n(S) = 200$$

Let Y be the event of selecting a yellow flower, and R be the event of selecting a red flower.

$$P(Y) = \frac{n(Y)}{n(S)} = \frac{80}{200}$$

$$P(R) = \frac{n(R)}{n(S)} = \frac{70}{200}$$

Since selecting a yellow flower and a red flower are mutually exclusive events, $P(Y \cap R) = 0$.

Probability of selecting a yellow or red flower:

$$\begin{aligned} P(Y \cup R) &= P(Y) + P(R) \\ &= \frac{80}{200} + \frac{70}{200} \\ &= \frac{150}{200} \\ P(Y \cup R) &= \frac{3}{4} \end{aligned}$$

21. A and B are two candidates seeking admission to IIT. The probability that A getting selected is 0.5 and the probability that both A and B getting selected is 0.3. Prove that the probability of B being selected is at most 0.8. **APR-23, PTA-6**

solution

Given data:

$$P(A) = 0.5 \quad P(A \cap B) = 0.3$$

The maximum probability of any event ($A \cup B$) cannot exceed 1.

$$P(A \cup B) \leq 1$$

The addition theorem of probability:

$$\begin{aligned} P(A) + P(B) - P(A \cap B) &\leq 1 \\ 0.5 + P(B) - 0.3 &\leq 1 \\ 0.2 + P(B) &\leq 1 \\ P(B) &\leq 1 - 0.2 \\ P(B) &\leq 0.8 \end{aligned}$$

The probability of B being selected is at most 0.8.

Hence proved.

22. If $P(A) = \frac{2}{3}$, $P(B) = \frac{2}{5}$, $P(A \cup B) = \frac{1}{3}$ then find $P(A \cap B)$. **PTA-1**

solution

Given data:

$$P(A) = \frac{2}{3}, \quad P(B) = \frac{2}{5}, \quad P(A \cup B) = \frac{1}{3}$$

The Addition Theorem of probability:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\ &= \frac{2}{3} + \frac{2}{5} - \frac{1}{3} \\ &= \left(\frac{2}{3} - \frac{1}{3}\right) + \frac{2}{5} \\ &= \frac{1}{3} + \frac{2}{5} \\ &= \frac{5+6}{15} \\ P(A \cap B) &= \frac{11}{15} \end{aligned}$$

23. If A and B are two mutually exclusive events of a random experiment and $P(\text{not } A) = 0.45$, $P(A \cup B) = 0.65$ then find $P(B)$. **PTA-2**

solution

Given data:

$$P(\text{not } A) \text{ or } P(\bar{A}) = 0.45, \quad P(A \cup B) = 0.65$$

WKT $P(A) = 1 - P(\bar{A})$:

$$P(A) = 1 - 0.45$$

$$P(A) = 0.55$$

Since A and B are mutually exclusive events, $P(A \cap B) = 0$. The Addition Theorem of probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.65 = 0.55 + P(B)$$

$$P(B) = 0.65 - 0.55$$

$$P(B) = 0.10$$

24. The probability that atleast one of A and B occur is 0.6. If A and B occur simultaneously with probability 0.2, then find $P(\bar{A}) + P(\bar{B})$

solution Given data:

$$P(A \cup B) = 0.6, \quad P(A \cap B) = 0.2$$

To find $P(\bar{A}) + P(\bar{B})$:

$$P(\bar{A}) + P(\bar{B}) = (1 - P(A)) + (1 - P(B))$$

$$= 2 - (P(A) + P(B))$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Rearranging to find $P(A) + P(B)$:

$$P(A) + P(B) = P(A \cup B) + P(A \cap B)$$

$$P(A) + P(B) = 0.6 + 0.2$$

$$P(A) + P(B) = 0.8$$

$$P(\bar{A}) + P(\bar{B}) = 2 - 0.8$$

$$= 1.2$$

25. If the range and coefficient of range of the data are 20 and 0.2 respectively, then find the largest and smallest values of the data.

solution

Given data: Range (R) = 20

Coefficient of Range ($C.R.$) = 0.2

To Use Range Formula:

$$R = L - S$$

$$L - S = 20 \quad \text{--- (1)}$$

To use Coefficient of Range Formula:

$$C.R. = \frac{L - S}{L + S}$$

$$0.2 = \frac{20}{L + S}$$

$$L + S = \frac{20}{0.2}$$

$$L + S = 100 \quad \text{--- (2)}$$

$$(1 + 2)(L - S) + (L + S) = 20 + 100$$

$$2L = 120$$

$$L = 60$$

$$60 + S = 100$$

$$S = 100 - 60$$

$$S = 40$$

The largest value (L) = 60, The smallest value (S) = 40.

26. In a two children family, find the probability that there is at least one girl in a family.

solution

Let B denote a boy and G denote a girl.

Sample space (S):

$$S = \{(B, B), (B, G), (G, B), (G, G)\}$$

$$n(S) = 4$$

Let A be the event of having at least one girl in the family.

$$A = \{(B, G), (G, B), (G, G)\}$$

$$n(A) = 3$$

$$\text{Probability } P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{3}{4}$$

27. A bag contains 5 white and some black balls. If the probability of drawing a black ball from the bag is twice the probability of drawing a white ball then find the number of black balls.

solution

Let the number of black balls be x .

Number of white balls = 5

Total number of balls in the bag, $n(S) = 5 + x$.

Probability of drawing a white ball, $P(W)$, and a black ball, $P(B)$:

$$P(W) = \frac{5}{5+x}, \quad P(B) = \frac{x}{5+x}$$

According to the given condition:

$$P(B) = 2 \times P(W)$$

$$\frac{x}{5+x} = 2 \times \left(\frac{5}{5+x} \right)$$

$$\frac{x}{5+x} = \frac{10}{5+x}$$

$$x = 10$$

The number of black balls in the bag is 10.

28. In a box there are 20 non-defective and some defect bulbs. If the probability that a bulb selected at random from the box found to be defective is $\frac{3}{8}$ then, find the number of defective bulbs.

solution

Let the number of defective bulbs be x .

Number of non-defective bulbs = 20

Total number of bulbs in the box, $n(S) = 20 + x$.

Let D be the event of selecting a defective bulb.

$$P(D) = \frac{x}{20+x}$$

$$\text{Given } P(D) = \frac{3}{8} :$$

$$\frac{x}{20+x} = \frac{3}{8}$$

$$8x = 3(20+x)$$

$$8x = 60 + 3x$$

$$8x - 3x = 60$$

$$5x = 60$$

$$x = \frac{60}{5}$$

$$x = 12$$

The number of defective bulbs is 12.

CHAPTER-8. STATISTICS & PROBABILITY

5 MARK IMPORTANT QUESTIONS

1. The number of televisions sold in each day of a week are 13, 8, 4, 9, 7, 12, 10. Find its standard deviation.

solution

Given data: 13, 8, 4, 9, 7, 12, 10

Number of observations (n) = 7

Using the direct method, the formula for standard deviation is:

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

x	x^2
13	169
8	64
4	16
9	81
7	49
12	144
10	100
$\sum x = 63$	$\sum x^2 = 623$

Standard Deviation:

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2} \\ &= \sqrt{\frac{623}{7} - \left(\frac{63}{7}\right)^2} \\ &= \sqrt{89 - (9)^2} \\ &= \sqrt{89 - 81} \\ &= \sqrt{8} \\ \sigma &\approx 2.828\end{aligned}$$

The standard deviation of the number of televisions sold is approximately 2.83.

2. The marks scored by 10 students in a class test are 25, 29, 30, 33, 35, 37, 38, 40, 44, 48. Find the standard deviation.

solution

Given marks (x): 25, 29, 30, 33, 35, 37, 38, 40, 44, 48

Number of students (n) = 10 Let the assumed mean (A) be 35.

Using the Assumed Mean method, the formula for standard deviation is:

$$\sigma = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2}$$

where $d = x - A$.

x	$d = x - 35$	d^2
25	-10	100
29	-6	36
30	-5	25
33	-2	4
35	0	0
37	2	4
38	3	9
40	5	25
44	9	81
48	13	169
Total	$\sum d = 9$	$\sum d^2 = 453$

Standard Deviation:

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2} \\ &= \sqrt{\frac{453}{10} - \left(\frac{9}{10}\right)^2} \\ &= \sqrt{45.3 - (0.9)^2} \\ &= \sqrt{45.3 - 0.81} \\ &= \sqrt{44.49} \\ \sigma &\approx 6.67\end{aligned}$$

The standard deviation of the marks scored by the students is approximately 6.67.

3. Find the mean and variance of the first n natural numbers.

solution

$$\begin{aligned}\text{Mean } (\bar{x}) &= \frac{1}{n} \sum_{i=1}^n i \\ &= \frac{1}{n} \left[\frac{n(n+1)}{2} \right] \\ &= \frac{n+1}{2}\end{aligned}$$

$$\begin{aligned}\text{Variance } (\sigma^2) &= \frac{1}{n} \sum_{i=1}^n i^2 - (\bar{x})^2 \\ &= \frac{1}{n} \left[\frac{n(n+1)(2n+1)}{6} \right] - \left(\frac{n+1}{2} \right)^2 \\ &= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} \\ &= \frac{n+1}{2} \left(\frac{2n+1}{3} - \frac{n+1}{2} \right) \\ &= \frac{n+1}{2} \left(\frac{4n+2-3n-3}{6} \right) \\ &= \frac{n+1}{2} \left(\frac{n-1}{6} \right) \\ \sigma^2 &= \frac{n^2-1}{12}\end{aligned}$$

4. The marks scored by the students in a slip test are given below. Find the standard deviation of their marks. **PTA-1, JUL-25**

x	4	6	8	10	12
f	7	3	5	9	5

solution

Using Assumed Mean (A) is 8. where $d = x - A$.

x	f	$d = x - 8$	d^2	fd	fd^2
4	7	-4	16	-28	112
6	3	-2	4	-6	12
8	5	0	0	0	0
10	9	2	4	18	36
12	5	4	16	20	80
Total	$N = 29$			$\sum fd = 4$	$\sum fd^2 = 240$

Standard Deviation formula using the Assumed Mean method is:

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \\ &= \sqrt{\frac{240}{29} - \left(\frac{4}{29}\right)^2} \\ &= \sqrt{8.2758 - (0.1379)^2} \\ &= \sqrt{8.2758 - 0.0190} \\ &= \sqrt{8.2568} \\ &\approx 2.87\end{aligned}$$

The standard deviation of the marks scored by the students is approximately 2.87.

5. Marks of the students in a particular subject of a class are given below. Find its standard deviation.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No.of students	8	12	17	14	9	7	4

solution

The Assumed Mean (A) is 35 and the class width (c) be 10.

Use **Step-Deviation Method**: $d = \frac{x-35}{10}$.

Marks	Midpoint (x)	f	$d = \frac{x-35}{10}$	d^2	fd	fd^2
0-10	5	8	-3	9	-24	72
10-20	15	12	-2	4	-24	48
20-30	25	17	-1	1	-17	17
30-40	35	14	0	0	0	0
40-50	45	9	1	1	9	9
50-60	55	7	2	4	14	28
60-70	65	4	3	9	12	36
Total		$N = 71$			$\sum fd = -30$	$\sum fd^2 = 210$

S.D Step-Deviation Method Formula:

$$\begin{aligned}\sigma &= c \times \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \\ &= 10 \times \sqrt{\frac{210}{71} - \left(\frac{-30}{71}\right)^2} \\ &= 10 \times \sqrt{2.9577 - (-0.4225)^2} \\ &= 10 \times \sqrt{2.9577 - 0.1785} \\ &= 10 \times \sqrt{2.7792} \\ &= 10 \times 1.667 \\ \sigma &\approx 16.67\end{aligned}$$

The standard deviation of the marks is approximately 16.67.

6. A teacher asked the students to complete 60 pages of a record note book. Eight students have completed only 32, 35, 37, 30, 33, 36, 35 and 37 pages. Find the standard deviation of the pages completed by them. **JUL-24**

solution

The total number of pages assigned (60) . To calculate the standard deviation for the pages actually completed by the 8 students.

Given data (x): 32, 35, 37, 30, 33, 36, 35, 37

Number of students (n) = 8 Assumed Mean A = 35.

The formula for standard deviation using the Assumed Mean method is:

$$\sigma = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2}$$

where $d = x - A$.

Pages completed (x)	$d = x - 35$	d^2
32	-3	9
35	0	0
37	2	4
30	-5	25
33	-2	4
36	1	1
35	0	0
37	2	4
Total	$\sum d = -5$	$\sum d^2 = 47$

Standard Deviation formula:

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2} \\ &= \sqrt{\frac{47}{8} - \left(\frac{-5}{8}\right)^2}\end{aligned}$$

$$\begin{aligned}&= \sqrt{5.875 - (-0.625)^2} \\ &= \sqrt{5.875 - 0.390625} \\ &= \sqrt{5.484375} \\ \sigma &\approx 2.34\end{aligned}$$

The Standard Deviation of the Student Completed Pages is approximately 2.34

7. The rainfall recorded in various places of five districts in a week are given below. Find its standard deviation.

Rainfall (in mm)	45	50	55	60	65	70
Number of places	5	13	4	9	5	4

solution

The Assumed Mean (A) is 55 and the common width (c) is 5.

Using step-deviation $d = \frac{x-A}{c} = \frac{x-55}{5}$.

(x)	(f)	$d = \frac{x-55}{5}$	d^2	fd	fd^2
45	5	-2	4	-10	20
50	13	-1	1	-13	13
55	4	0	0	0	0
60	9	1	1	9	9
65	5	2	4	10	20
70	4	3	9	12	36
Total	$N = 40$			$\sum fd = 8$	$\sum fd^2 = 98$

S.D using the step-deviation method is:

$$\begin{aligned}\sigma &= c \times \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \\ &= 5 \times \sqrt{\frac{98}{40} - \left(\frac{8}{40}\right)^2} \\ &= 5 \times \sqrt{2.45 - (0.2)^2} \\ &= 5 \times \sqrt{2.45 - 0.04} \\ &= 5 \times \sqrt{2.41} \\ &= 5 \times 1.5524 \\ \sigma &\approx 7.76\end{aligned}$$

The Standard Deviation of the rainfall recorded is approximately 7.76 mm

8. The time taken by 50 students to complete a 100 meter race are given below. Find its standard deviation. **PTA-5**

Time taken (sec)	8.5-9.5	9.5-10.5	10.5-11.5	11.5-12.5	12.5-13.5
No.of students	6	8	17	10	9

solution

The Assumed Mean (A) is 11. $d = x - A = x - 11$.

Time (sec)	Mid point (x)	f	$d = x - 11$	d^2	fd	fd^2
8.5-9.5	9	6	-2	4	-12	24
9.5-10.5	10	8	-1	1	-8	8
10.5-11.5	11	17	0	0	0	0
11.5-12.5	12	10	1	1	10	10
12.5-13.5	13	9	2	4	18	36
Total		$N = 50$			$\sum fd = 8$	$\sum fd^2 = 78$

S.D using the Assumed Mean method is:

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \\ &= \sqrt{\frac{78}{50} - \left(\frac{8}{50}\right)^2} \\ &= \sqrt{1.56 - (0.16)^2} \\ &= \sqrt{1.56 - 0.0256} \\ &= \sqrt{1.5344} \\ &\approx 1.24\end{aligned}$$

9. The following table gives the values of mean and variance of heights and weights of the 10th standard students of a school. **PTA-6**

	Height	Weight
Mean	155 cm	46.50 kg
Variance	72.25 cm ²	28.09 kg

Which is more varying than the other?

solution

To Find the Coefficient of Variation (C.V.):

$$C.V. = \frac{\sigma}{\bar{x}} \times 100$$

1. For Height:

$$\text{Mean } (\bar{x}_1) = 155$$

$$\text{Variance } (\sigma_1^2) = 72.25$$

$$\text{S.D. } (\sigma_1) = \sqrt{72.25} = 8.5$$

$$C.V._1 = \frac{8.5}{155} \times 100 = \frac{850}{155}$$

$$C.V._1 \approx 5.48\%$$

2. For Weight:

$$\text{Mean } (\bar{x}_2) = 46.50$$

$$\text{Variance } (\sigma_2^2) = 28.09$$

$$\text{S.D. } (\sigma_2) = \sqrt{28.09} = 5.3$$

$$C.V._2 = \frac{5.3}{46.50} \times 100 = \frac{530}{46.5}$$

$$C.V._2 \approx 11.40\%$$

Since $C.V._2 > C.V._1$ ($11.40\% > 5.48\%$), The weight is more varying than the height.

10. Find the coefficient of variation of 24, 26, 33, 37, 29, 31.

APR-24, JUL-23, MDL-19, MAR-26

solution

Given data: 24, 26, 29, 31, 33, 37

Number of observations (n) = 6

$$\begin{aligned}\bar{x} &= \frac{\sum x}{n} = \frac{24 + 26 + 29 + 31 + 33 + 37}{6} = \frac{180}{6} \\ \bar{x} &= 30\end{aligned}$$

Using Direct deviation method where $d = x - \bar{x} = x - 30$.

x	$d = x - 30$	d^2
24	-6	36
26	-4	16
29	-1	1
31	1	1
33	3	9
37	7	49
Total	$\sum d = 0$	$\sum d^2 = 112$

Standard Deviation:

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum d^2}{n}} \\ &= \sqrt{\frac{112}{6}} \\ &= \sqrt{18.6667} \\ \sigma &\approx 4.32\end{aligned}$$

The Coefficient of Variation (C.V.):

$$\begin{aligned}C.V. &= \frac{\sigma}{\bar{x}} \times 100\% \\ &= \frac{4.32}{30} \times 100\% \\ &= 0.144 \times 100\% \\ &= 14.4\%\end{aligned}$$

The coefficient of variation of the given data is 14.4%.

11. The time taken (in minutes) to complete a homework by 8 students in a day are given by 38, 40, 47, 44, 46, 43, 49, 53. Find the coefficient of variation.

solution

Given data: 38, 40, 43, 44, 46, 47, 49, 53

Number of observations (n) = 8

$$\begin{aligned}\bar{x} &= \frac{\sum x}{n} = \frac{38 + 40 + 43 + 44 + 46 + 47 + 49 + 53}{8} = \frac{360}{8} \\ &= 45\end{aligned}$$

The direct deviation method where $d = x - \bar{x} = x - 45$.

Time (x)	$d = x - 45$	d^2
38	-7	49
40	-5	25
43	-2	4
44	-1	1
46	1	1
47	2	4
49	4	16
53	8	64
Total	$\sum d = 0$	$\sum d^2 = 164$

The Coefficient of Variation (C.V.):

$$\begin{aligned}C.V. &= \frac{\sigma}{\bar{x}} \times 100\% \\ &= \frac{4.53}{45} \times 100\% \\ &= 0.10066... \times 100\% \\ &\approx 10.07\%\end{aligned}$$

The coefficient of variation of the time taken is approximately 10.07%.

12. The total marks scored by two students Sathya and Vidhya in 5 subjects are 460 and 480 with standard deviation 4.6 and 2.4 respectively. Who is more consistent in performance? **APR-25**

solution

For Sathya (C.V):

$$\begin{aligned}n &= 5 \\ \bar{x} &= \frac{\sum x}{n} \\ \text{Mean } (\bar{x}_S) &= \frac{460}{5} = 92 \\ \text{S.D. } (\sigma_S) &= 4.6 \\ C.V._S &= \frac{\sigma_S}{\bar{x}_S} \times 100 \\ &= \frac{4.6}{92} \times 100 = 5\%\end{aligned}$$

For Vidhya (C.V):

$$\begin{aligned}n &= 5 \\ \bar{x} &= \frac{\sum x}{n} \\ \text{Mean } (\bar{x}_V) &= \frac{480}{5} = 96 \\ \text{S.D. } (\sigma_V) &= 2.4 \\ C.V._V &= \frac{\sigma_V}{\bar{x}_V} \times 100 \\ &= \frac{2.4}{96} \times 100 = 2.5\%\end{aligned}$$

Since $C.V._V < C.V._S$ ($2.5\% < 5\%$), **Vidhya** is more consistent in her performance.

13. Two dice are rolled. Find the probability that the sum of outcomes is

JUL-21, 24, 25

- (i) equal to 4 (ii) greater than 10 (iii) less than 13

solution

Two Dice are rolled, The sample space S is:

$$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), \\ (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), \\ (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

$n(S) = 36$. **(i) Sum is equal to 4:** Let A be the event of getting a sum of 4.

$$A = \{(1, 3), (2, 2), (3, 1)\} \\ n(A) = 3 \\ P(A) = \frac{n(A)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

(ii) Sum is greater than 10: Let B be the event of getting a sum > 10 (i.e., 11 or 12).

$$B = \{(5, 6), (6, 5), (6, 6)\} \\ n(B) = 3 \\ P(B) = \frac{n(B)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

(iii) Sum is less than 13: Let C be the event of getting a sum < 13 . Since the maximum possible sum is 12, all outcomes are favorable.

$$n(C) = 36 \\ P(C) = \frac{n(C)}{n(S)} = \frac{36}{36} = 1$$

14. A bag contains 6 green balls, some black and red balls. Number of black balls is as twice as the number of red balls. Probability of getting a green ball is thrice the probability of getting a red ball. Find (i) number of black balls (ii) total number of balls.

solution

Let the number of red balls be x , $n(R) = x$. Given number of green balls, $n(G) = 6$. Since the number of black balls is twice the number of red balls:

$$n(B) = 2x$$

The total number of balls in the bag, $n(S)$, is:

$$n(S) = n(G) + n(R) + n(B) \\ = 6 + x + 2x \\ = 6 + 3x$$

Given $P(\text{Green}) = 3 \times P(\text{Red})$.

$$\frac{n(G)}{n(S)} = 3 \times \frac{n(R)}{n(S)} \\ \frac{6}{6 + 3x} = 3 \times \left(\frac{x}{6 + 3x} \right) \\ 6 = 3x \\ x = 2$$

The No. of red balls is 2.

(i) Number of black balls:

$$n(B) = 2x = 2(2) = 4$$

(ii) Total number of balls:

$$n(S) = 6 + 3x = 6 + 3(2) = 12$$

15. If A is an event of a random experiment such that $P(A) : P(\bar{A}) = 17 : 15$ and $n(S) = 640$ then find (i) $P(\bar{A})$ (ii) $n(A)$. **PTA-3**

solution

Given $P(A) : P(\bar{A}) = 17 : 15$ and $n(S) = 640$.

$$P(A) + P(\bar{A}) = 1$$

Let $P(A) = 17k$ and $P(\bar{A}) = 15k$.

$$17k + 15k = 1 \\ 32k = 1 \\ k = \frac{1}{32}$$

(i) Find $P(\bar{A})$:

$$P(\bar{A}) = 15k = 15 \left(\frac{1}{32} \right) = \frac{15}{32}$$

(ii) Find $n(A)$:

$$P(A) = 17k = \frac{17}{32}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$\frac{17}{32} = \frac{n(A)}{640}$$

$$n(A) = \frac{17 \times 640}{32}$$

$$n(A) = 17 \times 20$$

$$n(A) = 340$$

16. Two unbiased dice are rolled once. Find the probability of getting

APR-23, JUL-20, 22

- (i) a doublet (equal numbers on both dice)
(ii) the product as a prime number
(iii) the sum as a prime number
(iv) the sum as 1

solution

Two unbiased dice are rolled, the Sample Space S .

$$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), \\ (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), \\ (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

The total No. of possible outcomes is $n(S) = 36$.

(i) A doublet: Let A be the event of getting equal numbers on both dice.

$$A = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\} \\ n(A) = 6 \\ P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

(ii) Product as a prime number: Let B be the event where the product is a prime number (2, 3, or 5).

$$B = \{(1, 2), (2, 1), (1, 3), (3, 1), (1, 5), (5, 1)\} \\ n(B) = 6 \\ P(B) = \frac{n(B)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

(iii) Sum as a prime number: Let C be the event where the sum is a prime number (2, 3, 5, 7, or 11).

$$C = \{(1, 1), (1, 2), (2, 1), (1, 4), (2, 3), \\ (3, 2), (4, 1), (1, 6), (2, 5), (3, 4), \\ (4, 3), (5, 2), (6, 1), (5, 6), (6, 5)\}$$

$$n(C) = 15$$

$$P(C) = \frac{n(C)}{n(S)} = \frac{15}{36} = \frac{5}{12}$$

(iv) Sum as 1: Let D be the event of getting a sum of 1. Since the minimum possible sum is $1 + 1 = 2$, this is an impossible event.

$$n(D) = 0$$

$$P(D) = \frac{n(D)}{n(S)} = \frac{0}{36} = 0$$

17. Three fair coins are tossed together. Find the probability of getting

PTA-5

- (i) all heads (ii) atleast one tail (iii) atmost one head (iv) atmost two tails

solution Three fair coins are tossed, the sample space S is:

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Total number of possible outcomes, $n(S) = 8$.

(i) All heads: Let A be the event of getting all heads.

$$\begin{aligned} A &= \{HHH\} \\ n(A) &= 1 \\ P(A) &= \frac{n(A)}{n(S)} = \frac{1}{8} \end{aligned}$$

(ii) At least one tail: Let B be the event of getting at least one tail (1, 2, or 3 tails).

$$\begin{aligned} B &= \{HHT, HTH, THH, HTT, THT, TTH, TTT\} \\ n(B) &= 7 \\ P(B) &= \frac{n(B)}{n(S)} = \frac{7}{8} \end{aligned}$$

(iii) At most one head: Let C be the event of getting at most one head (0 or 1 head).

$$\begin{aligned} C &= \{HTT, THT, TTH, TTT\} \\ n(C) &= 4 \\ P(C) &= \frac{n(C)}{n(S)} = \frac{4}{8} = \frac{1}{2} \end{aligned}$$

(iv) At most two tails: Let D be the event of getting at most two tails (0, 1, or 2 tails).

$$D = \{HHH, HHT, HTH, THH, HTT, THT, TTH\}$$

$$\begin{aligned} n(D) &= 7 \\ P(D) &= \frac{n(D)}{n(S)} = \frac{7}{8} \end{aligned}$$

18. A bag contains 5 red balls, 6 white balls, 7 green balls, 8 black balls. One ball is drawn at random from the bag. Find the probability that the ball drawn is **APR-25, JUL-22**

- (i) white (ii) black or red (iii) not white (iv) neither white nor black

solution

The No.of Red, White, Green, and Black balls is denoted by $n(R)$, $n(W)$, $n(G)$, and $n(B)$.

$$n(R) = 5, \quad n(W) = 6 \quad n(G) = 7 \quad n(B) = 8$$

Total number of balls, $n(S) = 5 + 6 + 7 + 8 = 26$.

(i) White ball:

$$\begin{aligned} P(W) &= \frac{n(W)}{n(S)} \\ &= \frac{6}{26} = \frac{3}{13} \end{aligned}$$

(ii) Black or red ball: Let A be the event of getting a black or red ball.

$$\begin{aligned} n(A) &= n(B) + n(R) \\ &= 8 + 5 = 13 \\ P(A) &= \frac{n(A)}{n(S)} = \frac{13}{26} = \frac{1}{2} \end{aligned}$$

(iii) Not a white ball: Let $\bar{W}$ be the event of not getting a white ball.

$$\begin{aligned} P(\bar{W}) &= 1 - P(W) \\ &= 1 - \frac{3}{13} \\ &= \frac{10}{13} \end{aligned}$$

(Alternatively, $n(\text{not } W) = 26 - 6 = 20$, so $P = \frac{20}{26} = \frac{10}{13}$.)

(iv) Neither white nor black ball: Let C be the event of getting neither white nor black (which means getting either red or green).

$$\begin{aligned} n(C) &= n(R) + n(G) \\ &= 5 + 7 = 12 \\ P(C) &= \frac{n(C)}{n(S)} = \frac{12}{26} = \frac{6}{13} \end{aligned}$$

19. Two dice are rolled together. Find the probability of getting a doublet or sum of faces as 4. **MAR-26**

solution

Two Dice are rolled together, the Sample Space S

$$\begin{aligned} S &= \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ &\quad (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ &\quad (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), \\ &\quad (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ &\quad (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), \\ &\quad (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\} \end{aligned}$$

The Total No.of outcomes is $n(S) = 36$.

Let A be the event of getting a doublet.

$$\begin{aligned} A &= \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\} \\ n(A) &= 6 \\ P(A) &= \frac{n(A)}{n(S)} = \frac{6}{36} \end{aligned}$$

Let B be the event of getting a sum of 4.

$$\begin{aligned} B &= \{(1, 3), (2, 2), (3, 1)\} \\ n(B) &= 3 \\ P(B) &= \frac{n(B)}{n(S)} = \frac{3}{36} \end{aligned}$$

Let $A \cap B$ be the event of getting a doublet and a sum of 4.

$$\begin{aligned} A \cap B &= \{(2, 2)\} \\ n(A \cap B) &= 1 \\ P(A \cap B) &= \frac{n(A \cap B)}{n(S)} = \frac{1}{36} \end{aligned}$$

Addition Theorem of Probability:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{6}{36} + \frac{3}{36} - \frac{1}{36} \\ &= \frac{8}{36} \\ &= \frac{2}{9} \end{aligned}$$

20. In a class of 50 students, 28 opted for NCC, 30 opted for NSS and 18 opted both NCC and NSS. One of the students is selected at random. Find the probability that **APR-22, PTA-1, 4**

- (i) The student opted for NCC but not NSS.
(ii) The student opted for NSS but not NCC.
(iii) The student opted for exactly one of them.

solution

Let S be the sample space (total students).

$$n(S) = 50$$

Let A be the event that a student opted for NCC, and B be the event that a student opted for NSS.

$$n(A) = 28, \quad n(B) = 30, \quad n(A \cap B) = 18$$

(i) Opted for NCC but not NSS: This represents the event $A \cap B'$.

$$\begin{aligned} n(A \cap B') &= n(A) - n(A \cap B) \\ &= 28 - 18 = 10 \\ P(A \cap B') &= \frac{n(A \cap B')}{n(S)} \\ &= \frac{10}{50} = \frac{1}{5} \end{aligned}$$

(ii) **Opted for NSS but not NCC:** This represents the event $B \cap A'$.

$$\begin{aligned} n(B \cap A') &= n(B) - n(A \cap B) \\ &= 30 - 18 = 12 \\ P(B \cap A') &= \frac{n(B \cap A')}{n(S)} \\ &= \frac{12}{50} = \frac{6}{25} \end{aligned}$$

(iii) **Opted for exactly one of them:** This is the combination of students who opted for *only* NCC and *only* NSS.

$$\begin{aligned} n(\text{Exactly one}) &= n(A \cap B') + n(B \cap A') \\ &= 10 + 12 = 22 \\ P(\text{Exactly one}) &= \frac{22}{n(S)} \\ &= \frac{22}{50} = \frac{11}{25} \end{aligned}$$

21. In an apartment, in selecting a house from door numbers 1 to 100 randomly, find the probability of getting the door number of the house to be an even number or a perfect square number or a perfect cube number.

solution

Let the sample space $S = \{1, 2, 3, \dots, 100\}$.

$$n(S) = 100$$

Let A , B , and C be the events of getting an even number, a perfect square, and a perfect cube.

$$\begin{aligned} n(A) &= 50 \\ B &= \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100\} \\ n(B) &= 10 \\ C &= \{1, 8, 27, 64\} \\ n(C) &= 4 \end{aligned}$$

To find the intersections:

$$\begin{aligned} A \cap B &= \{4, 16, 36, 64, 100\} \\ n(A \cap B) &= 5 \\ B \cap C &= \{1, 64\} \\ n(B \cap C) &= 2 \\ A \cap C &= \{8, 64\} \\ n(A \cap C) &= 2 \\ A \cap B \cap C &= \{64\} \\ n(A \cap B \cap C) &= 1 \end{aligned}$$

Addition Theorem for three events To find $n(A \cup B \cup C)$:

$$\begin{aligned} n(A \cup B \cup C) &= n(A) + n(B) + n(C) \\ &\quad - n(A \cap B) - n(B \cap C) \\ &\quad - n(A \cap C) + n(A \cap B \cap C) \\ &= 50 + 10 + 4 - 5 - 2 - 2 + 1 \\ &= 64 - 9 + 1 \\ &= 56 \end{aligned}$$

To find Probability

$$\begin{aligned} P(A \cup B \cup C) &= \frac{n(A \cup B \cup C)}{n(S)} \\ &= \frac{56}{100} \\ &= \frac{14}{25} \end{aligned}$$

22. Two dice are rolled once. Find the probability of getting an even number on the first die or a total of face sum 8. **APR-24, JUL-23**

solution

Two Dice are rolled, the total number of outcomes is $n(S) = 36$

$$\begin{aligned} S &= \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ &\quad (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ &\quad (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), \\ &\quad (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ &\quad (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), \\ &\quad (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\} \end{aligned}$$

Let A be the event of getting an even number on the first die.

$$\begin{aligned} A &= \{(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ &\quad (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ &\quad (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\} \\ n(A) &= 3 \times 6 = 18 \\ P(A) &= \frac{n(A)}{n(S)} = \frac{18}{36} \end{aligned}$$

Let B be the event of getting a sum of 8.

$$\begin{aligned} B &= \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\} \\ n(B) &= 5 \\ P(B) &= \frac{n(B)}{n(S)} = \frac{5}{36} \end{aligned}$$

Let $A \cap B$ be the event of getting an even number on the first die *and* a sum of 8.

$$\begin{aligned} A \cap B &= \{(2, 6), (4, 4), (6, 2)\} \\ n(A \cap B) &= 3 \\ P(A \cap B) &= \frac{n(A \cap B)}{n(S)} = \frac{3}{36} \end{aligned}$$

By Addition Theorem of Probability:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{18}{36} + \frac{5}{36} - \frac{3}{36} \\ &= \frac{20}{36} \\ &= \frac{5}{9} \end{aligned}$$

23. A coin is tossed thrice. Find the probability of getting exactly two heads or atleast one tail or two consecutive heads. **PTA-2, 6**

solution

When a coin is tossed thrice, the sample space S is:

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Total No.of possible outcomes, $n(S) = 8$.

Let A , B , and C be the events of getting exactly two heads, at least one tail, and two consecutive heads, respectively.

$$\begin{aligned} A &= \{HHT, HTH, THH\} \\ n(A) &= 3 \Rightarrow P(A) = \frac{3}{8} \\ B &= \{HHT, HTH, THH, HTT, THT, TTH, TTT\} \end{aligned}$$

$$\begin{aligned} n(B) &= 7 \Rightarrow P(B) = \frac{7}{8} \\ C &= \{HHH, HHT, THH\} \\ n(C) &= 3 \Rightarrow P(C) = \frac{3}{8} \end{aligned}$$

To find intersections:

$$\begin{aligned} A \cap B &= \{HHT, HTH, THH\} \\ n(A \cap B) &= 3 \Rightarrow P(A \cap B) = \frac{3}{8} \\ B \cap C &= \{HHT, THH\} \\ n(B \cap C) &= 2 \Rightarrow P(B \cap C) = \frac{2}{8} \\ A \cap C &= \{HHT, THH\} \\ n(A \cap C) &= 2 \Rightarrow P(A \cap C) = \frac{2}{8} \\ A \cap B \cap C &= \{HHT, THH\} \\ n(A \cap B \cap C) &= 2 \Rightarrow P(A \cap B \cap C) = \frac{2}{8} \end{aligned}$$

Addition Theorem for three events:

$$\begin{aligned}
 P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\
 &\quad - P(A \cap B) - P(B \cap C) \\
 &\quad - P(A \cap C) + P(A \cap B \cap C) \\
 &= \frac{3}{8} + \frac{7}{8} + \frac{3}{8} - \frac{3}{8} - \frac{2}{8} - \frac{2}{8} + \frac{2}{8} \\
 &= \frac{3+7+3-3-2-2+2}{8} \\
 &= \frac{8}{8} = 1
 \end{aligned}$$

24. Three unbiased coins are tossed once. Find the probability of getting atmost 2 tails or atleast 2 heads.

solution

Three Unbiased coins are tossed, the sample space S is:

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Total number of possible outcomes, $n(S) = 8$.

Let A be the event of getting at most 2 tails This includes all outcomes except TTT .

$$\begin{aligned}
 A &= \{HHH, HHT, HTH, THH, HTT, THT, TTH\} \\
 n(A) &= 7 \Rightarrow P(A) = \frac{7}{8}
 \end{aligned}$$

Let B be the event of getting at least 2 heads

$$\begin{aligned}
 B &= \{HHH, HHT, HTH, THH\} \\
 n(B) &= 4 \Rightarrow P(B) = \frac{4}{8}
 \end{aligned}$$

Let $A \cap B$ be the event of getting both at most 2 tails and at least 2 heads.

$$\begin{aligned}
 A \cap B &= \{HHH, HHT, HTH, THH\} \\
 n(A \cap B) &= 4 \Rightarrow P(A \cap B) = \frac{4}{8}
 \end{aligned}$$

By Addition Theorem of Probability:

$$\begin{aligned}
 P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\
 &= \frac{7}{8} + \frac{4}{8} - \frac{4}{8} \\
 &= \frac{7}{8}
 \end{aligned}$$

25. A box contains cards numbered 3, 5, 7, 9, ...35, 37. A card is drawn at random from the box. Find the probability that the drawn card have either multiples of 7 or a prime number.

solution

Given $S = \{3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37\}$
The sample space S consists of odd numbers from 3 to 37. In Arithmetic Progression with $a = 3$, $d = 2$, and $l = 37$.

$$\begin{aligned}
 n(S) &= \frac{l-a}{d} + 1 \\
 &= \frac{37-3}{2} + 1 \\
 &= 17 + 1 = 18
 \end{aligned}$$

Let A be the event of drawing a multiple of 7.

$$\begin{aligned}
 A &= \{7, 21, 35\} \\
 n(A) &= 3 \Rightarrow P(A) = \frac{3}{18}
 \end{aligned}$$

Let B be the event of drawing a prime number.

$$\begin{aligned}
 B &= \{3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\} \\
 n(B) &= 11 \Rightarrow P(B) = \frac{11}{18}
 \end{aligned}$$

Let $A \cap B$ be the event of drawing a card that is both a multiple of 7 and a prime number.

$$\begin{aligned}
 A \cap B &= \{7\} \\
 n(A \cap B) &= 1 \Rightarrow P(A \cap B) = \frac{1}{18}
 \end{aligned}$$

By Addition Theorem of Probability:

$$\begin{aligned}
 P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\
 &= \frac{3}{18} + \frac{11}{18} - \frac{1}{18} \\
 &= \frac{13}{18}
 \end{aligned}$$

26. In a class of 35, students are numbered from 1 to 35. The ratio of boys to girls is 4:3. The roll numbers of students begin with boys and end with girls. Find the probability that a student selected is either a boy with prime roll number or a girl with composite roll number or an even roll number.

solution

Total number of students, $n(S) = 35$. The ratio of boys to girls is 4 : 3.
Total parts = 7.

$$\begin{aligned}
 \text{Number of boys} &= \frac{4}{7} \times 35 = 20 \\
 \text{Number of girls} &= \frac{3}{7} \times 35 = 15
 \end{aligned}$$

Roll numbers for boys: 1 to 20. Roll numbers for girls: 21 to 35. Let A be the event of selecting a boy with a prime roll number.

$$\begin{aligned}
 A &= \{2, 3, 5, 7, 11, 13, 17, 19\} \\
 n(A) &= 8 \Rightarrow P(A) = \frac{8}{35}
 \end{aligned}$$

Let B be the event of selecting a girl with a composite roll number. (Primes from 21 to 35 are 23, 29, 31).

$$\begin{aligned}
 B &= \{21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35\} \\
 n(B) &= 12 \Rightarrow P(B) = \frac{12}{35}
 \end{aligned}$$

Let C be the event of selecting an even roll number (from 1 to 35).

$$n(C) = 17 \Rightarrow P(C) = \frac{17}{35}$$

To find the intersections:

$$\begin{aligned}
 A \cap B &= \emptyset \Rightarrow P(A \cap B) = 0 \\
 B \cap C &= \{22, 24, 26, 28, 30, 32, 34\} \\
 n(B \cap C) &= 7 \Rightarrow P(B \cap C) = \frac{7}{35} \\
 A \cap C &= \{2\} \\
 n(A \cap C) &= 1 \Rightarrow P(A \cap C) = \frac{1}{35} \\
 A \cap B \cap C &= \emptyset \Rightarrow P(A \cap B \cap C) = 0
 \end{aligned}$$

By Addition Theorem for three events:

$$\begin{aligned}
 P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\
 &\quad - P(A \cap B) - P(B \cap C) \\
 &\quad - P(A \cap C) + P(A \cap B \cap C) \\
 &= \frac{8}{35} + \frac{12}{35} + \frac{17}{35} - 0 - \frac{7}{35} - \frac{1}{35} + 0 \\
 &= \frac{37-8}{35} \\
 &= \frac{29}{35}
 \end{aligned}$$

27. Prices of peanut packets in various places of two cities are given below. In which city, prices were more stable?

Prices in city A	20	22	19	23	16
Prices in city B	10	20	18	12	15

solution

To find which city has more stable prices, we need to calculate the

Coefficient of Variation (C.V.) for both cities. **1. Calculation for City A:**

Given prices (x_A): 16, 19, 20, 22, 23 (arranged in ascending order)

Number of observations (n) = 5 To find the arithmetic mean ($\bar{x}_A$):

$$\bar{x}_A = \frac{\sum x_A}{n} = \frac{16 + 19 + 20 + 22 + 23}{5} = \frac{100}{5} = 20$$

using the direct method $d = x - 20$:

Price (x_A)	$d = x - 20$	d^2
16	-4	16
19	-1	1
20	0	0
22	2	4
23	3	9
Total	$\sum d = 0$	$\sum d^2 = 30$

standard deviation (σ_A):

$$\begin{aligned}\sigma_A &= \sqrt{\frac{\sum d^2}{n}} \\ &= \sqrt{\frac{30}{5}} \\ &= \sqrt{6} \\ &\approx 2.45\end{aligned}$$

Coefficient of Variation for City A:

$$\begin{aligned}\text{C.V. (A)} &= \frac{\sigma_A}{\bar{x}_A} \times 100\% \\ &= \frac{2.45}{20} \times 100\% \\ &= 2.45 \times 5\% \\ &= 12.25\%\end{aligned}$$

2. Calculation for City B:

Given prices (x_B): 10, 12, 15, 18, 20 (arranged in ascending order)

Number of observations (n) = 5 To find the arithmetic mean ($\bar{x}_B$):

$$\begin{aligned}\bar{x}_B &= \frac{\sum x_B}{n} \\ &= \frac{10 + 12 + 15 + 18 + 20}{5} \\ &= \frac{75}{5} \\ &= 15\end{aligned}$$

Using Direct Method find deviations $d = x - 15$:

Price (x_B)	$d = x - 15$	d^2
10	-5	25
12	-3	9
15	0	0
18	3	9
20	5	25
Total	$\sum d = 0$	$\sum d^2 = 68$

Standard Deviation (σ_B):

$$\begin{aligned}\sigma_B &= \sqrt{\frac{\sum d^2}{n}} \\ &= \sqrt{\frac{68}{5}} \\ &= \sqrt{13.6} \\ &\approx 3.69\end{aligned}$$

Coefficient of Variation for City B:

$$\begin{aligned}\text{C.V. (B)} &= \frac{\sigma_B}{\bar{x}_B} \times 100\% \\ &= \frac{3.69}{15} \times 100\% \\ &= 0.246 \times 100\% \\ &= 24.6\%\end{aligned}$$

C.V. of City A (12.25%) < C.V. of City B (24.6%). **Since the coefficient of variation for City A is lesser, the prices in City A are more stable.**

CHAPTER - 3, 4. GEOMETRY & GRAPH

8 MARK IMPORTANT QUESTIONS (GEOMETRY & GRAPH)

- Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{3}{5}$ of the corresponding sides of the triangle PQR (Scale Factor $\frac{3}{5} < 1$). **JUL-23**
- Construct a triangle similar to a given triangle LMN with its sides equal to $\frac{4}{5}$ of the corresponding sides of the triangle LMN (Scale Factor $\frac{4}{5} < 1$).
- Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{7}{4}$ of the corresponding sides of the triangle PQR (Scale Factor $\frac{7}{4} > 1$). **JUL-25**
- Construct a triangle similar to a given triangle ABC with its sides equal to $\frac{6}{5}$ of the corresponding sides of the triangle ABC (Scale Factor $\frac{6}{5} > 1$). **JUL-20**
- Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{7}{3}$ of the corresponding sides of the triangle PQR (Scale Factor $\frac{7}{3} > 1$). **JUL-22**
- Construct a $\triangle PQR$ in which $PQ = 8$ cm, $\angle R = 60^\circ$ and the Median RG from R to PQ is 5.8 cm. Find the length of the altitude from R to PQ. **PTA-3**
- Construct a $\triangle PQR$ in which base $PQ = 4.5$ cm, $\angle R = 35^\circ$ and the Median RG from R to PQ is 6 cm. **APR-24, JUL-21**
- Construct a $\triangle PQR$ in which $QR = 5$ cm, $\angle P = 40^\circ$ and the Median PG from P to QR is 4.4 cm. Find the length of the Altitude from P to QR. **JUL-24**
- Construct a $\triangle PQR$ in which such that $QR = 5$ cm, $\angle P = 30^\circ$ and the Altitude from P to QR is of length 4.2 cm. **PTA-5**
- Construct a $\triangle PQR$ in which such that $QR = 5.5$ cm, $\angle P = 25^\circ$ and the Altitude from P to QR is of length 4 cm. **APR-22**
- Draw a triangle $\triangle ABC$ of base $BC = 8$ cm, $\angle A = 60^\circ$ and the Bisector of $\angle A$ meets BC at D such that $BD = 6$ cm. **APR-23, 25, MDL-19**
- Draw a triangle $\triangle ABC$ of base $BC = 5.6$ cm, $\angle A = 40^\circ$ and the Bisector of $\angle A$ meets BC at D such that $BD = 4$ cm.
- Draw a circle of diameter 6 cm from a point P, which is 8 cm away from its centre. Draw the two tangents PA and PB to the circle and measure their lengths. **JUL-21, 22, PTA-6, APR-24**
- Draw the two tangents from a point which is 10 cm away from the centre of a circle of radius 5 cm. Also, measure the lengths of the tangents. **JUL-20, 23, 24**
- Take a point which is 11 cm away from the centre of a circle of radius 4 cm and draw the two tangents to the circle from that point. **APR-23, 25, PTA-2**
- Draw the two tangents from a point which is 5 cm away from the centre of a circle of diameter 6 cm. Also, measure the lengths of the tangents. **APR-22, JUL-25**
- Draw a circle of radius 4.5 cm. Take a point on the circle. Draw the tangent at that point using the Alternate Segment Theorem.
- Draw a circle of radius 3 cm. Take a point P on this circle and draw a tangent at P.

Diameter (x) cm	1	2	3	4	5
Circumference (y) cm	3.1	6.2	9.3	12.4	15.5

GRAPH QUESTIONS

- Varshika drew 6 circles with different sizes. Draw a graph for the relationship between the diameter and circumference (approximately related) of each circle as shown in the table and use it to find the circumference of a circle when its diameter is 6 cm. **APR-23**
- A bus is travelling at a uniform speed of 50 km/hr. Draw the distance time graph and hence find **APR-25**
 - The constant of variation.
 - How far will it travel in 90 minutes or $1\frac{1}{2}$ hrs?.
 - The time required to cover a distance of 300 km from the graph.
- A garment shop announces a flat 50% discount on every purchase of items for their customers. Draw the graph for the relation between the Marked Price and the Discount. Hence find
 - The marked price when a customer gets a discount of 3250 (from graph).
 - The discount when the marked price is 2500.
- Graph the following linear function $y = \frac{1}{2}x$. Identify the constant of variation and verify it with the graph. Also find **JUL-23**
 - y when $x = 9$ and
 - x when $y = 7.5$.
- A two wheeler parking zone near bus stand charges as below. Check

Time x (in hours)	4	8	12	24
Amount y (in)	60	120	180	360

if the amount charged are in direct variation or in inverse variation to the parking time. Graph the data. Also

- Find the amount to be paid when parking time is 6 hrs.
 - Find the parking duration when the amount paid is 150.
- Nishanth is the winner in a Marathon race 12 km distance. He ran at the uniform speed of 12 km/hr and reached the destination in 1 hour. He was followed by Aradhana, Jeyanth, Sathya and Swetha with their respective speed of 6 km/hr, 4 km/hr, 3 km/hr and 2 km/hr. And, they covered the distance in 2 hrs, 3 hrs, 4 hrs and 6 hours respectively. Draw the Speed- time graph and use it to find time taken to Kaushik with his speed 2.4 km/hr.
 - Draw the graph of $xy = 24$, $x, y > 0$. Using the graph find, **APR-24**
 - y when $x = 3$ and
 - x when $y = 6$.
 - The following table shows the data about the number of pipes and the time taken to fill the same tank. (i) Draw the graph for the above data

No. of. pipes x	2	3	6	9
Time Taken y (in mins)	45	30	15	10

and hence

- Find the time taken to fill the tank when five pipes are used.
- Discuss the nature of solutions of the following quadratic equations.
 - $x^2 + x - 12 = 0$
 - $x^2 - 8x + 16 = 0$
 - Graph the following quadratic equations and state the nature of solutions. **APR-25, JUL-22**
 - $x^2 - 9x + 20 = 0$
 - $x^2 - 4x + 4 = 0$
 - $x^2 - 6x + 9 = 0$
 - Draw the graph of $y = 2x^2$ and hence solve $2x^2 - x - 6 = 0$. **APR-22**
 - Draw the graph of $y = x^2 + x - 2$ and hence solve $x^2 + x - 2 = 0$. **PTA-4**
 - Draw the graph of $y = x^2 - 4x + 3$ and hence solve $x^2 - 6x + 9 = 0$. **JUL-24, PTA-1**
 - Draw the graph of $y = x^2 - 4$ and hence solve $x^2 - x - 12 = 0$. **PTA-5**
 - Draw the graph of $y = x^2 + 3x + 2$ and hence solve $x^2 + 2x + 1 = 0$. **JUL-22, APR-22**
 - Draw the graph of $y = x^2 + 3x - 4$ and hence solve $x^2 + 3x - 4 = 0$. **JUL-21**

- Draw the graph of $y = x^2 - 5x - 6$ and hence solve $x^2 - 5x - 14 = 0$. **JUL-24**

- Draw the graph of $y = 2x^2 - 3x - 5$ and hence solve $2x^2 - 4x - 6 = 0$. **APR-23, PTA-2, 6**